

CONVEX NON-BALL SCHIFFER DOMAINS IN INFINITELY MANY DIMENSIONS

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ABSTRACT. We construct bounded strictly convex non-ball Schiffer domains with real-analytic boundary in two arithmetic families of dimensions $2p$, with p integral or half-integral. The bounded-detuning families use a cofactor inverse and include infinitely many dimensions in every residue class, with a detuning bound depending on the modulus. A second family contains every sufficiently large half-lattice input within $\log p/(8p)$ of a primary crossing and has unbounded detuning in both parities. Its mixed conditioning estimate closes finite-order centres at accuracy twenty. In each family all but a sublinear number of interior integer block splits yield pairwise non-similar domains. The constructions use one graded Jacobi coefficient space and a quartic angular seed. The thresholds and splits are implicit. Each fixed-width candidate family and the logarithmic family have density zero. Green's identity gives failure of the Pompeiu property for the resulting domains.

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1. RESULTS AND SCOPE

In the smooth sphere-boundary formulation, the Schiffer conjecture states that a bounded domain $\Omega \subset \mathbb{R}^n$ with boundary homeomorphic to S^{n-1} must be a ball if it supports a non-constant solution of $\Delta u + \lambda u = 0$, for some $\lambda > 0$, with constant boundary value and zero normal derivative [15, 16]. We normalize the frequency to one and the boundary value to one.

We construct domains with these overdetermined data in unbounded families of both odd and even ambient dimensions, while retaining strict convexity.

Write J_ν for the Bessel function of the first kind of order ν . The real primary crossing orders $\widehat{p}_m$, for all sufficiently large indices $m \geq m_0$, satisfy

$$J_{\widehat{p}_m}(2\sqrt{(\widehat{p}_m + 1)(\widehat{p}_m + 2)}) = 0.$$

They are specified uniquely by the expansion in Proposition 4.2. The arithmetic construction uses the fixed irrational number

$$\alpha_{\text{ar}} = \frac{\pi}{2(\sqrt{3} - \pi/3)}$$

and continued-fraction approximations with denominator one modulo four. Define $\mathcal{P}_{\text{bd}}$ to consist of all positive $p \in \frac{1}{2}\mathbb{Z}$ with $|p - \widehat{p}_m| < 31/(10\widehat{p}_m)$ for some $m \geq m_0$. The detuning is (4.10) at the Neumann zero continued from the nearby crossing. A domain carries the frequency-one Schiffer data (1.1) if it admits a classical solution u of

$$(1.1) \quad \Delta u + u = 0 \quad \text{in } \Omega, \quad u = 1, \quad \nabla u = 0 \quad \text{on } \partial\Omega.$$

Theorem A (Bounded detuning). *For every sufficiently large $p \in \mathcal{P}_{\text{bd}}$, there is an integer $a \in [p/2, 3p/4]$ and a bounded strictly convex non-ball domain $\Omega \subset \mathbb{R}^{2p}$ with real-analytic boundary, invariant under $O(a) \times O(2p - a)$ and carrying (1.1). Its detuning satisfies $0 < |\theta| \leq 20$. There are infinitely many such inputs in each of $\mathbb{Z}$ and $\mathbb{Z} + 1/2$.*

For every fixed positive integer L_{ap} and integer b_{ap} , there is also an unbounded sequence of dimensions (n_k) , indexed by the continued-fraction convergent index k . Its sufficiently large terms $n = n_k$ satisfy $n \equiv b_{\text{ap}} \pmod{4L_{\text{ap}}}$, $|n/2 - \widehat{p}_m| < 32L_{\text{ap}}^2/\widehat{p}_m$ for some $m = m_k \geq m_0$, and $0 < |\theta| < 256L_{\text{ap}}^2$, and have the same domain conclusion. The starting index and thresholds may depend on $L_{\text{ap}}, b_{\text{ap}}$ and, for the split-counting conclusion below, on the fixed $\varepsilon_{\text{sp}}, \eta_{\text{cnt}}$. Hence every residue modulo L_{ap} contains infinitely many such dimensions. For even L_{ap} a residue modulo L_{ap} permits only its compatible parity; for odd L_{ap} each residue modulo L_{ap} contains infinitely many dimensions of both parities.

In each of these bounded families, fix $0 < \varepsilon_{\text{sp}} < 1/4$ and $0 < \eta_{\text{cnt}} < 1/3$. For every sufficiently large member $n = 2p$, all but $O(n^{2/3+\eta_{\text{cnt}}})$ integer splits $\varepsilon_{\text{sp}}n \leq a \leq (1/2 - \varepsilon_{\text{sp}})n$ give such domains, pairwise non-similar for distinct retained splits, where retained means not among the excluded splits. The constant and threshold may depend on the family and these fixed parameters. Let $\mathcal{N}_{\text{sim}}(n)$ be the number of integer splits $0 < a < n/2$ for which there exists a bounded strictly convex non-ball domain in $\mathbb{R}^n$ with real-analytic boundary, invariant under $O(a) \times O(n - a)$ and carrying (1.1). Choosing one domain at each such split gives this many similarity classes, independently of the choices. Then along each family's dimensions

$$\liminf_{n \rightarrow \infty} \mathcal{N}_{\text{sim}}(n)/n \geq 1/2.$$

Since $\mathcal{N}_{\text{sim}}(n) \leq n/2$, this ratio tends to $1/2$; the number of interior integer splits carrying no such domain is $o(n)$ along each family, with the rate depending on the family. The closure uses the cofactor estimate and a fixed centre order $N = 4M + 40$, with M fixed after the bounded family and compact split interval have been chosen and before N .

Theorem B (Logarithmic window). *Let*

$$\mathcal{P}_{\text{log}} = \{p \in \frac{1}{2}\mathbb{Z} : p > 1, \quad \exists m \geq m_0 : |p - \widehat{p}_m| \leq \log p/(8p)\}.$$

For every sufficiently large $p \in \mathcal{P}_{\text{log}}$, with $n = 2p$, there is an integer split $0 < a < n/2$ and a bounded strictly convex non-ball domain $\Omega \subset \mathbb{R}^{2p}$ with real-analytic boundary, invariant under $O(a) \times O(2p - a)$ and carrying (1.1). For every fixed $0 < \varepsilon_{\text{sp}} < 1/4$ and $0 < \eta_{\text{cnt}} < 1/3$, all

but $O_{\text{sp}, \eta_{\text{cnt}}}(n^{2/3+\eta_{\text{cnt}}})$ integer splits $\varepsilon_{\text{sp}}n \leq a \leq (1/2 - \varepsilon_{\text{sp}})n$ give pairwise non-similar domains at the retained splits. The threshold and constant depend on the fixed $\varepsilon_{\text{sp}}, \eta_{\text{cnt}}$. The same similarity-class lower limit is at least $1/2$. This family contains sequences of both parities with $|\theta| \rightarrow \infty$. The closure uses mixed conditioning and the fixed centre order $N = 20$.

Proofs of Theorems A and B. Corollary 15.6, with the inputs supplied by Lemma 5.2 and Theorem 4.5, proves the first bounded construction. Lemmas 17.4 and 17.6, Proposition 17.5, and Theorem 17.7 give all the further assertions of Theorem A through the cofactor route. The wider windows and interior split intervals are supplied by Lemmas 17.3, 17.2 and 17.1. Proposition 21.4, using Corollary 20.3 and Theorem 21.3, gives the logarithmic construction and multiplicity, and Lemma 21.2 gives unbounded detuning in both parities. These prove Theorem B. $\square$

Both assertions concern sufficiently large inputs, with implicit thresholds and splits. Each fixed-width family and the logarithmic family have density zero by Proposition 22.3. The quartic profile and its Hermite limit are described in Section 22.

The Pompeiu property means that a continuous function on $\mathbb{R}^n$ whose integral over every rigid copy of a domain is zero must itself be zero. The Pompeiu conjecture states that, among bounded domains with Lipschitz boundary homeomorphic to S^{n-1} , failure of this property forces the domain to be a ball [15, 16]. The domains in both theorems fail this property because their indicator transforms vanish on the unit sphere. A single nonzero plane wave, or its real part, then gives the required nonzero function. This implication is proved directly by Green's identity in Proposition 15.5. Consequently the Schiffer and Pompeiu conjectures fail within the class of convex domains in infinitely many dimensions.

Brown, Schreiber and Taylor developed the spectral-synthesis approach to the Pompeiu problem and its Fourier–Laplace characterization [14]. Williams related failure of the Pompeiu property on simply connected Lipschitz domains to the overdetermined eigenvalue problem [15]; Zalcman's survey gives the classical bibliography [16]. Recent planar counterexamples were constructed by Cao-Labora and de Dios Pont [17] and by Colbrook and Stepaniants [18]. Convex counterexamples in the plane, in every dimension from three to eighteen, and in dimensions twenty and twenty-one were obtained by computer-assisted proofs in [19]. The construction below uses a two-block split parameter at fixed ambient dimension and closes finite-order centres in a graded Jacobi coefficient space.

1.1. The construction and its estimates. The block split supplies a parameter while the ambient dimension and the ball radial spectrum remain fixed. Write an ambient point as $(X, Y) \in \mathbb{R}^a \times \mathbb{R}^{2p-a}$. With $n = 2p$ and $r = a/(2p)$, invariant functions depend on radius and $x = |X|^2/(|X|^2 + |Y|^2)$. Their angular basis is Jacobi orthogonal for the beta measure of parameters $pr, p(1-r)$. The monic quadratic $H_2(x)$ has spherical degree four and is the shape used to start the construction. At a continued radial Neumann zero $J_p(R) = 0$, define

$$\chi = \frac{J_{p+3}(R)}{J'_{p+3}(R) - (p-1)J_{p+3}(R)/R}, \quad \theta = 4Rp\chi.$$

For all sufficiently large members of the bounded base family, $R = 2p + 3 + O(p^{-1})$ and $0 < |\theta| < 20$. The estimates use the fixed parameter range $0 < |\theta| \leq T = 20$.

The normalized quartic radius is $1 - \theta H_2/(2R^2 p g_2)$, where $g_2 = [H_2]H_2^2 \asymp p^{-1}$. Vary the real split through $\varsigma = (r - 1/2)^2 \in [1/64, 1/16]$. The exact weighted Dirichlet eigenvalue curves have signed speed at least $c|\theta|/p$ when their offsets from R^2 lie in $(-|\theta|/(16p), |\theta|/(16p))$. Only $O(p^{2/3})$ ordered curves can enter this band. Avoiding their small intervals on the integer split grid gives an actual invariant Dirichlet gap $c|\theta|p^{-5/3}$ for at least one admissible a .

For each fixed accuracy order N , finite-order centre construction at that split solves the interior Helmholtz equation exactly and makes both boundary defects $O_N(\theta^2 p^{-N})$ on every prescribed finite complex disc, with the constant and threshold allowed to depend on that disc. The recurrence uses only fixed nonzero leading divisors, including the seed divisor $-1/8$ after its natural scaling. The centre differs from the quartic radius by $O_N(|\theta|p^{-4})$ in normalized coordinates, so it inherits half the Dirichlet gap by domain monotonicity.

The nonlinear argument uses one fixed coefficient space $\mathcal{C}_{p,2}$ for sources and a boundary space $\mathcal{E}_1$ with one angular degree moment. The clamped source subspace $\mathcal{G}_p$ contains exactly the sources whose Poisson fields have zero Dirichlet and normal traces. Positive Jacobi multiplication and a parameter-raising coupling give a high-shape-degree gain on these clamped fields. This makes the scalar pullback analytic on the fixed small-shape chart in $\mathcal{G}_p \times \mathcal{E}_1$, with a second-derivative bound $C_N p^6$ on the stated centre neighbourhood. An exact material change of variables has inverse bounded by $C_N p^6$ in those same spaces.

The remaining inverse estimate passes through $\mathcal{M}$, which has radial L^2 norms and angular ℓ^1 coefficients. In a ball window of width $E_{\text{inv}}|\theta|$, the exact Schur matrix has entries of size $O(|\theta|)$, exponential angular decay in two matrix norms, and the actual Hilbert gap. Its logarithmic angular clusters have bounded *cardinality* M , independent of p and of the separation constant. A determinant and cofactor estimate on each cluster therefore gives

$$\|F_{\text{eff}}^{-1}\|_{\ell^1(2^j)} \leq C|\theta|^{-1}p^{5M/3}.$$

Radial graph reconstruction and a finite-head/infinite-tail argument give the full coefficient inverse for the scalar Dirichlet graph operator $J_D = B_{\Phi_N,D}K_D$, with $B_{\Phi,D}$ defined after (10.2),

$$(1.2) \quad \|J_D^{-1}\|_{\mathcal{C}_{p,2} \rightarrow \mathcal{C}_{p,2}} \leq C|\theta|^{-1}p^{5/2+5M/3}.$$

The constants determining M and the inverse exponent are fixed before the centre accuracy is chosen. With $N = 4M + 40$, the residual, corrected derivative and second derivative satisfy the three gates of Theorem 14.1 on a ball of radius $|\theta|p^{-2M-62/3}$. The exact zero remains strictly convex and keeps a nonzero physical monic seed coefficient. Its coefficient equation is identified with a weak equation on the completed spaces, then with a classical solution by elliptic regularity.

The logical order of the main steps is

$$\begin{aligned} \text{uniform Bessel expansion and congruence approximation} &\implies \mathcal{P}_{\text{bd}}, R, \theta, \\ \text{angular positivity and exact local spectral flow} &\implies \text{integer-split gap}, \\ \text{finite centres and exact clamping} &\implies \text{residuals and material map}, \\ \text{relative graph bounds and bounded-cardinality clusters} &\implies \text{coefficient inverse}, \\ \text{Newton threshold} &\implies \text{classical convex non-ball domain}. \end{aligned}$$

For Theorem B, one radial pole per angular degree is retained first. A finite-shift Lommel recurrence isolates four critical phase ratios. Their short components have subpolynomial coefficient-to-Hilbert conditioning, while the complementary couplings have two inverse powers of p , up to a chosen small loss. The exact centre is restored in the mixed norm before the single radial conversion factor $\sqrt{p}$. Section 18 gives material and nonlinear exponents two and three; Section 20 records their dependence on $\tau \leq \log p$. The three gates then close at order twenty and radius τp^{-11} as in Theorem 21.3.

The notation index in Appendix A distinguishes the fixed flow window, the local-count window, and the shrinking inverse window.

1.2. Classical inputs. The proof uses the following classical results; the specialized estimates needed in the construction are proved in the body of the paper.

- (1) The integer-order common-zero theorem of Bourget, as a consequence of Siegel's theorem, excludes integer primary crossings [2, §15.28]; see also [5]. At half-integer orders the sine–cosine representation and Hermite–Lindemann theorem [3, §1.2] give non-coincidence as proved in Lemma 5.1. The finite Lommel polynomials [2] give the shift resonance estimate in Lemma 19.4. The transcendence of π [6] makes α_{ar} irrational.
- (2) Bessel power series and recurrences, the fixed-order outgoing Hankel normalization and Watson's integral for the derivative of a positive zero with respect to order are used in their classical forms [1, §§10.2, 10.6, 10.17; Eq. 10.21.17]. The uniform large-order remainders, derivative bounds, spacing, and radial normalization used here are then proved in Section 3.
- (3) Gasper's positive linearization theorem is used with (9.7) and normalization at the endpoint of the larger Jacobi exponent [4, Theorem 1]. The parameter substitution is given before Lemma 9.3.
- (4) Closed Dirichlet forms, relatively bounded perturbations, compact resolvents, the min–max principle and finite Riesz reductions are used as in [7, Chs. II and IV–VII]. The common domains, norm differentiability, eigenvalue derivative argument and Schur reconstructions for this family are written out in Section 7.
- (5) Dirichlet H^2 and higher regularity and Sobolev traces are used on smooth bounded domains [8, Chs. 6–9]; analytic boundary regularity is used for analytic boundaries and constant Dirichlet data [9, 10]. The convex Dirichlet Hessian inequality is proved by its boundary identity in (11.9). Polynomial core density and the zero-capacity cutoff statements at the axes and origin are included in Proposition 7.3.
- (6) Hadamard's convexity theorem for a compact hypersurface with positive second fundamental form is used in Lemma 15.1; see [11, 20]. Bernstein's inequality for trigonometric polynomials, applied on great circles, gives the dimension-independent shape derivative bounds [12, Vol. II, Ch. X, Theorem 3.13]. The Jacobi endpoint bound itself is proved in Lemma 2.3. The rotation argument in Proposition 22.3 uses trigonometric polynomial approximation [12].
- (7) Banach's contraction theorem [13] is applied to a closed ball in the stated complete spaces. Taylor's formula, continued-fraction determinant identities and elementary lattice determinant facts are used in the explicit proofs of their specialized forms.
- (8) The analytic implicit function theorem is used for the scaled radius and seed equations in Theorem 6.2. Cauchy's estimates and the Hermite–Genocchi integral formula for divided differences give the analytic coefficient bounds in Lemma 6.3; Cauchy's estimates also give the centre remainder bounds. The Dirichlet–Poincaré (Wirtinger) inequality on an interval is used in Lemma 3.1 and Theorem 12.1.

2. TWO-BLOCK REDUCTION AND NORMALIZATIONS

Fix $p \geq 64$ with $2p \in \mathbb{Z}$ and an integer a with $p/2 \leq a \leq 3p/4$. Write

$$(2.1) \quad n = 2p, \quad b = 2p - a, \quad \alpha_b = \frac{a}{2} = pr, \quad \beta_b = \frac{b}{2} = p(1 - r), \quad \frac{1}{4} \leq r \leq \frac{3}{8}.$$

For the spectral deformation argument the parameters α_b, β_b will also be allowed to be positive real numbers with sum p . All constructions that define Euclidean domains are made at integer block sizes. In particular $a, b \geq 3$ and $\alpha_b, \beta_b > 1$ in this range. We use $y = (X, Y) \in \mathbb{R}^a \times \mathbb{R}^b$

and, away from the origin, set

$$(2.2) \quad \varrho = |y|, \quad t = |y|^2, \quad x = \frac{|X|^2}{|X|^2 + |Y|^2}.$$

Here t always denotes squared radius. A nonsquared reference radial coordinate used later will be denoted by ζ .

Lemma 2.1 (Invariant differential operator). *For a twice differentiable invariant function $u(\varrho, x)$,*

$$(2.3) \quad \Delta u = u_{\varrho\varrho} + \frac{2p-1}{\varrho} u_{\varrho} + \frac{4}{\varrho^2} \{x(1-x)u_{xx} + (\alpha_b - px)u_x\}.$$

Normalized spherical measure induces on x the probability measure

$$(2.4) \quad d\mu_r(x) = \frac{x^{\alpha_b-1}(1-x)^{\beta_b-1}}{B(\alpha_b, \beta_b)} dx.$$

Normalized invariant volume measure on the unit ball is $pt^{p-1} dt d\mu_r(x)$. Each angular polynomial occurs with multiplicity one in this invariant sector.

Proof. Put $u(y) = v(q_X, q_Y)$, where $q_X = |X|^2$ and $q_Y = |Y|^2$. Differentiating in each Euclidean coordinate and summing gives

$$\Delta u = 4q_X v_{q_X q_X} + 2av_{q_X} + 4q_Y v_{q_Y q_Y} + 2bv_{q_Y}.$$

Use $q_X = \varrho^2 x$ and $q_Y = \varrho^2(1-x)$ in this identity. The mixed second derivatives in ϱ, x cancel, and the remaining coefficients are those in (2.3). Alternatively, the coefficients of u_{xx} and u_x follow directly from $|\nabla_S x|^2 = 4x(1-x)$ and $\Delta_S x = 4(\alpha_b - px)$.

Polar integration in the two blocks has volume element proportional to $q_X^{\alpha_b-1} q_Y^{\beta_b-1} dq_X dq_Y$. The determinant of the change $(t, x) \mapsto (tx, t(1-x))$ has absolute value t . Its angular integral is $B(\alpha_b, \beta_b)$; its radial integral over $0 < t < 1$ is $1/p$. Division by these two integrals proves the measure assertions. An invariant function is a function of ϱ, x , so each angular polynomial below occurs once. $\square$

2.1. The angular polynomials. Let $H_j(x)$ be the monic degree- j orthogonal polynomial for μ_r , and define

$$(2.5) \quad G_j(x) = \frac{H_j(x)}{H_j(1)}, \quad \eta_j^2 = \int_0^1 G_j(x)^2 d\mu_r(x), \quad P_j(x) = \eta_j^{-1} G_j(x).$$

The sign of P_j is chosen to give a positive leading coefficient. The two uses of Jacobi polynomials in this paper are distinguished by letters: G_j is angular, whereas J_s^β below is radial. The symbol J_ν without an upper index denotes the Bessel function.

Lemma 2.2 (Angular formulas). *For $j \geq 0$,*

$$(2.6) \quad G_j(x) = \frac{P_j^{(\beta_b-1, \alpha_b-1)}(2x-1)}{(\beta_b)_j/j!}, \quad H_j(1) = \frac{(\beta_b)_j}{(p+j-1)_j},$$

$$(2.7) \quad \eta_j^2 = \frac{(p-1)(\alpha_b)_j j!}{(2j+p-1)(p-1)_j (\beta_b)_j}, \quad \Lambda_r G_j = \lambda_j G_j,$$

$$\Lambda_r = -4x(1-x)\partial_x^2 - 4(\alpha_b - px)\partial_x, \quad \lambda_j = 4j(j+p-1).$$

The convention for a rising factorial of length zero is $(z)_0 = 1$. In the orthonormal basis multiplication by x is

$$(2.8) \quad xP_j = \mathbf{a}_{j+1}P_{j+1} + \mathbf{b}_jP_j + \mathbf{a}_jP_{j-1},$$

where $\mathfrak{a}_0 = 0$ and

$$(2.9) \quad \mathfrak{a}_j^2 = \frac{j(j + \alpha_b - 1)(j + \beta_b - 1)(j + p - 2)}{(2j + p - 2)^2(2j + p - 1)(2j + p - 3)}, \quad j \geq 1,$$

$$(2.10) \quad \mathfrak{b}_j = \frac{1}{2} + \frac{(r - 1/2)p(p - 2)}{(2j + p - 2)(2j + p)}.$$

All these identities hold also for real positive α_b, β_b with $p \geq 64$.

Proof. The shifted Rodrigues identity is

$$x^{\alpha_b - 1}(1 - x)^{\beta_b - 1}P_j^{(\beta_b - 1, \alpha_b - 1)}(2x - 1) = \frac{(-1)^j}{j!} \frac{d^j}{dx^j} \{x^{\alpha_b + j - 1}(1 - x)^{\beta_b + j - 1}\}.$$

Repeated integration by parts against a polynomial of degree less than j proves orthogonality. The endpoint value is $(\beta_b)_j/j!$ and the leading coefficient is $(p + j - 1)_j/j!$. Their quotient proves the second identity in (2.6). The integrations by parts are valid: at each intermediate stage the endpoint powers in the boundary terms have positive exponent. Differentiation of the Rodrigues identity gives the displayed angular differential equation.

A further integration by parts against the degree- j polynomial gives

$$\int_0^1 (P_j^{(\beta_b - 1, \alpha_b - 1)}(2x - 1))^2 d\mu_r = \frac{(p - 1)(\alpha_b)_j(\beta_b)_j}{(2j + p - 1)(p - 1)_j j!}.$$

Dividing by the squared endpoint value proves (2.7). Orthogonality makes multiplication by x tridiagonal and symmetric. The ratio of the leading coefficients of P_{j-1} and P_j , followed by the norm formula, gives (2.9). Comparing the coefficients of x^j in the monic recurrence gives (2.10). These computations hold for these positive real parameters. $\square$

Lemma 2.3 (Endpoint bound and angular derivatives). *For the parameters in (2.1),*

$$|G_j(x)| \leq 1 \quad (0 \leq x \leq 1), \quad 0 < \eta_j \leq 1.$$

On the unit sphere the homogeneous extension $S_j(y) = |y|^{2j} G_j(x)$ satisfies

$$(2.11) \quad \|G_j\|_{L^\infty(S^{2p-1})} \leq 1, \quad \|\nabla_S G_j\|_\infty \leq 2j, \quad \|\nabla_S^2 G_j\|_{\text{op}, \infty} \leq 4j^2.$$

The constants are independent of p and of the admissible split.

Proof. For a Jacobi polynomial $v(z) = P_j^{(\alpha, \beta)}(z)$ with $\alpha = \beta_b - 1 \geq \beta = \alpha_b - 1 \geq 0$ and $j > 0$, write $\kappa = j(j + \alpha + \beta + 1)$. Its equation implies

$$\frac{d}{dz} \left(v^2 + \frac{1 - z^2}{\kappa} (v')^2 \right) = \frac{2}{\kappa} \{ \alpha - \beta + (\alpha + \beta + 1)z \} (v')^2.$$

The factor in braces is increasing. The expression in parentheses therefore first decreases and then increases, with either interval possibly empty. Its maximum is attained at an endpoint. The absolute endpoint values are $(\alpha + 1)_j/j!$ and $(\beta + 1)_j/j!$; the former is at least the latter. Division by the former proves the supremum bound. The case $j = 0$ is constant. The bound for η_j follows by integration against a probability measure.

The restriction to any great circle of the polynomial S_j is a trigonometric polynomial of degree at most $2j$. The first and second Bernstein inequalities give bounds $2j$ and $4j^2$ for its derivatives along that circle [12, Vol. II, Ch. X, Theorem 3.13]. Great circles with prescribed unit tangent vector compute the spherical gradient and the diagonal values of the spherical Hessian. A real symmetric bilinear form has operator norm equal to the supremum of its absolute diagonal values on unit vectors. This proves (2.11). $\square$

Lemma 2.4 (Solid harmonics and radial profiles). *The function*

$$(2.12) \quad S_j(y) = t^j G_j(x)$$

is an invariant homogeneous harmonic polynomial of degree $2j$. For a twice differentiable radial profile v ,

$$(2.13) \quad \Delta\{S_j v(t)\} = 4S_j\{tv'' + (2j + p)v'\}.$$

The regular solutions of $(\Delta + 1)u = 0$ in angular degree $2j$ are multiples of

$$(2.14) \quad \varrho^{1-p} J_{\nu_j}(\varrho) G_j(x), \quad \nu_j = p - 1 + 2j.$$

Their apparent singularities at the origin are removable.

Proof. As G_j is a polynomial of degree j , multiplication by t^j expresses S_j as a polynomial in q_X, q_Y . Applying (2.3) and using its angular eigenvalue proves harmonicity. For the product with $v(t)$, use $\nabla t = 2y$, $\Delta t = 4p$, and $y \cdot \nabla S_j = 2j S_j$. The product rule gives (2.13). Substituting $\varrho^{1-p} w(\varrho)$ in the radial Helmholtz equation gives Bessel's equation of order ν_j . Its regular solution is J_{ν_j} . Its convergent expansion is a constant times ϱ^{2j} multiplied by a power series in ϱ^2 . The factor $\varrho^{2j} G_j$ is the polynomial just constructed, which proves regularity at the origin and across both block axes. $\square$

2.2. The ball field and the invariant Dirichlet spectrum. Write $j_{\nu,k}$ for the k th positive zero of J_ν and set $z_{j,k} = j_{\nu_j,k}$. All radial zero indices k start at 1. Choose a positive zero $R = j_{p,k_0}$, where k_0 will be specified by the arithmetic construction. The radial field with boundary value one is

$$(2.15) \quad u_0(\varrho) = \frac{\varrho^{1-p} J_{p-1}(\varrho)}{R^{1-p} J_{p-1}(R)}.$$

Consecutive orders have no common positive zero: their recurrences would give both the value and derivative of one solution equal to zero, contradicting uniqueness for the Bessel equation. Thus the denominator in (2.15) is nonzero. The derivative identity $(\varrho^{1-p} J_{p-1}(\varrho))' = -\varrho^{1-p} J_p(\varrho)$ proves

$$(2.16) \quad u_0(R) = 1, \quad u_0'(R) = 0, \quad u_0''(R) = -1.$$

The last equality follows from the radial equation.

Proposition 2.5 (Invariant Dirichlet realization). *The Friedrichs Dirichlet Laplacian $H_0 = -\Delta_D$ in the invariant subspace of the unit ball has eigenvalues $z_{j,k}^2$, one radial sequence for each $j \geq 0$, with repetitions retained. It has compact resolvent. After the unitary radial substitution $v(\zeta) = \sqrt{2p} \zeta^{p-1/2} u(\zeta)$ and the angular basis P_j , its j th radial operator is*

$$(2.17) \quad -\frac{d^2}{d\zeta^2} + \frac{\nu_j^2 - 1/4}{\zeta^2}, \quad 0 < \zeta < 1,$$

with Friedrichs condition at zero and Dirichlet condition at one. The spectrum does not depend on the split r .

Proof. The orthogonal group action commutes with the Dirichlet form. Averaging over it is an orthogonal projection preserving H_0^1 , so restriction to the invariant subspace gives a closed Dirichlet form. Compactness of $H_0^1(B) \hookrightarrow L^2(B)$ implies compact resolvent for this restriction. The polynomials P_j form a complete angular orthonormal basis; this follows from density of polynomials in the continuous functions on $[0, 1]$ and density of continuous functions in $L^2(\mu_r)$. Separation of variables gives (2.17). Its regular solution at energy z^2 is a multiple of $\sqrt{\zeta} J_{\nu_j}(z\zeta)$. The other independent solution is not in the Friedrichs form domain, as $\nu_j \geq p - 1 > 1$. The

boundary condition at one is $J_{\nu_j}(z) = 0$. Neither ν_j nor the separated eigenvalue equation contains r . $\square$

Lemma 2.6 (Radial polynomial orthogonality). *Set $\beta_j = 2j + p - 1$ and*

$$(2.18) \quad J_s^\beta(t) = P_s^{(0,\beta)}(2t-1), \quad \Psi_{j,s}(y) = S_j(y)J_s^{\beta_j}(t), \quad s \geq 0.$$

Then $J_s^\beta(1) = 1$ and

$$(2.19) \quad J_s^\beta(t) = \sum_{h=0}^s (-1)^{s-h} \binom{s}{h} \binom{s+\beta+h}{s} t^h,$$

$$(2.20) \quad \int_0^1 t^\beta J_s^\beta J_v^\beta dt = \frac{\delta_{sv}}{2s+\beta+1}, \quad \|\Psi_{j,s}\|_{L^2(B)}^2 = \eta_j^2 \frac{p}{p+2j+2s}.$$

Here volume is normalized as in Lemma 2.1. All invariant polynomials are finite linear combinations of these basis elements.

Proof. Apply the Rodrigues computation of Lemma 2.2 with Jacobi parameters $0, \beta$. It gives the coefficient formula, endpoint value, and integral norm. Multiplication by $S_j = t^j G_j$ changes the radial weight pt^{p-1} to pt^{β_j} ; angular orthogonality then proves the last identity. A monomial in q_X, q_Y is t^d times a polynomial in x of degree at most d . Expand that polynomial in the G_j , $j \leq d$, and expand the remaining power t^{d-j} in the $J_s^{\beta_j}$. This proves the spanning assertion. $\square$

3. BESSEL ESTIMATES WITH UNIFORM REMAINDERS

The large-order estimates needed here are uniform as the ratio of argument to order varies. We derive them from an outgoing integral equation. The only asymptotic normalization used in that derivation is the fixed-order, large-argument normalization of the Hankel function. The recurrence identities and that normalization are as in [1, §§10.6, 10.17]; the order derivative formula used below is [1, Eq. 10.21.17].

We write Y_ν for the Bessel function of the second kind and $H_\nu^{(1)} = J_\nu + iY_\nu$ for the outgoing Hankel function. The modified Bessel functions of the second kind are denoted by K_ν .

3.1. Zeros, radial norms, and differentiation in the order.

Lemma 3.1 (Spacing and boundary normalization). *Let $\nu > 1/2$. Every positive zero of J_ν is simple, $j_{\nu,k} > \nu$, and*

$$(3.1) \quad j_{\nu,k+1} - j_{\nu,k} > \pi.$$

At a zero $z = j_{\nu,k}$ one has

$$(3.2) \quad \int_0^1 \zeta J_\nu(z\zeta)^2 d\zeta = \frac{1}{2} J_\nu'(z)^2.$$

Consequently a radial Dirichlet eigenfunction, normalized in its radial L^2 space and with its sign chosen appropriately, has boundary normal derivative $\sqrt{2}z$.

Proof. A simultaneous zero of J_ν and J_ν' would give the zero solution of its nonsingular ordinary differential equation at that point. Let $v(x) = \sqrt{x}J_\nu(x)$. On an interval between two positive zeros,

$$\int |v'|^2 = \int \left(1 - \frac{\nu^2 - 1/4}{x^2}\right) |v|^2 < \int |v|^2.$$

The Dirichlet Poincaré inequality on an interval of length L gives $\pi^2/L^2 < 1$, proving (3.1). On $(0, 1)$ the lowest eigenvalue of $-\partial_\zeta^2 + (\nu^2 - 1/4)/\zeta^2$ is at least $\pi^2 + \nu^2 - 1/4 > \nu^2$, by the same Poincaré inequality and $\zeta^{-2} \geq 1$. Every eigenvalue is at least this lower bound, so $j_{\nu,k} > \nu$.

For the integral identity put $w(\zeta, z) = J_\nu(z\zeta)$. Subtract the equation for w multiplied by $\partial_z w$ from its z derivative multiplied by w . Integration gives

$$[\zeta\{w(\partial_z w)_\zeta - w_\zeta \partial_z w\}]_0^1 = -2z \int_0^1 \zeta w^2 d\zeta.$$

At a zero z , $w(1, z) = 0$, $w_\zeta(1, z) = zJ'_\nu(z)$ and $\partial_z w(1, z) = J'_\nu(z)$. The term at zero vanishes by the regular Bessel series. This proves (3.2). The radial mass-unitary function is $\sqrt{\zeta}J_\nu(z\zeta)$, whose derivative at one is $zJ'_\nu(z)$ and whose squared norm is the integral in (3.2). Their ratio proves the last assertion. $\square$

Lemma 3.2 (Order derivative). *For $\nu \geq 1/2$ and $x = j_{\nu,k}$,*

$$(3.3) \quad 0 \leq \frac{\arccos(\nu/x)}{\sqrt{1 - (\nu/x)^2}} - \frac{\partial j_{\nu,k}}{\partial \nu} \leq \frac{\pi}{16x^2}.$$

In particular,

$$(3.4) \quad 1 - \frac{\pi}{16x^2} \leq \frac{\partial j_{\nu,k}}{\partial \nu} \leq \frac{\pi}{2}.$$

The estimates are independent of k . If $x/\nu \rightarrow 2$ and $x \rightarrow \infty$, then $\partial_\nu j_{\nu,k} \rightarrow 2\pi/(3\sqrt{3})$, also when k varies.

Proof. Watson's order-derivative identity uses the modified Bessel function K_0 and reads

$$\frac{\partial j_{\nu,k}}{\partial \nu} = 2x \int_0^\infty K_0(2x \sinh v) e^{-2\nu v} dv.$$

It applies to the simple positive zeros and positive orders in question. At $\nu = 1/2$ the regular Schrödinger solution is a multiple of $\sin x$, so its positive zeros also satisfy $x > \nu$. For larger orders this follows from Lemma 3.1. Replacing $\sinh v$ by v increases the integral. The replacement integral is

$$2x \int_0^\infty K_0(2xv) e^{-2\nu v} dv = \frac{\arccos(\nu/x)}{\sqrt{1 - (\nu/x)^2}}.$$

Insert the integral representation

$$K_0(u) = \int_0^\infty e^{-u \cosh w} dw$$

and integrate first in v . The result is $\int_0^\infty (\cosh w + \nu/x)^{-1} dw$. The substitution $z = \tanh(w/2)$ evaluates this integral to the quotient above.

Since $-K'_0 = K_1$ and K_1 is positive and decreasing,

$$0 \leq K_0(2xv) - K_0(2x \sinh v) \leq 2x(\sinh v - v)K_1(2xv).$$

Taylor's integral formula gives $\sinh v - v \leq v^3 \cosh v/6$. The hypothesis $\nu \geq 1/2$ implies $e^{-2\nu v} \cosh v \leq 1$. The error in the order derivative is therefore at most

$$\frac{4x^2}{6} \int_0^\infty v^3 K_1(2xv) dv = \frac{1}{24x^2} \int_0^\infty u^3 K_1(u) du = \frac{\pi}{16x^2}.$$

To evaluate the last integral, use $K_1(u) = \int_0^\infty e^{-u \cosh w} \cosh w dw$. Tonelli's theorem and the elementary gamma integral give $6 \int_0^\infty \operatorname{sech}^3 w dw = 3\pi/2$. The quotient in (3.3), written as

$\beta/\sin \beta$ for $0 < \beta < \pi/2$, lies in $[1, \pi/2]$. This proves (3.4). If $x/\nu \rightarrow 2$ and $x \rightarrow \infty$, the error $\pi/(16x^2)$ tends to zero, and the quotient in (3.3) tends to $2\pi/(3\sqrt{3})$, proving the limit. $\square$

3.2. The outgoing integral equation. For $z > 1$ define

$$(3.5) \quad \begin{aligned} \mathfrak{s}(z) &= \sqrt{z^2 - 1} - \arccos(1/z), & A_{\text{Deb}}(z) &= (z^2 - 1)^{-1/4}, \\ I(z) &= \frac{1}{4\sqrt{z^2 - 1}} + \frac{5}{12(z^2 - 1)^{3/2}}, & V(z) &= \frac{z^2(z^2 + 4)}{4(z^2 - 1)^3}. \end{aligned}$$

The letters I, V in this subsection denote scalar functions only. The perturbation operator in the inverse argument will be denoted by $\mathcal{V}$.

Lemma 3.3 (First uniform correction). *Let $\nu \geq 1$, $z \geq 3/2$, and set*

$$\phi = \nu \mathfrak{s}(z) - \frac{\pi}{4}, \quad \mathcal{A}_\nu(z) = \sqrt{\frac{2}{\pi\nu}} A_{\text{Deb}}(z), \quad b_\nu(z) = \frac{I(z)}{2\nu}.$$

Then

$$(3.6) \quad \left| \frac{J_\nu(\nu z)}{\mathcal{A}_\nu(z)} - \cos \phi - b_\nu(z) \sin \phi \right| \leq R_\nu^{\text{Deb}}(z),$$

$$(3.7) \quad \left| \frac{Y_\nu(\nu z)}{\mathcal{A}_\nu(z)} - \sin \phi + b_\nu(z) \cos \phi \right| \leq R_\nu^{\text{Deb}}(z),$$

$$(3.8) \quad R_\nu^{\text{Deb}}(z) = \frac{1}{2\nu^2} \left\{ V(z) + \frac{I(z)^2}{1 - I(z)/\nu} \right\} < \frac{6}{5\nu^2}.$$

Thus the same bound holds on every compact interval about $z = 2$, including a value of z depending on ν .

Proof. Put $\xi = \mathfrak{s}(z)$ and write $H_\nu^{(1)}(\nu z) = A_{\text{Deb}}(z)w(\xi)$. Here $\mathfrak{s}'(z) = \sqrt{z^2 - 1}/z$ and

$$2 \frac{A'_{\text{Deb}}}{A_{\text{Deb}}} + \frac{\mathfrak{s}''}{\mathfrak{s}'} + \frac{1}{z} = 0, \quad A''_{\text{Deb}} + z^{-1} A'_{\text{Deb}} = \frac{z^2 + 4}{4(z^2 - 1)^{9/4}}.$$

Thus substitution in Bessel's equation cancels the first derivative and gives

$$w_{\xi\xi} + \nu^2 w = \psi(\xi)w, \quad \psi(\mathfrak{s}(z)) = -V(z).$$

The outgoing Hankel normalization for fixed ν at infinity is $w \sim \sqrt{2/(\pi\nu)} e^{i\nu\xi - i\pi/4}$, with the differentiated normalization as well. This fixes the outgoing solution integrated from infinity. Define g by $w = \sqrt{2/(\pi\nu)} e^{i\nu\xi - i\pi/4} g$. Variation of constants yields

$$(3.9) \quad g(\xi) = 1 + \frac{1}{2i\nu} \int_\xi^\infty \{e^{2i\nu(v-\xi)} - 1\} \psi(v) g(v) dv.$$

The potential is integrable. Indeed, with $u = \sqrt{z^2 - 1}$,

$$(3.10) \quad \int_{\mathfrak{s}(z)}^\infty |\psi(v)| dv = \int_z^\infty \frac{y(y^2 + 4)}{4(y^2 - 1)^{5/2}} dy = I(z).$$

The iterated integral of order k in (3.9) has norm at most $(I(z)/\nu)^k/k!$. The series converges absolutely and gives the unique bounded outgoing amplitude. Integrating first to a finite endpoint and then using the Hankel normalization identifies that solution with the given Hankel function. It follows that $|g - 1| \leq e^{I/\nu} - 1$.

The first iterate has nonoscillatory term $-iI/(2\nu)$. Its oscillatory term has absolute value at most $V/(2\nu^2)$. To see this, integrate by parts once and use

$$V'(z) = -\frac{z(z^4 + 10z^2 + 4)}{2(z^2 - 1)^4} < 0, \quad \int_{\xi}^{\infty} |\psi'(v)| dv = V(z).$$

The contribution of the remaining iterates is bounded by

$$\frac{1}{\nu} \int_{\xi}^{\infty} |\psi(v)|(e^{I(v)/\nu} - 1) dv = e^{I/\nu} - 1 - I/\nu \leq \frac{I^2}{2\nu^2(1 - I/\nu)}.$$

The final inequality follows by expanding the exponential and bounding $1/k! \leq 1/2$ for $k \geq 2$. Thus $|g - (1 - iI/(2\nu))| \leq R_{\nu}^{\text{Deb}}$. Taking real and imaginary parts after multiplication by $e^{i\phi}$ proves (3.6) and (3.7).

For $z \geq 3/2$, $I(z) \leq 7/(6\sqrt{5}) < 21/40$ and $V(z) \leq 9/5$. In particular $I(z)/\nu < 1$. Substitution gives

$$R_{\nu}^{\text{Deb}}(z) < \frac{1}{2\nu^2} \left(\frac{9}{5} + \frac{(21/40)^2}{1 - 21/40} \right) = \frac{1809}{1520\nu^2} < \frac{6}{5\nu^2}.$$

□

Lemma 3.4 (Any fixed order). *Let $z > 1$, $\nu > 0$, and $v = (z^2 - 1)^{-1/2}$. Define polynomials $U_k(v)$ by $U_0 = 1$ and*

$$(3.11) \quad U_{k+1}(v) = \frac{v^2(1+v^2)}{2} U_k'(v) + \frac{1}{8} \int_0^v (1+5w^2) U_k(w) dw.$$

For every fixed integer $L \geq 0$, the amplitude in (3.9) satisfies

$$(3.12) \quad \left| g - \sum_{k=0}^L \frac{U_k(v)}{(i\nu)^k} \right| \leq \frac{2U_{L+1}(v)}{\nu^{L+1}} e^{I(z)/\nu}.$$

If $I(z)/\nu < 1$, the exponential can be replaced by $(1 - I(z)/\nu)^{-1}$. In particular,

$$(3.13) \quad U_1(v) = \frac{v}{8} + \frac{5v^3}{24}, \quad U_2(v) = \frac{81v^2 + 462v^4 + 385v^6}{1152}.$$

Proof. One has $dv/d\xi = -v^2(1+v^2)$ and $V = v^2(1+v^2)(1+5v^2)/4$. The recursion is equivalent to

$$2 \frac{dU_{k+1}}{d\xi} = \psi U_k - \frac{d^2 U_k}{d\xi^2}, \quad U_{k+1}(\infty) = 0.$$

Every coefficient in (3.11) is nonnegative. It follows that $U_k \geq 0$, $dU_k/d\xi \leq 0$ and $d^2 U_k/d\xi^2 \geq 0$. Substitution of the truncated sum g_L into $g'' + 2i\nu g' - \psi g = 0$ leaves only $(U_L'' - \psi U_L)/(i\nu)^L$. This numerator is nonnegative and

$$\int_{\xi}^{\infty} (U_L'' + V U_L) d\xi = 2U_{L+1}(v).$$

The integral equation for $g - g_L$ has a forcing term bounded by $2U_{L+1}/\nu^{L+1}$. In each iterated potential integral, its value at a later point is no greater than its value at the initial point. The same ordered-simplex estimate as for (3.9) therefore multiplies this bound by at most $e^{I/\nu}$. This proves (3.12). The inequality $e^w \leq (1-w)^{-1}$ for $0 \leq w < 1$ proves the alternative bound. The two polynomials follow by integrating the recursion twice. □

3.3. Fixed spectral windows.

Lemma 3.5 (One radial state and no adjacent angular states). *For each fixed $W > 0$ there is P_W such that for real $p \geq P_W$ and $R \in [2p - 1, 2p + 4]$, the set*

$$(3.14) \quad \mathcal{W}_p(W) = \{(j, k) : |z_{j,k}^2 - R^2| \leq W\}$$

contains at most one state for each angular index j and has no pair of angular indices differing by one. The threshold is independent of the split.

Proof. The argument interval has length $O_W(p^{-1})$, which is less than π for large p . Lemma 3.1 proves the first assertion. For the second we prove an explicit elementary separation. Let $z \geq 8$ be a zero of J_ν , with $1/3 \leq \nu/z < 1$, and set $y(v) = J_{\nu+2}(z+v)/J_{\nu+1}(z)$. Consecutive orders have no common positive zero, so this normalization is valid. The recurrences give

$$y(0) = \frac{2(\nu+1)}{z} \in [2/3, 9/4], \quad y'(0) = 1 - \frac{2(\nu+1)(\nu+2)}{z^2}, \quad |y'(0)| \leq 2.$$

For $|v| \leq 1/10$, both coefficients $(z+v)^{-1}$ and $1 - (\nu+2)^2/(z+v)^2$ of the differential equation have absolute value below one. In fact $(\nu+2)/(z-1/10) \leq (z+2)/(z-1/10) \leq 100/79 < \sqrt{2}$. The first-order system for (y, y') has maximum row norm at most two. Its integral equation gives $\max(|y(v)|, |y'(v)|) \leq (9/4)e^{1/5} < 3$. Therefore $y(v) \geq 2/3 - 3/10 > 0$ on this interval.

Two roots in (3.14) differ in argument by at most $2W/R$ for large p , which is below $1/10$ after increasing P_W . Their orders are at least $p-1$ and smaller than their roots. Thus the smaller order and its root satisfy $1/3 \leq \nu/z < 1$. The separation just proved excludes adjacent angular indices. $\square$

Lemma 3.6 (Thin phase strip). *For each fixed $W > 0$ there are $C_W, P_W < \infty$ with the following property. Suppose $p \geq P_W$, $R \in [2p - 1, 2p + 4]$, and $p - 1 \leq \nu \leq R - p^{2/3}/2$. Define*

$$(3.15) \quad \begin{aligned} \kappa(\nu, z) &= \sqrt{z^2 - \nu^2}, \\ \Xi(\nu, z) &= \kappa(\nu, z) - \nu \arccos(\nu/z) - \pi/4, \\ b(\nu, z) &= \frac{1}{8\kappa(\nu, z)} + \frac{5\nu^2}{24\kappa(\nu, z)^3}. \end{aligned}$$

If $J_\nu(z) = 0$ and $|z^2 - R^2| \leq W$, then

$$(3.16) \quad \text{dist}(\Xi(\nu, R) - \arctan b(\nu, R), \pi(\mathbb{Z} + 1/2)) \leq C_W/p.$$

For $\nu = p - 1 + 2j$, put

$$(3.17) \quad \varphi_p(j) = \pi^{-1} \{ \Xi(p - 1 + 2j, R) - \arctan b(p - 1 + 2j, R) \} - \frac{1}{2}.$$

On a band $D_0 \leq R - \nu \leq 2D_0$ with $D_0 \geq p^{2/3}/2$ one has

$$(3.18) \quad \frac{c}{\sqrt{pD_0}} \leq \varphi_p''(j) \leq \frac{C}{\sqrt{pD_0}},$$

where $c, C > 0$ are absolute and the threshold may depend on W .

Proof. Write $D = R - \nu$. In the stated region and for $|z - R| \leq W/R$, one has $\nu \asymp p$, $z + \nu \asymp p$, and $\kappa^2 \asymp pD$, with absolute comparison constants. The case $L = 1$ of Lemma 3.4 gives

$$\frac{J_\nu(z)}{\sqrt{2/\pi} \kappa^{-1/2}} = \cos \Xi + b \sin \Xi + e, \quad |e| \leq \frac{2U_2(\nu/\kappa)}{\nu^2(1 - I(z/\nu)/\nu)}.$$

Here $I(z/\nu)/\nu = 1/(4\kappa) + 5\nu^2/(12\kappa^3) = O(p^{1/2}D^{-3/2}) = O(p^{-1/2})$. The explicit polynomial U_2 gives $|e| = O(pD^{-3}) = O(p^{-1})$. Thus the denominator is bounded away from zero uniformly.

At a zero, divide by $\sqrt{1+b^2}$ and use $\arcsin u \leq 2u$ for $0 \leq u \leq 1/2$ to obtain a distance $O(p^{-1})$ from $\Xi - \arctan b$ to $\pi(\mathbb{Z} + 1/2)$. Now $\Xi_z = \kappa/z \leq 1$ and

$$b_z = -\frac{z}{8\kappa^3} - \frac{5\nu^2 z}{8\kappa^5} = o(1).$$

Replacing z by R changes the phase by $O_W(p^{-1})$, proving (3.16).

At fixed R , $\Xi_{\nu\nu} = 1/\kappa$ and direct differentiation gives

$$b_\nu = \frac{13\nu}{24\kappa^3} + \frac{5\nu^3}{8\kappa^5}, \quad b_{\nu\nu} = \frac{13}{24\kappa^3} + \frac{7\nu^2}{2\kappa^5} + \frac{25\nu^4}{8\kappa^7}.$$

It follows that

$$\kappa |(\arctan b)_{\nu\nu}| \leq \kappa\{|b_{\nu\nu}| + 2|b||b_\nu|^2\} = O(pD^{-3}) + O(p^2D^{-6}) = O(p^{-1}).$$

The constants are absolute throughout the region. Therefore $\varphi_p'' = (4/\pi)\kappa^{-1}(1 + O(p^{-1}))$. Since $\kappa \asymp \sqrt{pD_0}$ on the band, this proves (3.18). $\square$

Lemma 3.7 (Convex lattice strip). *Let $f \in C^2(I)$, where I has length at most L , and suppose $0 < m_0 \leq f'' \leq M_0$. Let $\delta > 0$ satisfy*

$$\frac{1}{2}M_0L^3 + 2\delta L < 1.$$

Then the integer points (j, k) with $j \in I$ and $|k - f(j)| \leq \delta$ are collinear. If there is at most one such point at each j , their number is at most $2 + 4\sqrt{\delta/m_0}$, with an integer upper bound understood.

Proof. For three such points, the determinant with rows $(1, j, k)$ is an integer. The determinant of the three graph values is $(x_2 - x_1)(x_3 - x_1)(x_3 - x_2)f[x_1, x_2, x_3]$ when the abscissae are distinct. The integral formula for the second divided difference bounds its absolute value by $M_0/2$, so this contribution is at most $M_0L^3/2$. If two abscissae coincide, the graph determinant is zero and the same bound holds. The three vertical errors contribute at most $2\delta L$: for ordered abscissae their coefficient absolute values sum to twice the largest separation. Thus the determinant has absolute value less than one and is zero. If there are at least three points, fixing the two extreme ones shows that every point lies on their line.

Take a middle point with abscissa x_2 and extreme abscissae x_1, x_3 . The chord of f satisfies

$$\text{chord}(x_2) - f(x_2) \geq \frac{m_0}{2}(x_2 - x_1)(x_3 - x_2).$$

One proof subtracts $m_0x^2/2$ from f and uses convexity of the result. Since the three integer points are collinear, their errors bound the same chord deficit by 2δ . Among n distinct integer abscissae choose a middle one; its two separations have product at least $(n-2)^2/4$. Therefore $(n-2)^2 \leq 16\delta/m_0$. The assertions for zero, one or two points need no determinant. $\square$

Theorem 3.8 (Sublinear window count). *For every fixed $W > 0$ there are $C_W, P_W < \infty$ such that, for $p \geq P_W$ and $R \in [2p-1, 2p+4]$,*

$$(3.19) \quad \#\mathcal{W}_p(W) \leq C_W p^{2/3}.$$

The bound holds for real p , and is independent of the split.

Proof. There is at most one radial state per angular index by Lemma 3.5. Also $z_{j,k} > \nu_j$, so the region $\nu_j > R - p^{2/3}$ has only $O_W(p^{2/3})$ possible indices. In the remaining region Lemma 3.6 assigns to each state a lattice point (j, m) with $|m - \varphi_p(j)| \leq \delta = C'_W/p$. This assignment is injective: there is at most one radial state at a given j , and $2\delta < 1$.

Divide $D = R - \nu_j$ into dyadic bands $D_0 \leq D \leq 2D_0$, from $p^{2/3}$ up to $O(p)$. On each band $f'' \asymp (pD_0)^{-1/2}$ for $f = \varphi_p$. Divide its horizontal interval into intervals of length $L = c_*(pD_0)^{1/6}$, with $c_* > 0$ fixed so small that $M_0 L^3/2 \leq 1/4$. Since $D_0 = O(p)$, the other term $2\delta L = O_W(p^{-2/3})$ is below $1/4$ for large p . Lemma 3.7 bounds the number of points per interval by an absolute multiple of $1 + \sqrt{C'_W}$, because $\delta/m_0 = O_W(\sqrt{D_0/p}) = O_W(1)$.

The band has horizontal length $O(D_0)$, so it requires $O(p^{-1/6}D_0^{5/6} + 1)$ such intervals. The sum of $D_0^{5/6}$ over the dyadic bands is $O(p^{5/6})$; the sum of the additive ones is $O(\log p)$. This proves (3.19) after adding the turning region. Every constant and threshold in this proof is fixed once W is fixed. $\square$

Corollary 3.9 (Domains in a thin shell). *Fix $W, C_{\text{shell}} > 0$ and integer blocks a, b with $a+b = 2p$ in the admissible range. If a two-block invariant domain in the unit-ball normalization satisfies $B_{1-C_{\text{shell}}/p^2} \subset \Omega \subset B_{1+C_{\text{shell}}/p^2}$, then its invariant Dirichlet spectrum has at most $C_{W, C_{\text{shell}}} p^{2/3}$ eigenvalues in $[R^2 - W, R^2 + W]$ for large p and $R \in [2p-1, 2p+4]$. The constants are uniform in the admissible integer splits.*

Proof. Zero extension preserves invariance. The min-max principle and radial dilation give, for each ordered eigenvalue index i ,

$$(1 + C_{\text{shell}}/p^2)^{-2} \lambda_i(B) \leq \lambda_i(\Omega) \leq (1 - C_{\text{shell}}/p^2)^{-2} \lambda_i(B).$$

Any index counted for Ω has its ball eigenvalue in $[R^2 - W', R^2 + W']$, where W' depends only on W, C_{shell} since $R^2 = O(p^2)$. Apply Theorem 3.8. $\square$

4. PRIMARY CROSSINGS AND A CONTINUED-FRACTION SEQUENCE

The real crossing orders and the integer orders are different objects. We denote a real crossing by $\hat{p}_m$ and use $\mathcal{P}$ for the selected set of integers.

4.1. A congruence approximation lemma.

Lemma 4.1 (Congruence approximation). *For every positive irrational number α there are infinitely many integer pairs (P, Q) with*

$$(4.1) \quad P \equiv 1 \pmod{2}, \quad Q \equiv 1 \pmod{4}, \quad Q > 0, \quad Q|Q\alpha - P| < \frac{9}{2}.$$

They can be selected by the finite case rule in the proof from every sufficiently late pair of consecutive continued-fraction convergents.

Proof. Write the convergents as $P_j^{\text{cf}}/Q_j^{\text{cf}}$ and put $v_j = (P_j^{\text{cf}}, Q_j^{\text{cf}})$. Use indices for which all the three preceding denominators are positive. The determinant of consecutive vectors has absolute value one. Set

$$\rho_j = Q_{j-1}^{\text{cf}}/Q_j^{\text{cf}} \in (0, 1), \quad T_j^{\text{cf}} = [a_{j+1}; a_{j+2}, \dots] > 1.$$

The identity $\alpha = (T_j^{\text{cf}} P_j^{\text{cf}} + P_{j-1}^{\text{cf}})/(T_j^{\text{cf}} Q_j^{\text{cf}} + Q_{j-1}^{\text{cf}})$ gives, for $(P, Q) = Av_j + Bv_{j-1}$ and nonnegative integers A, B ,

$$(4.2) \quad Q|Q\alpha - P| = \frac{(A + B\rho_j)|A - BT_j^{\text{cf}}|}{T_j^{\text{cf}} + \rho_j}.$$

Denote this expression by $E(A, B)$ when the index is understood. Directly from (4.2),

$$(4.3) \quad E(1, 0) < 1, \quad E(1, 1) < 2, \quad E(2, 1) < 3, \quad E(1, 2) < 2(1 + 2\rho_j).$$

For $E(2, 1)$, split at $T_j^{\text{cf}} = 2$. On $1 < T_j^{\text{cf}} < 2$ the quotient has maximum at $T_j^{\text{cf}} = 1$ and is at most $(2 + \rho_j)/(1 + \rho_j) < 2$. On $T_j^{\text{cf}} \geq 2$ it is less than $2 + \rho_j < 3$. The other three inequalities follow by bounding respectively $1/(T_j^{\text{cf}} + \rho_j) < 1$, $(T_j^{\text{cf}} - 1)/(T_j^{\text{cf}} + \rho_j) < 1$, and $(2T_j^{\text{cf}} - 1)/(T_j^{\text{cf}} + \rho_j) < 2$.

We first give a helper rule. Suppose v_k has even numerator and odd denominator, v_{k-1} has odd numerator and even denominator, and $Q_k^{\text{cf}} + Q_{k-1}^{\text{cf}} \equiv 3 \pmod{4}$. The vector $S = 3v_k + v_{k-1}$ has the target residues. Put $T = T_k^{\text{cf}}$ and $\rho = \rho_k$. If $T \geq 2$, its error is less than four: for $2 \leq T \leq 3$ it is at most $(3 + \rho)/(2 + \rho) < 3/2$, and for $T \geq 3$ it is less than $3 + \rho < 4$. If $1 < T < 2$, then $a_{k+1} = 1$ and $v_{k+1} = v_k + v_{k-1}$. The alternative $D = 3v_{k+1}$ also has the target residues. Their error products are

$$E_S = \frac{(3 + \rho)(3 - T)}{T + \rho}, \quad E_D = \frac{9(1 + \rho)(T - 1)}{T + \rho}.$$

The inequalities $E_S \leq 3$ and $E_D \leq 3$ hold respectively when

$$T \geq \frac{9}{6 + \rho}, \quad T \leq \frac{3 + 4\rho}{2 + 3\rho}.$$

These two ranges cover $(1, 2)$, because $(3 + 4\rho)(6 + \rho) - 9(2 + 3\rho) = 4\rho^2 \geq 0$. Choose the candidate with smaller error. This fixes the helper rule and always gives error less than four, with positive denominator at least Q_{k-1}^{cf} .

Consider now v_j, v_{j-1} . There are two cases.

Case I: one vector has both entries odd. Call that vector v_k , where $k = j$ or $j - 1$. If $Q_k^{\text{cf}} \equiv 1 \pmod{4}$, take v_k itself. Otherwise $Q_k^{\text{cf}} \equiv 3 \pmod{4}$. If Q_{k-1}^{cf} is even, its numerator is odd by the determinant condition. For even a_{k+1} , take $3v_k$; here $T_k^{\text{cf}} > 2$ and its error is $9/(T_k^{\text{cf}} + \rho_k) < 9/2$. For odd a_{k+1} , the vector v_{k+1} has even numerator and odd denominator. Take $2v_{k+1} + v_k$. Its denominator is 1 modulo four, its numerator is odd, and (4.3) at index $k + 1$ gives error less than three.

If Q_{k-1}^{cf} is odd, its numerator is even. When $a_k \geq 2$ one has $\rho_k < 1/2$; take $v_k + 2v_{k-1}$. Its denominator is 1 modulo four and its error is less than $2(1 + 2\rho_k) < 4$. When $a_k = 1$, the vector $v_{k-2} = v_k - v_{k-1}$ has odd numerator and even denominator. The pair v_{k-1}, v_{k-2} satisfies the helper hypotheses, since its denominator sum is $Q_k^{\text{cf}} \equiv 3 \pmod{4}$. Apply the helper rule at index $k - 1$.

Case II: neither vector has both entries odd. Their types modulo two must be (even, odd) and (odd, even): neither vector is zero modulo two, and their determinant is odd. If $Q_j^{\text{cf}} + Q_{j-1}^{\text{cf}} \equiv 1 \pmod{4}$, use their sum, whose error is less than two. If that sum is 3 modulo four and Q_j^{cf} is odd, apply the helper rule directly.

It remains that v_j has odd numerator and even denominator Q_e , v_{j-1} has even numerator and odd denominator Q_o , and $Q_e + Q_o \equiv 3 \pmod{4}$. Thus $Q_e - Q_o \equiv 1 \pmod{4}$. Use $v_{j-2} = v_j - a_j v_{j-1}$ and distinguish the residue of a_j . If $a_j \equiv 1 \pmod{4}$, take v_{j-2} . If $a_j \equiv 2 \pmod{4}$, take $v_{j-1} + v_{j-2}$; its error is less than two by (4.3) at index $j - 1$. If $a_j \equiv 3 \pmod{4}$, the denominator of v_{j-2} is 3 modulo four; take $2v_{j-1} + v_{j-2}$, whose error is less than three. In these three choices the numerator is odd and the denominator is 1 modulo four. If $a_j \equiv 0 \pmod{4}$, the pair v_{j-1}, v_{j-2} has the helper types and denominator sum 3 modulo four, so apply that rule at index $j - 1$.

Every selected denominator is positive and at least Q_{j-3}^{cf} . These lower bounds tend to infinity. Thus infinitely many distinct pairs have been constructed, and all their errors are strictly below $9/2$. $\square$

4.2. The crossing expansion. Define the fixed constants

$$(4.4) \quad c_{\text{ar}} = \sqrt{3} - \frac{\pi}{3}, \quad \alpha_{\text{ar}} = \frac{\pi}{2c_{\text{ar}}}, \quad d_{\text{ar}} = \frac{5}{9\sqrt{3}c_{\text{ar}}}, \quad \delta_0 = 1 - \frac{\pi}{3\sqrt{3}}.$$

Also write

$$(4.5) \quad R_{\text{alg}}(v) = 2\sqrt{(v+1)(v+2)}, \quad A_m = \frac{\alpha_{\text{ar}}(4m+1) - 3}{2}.$$

These constants are unrelated to the block parameters α_b, β_b .

Proposition 4.2 (Primary-crossing expansion). *For every sufficiently large integer m , there is a unique primary crossing $\hat{p}_m$ in a fixed $O(A_m^{-1})$ neighbourhood of A_m , meaning*

$$J_{\hat{p}_m}(R_{\text{alg}}(\hat{p}_m)) = J_{\hat{p}_m+3}(R_{\text{alg}}(\hat{p}_m)) = 0.$$

It satisfies

$$(4.6) \quad \hat{p}_m = A_m - \frac{d_{\text{ar}}}{A_m} + O(A_m^{-2}),$$

with an absolute remainder constant. No positive integer or half-integer is a primary crossing.

Proof. At a zero R of J_v , the three-term recurrence gives

$$(4.7) \quad \frac{J_{v+3}(R)}{J_{v-1}(R)} = 1 - \frac{4(v+1)(v+2)}{R^2}.$$

The denominator is nonzero. Thus the crossing equation is precisely $J_v(R_{\text{alg}}(v)) = 0$.

At $R = R_{\text{alg}}(v)$, direct Taylor expansion gives

$$\begin{aligned} \sqrt{R^2 - v^2} &= \sqrt{3}v + 2\sqrt{3} - \frac{2}{\sqrt{3}v} + O(v^{-2}), \\ v \arccos(v/R) &= \frac{\pi v}{3} + \frac{\sqrt{3}}{2} - \frac{11}{4\sqrt{3}v} + O(v^{-2}). \end{aligned}$$

Consequently the phase of Lemma 3.3 is

$$\Xi(v, R) = c_{\text{ar}}v + \frac{3\sqrt{3}}{2} - \frac{\pi}{4} + \frac{\sqrt{3}}{4v} + O(v^{-2}),$$

and its first sine coefficient is $b(v, R) = 7/(36\sqrt{3}v) + O(v^{-2})$. Lemma 3.3 applies uniformly since $R/v \rightarrow 2$. In particular, the remainder is $O(v^{-2})$ for these varying arguments.

The identity $c_{\text{ar}}A_m + 3\sqrt{3}/2 - \pi/4 = \pi(m + 1/2)$ is exact. At $v = A_m \pm K/A_m$, for a sufficiently large fixed K , the leading cosine changes sign with magnitude exceeding the first correction and its remainder. The intermediate value theorem gives a zero. Every such zero obeys

$$\Xi(v, R) = \pi(m + 1/2) + \frac{7}{36\sqrt{3}v} + O(v^{-2}).$$

Substitution of the phase expansion proves (4.6), since $\{7/(36\sqrt{3}) - \sqrt{3}/4\}/c_{\text{ar}} = -d_{\text{ar}}$.

For uniqueness, the zero lies on one of the analytic curves $R = j_{v,k}$, because the zeros are simple. On the interval under consideration, Lemma 3.2 gives $\partial_v j_{v,k} \leq \pi/2$, whereas $R'_{\text{alg}}(v) = (2v+3)/\sqrt{(v+1)(v+2)} > 2$. Thus each difference $R_{\text{alg}}(v) - j_{v,k}$ is strictly increasing. Two different indices cannot cross the algebraic radius in an interval of length $O(A_m^{-1})$: their roots are separated by more than π , while both the algebraic radius and the zero curves move by $O(A_m^{-1})$ there. This proves uniqueness for large m .

Finally, for integer v , two distinct nonnegative integer-order Bessel functions have no common positive zero (Bourget's theorem). This classical consequence of Siegel's theorem is stated

in [1, §10.21(i)], with the source [2, §15.28]. Applying it to orders v and $v + 3$ excludes an integer primary crossing. For positive half-integers, Lemma 5.1 gives the same conclusion. $\square$

Definition 4.3 (Selected integer sequence). Apply the case rule in Lemma 4.1 to $\alpha = \alpha_{\text{ar}}$, with convergents indexed starting at 0, for each index $j \geq 6$. In its helper choose the candidate with smaller error. Remove repeated pairs and list the resulting integers $(P - 3)/2$ in increasing order. Their set is $\mathcal{P}$; its increasing enumeration is denoted by (p_m) . The integer m in (p_m) is an enumeration index, whereas the primary crossing associated to a selected pair has index $(Q - 1)/4$.

Definition 4.4 (Near-primary half-lattice inputs). Fix an integer m_0 beyond which Proposition 4.2 holds. For $C_{\text{ap}} > 0$ set

$$\mathcal{P}_{\text{fix}}(C_{\text{ap}}) = \left\{ p \in \frac{1}{2}\mathbb{Z} : p > 0, \exists m \geq m_0 : |p - \hat{p}_m| < C_{\text{ap}}/\hat{p}_m \right\},$$

$$\mathcal{P}_{\text{bd}} = \mathcal{P}_{\text{fix}}(31/10).$$

The continued-fraction set $\mathcal{P}$ remains the selected integer subsequence of Definition 4.3. The symbol p_0 denotes the crossing associated to the current input. It is unique for sufficiently large inputs in each fixed-width family, since successive crossings have separation $2\alpha_{\text{ar}} + o(1)$.

Theorem 4.5 (Near-integer primary crossings). *The set $\mathcal{P}$ is unbounded. For every sufficiently large selected pair (P, Q) , the integer $p = (P - 3)/2$ and the primary crossing $p_0 = \hat{p}_{(Q-1)/4}$ satisfy*

$$(4.8) \quad 0 < |p - p_0| < \frac{31}{10p_0}.$$

Thus the assertion holds for every sufficiently large member of $\mathcal{P}$, with its associated crossing. The threshold is implicit.

Proof. The number α_{ar} is irrational. Indeed, a rational value would imply $\pi(1 + 2\alpha_{\text{ar}}/3) = 2\alpha_{\text{ar}}\sqrt{3}$, contradicting the transcendence of π . For $Q = 4m + 1$ and $p = (P - 3)/2$, the expansion (4.6) gives

$$p_0|p_0 - p| \leq \frac{p_0}{2}|Q\alpha_{\text{ar}} - P| + d_{\text{ar}}\frac{p_0}{A_m} + O(A_m^{-1}).$$

Here $p_0/Q \rightarrow \alpha_{\text{ar}}/2$ and $p_0/A_m \rightarrow 1$. The congruence error bound therefore yields

$$(4.9) \quad \limsup p_0|p_0 - p| \leq \frac{9}{8}\alpha_{\text{ar}} + d_{\text{ar}} < \frac{31}{10}.$$

The final comparison follows from rational bounds. From $\pi < 22/7$ and $\sqrt{3} > 19/11$ one obtains $c_{\text{ar}} > 157/231$, $\alpha_{\text{ar}} < 363/157$, and

$$\frac{9}{8}\alpha_{\text{ar}} + d_{\text{ar}} < \frac{220099}{71592} < \frac{31}{10}, \quad 31 \cdot 71592 - 10 \cdot 220099 = 18362 > 0.$$

Bourget's theorem gives $p \neq p_0$. The denominators tend to infinity and $P/Q \rightarrow \alpha_{\text{ar}} > 0$, so the selected integers are unbounded. Distinct sufficiently large Q cannot give the same integer p : the corresponding numerators would be equal, while $\alpha_{\text{ar}}|Q - Q'| \leq 9/(2Q) + 9/(2Q')$ would tend to zero. This also removes any ambiguity in the associated crossing for large members of the sequence. $\square$

Corollary 4.6 (Sparsity). *The selected integer set satisfies $\#(\mathcal{P} \cap [1, X]) = O(\log X)$.*

Proof. The rule selects at most one pair at each continued-fraction index j , and its denominator is at least Q_{j-3}^{cf} . These denominators grow at least as the Fibonacci sequence. If the selected integer is at most X , its error bound gives $Q = O(X)$, since $P = 2p+3 = \alpha_{\text{ar}}Q + O(Q^{-1})$. Hence only $O(\log X)$ indices can contribute. Removing duplicates cannot increase the count. $\square$

4.3. The detuning parameter. Continue the zero at a primary crossing p_0 as $R(p) = j_{p,k_0}$. In this paragraph p is real on the continuation interval; its selected endpoint belongs to $\frac{1}{2}\mathbb{Z}$. Put

$$(4.10) \quad \chi = \frac{J_{p+3}(R)}{J'_{p+3}(R) - (p-1)J_{p+3}(R)/R}, \quad \theta = 4Rp\chi, \quad \tau = |\theta|.$$

Lemma 4.7 (Uniform bounded detuning). *At every sufficiently large $p \in \mathcal{P}_{\text{bd}}$,*

$$(4.11) \quad R = 2p + 3 + O(p^{-1}), \quad 0 < |\theta| < 20.$$

More precisely, uniformly for $|p - p_0| \leq 31/(10p_0)$,

$$(4.12) \quad \theta = (16\delta_0 + o(1))p_0(p - p_0).$$

The denominator in (4.10) is nonzero in this interval for large p_0 .

Proof. Repeated use of the Bessel recurrences at $J_p(R) = 0$ gives the exact identity

$$(4.13) \quad \chi = -\frac{R}{2(p+1)} \frac{R^2 - 4(p+1)(p+2)}{2R^2 - 4(p+1)(p+2)}.$$

The omitted common factor in numerator and denominator is $J_{p-1}(R)$, which is nonzero. The denominator in the rational expression is $4p^2 + O(p)$ throughout the interval, so the physical normal-trace denominator is also nonzero. Lemma 3.2, $R/p \rightarrow 2$, and $R \rightarrow \infty$ give

$$R'(p) = \frac{2\pi}{3\sqrt{3}} + o(1), \quad R'_{\text{alg}}(p) = 2 + o(1),$$

uniformly on that interval. Since the two radii agree at p_0 ,

$$R(p) - R_{\text{alg}}(p) = -(2\delta_0 + o(1))(p - p_0).$$

Substitute this in (4.13) and multiply by $4Rp$ to prove (4.12). The radius assertion follows from the same comparison and $R_{\text{alg}}(p) = 2p + 3 + O(p^{-1})$. The inequality $\delta_0 < 2/5$ and the defining distance bound give eventually $\tau < 16(2/5)(31/10) + o(1) = 19.84 + o(1) < 20$. The difference of the two radii has a strictly negative derivative, because $R' \leq \pi/2 < 2 < R'_{\text{alg}}$. It vanishes only at $p = p_0$. Proposition 4.2 excludes that equality on $\frac{1}{2}\mathbb{Z}$. Hence $\chi \neq 0$ and $\theta \neq 0$ at every input. $\square$

4.4. Low angular degrees at near-primary inputs.

Lemma 4.8 (A radial pole estimate). *Let $p \geq 64$, $R = R_{\text{alg}}(p)$, and $\nu = p + \ell - 1 < 2p + 3$. If $J_\nu(R) \neq 0$, set $M_\ell = J'_\nu(R)/J_\nu(R) - (p-1)/R$. Then every positive zero z of J_ν satisfies*

$$(4.14) \quad |z^2 - R^2| \geq \frac{2(R-1)}{|M_\ell| + 2}.$$

Proof. For positive λ away from poles define $w(\lambda) = \sqrt{\lambda}J'_\nu(R\sqrt{\lambda})/J_\nu(R\sqrt{\lambda})$. Bessel's equation implies

$$w'(\lambda) = -\frac{R}{2\lambda} \left(w(\lambda)^2 + \lambda - \frac{\nu^2}{R^2} \right).$$

Put $\eta = 1/R$ and $A = \sqrt{1+\eta}$. On $[1-\eta, 1+\eta]$ one has $|\lambda - \nu^2/R^2| \leq 1+\eta$, because $\nu^2 < R^2 + 1$. Consequently

$$\left| \frac{d}{d\lambda} \arctan \frac{w(\lambda)}{A} \right| \leq \frac{RA}{2(1-\eta)}.$$

At the first pole on either side of one the arctangent tends to an endpoint $\pm\pi/2$. Its distance to that endpoint at $\lambda = 1$ is at least $\arctan(A/|w(1)|) \geq A/(|w(1)| + A)$. (The latter inequality follows by differentiating $\arctan u - u/(1+u)$.) A pole in this interval is therefore at λ -distance at least $2(1-\eta)/(R(|w(1)| + A))$ from one. Now $|w(1)| + A < |M_\ell| + 2$, since $(p-1)/R < 1/2$ and $A < 3/2$. Multiplication by R^2 proves (4.14) for these poles. A pole outside the interval gives $|z^2 - R^2| \geq R$, which is at least the claimed right-hand side. Poles beyond the first on either side are farther away. This proves the bound for the entire radial spectrum. $\square$

Lemma 4.9 (Uniform recurrence remainder). *At a real primary crossing $p \geq 10^6$, put $s_0 = R_{\text{alg}}(p)/2$ and define*

$$q_{-1} = 1, \quad q_0 = 0, \quad q_{k+1} = \frac{p+k}{s_0} q_k - q_{k-1}.$$

Let $\mathbf{a}_{-1} = 1, \mathbf{a}_0 = 0, \mathbf{a}_{k+1} = \mathbf{a}_k - \mathbf{a}_{k-1}$, and $\mathbf{b}_{-1} = \mathbf{b}_0 = 0, \mathbf{b}_{k+1} = \mathbf{b}_k - \mathbf{b}_{k-1} + (k-3/2)\mathbf{a}_k$. Set $\mathbf{e}_k = q_k - \mathbf{a}_k - \mathbf{b}_k/p$ and $\|(u, v)\|_\Delta = \max(|u|, |v|, |u-v|)$. Then

$$(4.15) \quad \|(\mathbf{e}_k, \mathbf{e}_{k-1})\|_\Delta \leq e^{k(k+3)/p} \frac{(k+2)^4}{4p^2}, \quad \mathbf{b}_{3h} = (-1)^h 3h(h-1).$$

Proof. The transformation $(u, v) \mapsto (u-v, u)$ preserves the norm $\max(|u|, |v|, |u-v|)$. The sequence $\mathbf{a}_k$ has period six and norm one. Write $\delta_k = (p+k)/s_0 - 1$. From $p + 3/2 - s_0 = 1/[4(p + 3/2 + s_0)]$ one obtains

$$\left| \delta_k - \frac{k-3/2}{p} \right| \leq \frac{3(k+2)}{2p^2}, \quad |\delta_k| \leq \frac{2(k+2)}{p}.$$

Summing the first-order recurrence in the invariant norm bounds its vector by $k(k+3)/2$. Subtracting the first-order approximation from the exact recurrence then bounds the next remainder vector by $(1+|\delta_k|)$ times the current vector plus $(k+2)^3/p^2$. Iteration and $1+u \leq e^u$ give an exponential no greater than $e^{k(k+3)/p}$; the sum of the cubics is at most $(k+2)^4/4$. This proves the first bound. Three successive steps of the recurrence for $\mathbf{b}_k$ give $\mathbf{b}_{3(h+1)} = -\mathbf{b}_{3h} + 6h(-1)^{h+1}$. Its initial value is zero, so induction gives the second formula. $\square$

Proposition 4.10 (Low-degree exclusion). *Let $p_0 \geq 10^6$ be a sufficiently large primary crossing and $R_0 = R_{\text{alg}}(p_0)$. Apart from the degree-four state at R_0 , every even degree $\ell \geq 0$ with $\ell-1 \leq \sqrt{p_0}/4$ has its ball Dirichlet spectrum at distance greater than 64 from R_0^2 . At any real p with $|p-p_0| \leq 31/(10p_0)$ and continued radius $R = j_{p,k_0}$, those nonseed degrees have their spectrum outside $[R^2 - 32, R^2 + 32]$. The seed's other radial states are outside this window as well.*

Proof. At the crossing the recurrence is exactly $q_k = J_{p_0+k}(R_0)/J_{p_0-1}(R_0)$. Put $k = \ell - 1$, initially $k \geq 7$ with $k^2 \leq p_0/16$. The first-order contribution in the invariant vector norm is at most $5/112$ and the remainder in (4.15) is at most $81/8192$; their sum is less than $1/16$. If $3 \nmid k$, this gives $|q_k| > 15/16$, $|q_{k-1}| < 17/16$, and hence

$$|M_{k+1}| = \left| \frac{q_{k-1}}{q_k} - \frac{2p_0 + k - 1}{R_0} \right| < 4.$$

If $3 \mid k$ and $k \geq 15$, then $|\mathbf{b}_k| = k(k-3)/3 \geq 4k^2/15$. The remainder is at most $648k^4/(625p_0^2)$, which is less than $|\mathbf{b}_k|/(4p_0)$. It follows that

$$|q_k| \geq \frac{k^2}{5p_0}, \quad |q_k| < \frac{1}{32}, \quad |M_{k+1}| \leq \frac{10p_0}{k^2}.$$

For the last estimate use $|q_{k-1}| < 17/16$ and $(2p_0 + k - 1)/R_0 < 2$. Lemma 4.8 now bounds the inverse distance by

$$\frac{|M_{k+1}| + 2}{2(R_0 - 1)} \leq \frac{27}{8k^2} \leq \frac{27}{1800} < \frac{1}{64}$$

in the resonant range. In the nonresonant range it gives a smaller bound for $p_0 \geq 10^6$.

The remaining resonant index is $k = 9$. Equation (4.15) gives $|q_9| \geq (18 - 3661/p_0)/p_0 > 17.99/p_0$ and $|q_8| \leq 1 + 44/p_0 + 2501/p_0^2 < 1.001$. Thus $|M_{10}| < p_0/17 + 2$, and its inverse distance is less than $(p_0/17 + 4)/(4p_0 - 2) < 1/64$. For $\ell = 2$ use $q_1 = -1, q_0 = 0$, giving $|M_2| < 1$. For $\ell = 6$ the exact values $q_4 = (p_0 + 1)/s_0$ and $q_5 = (p_0 + 4)/(p_0 + 2)$ give $|M_6| < 2$. For $\ell = 0$ one has $M_0 = 0$ by (2.16). These exhaust the low even degrees except $\ell = 4$. Its remaining radial states are separated by Lemma 3.1.

To pass to p , continue each radial zero in its order. Both relevant orders exceed eight along the interval, so (3.4) bounds the difference of any two such order derivatives by less than $3/5$. Consequently $j_{p+\ell-1,k} - R(p)$ changes by less than $93/(50p_0) < 2/p_0$. If its squared-energy offset at p had absolute value at most 32, then $R(p) > 2p_0$ and $j_{p+\ell-1,k} + R(p) \geq (39/10)p_0$ for large p_0 . Its radius offset would have absolute value below $9/p_0$, and at p_0 below $11/p_0$. Its squared offset at the crossing would then be less than

$$\frac{11}{p_0} \left(4p_0 + 6 + \frac{11}{p_0} \right) < 55,$$

contradicting the bound 64. The other radial seed states remain separated by the spacing and the same Lipschitz estimate. $\square$

5. HALF-INTEGER ORDERS AND THE EUCLIDEAN CHAIN

The order parameter belongs to $\frac{1}{2}\mathbb{Z}$ at a Euclidean input. The block dimensions a, b and the angular and radial degree indices are integers; the Jacobi parameters and the Bessel orders may be half-integers.

Lemma 5.1 (Half-integer non-coincidence). *For $\nu \in \{1/2, 3/2, \dots\}$, J_ν has no positive algebraic zero. Two distinct positive half-integer orders have no common positive zero. In particular J_p and J_{p+3} have no common positive zero for $p \in \mathbb{Z} + 1/2$, $p > 0$.*

Proof. Write $\nu = l + 1/2$, $l \geq 0$. The half-order identities and recurrence [1, §§10.6, 10.16] give

$$\sqrt{\pi x/2} J_{l+1/2}(x) = F_l(x) \sin x + G_l(x) \cos x,$$

where $F_l, G_l \in \mathbb{Q}[x^{-1}]$ and

$$(F_{-1}, G_{-1}) = (0, 1), \quad (F_0, G_0) = (1, 0), \quad (F_{l+1}, G_{l+1}) = \frac{2l+1}{x}(F_l, G_l) - (F_{l-1}, G_{l-1}).$$

The determinant of the successive rows $(l, l-1)$ is one: the recurrence preserves it and the initial determinant is one. Thus the two entries in row l never vanish together at $x \neq 0$. If $x > 0$ were algebraic and a zero, the real algebraic numbers $F_l(x), G_l(x)$ would satisfy

$$e^{2ix} = -\frac{G_l(x) + iF_l(x)}{G_l(x) - iF_l(x)}.$$

The denominator is nonzero. This contradicts the Hermite–Lindemann theorem, which states that e^z is transcendental for each nonzero algebraic complex number z [3, §1.2].

At a zero x of J_ν , $J_{\nu+1}(x) \neq 0$: otherwise the derivative recurrence and uniqueness for Bessel's equation give $J_\nu = 0$ identically. Define the Lommel polynomials by

$$\mathbb{L}_{0,\mu}^{\text{Lom}}(x) = 1, \quad \mathbb{L}_{1,\mu}^{\text{Lom}}(x) = 2\mu/x, \quad \mathbb{L}_{k+1,\mu}^{\text{Lom}}(x) = \frac{2(\mu+k)}{x} \mathbb{L}_{k,\mu}^{\text{Lom}}(x) - \mathbb{L}_{k-1,\mu}^{\text{Lom}}(x).$$

The Bessel recurrence implies $J_{\nu+k}(x) = \mathbb{L}_{k-1,\nu+1}^{\text{Lom}}(x) J_{\nu+1}(x)$ for $k \geq 1$; see [2]. For $k \geq 2$ this is a nonzero polynomial in x^{-1} with rational coefficients and leading coefficient $\prod_{h=1}^{k-1} 2(\nu+h)$. A common zero would therefore make x algebraic, contrary to the first assertion. For $k = 1$ the non-coincidence was already proved. For the primary pair the identity specializes to

$$J_{p+3}(x) = \left\{ \frac{4(p+1)(p+2)}{x^2} - 1 \right\} J_{p+1}(x) \quad \text{when } J_p(x) = 0.$$

□

Lemma 5.2 (Shifted congruence approximation). *The set $\mathcal{P}_{\text{bd}}$ has infinitely many members in each of $\mathbb{Z}$ and $\mathbb{Z} + 1/2$. For the pairs produced by Lemma 4.1 applied to $\alpha_{\text{ar}} + 1$, set $\tilde{P} = P - Q$ and $p = (\tilde{P} - 3)/2$. Along this particular half-integer selection,*

$$\limsup \hat{p}_{(Q-1)/4} |p - \hat{p}_{(Q-1)/4}| \leq \frac{9}{8} \alpha_{\text{ar}} + d_{\text{ar}} < 31/10.$$

Proof. Both P, Q are odd, so $\tilde{P}$ is even and $2p$ is odd. Moreover $Q \equiv 1 \pmod{4}$ and

$$Q|Q\alpha_{\text{ar}} - \tilde{P}| = Q|Q(\alpha_{\text{ar}} + 1) - P| < 9/2.$$

The denominators tend to infinity. Put $m = (Q - 1)/4$ and use (4.6); multiplication by $\hat{p}_m$ gives

$$\hat{p}_m |p - \hat{p}_m| \leq \frac{1}{2} \hat{p}_m |Q\alpha_{\text{ar}} - \tilde{P}| + d_{\text{ar}} \hat{p}_m / A_m + O(A_m^{-1}).$$

Now $\hat{p}_m / Q \rightarrow \alpha_{\text{ar}}/2$ and $\hat{p}_m / A_m \rightarrow 1$. The rational comparison in (4.9) proves the bound. Positivity of p and $m \geq m_0$ hold eventually. Lemma 5.1 excludes zero distance. The integer selection is Theorem 4.5. □

Proposition 5.3 (The half-lattice analytic chain). *The analytic statements listed in the table below that use a near-primary input hold for $p \in \mathcal{P}_{\text{bd}}$, uniformly for $0 < \tau \leq 20$ and $r \in [1/4, 3/8]$, with integer a, b for each Euclidean conclusion. The remaining analytic statements hold under their displayed real parameter hypotheses.*

Proof. The chain is given below, with its integer objects and the identities that determine the permissible parameters. Each row is applied under the hypotheses of its indicated statements.

Statements	Parameter dependence
Lemma 2.1	The coordinate Laplacian and beta measure use integer a, b and $p = (a + b)/2$. The invariant angular multiplicity is one.
Lemma 2.2	Rodrigues integration and rising factorials use positive real α_b, β_b ; j is the integer polynomial degree.
Lemma 2.3	The endpoint comparison needs $\beta_b - 1 \geq \alpha_b - 1 \geq 0$. Bernstein applies on great circles in every integer dimension.
Lemma 2.4	$t^j G_j$ is a polynomial in q_X, q_Y . The regular Bessel mode is this polynomial times an entire power series in ϱ^2 , including the origin and axes.

Statements	Parameter dependence
Proposition 2.5	The radial order $p - 1 + 2j > 1$ is real; its singular solution is outside the Friedrichs form domain. Integer blocks identify the Euclidean invariant realization.
Lemma 2.6	The radial parameter $2j + p - 1$ is real. Rodrigues and generalized binomial identities require only integer polynomial indices.
Lemmas 3.1, 3.2	The radial ODE, Poincaré inequality and Watson integral have their stated real-order hypotheses.
Lemmas 3.3, 3.4	The outgoing Volterra equation is for real $\nu > 0$; only its finite expansion order is integral.
Lemma 3.5	The recurrence shifts the real Bessel order by two. Its frequency spacing concerns all radial indices.
Lemma 3.6, Theorem 3.8	The lattice coordinates are the integer angular index and integer phase index; their offsets may depend on real p .
Lemma 4.1, Theorem 4.5, Lemma 5.2	Apply the same irrational approximation to α_{ar} or $\alpha_{\text{ar}} + 1$. The latter changes P to $P - Q$ and preserves the error product.
Proposition 4.2	The root expansion and uniqueness use real-order continuation. Bourget and Lemma 5.1 exclude both half-lattice coincidences.
Lemma 4.7	The exact rational trace identity and the real-order derivative give the same multiplicative detuning relation.
Lemmas 4.8, 4.9	The finite integer order shifts leave the base order real. The period-six sequence comes from the limiting recurrence.
Proposition 4.10	All relevant zeros are continued on a real interval in their orders; the selected endpoint may be a half-integer.
Lemma 7.1	The entry and derivative formulas are rational identities for positive real beta parameters with sum p .
Lemma 7.2	The coefficient certificates hold for all real $p \geq 64$. The recurrence-ratio bounds use $\alpha_b, \beta_b \geq 16$.
Proposition 7.3	Axis cutoffs use $\alpha_b, \beta_b > 1$ and the origin cutoff $p > 1$. Integer blocks identify the weighted and Euclidean closures.
Lemmas 7.4, 7.5	The band estimates and Green identity use real Jacobi parameters and radial Bessel norms; both angular parities are retained.
Lemma 7.6	The complementary inverse is radial spectral calculus and a relative-form Neumann series on every complementary radial state.
Lemma 7.7	The finite recurrence gives seed offset $-\theta/(2p)$ and index-five offset -72 in either ambient parity.
Lemma 7.8, Theorem 7.9	Finite Riesz reduction and the weighted derivative margin apply on the common real-parameter form domain, in the stated narrow band.
Theorem 8.3	Integers in $[p/2, 3p/4]$ number at least $p/4 - 1$ and have ς spacing at least $1/(8p)$. $2p \in \mathbb{Z}$ ensures integer $b = 2p - a$.
Theorem 6.2	The finite recursion uses $\varepsilon = 1/p$, fixed degree indices, the divisor $-1/8$, positive c_j and powers of $r(1 - r)$.
Lemma 6.3, Corollary 6.4	The analytic coefficient transfer uses any positive measure and $0 < \mathbf{H}_j(1) = (\beta_b)_j / (p + j - 1)_j \leq 1$.

Statements	Parameter dependence
Proposition 6.5	Indexwise min–max and radial dilation hold in every integer dimension; the normalized error is $O_N(\tau p^{-4})$.
Definition 9.1, Lemma 9.3, Proposition 9.4	The grade exponent is real. Gasper uses real Jacobi parameters; the integer raising/lowering shifts are angular degree differences.
Lemma 9.5, Proposition 9.6	The radial columns are rational in the real parameter β ; the separate harmonic source column remains included.
Lemmas 9.7, 9.8	Positive radial stencils and their coupling use real $\beta \geq 0$; the smoothing gain acts on the clamped layers $s \geq 1$.
Lemma 9.9, Proposition 9.10	The solid product identity uses integer solid degrees and real p ; the grouped high-shape estimates have clamped source domains.
Lemmas 9.11, 9.12	The two lift columns and derivative columns use real radial parameters; their reference normal factor is $R = O(p)$.
Lemma 10.3, Proposition 10.4	Rodrigues–Sonine, Lommel–Parseval and the positive Bessel ratio hold for real orders. The head cutoff satisfies $\nu + 1 + 2S \geq 2R$; composition uses the finite list of known centre modes.
Lemma 10.1, Proposition 10.2	The Jacobian is $q^{2p-1}d = q^{n-1}d$ at integer n . The nonlinear map has clamped source domain $\mathcal{G}_p \times \mathcal{E}_1$.
Proposition 10.5, Theorem 10.6	The double lift, normal Hessian and material identities use the same real parameters; the L^2 completion is at fixed integer n .
Lemmas 11.2, 11.4, 11.5, Theorem 11.8	The radial Hilbert blocks use real weights; density uses real binomial powers. Hilbert and coefficient estimates retain their respective graph domains and all radial layers.
Theorem 12.1, Corollary 12.2	The integer phase lattice and real turning-point ODE give the same local count and cluster cardinality.
Lemma 12.3, Proposition 13.4	The cofactor argument is finite linear algebra, followed by the radial graph reconstruction in both spaces.
Lemma 13.5, Theorem 13.6	The head has at most $K_{\text{head}}p + 1$ radial indices; the infinite tail uses the same all-layer Poisson columns.
Theorem 14.1, Corollary 14.2	The contraction uses the displayed powers of p, τ in the same real Banach spaces; its centre order is fixed first.
Lemmas 15.1, 15.2, Propositions 15.3, 15.5	The sphere and elliptic arguments require integer n, a, b . The regular solid expansion covers the axes and origin in each parity.

The only lattice changes are thus the non-coincidence theorem, the arithmetic choice of p , the integer split grid, and the integer block realization. All estimates in the table retain their stated constants on this half-lattice. $\square$

6. THE QUARTIC DOMAIN AND FINITE-ORDER CENTRES

In this section r ranges over the fixed compact interval $[1/4, 3/8]$. All finite-order constants are uniform on this interval and for $|\theta| \leq 20$, including $\theta = 0$ when analytic extensions are discussed. The accuracy order is always fixed before p tends to infinity.

6.1. The quartic profile. The monic degree-two angular polynomial is

$$(6.1) \quad \mathbf{H}_2(x) = x^2 - \frac{2(\alpha_b + 1)}{p + 2}x + \frac{\alpha_b(\alpha_b + 1)}{(p + 1)(p + 2)}.$$

Its homogeneous spherical degree is four. Define its self-coupling by

$$(6.2) \quad g_2 = [\mathbf{H}_2]\mathbf{H}_2^2 = \frac{\int \mathbf{H}_2^3 d\mu_r}{\int \mathbf{H}_2^2 d\mu_r}.$$

The bracket denotes the coefficient in the monic orthogonal expansion.

Lemma 6.1 (Quartic normalization). *Put $\varsigma = (r - 1/2)^2 \in [1/64, 1/16]$. Then*

$$(6.3) \quad g_2 = \frac{p - 1}{(p + 1)(p + 5)} - \frac{4p^2(p^2 - 9p - 16)\varsigma}{(p + 1)(p + 2)^2(p + 4)(p + 5)}.$$

For $p \geq 64$,

$$(6.4) \quad \frac{9}{16} < pg_2 < 1, \quad \|\mathbf{H}_2\|_\infty < \frac{9}{16}, \quad \|T_\varsigma\|_\infty < 1, \quad T_\varsigma(x) = \frac{\mathbf{H}_2(x)}{pg_2}.$$

Also $g_2 = 4r(1 - r)p^{-1} + O(p^{-2})$ uniformly in r .

Proof. The beta moments are $\int x^h d\mu_r = (\alpha_b)_h / (p)_h$. They give the two orthogonality conditions for (6.1). Multiplying the three-term recurrence twice gives

$$\int \mathbf{H}_2 P_j^2 d\mu_r = \mathbf{a}_j^2 + \mathbf{a}_{j+1}^2 + \mathbf{b}_j^2 - \frac{2(\alpha_b + 1)}{p + 2} \mathbf{b}_j + \frac{\alpha_b(\alpha_b + 1)}{(p + 1)(p + 2)}.$$

At $j = 2$, substitute (2.9)–(2.10) and $\alpha_b = p/2 + p(r - 1/2)$. The terms linear in $r - 1/2$ cancel; the constant term is $(p - 1)/((p + 1)(p + 5))$, and the coefficient of $(r - 1/2)^2$ is the second coefficient in (6.3). Since P_2 is a multiple of $\mathbf{H}_2$, this gives g_2 .

Write $g_2 = c_2^{\text{diag}} + d_2^{\text{diag}}\varsigma$ with the coefficients in (6.3). For $p \geq 64$, $d_2^{\text{diag}} < 0$, $|pd_2^{\text{diag}}| < 4$, and $pc_2^{\text{diag}} \geq 4032/4485 > 13/16$. The last inequality persists for larger p because the derivative of $p(p - 1)/((p + 1)(p + 5))$ has numerator $7p^2 + 10p - 5 > 0$. Thus $pg_2 > 13/16 - 4/16 = 9/16$, while $pg_2 < pc_2^{\text{diag}} < 1$. The two positive endpoint values of $\mathbf{H}_2$ are less than $9/16$, using $\alpha_b, \beta_b \leq 3p/4$. Its negative minimum has magnitude $(\alpha_b + 1)(\beta_b + 1)/((p + 1)(p + 2)^2) \leq 1/(4(p + 1))$. This proves (6.4). Expansion of the rational formula gives $1 - 4\varsigma = 4r(1 - r)$ as its leading coefficient. $\square$

The physical quartic radius and its normalized version are

$$(6.5) \quad R + h_q(x), \quad h_q = -\frac{2\chi}{g_2}\mathbf{H}_2, \quad 1 + f_q(x) = 1 - \frac{\theta}{2R^2}T_\varsigma(x).$$

We denote the corresponding physical domain by Ω_q . For real block parameters the same formula denotes its weighted meridian domain. The scalar θ and the radius R stay fixed when ς varies.

6.2. An analytic finite-order construction.

Theorem 6.2 (Finite-order centres). *Fix an integer $N \geq 4$. For every prescribed finite radius $R_{\text{an}} > 0$, put $U = \{x \in \mathbb{C} : |x| \leq R_{\text{an}}\}$. There exist $C_{N,U}, P_{N,U} < \infty$ such that the following assertion holds. Suppose $p \in \frac{1}{2}\mathbb{Z}$, $p \geq P_{N,U}$, $J_p(R) = 0$, $R \in [2p - 1, 2p + 4]$, and $0 < |\theta| \leq 20$ with (4.10). For every $r \in [1/4, 3/8]$ one can choose a polynomial $h_N(x)$ and a finite sum u_N of regular Helmholtz modes. For real block parameters the equation means the weighted operator*

(2.3); at integer blocks it is the Euclidean equation. These choices satisfy $(\Delta + 1)u_N = 0$ exactly and

$$(6.6) \quad \sup_{x \in U} (|u_N(R + h_N(x), x) - 1| + |\partial_\varrho u_N(R + h_N(x), x)|) \leq C_{N,U} \theta^2 p^{-N}.$$

The number and degrees of the modes depend on N , not on p, r, θ . For every fixed polynomial coefficient norm and every fixed derivative order k on U , with constants $C_{N,U,k}$ and threshold $P_{N,U,k}$,

$$(6.7) \quad h_N = -\frac{2\chi}{g_2} H_2 + O_{N,U}(|\theta|p^{-3}),$$

$$(6.8) \quad [H_2]h_N = -\frac{\theta}{16r(1-r)p} + O_{N,U}(|\theta|p^{-2}).$$

The radius R_{an} can be chosen arbitrarily large before the constants and threshold are fixed. In particular the statement applies as follows: fix $\rho' > 2$ and $d > \rho'$, and then choose one $R_{\text{an}} > 1 + d$.

Proof. We construct a finite jet with more terms than the requested accuracy. Put $\varepsilon = 1/p$, write $R = \rho/\varepsilon$, and set $Z = \rho^2$ and $S_0 = 4(1 + 3\varepsilon + 2\varepsilon^2)$. Equation (4.13) is equivalent to

$$(6.9) \quad 2Z(Z - S_0) + \theta\varepsilon^2(1 + \varepsilon)(2Z - S_0) = 0.$$

At $(\varepsilon, Z) = (0, 4)$ its derivative in Z is eight. The analytic implicit-function theorem gives a unique root near four, uniformly for $|\theta| \leq 21$. The supplied radius satisfies $(R/p)^2 \rightarrow 4$ and, by (4.13), equation (6.9) with $Z = (R/p)^2$. Thus Z is this root for sufficiently large p . The other quadratic root is $O(\theta\varepsilon^2)$. Take the positive square root near two. Then

$$(6.10) \quad \rho = 2 + 3\varepsilon + O(\varepsilon^2), \quad \chi = \frac{\theta\varepsilon^2}{4\rho} = \frac{1}{8}\theta\varepsilon^2 + O(\theta\varepsilon^3).$$

Here and below analyticity means on an open complex neighbourhood of the stated compact real parameter set.

For finitely many angular indices, the boundary traces of regular Bessel modes are computed from

$$(6.11) \quad q_{-1} = 1, \quad q_0 = 0, \quad q_{k+1} = \frac{2(1 + k\varepsilon)}{\rho} q_k - q_{k-1}.$$

This finite recurrence is analytic in ε, θ . Its limiting zero indices give the resonant angular set $\mathcal{I}_{\text{res}} = \{2, 5, 8, \dots\}$. For $j \notin \mathcal{I}_{\text{res}}$ normalize the mode f_j by $f_j(R) = 1$; for $j \in \mathcal{I}_{\text{res}}$ normalize it by $f'_j(R) = 1$. The latter convention gives $f_2(R) = \chi$. For $j \geq 5$ in $\mathcal{I}_{\text{res}}$,

$$(6.12) \quad f_j(R) = c_j\varepsilon + O_j(\varepsilon^2), \quad c_j = \frac{(2j-1)(2j-4)}{3} > 0.$$

Indeed the first correction of (6.11) at $k = 3h$ is $(-1)^h 3h(h-1)\varepsilon$, while the corresponding normal trace tends to $(-1)^h$. Put $3h = 2j - 1$ to obtain (6.12). The change in ρ due to θ starts at order ε^2 , so this coefficient is independent of θ . The nonresonant normalization denominators tend to 1 or -1 and are units in the analytic coefficient ring.

All collar derivatives are supplied by

$$(6.13) \quad f''_j + \frac{(2-\varepsilon)}{\rho + \varepsilon y} f'_j + \left(1 - \frac{2j(2j\varepsilon^2 + 2\varepsilon - 2\varepsilon^2)}{(\rho + \varepsilon y)^2}\right) f_j = 0, \quad y = \varrho - R.$$

Its coefficients are analytic near $\varepsilon = 0$ on a fixed collar. The ball mode has traces $u_0(R) = 1$, $u'_0(R) = 0$, $u''_0(R) = -1$. The collar construction is a local analytic family. At the actual

inputs $J_p(R) = 0$ its traces are precisely those of the regular Bessel modes; uniqueness of the ordinary differential equation identifies them on the collar and then throughout their regular continuation.

We next record the angular filtration that makes the recursion finite. For fixed i, j write

$$(6.14) \quad \mathbf{H}_i \mathbf{H}_j = \sum_{k \leq i+j} \mathbf{c}_{ij}^k \mathbf{H}_k, \quad \mathbf{c}_{ij}^k = O\left(\varepsilon^{\lceil (i+j-k)/2 \rceil}\right).$$

To prove this, use the monic recurrence $\mathbf{H}_{j+1} = (x - \mathbf{b}_j) \mathbf{H}_j - \mathbf{a}_j^2 \mathbf{H}_{j-1}$. For fixed indices (2.9)–(2.10) give $\mathbf{b}_j - r = O(\varepsilon)$ and $\mathbf{a}_j^2 = O(\varepsilon)$. Induction shows that the coefficient of $(x - r)^k$ in $\mathbf{H}_j$ has valuation at least $\lceil (j - k)/2 \rceil$. Descending elimination proves the same valuation for the inverse monic triangular connection. Multiplying and composing these two connections proves (6.14), since a sum of ceilings is at least the ceiling of the sum. In particular,

$$(6.15) \quad \mathbf{H}_2^2 = \mathbf{H}_4 + O(\varepsilon) \mathbf{H}_3 + g_2 \mathbf{H}_2 + O(\varepsilon^2) \mathbf{H}_1 + O(\varepsilon^2) \mathbf{H}_0, \quad g_2 = 4r(1 - r)\varepsilon + O(\varepsilon^2).$$

Write the formal field and boundary as

$$u = u_0 + \sum_j a_j f_j \mathbf{H}_j, \quad \varrho = R + h(x).$$

Every coefficient of ε is a finite polynomial in x . Eliminate h from the radial Neumann condition first. Its derivative in h at zero is $u_0''(R) = -1$, so at each order the new coefficient of h is determined by the already known coefficients and the new field coefficients, using the unit divisor $u_0''(R) = -1$. In the Dirichlet equation eliminate a_j for $j \notin \mathcal{I}_{\text{res}}$. Its diagonal linear coefficient is $f_j(R) = 1$. Thus its new coefficients are also determined by unit divisions. This elimination includes the $j = 0$ field coefficient.

Parametrize the remaining coefficients by

$$(6.16) \quad a_2 = \theta \varepsilon \mathfrak{A}, \quad a_j = \theta^2 \varepsilon^3 Z_j \quad (j \in \mathcal{I}_{\text{res}}, j \geq 5).$$

After the two eliminations the residual Dirichlet coefficients F_j have the form

$$(6.17) \quad \frac{F_2}{\theta^2 \varepsilon^3} = \frac{\mathfrak{A}}{8} + 2r(1 - r)\mathfrak{A}^2 + O(\varepsilon),$$

$$(6.18) \quad \frac{F_j}{\theta^2 \varepsilon^4} = c_j Z_j + B_j(\mathfrak{A}, \theta, r) + O(\varepsilon), \quad j \in \mathcal{I}_{\text{res}}, j \geq 5.$$

Here B_j is a finite polynomial and is zero except possibly at $j = 5, 8$. These divisions are identities of formal series, including at $\theta = 0$.

The linear seed term is $a_2 \chi = \theta^2 \varepsilon^3 \mathfrak{A}/8 + O(\theta^2 \varepsilon^4)$. Eliminating h gives quadratic Dirichlet term $a_2^2 \mathbf{H}_2^2/2$; its seed coefficient is the second term of (6.17) by (6.15). A pure cubic term has angular degree at most six; reaching index two loses four degrees and contributes two further powers of ε . The leading nonresonant coefficient is $a_4 = -a_2^2/2 + O(\varepsilon^3)$, and its product with the seed has the same loss when projected to index two. Thus neither contributes to the leading seed equation. Quadratic terms have no nonseed resonant index. Cubic terms first reach index five after losing one angular degree, which supplies one extra ε . At total order four the only possible new resonant indices are five and eight. A new Z_j contributes linearly through (6.12), giving $c_j Z_j$. Its product with the seed has top index $j + 2$, which is nonresonant; reaching a resonant index at most j loses at least two degrees and therefore at least one extra ε . Contributions through a newly eliminated nonresonant coefficient have the same extra factor after its unit division. At order zero in (6.18), Z_j enters linearly as $c_j Z_j$. Every other term has at least two amplitudes or the factor $a_2 \chi$, so it is divisible by θ^2 .

Choose the nonzero root $\mathfrak{A}_0 = -1/(16r(1-r))$ of (6.17). The derivative of its left side with respect to $\mathfrak{A}$ at this root is $-1/8$. After choosing $\mathfrak{A}_0$, determine $Z_{j,0} = -B_j(\mathfrak{A}_0, \theta, r)/c_j$; these have finite support. Work in the ring of formal ε series whose coefficients are polynomials in x , with coefficients analytic in r, θ on the indicated compact sets. A series is required to have a finite polynomial coefficient at every order. For a fixed list of resonant amplitude coefficients the two unit eliminations have linearization, in the variables (h, a_{nr}) ,

$$(\dot{h}, \dot{a}_{\text{nr}}) \mapsto \left(-\dot{h} + \sum_{j \notin \mathcal{I}_{\text{res}}} f'_j(R)|_{\varepsilon=0} \dot{a}_j H_j|_{\varepsilon=0}, \dot{a}_{\text{nr}} \right).$$

Its inverse is triangular and sends a finite polynomial forcing to finite polynomial coefficients. At each order, the remaining terms involve only coefficients at lower orders. This proves existence and uniqueness of those unit eliminations by ordinary coefficient induction. It also proves preservation of finite support: multiplication and the monic connection use only finite polynomials, and the inverse displayed here adds no angular indices not present in its forcing. Dependence of the exact collar traces on ε contributes only previously determined lower-order coefficients in this calculation.

After these eliminations let $\mathcal{F}_2 = F_2/(\theta^2 \varepsilon^3)$ and $\mathcal{F}_j = F_j/(\theta^2 \varepsilon^4)$ for nonseed resonant j . Their constant terms are the right sides of (6.17)–(6.18) without the remainders. At $(\mathfrak{A}, Z) = (\mathfrak{A}_0, Z_0)$ their leading Jacobian is

$$(6.19) \quad \mathcal{J}_0(\dot{\mathfrak{A}}, \dot{Z}) = \left(-\frac{1}{8} \dot{\mathfrak{A}}, \{ \partial_{\mathfrak{A}} B_j(\mathfrak{A}_0, \theta, r) \dot{\mathfrak{A}} + c_j \dot{Z}_j \}_{j \geq 5, j \in \mathcal{I}_{\text{res}}} \right).$$

The off-diagonal column in this matrix has finite support, since $B_j = 0$ outside $j = 5, 8$. Each c_j is positive and fixed. Thus its inverse preserves finite support, using only the fixed nonzero divisors $-1/8$ and c_j .

Suppose all coefficients of $\mathfrak{A}, Z$ below order $v \geq 1$ have already been selected and the reduced equations vanish below that order. Substitute those truncated series with the unknown order- v coefficients set to zero, and denote the coefficients of ε^v in $\mathcal{F}_2, \mathcal{F}_j$ by $e_{2,v}, e_{j,v}$. They are explicitly defined by finite coefficient extraction from the collar Taylor formula, the polynomial product connection, and the two unit eliminations. Only finitely many $e_{j,v}$ are nonzero. The new coefficients occur only through (6.19): every other occurrence is multiplied by an additional ε , or by a coefficient of positive order of $\mathfrak{A} - \mathfrak{A}_0$ or $Z - Z_0$. Therefore the unique update is

$$(6.20) \quad \mathfrak{A}_v = 8e_{2,v}, \quad Z_{j,v} = -\frac{e_{j,v} + \partial_{\mathfrak{A}} B_j(\mathfrak{A}_0, \theta, r) \mathfrak{A}_v}{c_j}.$$

Set the other coefficients to zero. The update is finite, cancels exactly the order- v coefficient, and leaves every lower order unchanged. Recompute the unit-eliminated coefficients to that order. This completes an induction producing every finite jet.

All operations in (6.20) are additions, multiplications, coefficient extractions and divisions by the nonzero quantities already listed. The resulting coefficients are analytic in a common neighbourhood of the compact real parameter set for each finite jet. The neighbourhood may shrink with its order. At a fixed order only finitely many collar normalizations and monic connections occur, so a positive common neighbourhood exists for each finite order.

Through order ε^4 the seed equation is the expansion of $\chi a_2 + g_2 a_2^2/2$. All cubic seed projections start at order ε^5 by the degree loss just described. Hence the chosen root agrees with $-2\chi/g_2$ through order ε^2 . The order- $\theta^2 \varepsilon^2$ shape coefficient vanishes by the following cancellation. At that order the index-four amplitude is $-a_2^2/2$, whose physical normal derivative tends to -1 . The quadratic Neumann correction is $-(2p-1)a_2^2 H_2^2/(2R)$ and $(2p-1)/R \rightarrow 1$.

Their index-four contributions cancel. All lower indices of H_2^2 gain at least one ε , and the other resonant amplitudes start at order $\theta^2\varepsilon^3$. It follows that $h - a_2H_2 = O(\theta^2\varepsilon^3)$ at these orders, proving (6.7) for any sufficiently long finite jet with an $O_N(|\theta|\varepsilon^3)$ remainder. A finite truncation of the linear-in- θ part is included in this latter bound. Inserting (6.10) and (6.3) gives (6.8).

Choose the recursion order greater than $N + 4$ and retain its finite set of modes. Evaluate the truncated coefficients in the exact finite- p polynomials H_j and exact normalized radial functions. Restoring the factors $\theta^2\varepsilon^3$ and $\theta^2\varepsilon^4$ in (6.17) and (6.18) recovers the original resonant Dirichlet coefficients. The two unit eliminations preserve the chosen Taylor jet of the Neumann and nonresonant Dirichlet equations. Thus a jet of order greater than $N + 4$ makes every original defect coefficient through order N vanish. Choose the truncations so that $h - a_2H_2$ has a factor θ^2 , using the identical truncated a_2 in the field and the boundary. The higher powers of θ already present in a_2 are retained in this convention. For x in any fixed bounded complex neighbourhood $U' \supset U$, this polynomial has size $O_{N,U'}(|\theta|\varepsilon)$. For sufficiently small $|\varepsilon|$ its values lie in a fixed collar, where the integral equation for (6.13) gives jointly analytic solutions with uniform bounds. The denominators $\rho + \varepsilon y$ remain away from zero, and only finitely many mode normalizations occur. Thus both boundary defects are analytic on a common neighbourhood of the compact parameter set and U' . Their Taylor coefficients in ε vanish through order N by the finite induction, as polynomial identities in x .

Both defects have an analytic factor θ^2 . At $\theta = 0$ the field and domain are the ball. The linear Neumann defect is $-\partial_\theta h + (\partial_\theta a_2)H_2 = 0$ by the identical seed choice; the linear Dirichlet defect is zero since $u'_0(R) = 0$ and $f_2(R) = \chi$ itself has a factor θ . All other terms have at least two such factors. Divide the analytic defects by θ^2 and apply Taylor's formula with remainder in ε on a smaller fixed complex disk. Compactness in r, θ, x bounds the remainder by $C_{N,U}|\varepsilon|^N$. Cauchy's formula on U' gives every stipulated fixed angular derivative bound. This proves (6.6) and the remainder statements. At actual integer blocks the finite regular field is analytic across axes and origin by Lemma 2.4. $\square$

Lemma 6.3 (Analytic defects give weighted coefficients). *Let μ be any positive measure on $[0, 1]$ with infinite support, H_j its monic orthogonal polynomials, and suppose $0 < H_j(1) \leq 1$. For a complex-valued function F analytic on a neighbourhood of $U_d = \{z : \text{dist}(z, [0, 1]) \leq d\}$, its coefficients in $G_j = H_j/H_j(1)$ satisfy*

$$(6.21) \quad |F_j| \leq d^{-j} \sup_{U_d} |F|.$$

Consequently, for each fixed $k \geq 0$ and $1 < \rho' < d$,

$$(6.22) \quad \sum_{j \geq 0} (1+j)^k (\rho')^j |F_j| \leq \left(\sum_{j \geq 0} (1+j)^k (\rho'/d)^j \right) \sup_{U_d} |F|.$$

The constants do not depend on the measure. These assertions apply to all the beta measures used here.

Proof. For $j = 0$ the coefficient is the normalized integral of F , so (6.21) follows directly. Let $j \geq 1$. The zeros $x_1, \dots, x_j$ of a monic orthogonal polynomial are simple and lie in $(0, 1)$. Interpolate F at those nodes by a polynomial L of degree at most $j - 1$. The divided-difference identity gives

$$F(x) - L(x) = F[x_1, \dots, x_j, x]H_j(x).$$

The integral formula for divided differences, valid also for complex-valued functions, gives

$$|F[x_1, \dots, x_j, x]| \leq \frac{1}{j!} \sup_{[0,1]} |F^{(j)}|.$$

Orthogonality to L therefore gives

$$\frac{|\langle F, H_j \rangle|}{\|H_j\|_{L^2(\mu)}^2} \leq \frac{\sup_{[0,1]} |F^{(j)}|}{j!} \leq d^{-j} \sup_{U_d} |F|,$$

where the last inequality is Cauchy's estimate on each radius- d disk centred on the interval. Multiplication by $H_j(1) \leq 1$ proves (6.21); summation gives (6.22). For our measure (2.6) shows $H_j(1) = (\beta_b)_j / (p + j - 1)_j \leq 1$. $\square$

Corollary 6.4 (The coefficient form of the centre defects). *Fix N, k and $\rho' > 1$. The centre in Theorem 6.2 can be chosen so that its physical Dirichlet and radial Neumann boundary defects D_N, N_N^{phys} , expanded in G_j , obey*

$$(6.23) \quad \sum_j (1+j)^k (\rho')^j (|D_{N,j}| + |N_{N,j}^{\text{phys}}|) \leq C_{N,k,\rho'} \theta^2 p^{-N}.$$

The constants and threshold are uniform in r and $0 < |\theta| \leq 20$.

Proof. Choose $d > \rho'$ and then one fixed $R_{\text{an}} > 1 + d$. The set U_d lies in the disc of radius R_{an} . Use the analytic remainder established by the finite-jet construction in Theorem 6.2 on that disc, and apply Lemma 6.3 separately to the two defects. The radius choice changes only the constants and threshold for this fixed jet. For the Newton application choose $\rho' > 2$ once. $\square$

Proposition 6.5 (Transfer of an invariant spectral gap). *Fix the centre order N and suppose the actual quartic domain has invariant Dirichlet gap at least $c_q \tau p^{-5/3}$ about R^2 in unit-ball normalization, with fixed $c_q > 0$. Then, for all sufficiently large p depending on N, c_q , the centre domain Ω_N/R has gap at least $(c_q/2) \tau p^{-5/3}$. The threshold is independent of the admissible split and of $0 < \tau \leq 20$.*

Proof. By (6.7), the two normalized radius functions differ by $O_N(\tau p^{-4})$ and are both $1 + O_N(\tau p^{-2})$. Thus, for $\delta \leq C_N \tau p^{-4}$,

$$(1 - \delta) \Omega_q/R \subset \Omega_N/R \subset (1 + \delta) \Omega_q/R.$$

Zero extension preserves the invariant sector, so the min-max principle gives, for every ordered index i ,

$$(1 + \delta)^{-2} \lambda_i(\Omega_q/R) \leq \lambda_i(\Omega_N/R) \leq (1 - \delta)^{-2} \lambda_i(\Omega_q/R).$$

For any eigenvalue within a fixed window of R^2 , its counterpart is $O(p^2)$, and the difference is bounded by $C_N \tau p^{-2}$. This is smaller than $(c_q/2) \tau p^{-5/3}$ once $p^{1/3} > 2C_N/c_q$. An eigenvalue inside the proposed centre gap would then put its ordered quartic counterpart strictly inside the original gap, a contradiction. The factor τ cancels from the required threshold. $\square$

7. THE QUARTIC SPECTRAL FLOW

Throughout this section $p \geq 64$, $\varsigma \in I = [1/64, 1/16]$, $r = 1/2 - \sqrt{\varsigma}$, $R \in [2p - 1, 2p + 4]$, and $0 < \tau = |\theta| \leq 20$. Unless indicated otherwise, constants are absolute and independent of p, ς, R, θ . The fixed window used for the flow argument is W_{flow} . The coefficient inverse uses the separately defined windows in Appendix A.

7.1. Angular matrices and their weighted variation. Identify the angular Hilbert spaces for different ς by their orthonormal bases $P_j(\varsigma)$. In these coordinates the angular operator is the fixed diagonal matrix $\Lambda = \text{diag}(\lambda_j)$, and multiplication by x is the matrix X_ς in (2.8). Define

$$T_\varsigma = \frac{H_2(X_\varsigma)}{pg_2}, \quad w_j = \frac{j+1}{p+j+1}, \quad W = \text{diag}(w_j).$$

The same notation $T_\varsigma(x)$ denotes its scalar polynomial before identification. All estimates involving W^{-1} initially mean estimates on finite sequences, extended by boundedness.

Lemma 7.1 (Weighted angular bounds). *For $k = 0, 1, 2$, the matrix $\partial_\varsigma^k T_\varsigma$ has bandwidth two and*

$$(7.1) \quad |(\partial_\varsigma^k T_\varsigma)_{ij}| \leq C \min(w_i, w_j), \quad \|(\partial_\varsigma^k T_\varsigma)W^{-1}\| + \|W^{-1}(\partial_\varsigma^k T_\varsigma)\| \leq C.$$

The matrices X_ς and their first two derivatives are bounded uniformly. If

$$(7.2) \quad G_\varsigma = 4x(1-x)(\partial_x T_\varsigma)^2 = \frac{1}{2}[T_\varsigma, [\Lambda, T_\varsigma]],$$

then, for $k = 0, 1, 2$,

$$(7.3) \quad \|W^{-1}(\partial_\varsigma^k G_\varsigma)W^{-1}\| \leq Cp.$$

Proof. From (2.9), uniformly for $j \geq 1$ in the stated range,

$$\mathfrak{a}_j^2 \leq C \frac{j}{p+j}, \quad \frac{\partial_\varsigma \mathfrak{a}_j^2}{\mathfrak{a}_j^2} = -\frac{p^2}{(j+p/2-1)^2 - p^2\varsigma}.$$

For $j \geq 1$, write $\mathfrak{D}_j = (j+p/2-1)^2 - p^2\varsigma$. The exact derivatives are

$$(7.4) \quad \frac{\partial_\varsigma \mathfrak{a}_j}{\mathfrak{a}_j} = -\frac{p^2}{2\mathfrak{D}_j}, \quad \frac{\partial_\varsigma^2 \mathfrak{a}_j}{\mathfrak{a}_j} = -\frac{p^4}{4\mathfrak{D}_j^2}.$$

The factor $\mathfrak{D}_j$ is bounded below by $c(p+j)^2$, so $|\partial_\varsigma^k \mathfrak{a}_j| \leq C\mathfrak{a}_j$ for $k = 1, 2$. The coefficient $\mathfrak{a}_0$ and both its derivatives are identically zero. The diagonal entries $\mathfrak{b}_j$ have uniformly bounded first two derivatives: ς stays away from zero, so $\partial_\varsigma^k \sqrt{\varsigma}$ is bounded for $k \leq 2$. The two-step entry of $H_2(X_\varsigma)$ is $\mathfrak{a}_{j+1}\mathfrak{a}_{j+2}$ and has size, together with its two derivatives, at most Cw_j .

The one-step entry is

$$\mathfrak{a}_{j+1} \left(\mathfrak{b}_j + \mathfrak{b}_{j+1} - \frac{2(\alpha_b + 1)}{p+2} \right).$$

Its parenthesis equals $(r-1/2)$ times

$$p(p-2) \left\{ \frac{1}{(p+2j-2)(p+2j)} + \frac{1}{(p+2j)(p+2j+2)} \right\} - \frac{2p}{p+2}.$$

The braces with their prefactor have exactly the compensating value at $j = 0$. For $j \leq p$, subtract this value and differentiate the reciprocal products in j to bound the difference by Cj/p . For $j \geq p$ each term is bounded by a constant. Thus the parenthesis and both its ς derivatives are $O(j/(p+j))$. Multiplication by $\mathfrak{a}_{j+1}$ preserves the bound Cw_j .

The diagonal entries of $H_2(X_\varsigma)$ are $M_j = c_j^{\text{diag}} + d_j^{\text{diag}}\varsigma$, where

$$(7.5) \quad c_j^{\text{diag}} = \frac{j(p-1)(j+p-1)}{2(p+1)(2j+p-3)(2j+p+1)},$$

$$d_j^{\text{diag}} = -\frac{2jp^2(j+p-1)(p^2+7p-8j^2-8jp+8j)}{(2j+p)(p+1)(p+2)(2j+p-3)(2j+p-2)(2j+p+1)}.$$

These formulas follow by the multiplication calculation in the proof of Lemma 6.1, now at general j . The denominator factors are comparable to p or $p + j$; hence $|c_j^{\text{diag}}| + |d_j^{\text{diag}}| \leq Cj/(p + j)$. The scalar pg_2 is bounded away from zero and has bounded derivatives of every fixed order. Division by it proves the entry bounds. For $|i - j| \leq 2$ the weights have ratios between $1/3$ and 3 . The row and column sums over the five diagonals prove (7.1). The same row-sum argument for X_ζ and its derivatives proves their asserted bounds.

The identity in (7.2) follows by expanding $[\Lambda, T] = (\Lambda T)_{\text{mult}} - 8x(1 - x)T_x \partial_x$. The function G_ζ is a degree-four polynomial in X_ζ , with coefficients and their first two derivatives uniformly bounded. Its bandwidth is four and its unweighted norm is $O(1)$. When both indices exceed $p - 4$, the weights are bounded below and this already proves the required bound. For the other nonzero entries all indices are at most $p + 4$, and

$$(G_\zeta)_{ij} = \frac{1}{2} \sum_l (2\lambda_l - \lambda_i - \lambda_j)(T_\zeta)_{il}(T_\zeta)_{lj}.$$

Only finitely many neighbours occur; their eigenvalue differences are $O(p)$ and each product is at most $Cw_i w_j$ by (7.1). The product rule gives the same estimate for two derivatives. Summing over the nine diagonals proves (7.3). $\square$

Lemma 7.2 (Positive angular variation). *For every subset $\mathcal{I} \subset \{8, 9, \dots\}$ containing no adjacent indices, let $T_{\mathcal{I}}$ denote the principal angular submatrix with those indices. Then*

$$(7.6) \quad T'_{\mathcal{I}}(\zeta) \geq c_0 W_{\mathcal{I}}^2, \quad c_0 = 10^{-4},$$

for $p \geq 64$. The inequality holds for finite or infinite $\mathcal{I}$ as a bounded-operator inequality.

Proof. Differentiate $M_j/g_2 = (c_j^{\text{diag}} + d_j^{\text{diag}}\zeta)/(c_2^{\text{diag}} + d_2^{\text{diag}}\zeta)$. The numerator is

$$(7.7) \quad \begin{aligned} \mathcal{N}_j &= d_j^{\text{diag}} c_2^{\text{diag}} - c_j^{\text{diag}} d_2^{\text{diag}} \\ &= \frac{24jp^3(j-2)(p-1)(j+p-1)(j+p+1)}{(2j+p)(p+1)(p+2)^2(p+4)(p+5)(2j+p-3)(2j+p-2)(2j+p+1)}. \end{aligned}$$

Define also $\partial_\zeta(\mathfrak{a}_j^2/g_2) = \mathcal{E}_j/g_2^2$, where

$$(7.8) \quad \begin{aligned} \mathcal{E}_j &= \frac{4jp^2(j+p-2)F(p, j)}{(p+1)(p+2)^2(p+4)(p+5)(2j+p-3)(2j+p-2)^2(2j+p-1)}, \\ F(p, j) &= (p^2 - 9p - 16)j^2 + (p^3 - 11p^2 + 2p + 32)j - 5p^3 + 3p^2 + 8p - 12. \end{aligned}$$

These identities are obtained by substituting (7.5), (2.9), and (6.3) and clearing their positive denominators. For $u = p - 64$, $j_{\text{pos}} = j - 7$ one has the explicit identity

$$\begin{aligned} F(p, j) &= (u^2 + 119u + 3504)j_{\text{pos}}^2 \\ &\quad + (u^3 + 195u^2 + 12548u + 266304)j_{\text{pos}} \\ &\quad + 2u^3 + 359u^2 + 20957u + 394500. \end{aligned}$$

Thus $\mathcal{E}_j > 0$ for $p \geq 64, j \geq 7$. The polynomial identity in Appendix B gives

$$(7.9) \quad \frac{\mathcal{N}_j}{2} > \frac{2}{3}(\mathcal{E}_{j-1} + \mathcal{E}_j + \mathcal{E}_{j+1} + \mathcal{E}_{j+2}), \quad j \geq 8.$$

All denominators there are positive on the stated range.

The adjacent recurrence coefficients satisfy $3/4 < \mathbf{a}_{j+1}/\mathbf{a}_j < 4/3$ for $j \geq 7$. For the upper bound discard the denominator ratio in (2.9) to obtain

$$\frac{\mathbf{a}_{j+1}^2}{\mathbf{a}_j^2} \leq \frac{8}{7} \left(\frac{23}{22} \right)^2 \frac{70}{69} < \frac{16}{9}.$$

For the inverse ratio discard the increasing numerator factors and use $2j + p \geq 78$ to get

$$\frac{\mathbf{a}_j^2}{\mathbf{a}_{j+1}^2} \leq \frac{78^2 \cdot 79}{76^2 \cdot 75} < \frac{16}{9}.$$

The corresponding rational bound decreases with $2j + p$, factor by factor, so these estimates cover every larger j, p . On one parity the only off-diagonal entries of T_ζ are $\mathbf{a}_{j+1}\mathbf{a}_{j+2}/(pg_2)$. Their derivatives are positive and, for the upper neighbour, at most

$$\frac{2}{3pg_2^2}(\mathcal{E}_{j+1} + \mathcal{E}_{j+2}).$$

Indeed differentiate the product as one half the sum of the two square derivatives times the adjacent coefficient ratios. Equation (7.9) and $2|v_j v_k| \leq |v_j|^2 + |v_k|^2$ now give

$$\langle T'_\zeta v, v \rangle \geq \sum_j \frac{\mathcal{N}_j}{2pg_2^2} |v_j|^2$$

on either parity and every subset thereof. For $8 \leq j \leq p$, the numerator and denominator in (7.7) are respectively at least $9j^2 p^6$ and at most $3456p^9$. Since $g_2 < 1/p$, the margin is at least $j^2/(768p^2)$. For $j \geq p$, the corresponding estimates $6p^4 j^4$ and $3456p^5 j^4$ give a margin at least $1/1152$. Together with $w_j \leq (9/8)j/p$ for $j \geq 8$ and $w_j \leq 1$, these bounds imply (7.6) with $c_0 = 10^{-4}$.

A set with no adjacent indices has no opposite-parity off-diagonal entry: the matrix has bandwidth two. Its principal submatrix is therefore the direct sum of the two parity submatrices. The bounds extend from finite vectors by density and Lemma 7.1. $\square$

7.2. The common Dirichlet form. Let $\zeta \in (0, 1)$ be nonsquared reference radius and put

$$\epsilon_q = -\frac{\theta}{2R^2}, \quad \rho_\zeta(x) = 1 + \epsilon_q T_\zeta(x), \quad c_p = p - \frac{1}{2}.$$

Use the Hilbert space $\mathcal{H}_{\text{flow}} = L^2((0, 1), d\zeta) \otimes \ell^2(\mathbb{N}_0)$, with the Jacobi identification already made. The ball operator and its positive shift are

$$(7.10) \quad H_0 = -\partial_\zeta^2 + \frac{c_p(c_p - 1) + \Lambda}{\zeta^2}, \quad K_0 = H_0 + R^2, \quad D = \zeta \partial_\zeta + \frac{1}{2}.$$

This is the mass-unitary representation of the operator already denoted by H_0 . Its spectrum and all its spectral projections are unchanged by the unitary map.

Proposition 7.3 (Exact pullback form). *For the weighted quartic domain $0 < \varrho < \rho_\zeta(x)$, let $U(\zeta, x) = u(\zeta \rho_\zeta(x), x)$ and $v = \sqrt{2p} \zeta^{p-1/2} \rho_\zeta^p U$. Put $L_\zeta = \partial_x \log \rho_\zeta$, $\alpha_\zeta = \rho_\zeta^{-2}$, and $q_x = 4x(1 - x)$. Its Dirichlet form is*

$$(7.11) \quad \mathbf{q}_\zeta[v, w] = \int \alpha_\zeta \left\{ (v_\zeta - c_p v / \zeta) \overline{(w_\zeta - c_p w / \zeta)} + \frac{q_x}{\zeta^2} (v_x - L_\zeta Dv) \overline{(w_x - L_\zeta Dw)} \right\} d\zeta d\mu_r.$$

Define, initially as weak expressions on the common core,

$$(7.12) \quad \mathcal{L}(M) = MH_0 + \frac{1}{2\zeta^2}[\Lambda, M]D,$$

$$\mathfrak{g}_\gamma[v, w] = \int \zeta^{-2} \gamma(Dv) \overline{Dw}.$$

Write $\mathcal{G}(\gamma)$ for the weak form operator characterized on this core by $\langle \mathcal{G}(\gamma)v, w \rangle = \mathfrak{g}_\gamma[v, w]$. The operator associated with $\mathfrak{q}_\zeta$ is, in the sense of forms,

$$(7.13) \quad H_\zeta = H_0 + \mathcal{L}(\alpha_\zeta - I) + \mathcal{G}(\gamma_\zeta), \quad \gamma_\zeta = \epsilon_q^2(I + \epsilon_q T_\zeta)^{-4} G_\zeta.$$

The exact form has the same closed domain as H_0 for large p . The family is C^1 in the relative form norm and has compact resolvent. At integer block sizes it is unitarily equivalent to the invariant Euclidean Dirichlet Laplacian on Ω_q/R .

Proof. With the fixed normalization by the unit-ball volume, the radial change of variables has density $2p\zeta^{2p-1}\rho_\zeta^{2p}$, proving the mass normalization. Its angular derivative is $U_x - \zeta L_\zeta U_\zeta$. Substitution gives

$$U_x - \zeta L_\zeta U_\zeta = (2p)^{-1/2} \zeta^{-c_p} \rho_\zeta^{-p} \{v_x - L_\zeta(\zeta v_\zeta + v/2)\}.$$

The radial derivative gives $(2p)^{-1/2} \zeta^{-c_p} \rho_\zeta^{-p} (v_\zeta - c_p v/\zeta)$. These identities prove (7.11).

To integrate the form, radial integration of its first square gives $\alpha_\zeta(-\partial_\zeta^2 + c_p(c_p - 1)/\zeta^2)$. The purely angular square gives

$$\zeta^{-2} \{ \alpha_\zeta \Lambda + \frac{1}{2} [\Lambda, \alpha_\zeta] - \frac{1}{2} (\Lambda \alpha_\zeta)_{\text{mult}} \}.$$

For the cross terms use $\alpha_\zeta L_\zeta = -(\alpha_\zeta)_x/2$ and $D^* = -D + 2$ in radial measure $\zeta^{-2} d\zeta$. Their sum is

$$\zeta^{-2} \{ \frac{1}{2} [\Lambda, \alpha_\zeta] (D - 1) + \frac{1}{2} (\Lambda \alpha_\zeta)_{\text{mult}} \}.$$

The multiplication terms cancel. The remaining square has coefficient $\alpha_\zeta q_x L_\zeta^2$, giving (7.13) and (7.2). These are identities on finite angular sums with radial functions compactly supported in $(0, 1)$.

We specify the closure. Let $\mathcal{V}_0 = \text{Dom}(K_0^{1/2})$, defined as the completion of that core in its positive form norm. Angular spectral truncations commute with K_0 and converge in this norm. In each remaining radial block the Friedrichs definition gives approximation by compactly supported radial functions. Thus the stated core is form dense. Its angular integration by parts has no endpoint term: the boundary factor is $x^{\alpha_b}(1-x)^{\beta_b}$ and the test functions are polynomial there. Cutting off a smooth angular function near $x = 0$ costs $O(\delta^{\alpha_b-1})$ in energy, and near $x = 1$ costs $O(\delta^{\beta_b-1})$. Both tend to zero since $\alpha_b, \beta_b > 1$. At the physical origin a radial cutoff costs $O(\delta^{2p-2})$. Consequently the block axes and origin have zero capacity for this weighted energy. For integer blocks this is the invariant Euclidean H_0^1 realization, by averaging smooth Dirichlet approximations over the compact symmetry group. For real parameters define the original weighted Dirichlet realization as the closure of smooth functions compactly supported in the open meridian rectangle, using the norm from mass plus

$$2p \int_0^1 \int_0^1 (|U_\zeta|^2 + \zeta^{-2} q_x |U_x|^2) \zeta^{2p-1} d\zeta d\mu_r.$$

The polynomial angular core has the same closure. The cutoffs just estimated put every finite angular polynomial times a compactly supported smooth radial function in this closure. Conversely, for a smooth function compactly supported in the meridian rectangle, let U_J be its angular projection on $P_0, \dots, P_J$. Integration by parts in x gives $\langle \Lambda_r U, P_j \rangle = \lambda_j \langle U, P_j \rangle$. The

smooth compact support makes $\Lambda_r U$ square integrable, so Parseval gives square summability of these coefficients with weight λ_j^2 . It follows that $U_J \rightarrow U$ in angular energy as well as in angular L^2 . The same projection commutes with the radial derivative; Parseval applied to U_ζ gives convergence of its radial energy. The radial support stays in a compact subinterval of $(0, 1)$, so the factor ζ^{-2} is bounded there. Dominated convergence in the radial integral proves convergence in the full weighted form norm. Each projected radial coefficient is smooth with compact support. Thus both cores have the same closure, establishing the real-parameter domain.

For integer blocks, the identification with the Euclidean invariant form domain can also be read in both directions. Average a compactly supported smooth Euclidean approximation over $O(a) \times O(b)$; averaging is a contraction for mass and Dirichlet energy. The result is smooth and invariant. Cut it off at the origin and at each block axis. In squared angular distance the extra energy is bounded by $C\delta^{\alpha_b-1} + C\delta^{\beta_b-1}$; at the origin it is bounded by $C\delta^{2p-2}$. All tend to zero. Away from those sets it is a smooth function of the meridian coordinates with compact support, so it belongs to the weighted closure just described. In the reverse direction a meridian test function with compact support away from the axes and origin lifts to a smooth invariant Euclidean test function, extended by zero across those sets. This proves equality of the closed domains.

The estimates proved immediately below show that the difference of (7.11) and the ball form has relative norm below $1/2$ on $\mathcal{V}_0$. Hence its closed form norm is equivalent to that of K_0 , and its closure has exactly domain $\mathcal{V}_0$. The angular mass factor ρ_ζ^p also preserves the weighted first-order domain. The required control at the origin is Hardy's inequality in the radial weight:

$$(7.14) \quad \int_0^1 |U|^2 \zeta^{2p-3} d\zeta \leq \frac{1}{(p-1)^2} \int_0^1 |U_\zeta|^2 \zeta^{2p-1} d\zeta.$$

For compactly supported U this follows by integrating $(\zeta^{2p-2})'|U|^2$ and applying Cauchy-Schwarz; completion extends it. Integrate also against μ_r . Multiplication by $\rho_\zeta^{\pm p}$ differentiates in x by the extra term $\pm p L_\zeta U$. Its angular energy has the factor $p^2 |U|^2 / \zeta^2$, which is controlled by (7.14). Both multipliers and their inverses are bounded for fixed parameters. Thus the unitary mass transformation preserves the closure, including its origin behaviour. The radial graph map is bi-Lipschitz even at the origin: on the sphere its radial factor and first derivatives are bounded and it is bounded away from zero. Such a map transports Dirichlet energy and its closure. Therefore the just described closure is the physical one, including at the origin. The relative derivative bounds below prove norm C^1 dependence. Compactness of the embedding $\mathcal{V}_0 \hookrightarrow \mathcal{H}_{\text{flow}}$ persists under equivalent form norms and proves compact resolvent. $\square$

Lemma 7.4 (Relative form estimates). *Set $B_{1,\varsigma} = -2\mathcal{L}(\epsilon_q T_\varsigma)$ and $\mathcal{R}_{2,\varsigma} = H_\varsigma - H_0 - B_{1,\varsigma}$. For $k = 0, 1, 2$ and large p ,*

$$(7.15) \quad \|K_0^{-1/2}(\partial_\zeta^k B_{1,\varsigma})K_0^{-1/2}\mathcal{W}^{-1}\| \leq C\tau p^{-2},$$

$$(7.16) \quad \|\mathcal{W}^{-1}K_0^{-1/2}(\partial_\zeta^k \mathcal{R}_{2,\varsigma})K_0^{-1/2}\mathcal{W}^{-1}\| \leq C\theta^2 p^{-3}.$$

These are estimates on the full radial spaces. They also establish the closure and differentiability assertions in Proposition 7.3.

Proof. We first prove the band estimate used to sum the exact form. If an angular matrix M has bandwidth m and $|M_{ij}| \leq M_* w_i^a w_j^b$, $a, b \in \{0, 1\}$, then

$$(7.17) \quad \|\mathcal{W}^{-a}K_0^{-1/2}\mathcal{L}(M)K_0^{-1/2}\mathcal{W}^{-b}\| \leq C(m+1)^3 M_*.$$

Indeed the radial forms $K_{0,j} = -\partial_\zeta^2 + (\nu_j^2 - 1/4)/\zeta^2 + R^2$ are comparable with ratio at most $C(m+1)^2$ for $|i-j| \leq m$. This follows from ν_i/ν_j and its inverse being at most $C(m+1)$. Thus $\|K_{0,i}^{-1/2}K_{0,j}^{1/2}\| \leq C(m+1)$. The block $M_{ij}H_{0,j}$ in (7.12) is bounded after normalization since $\|H_{0,j}K_{0,j}^{-1}\| \leq 1$. For the commutator block use the two form bounds

$$\|\zeta^{-1}K_{0,i}^{-1/2}\| \leq C/(p+i), \quad \|\zeta^{-1}\mathbf{D}K_{0,j}^{-1/2}\| \leq C.$$

The first follows from the centrifugal potential; the second from $\zeta^{-1}\mathbf{D} = \partial_\zeta + 1/(2\zeta)$ and the same positive form. Also $|\lambda_i - \lambda_j| \leq Cm(p+i+j) \leq Cm(m+1)(p+i)$. Each normalized block is bounded by $C(m+1)^2M_*$ after its weights are removed. The row and column sums involve at most $2m+1$ blocks. The scalar Schur inequality proves (7.17) on the form core, then by closure.

For $n \geq 2$, Lemma 7.1 gives

$$|(\partial_\zeta^k T_\zeta^n)_{ij}| \leq n^2 C^n w_i w_j, \quad k \leq 2,$$

with the factor n^2 replaced by one when $k = 0$ if desired. Put one inverse weight on the first factor and one on the last; each differentiated endpoint has the same one-sided bound, and interior factors are uniformly bounded. The product rule has at most a fixed multiple of n^2 terms. The bandwidth is $2n$. The series

$$\alpha_\zeta = I - 2\epsilon_q T_\zeta + \sum_{n \geq 2} (-1)^n (n+1) \epsilon_q^n T_\zeta^n$$

therefore has a two-sided weighted remainder under (7.17) of size $C\epsilon_q^2$, including two derivatives. Its dominating series is $\sum n^6 (C|\epsilon_q|)^n$, which converges uniformly.

Equation (7.3) and the convergent series for $(I + \epsilon_q T_\zeta)^{-4}$ give

$$\|W^{-1}(\partial_\zeta^k \gamma_\zeta)W^{-1}\| \leq C p \epsilon_q^2, \quad k \leq 2.$$

To move the left weight through the inverse series, use boundedness of $W^{-1}T_\zeta W$ and its derivatives, which follows from (7.1). Since $\zeta^{-1}\mathbf{D}$ is controlled by $K_0^{1/2}$ and commutes with the angular weights, the same bound holds for the normalized form $\mathcal{G}(\gamma_\zeta)$. Now $|\epsilon_q| \leq C\tau/p^2$. The first-order band estimate proves (7.15); the two remainder estimates prove (7.16). Their unweighted sum is below $1/2$ for large p .

The $k = 2$ bounds provide a uniform modulus of continuity for the first derivative: Taylor's integral formula on the finite bands, followed by the uniformly convergent series, bounds the difference quotient remainder in relative form norm. Thus the family is norm C^1 on the common form space. These bounds also prove the closure and regularity assertions of Proposition 7.3. $\square$

7.3. The boundary matrix and the transport term. Fix a finite W_{flow} and set $\mathbf{P} = \mathbf{1}_{[-W_{\text{flow}}, W_{\text{flow}}]}(H_0 - R^2)$. For large p , Lemma 3.5 says its states have distinct, nonadjacent angular indices. Index its orthonormal states by i , let $j(i)$ be their angular indices and z_i their positive zeros, and write

$$D_{\mathbf{P}} = \text{diag}(z_i^2 - R^2), \quad Z_{\mathbf{P}} = \text{diag}(z_i/R), \quad T_{\mathbf{P}}^{\text{ang}} = [(T_\zeta)_{j(i), j(i')}], \quad \mathcal{S}_\zeta = \mathbf{P}(\epsilon_q T_\zeta \mathbf{D})\mathbf{P}.$$

The matrix $T_{\mathbf{P}}^{\text{ang}}$ is the *angular principal submatrix*, while $\mathcal{S}_\zeta$ is the actual Hilbert-space compression with radial overlaps.

Lemma 7.5 (Transport identity). *One has $\mathcal{S}_\zeta^* = -\mathcal{S}_\zeta$ and the exact identity*

$$(7.18) \quad \mathbf{P}B_{1,\zeta}\mathbf{P} = \theta Z_{\mathbf{P}}T_{\mathbf{P}}^{\text{ang}}Z_{\mathbf{P}} - [D_{\mathbf{P}}, \mathcal{S}_\zeta].$$

For $k = 0, 1$,

$$(7.19) \quad \|\mathbf{P}(\partial_\zeta^k B_{1,\varsigma})\mathbf{P}\mathbf{W}^{-1}\| \leq C\tau, \quad \|(\partial_\zeta^k \mathcal{S}_\varsigma)\mathbf{W}^{-1}\| \leq C\tau/p,$$

with constants independent of W_{flow} once $p^2 \geq W_{\text{flow}}$.

Proof. As interior differential expressions, $[H_0, \mathbf{D}] = 2H_0$ and $[H_0, h] = \zeta^{-2}[\Lambda, h]$ for an angular multiplier h . Thus $B_{1,\varsigma} = -[H_0, \epsilon_q T_\zeta \mathbf{D}]$. Green's identity applied to two ball Dirichlet eigenfunctions gives

$$\langle \phi_i, H_0(h\mathbf{D}\phi_{i'}) \rangle = z_i^2 \langle \phi_i, h\mathbf{D}\phi_{i'} \rangle + \int h \partial_\zeta \phi_i(1) \partial_\zeta \phi_{i'}(1) d\mu_r.$$

The origin term vanishes by the regular radial behaviour. The boundary derivatives are $\sqrt{2}z_i P_{j(i)}$ by Lemma 3.1. Substitution of $h = -\theta T_\zeta / (2R^2)$ gives (7.18). Radial integration on pairs of Dirichlet eigenfunctions gives $\mathbf{D}^* = -\mathbf{D}$; the angular multiplier commutes with $\mathbf{D}$. Hence $\mathcal{S}_\varsigma$ is skew-adjoint. On the fixed window $\|K_0^{1/2}\mathbf{P}\| \leq Cp$. Equation (7.15) proves the first bound in (7.19). The form estimate also gives $\|\mathbf{D}\mathbf{P}\| \leq Cp$; use (7.1) and $|\epsilon_q| \leq C\tau/p^2$ for the second bound. $\square$

7.4. Eliminating the full complementary space.

Lemma 7.6 (Complementary inverse and remainder). *There is an absolute lower bound on $W_{\text{flow}} \geq 100$ such that the following holds for p sufficiently large depending on that fixed window. For every $\varsigma \in I$ and real $|\lambda| \leq 1$, the form operator $\mathbf{P}^\perp(H_\varsigma - R^2 - \lambda)\mathbf{P}^\perp$ is invertible. Its exact Schur matrix is*

$$(7.20) \quad F_{\mathbf{P}}(\varsigma, \lambda) = D_{\mathbf{P}} - \lambda + \mathbf{P}B_{1,\varsigma}\mathbf{P} + \mathcal{R}_\varsigma(\lambda),$$

where, for $k = 0, 1$,

$$(7.21) \quad \|\mathbf{W}^{-1}(\partial_\zeta^k \mathcal{R}_\varsigma)\mathbf{W}^{-1}\| \leq C\theta^2(W_{\text{flow}}^{-1} + p^{-1}).$$

The constant C is independent of the chosen window provided $p^2 \geq W_{\text{flow}}$ and the window exceeds its fixed lower bound. A kernel vector is reconstructed uniquely from $v = \mathbf{P}u$, and its complementary part satisfies

$$(7.22) \quad \|\mathbf{P}^\perp u\| \leq C\tau W_{\text{flow}}^{-1} \|\mathbf{W}v\|.$$

Proof. Use the positive form normalization K_0 and abbreviate $Q = \mathbf{P}^\perp$, $\mathcal{B}_\varsigma = H_\varsigma - H_0$, and $\mathfrak{C}_\varsigma = K_0^{-1/2}\mathcal{B}_\varsigma K_0^{-1/2}$. Write $K_{0,\mathbf{P}}$ and $K_{0,Q}$ for the restrictions of K_0 to $\mathbf{P}\mathcal{H}_{\text{flow}}$ and $Q\mathcal{H}_{\text{flow}}$. The inverse $\mathfrak{U}_0$ of the normalized unperturbed complement is radially diagonal with entries

$$\frac{z_{j,k}^2 + R^2}{z_{j,k}^2 - R^2 - \lambda}, \quad |z_{j,k}^2 - R^2| > W_{\text{flow}}.$$

Their absolute supremum is at most Cp^2/W_{flow} . This includes the whole infinite tail, where the ratio tends to one; $p^2 \geq W_{\text{flow}}$ covers that limit. Lemma 7.4 gives $\|\mathfrak{C}_\varsigma \mathbf{W}^{-1}\| + \|\mathfrak{C}'_\varsigma \mathbf{W}^{-1}\| \leq C\tau/p^2$. Choose the fixed window large enough that $\|\mathfrak{U}_0 Q \mathfrak{C}_\varsigma Q\| \leq 1/2$ uniformly for $\tau \leq 20$. The normalized inverse

$$\mathfrak{U} = (I + \mathfrak{U}_0 Q \mathfrak{C}_\varsigma Q)^{-1} \mathfrak{U}_0$$

has norm at most Cp^2/W_{flow} , and $\mathfrak{U}'_\varsigma = -\mathfrak{U} Q \mathfrak{C}'_\varsigma Q \mathfrak{U}$ has norm at most $Cp^2\tau/W_{\text{flow}}^2$. The actual form inverse is $K_{0,Q}^{-1/2} \mathfrak{U} K_{0,Q}^{-1/2}$.

Eliminate Q using this inverse. The remainder in (7.20) is

$$\mathcal{R}_\varsigma = \mathbf{P}\mathcal{R}_{2,\varsigma}\mathbf{P} - \mathbf{P}\mathcal{B}_\varsigma Q K_{0,Q}^{-1/2} \mathfrak{U} K_{0,Q}^{-1/2} Q \mathcal{B}_\varsigma \mathbf{P}.$$

In the second term the two exterior $K_{0,\mathbf{P}}^{1/2}$ factors cost Cp each; each normalized perturbation with its exterior inverse weight costs $C\tau/p^2$. The weighted norm is therefore $C\theta^2/W_{\text{flow}}$. Its derivative either replaces one perturbation by $\mathfrak{E}'_\varsigma$ or replaces $\mathfrak{U}$ by $\mathfrak{U}'_\varsigma$. The added bound in the latter case is $C\tau^3/W_{\text{flow}}^2$, absorbed in the same bound. The first remainder term costs $C\theta^2/p$ by (7.16). This proves (7.21) with no dependence of C on the window size.

The eliminated component is $u_Q = -K_{0,Q}^{-1/2}\mathfrak{U}Q\mathfrak{E}_\varsigma\mathbf{P}K_{0,\mathbf{P}}^{1/2}v$. Since $K_0 \geq cp^2I$, the factors cost respectively C/p , Cp^2/W_{flow} , $C\tau/p^2$ after extracting Wv , and Cp . This proves (7.22). The reconstruction belongs to the common form domain. The eliminated equation is exactly zero there, so the reduced kernel equation is equivalent to the full weak eigenvalue equation. A weak solution of that equation is in the operator domain by the definition of the associated closed-form operator, completing the equivalence of the reduced and full eigenvalue equations. $\square$

7.5. The seed and the second low state. Assume now that R and θ satisfy (4.10) with $J_p(R) = 0$.

Lemma 7.7 (Low-block elimination). *For every fixed W_{flow} above the absolute lower bound in Lemma 7.6 and all sufficiently large p , the only angular indices below eight in the flow window are $j = 2, 5$, each with one radial state. Their ball offsets satisfy*

$$(7.23) \quad d_2 = -\frac{\theta}{2p} + O(\theta^2 p^{-3}), \quad d_5 = -72 + O(p^{-1}).$$

Let L be their two-dimensional span and $\mathbf{P}_H = \mathbf{P} - L$. Write F_L, W_L, u_L for the L blocks of $F_{\mathbf{P}}, W, u$, and $\mathcal{R}_{HL}, \mathcal{R}_{LH}$ for the corresponding off-diagonal blocks of $\mathcal{R}_\varsigma$ between L and $\mathbf{P}_H$. Write D_H, Z_H, W_H for the restrictions of $D_{\mathbf{P}}, Z_{\mathbf{P}}, W$ to this high block, and T_H^{ang} for its angular principal submatrix. For $|\lambda| \leq \omega_p = \tau/(16p)$, the L block of $F_{\mathbf{P}}(\varsigma, \lambda)$ is invertible. Eliminating it gives

$$(7.24) \quad F_H(\varsigma, \lambda) = D_H - \lambda + B_{1,HH} + \widehat{\mathcal{R}}_\varsigma(\lambda),$$

with

$$(7.25) \quad \|\mathbf{W}_H^{-1}(\partial_\varsigma^k \widehat{\mathcal{R}}_\varsigma)\mathbf{W}_H^{-1}\| \leq C\theta^2(W_{\text{flow}}^{-1} + p^{-1}), \quad k = 0, 1.$$

A nonzero full eigenvector in this band has $v = \mathbf{P}_H u \neq 0$ and $\|u\| \leq C\|v\|$, uniformly also as $\tau \downarrow 0$.

Proof. The trace ratio and (6.9) give $R = 2p+3+O(p^{-1})$. Normalize the seed radial function by $f'(R) = 1$, $f(R) = \chi = \theta/(4Rp)$. The collar equation has bounded coefficients and one bounded derivative for fixed angular index. On $|y| \leq 2|\chi|$ its integral equation yields $f'(R+y) = 1 + O(|\chi|)$. The endpoint values have opposite signs, and the derivative has positive sign. Its unique zero in the collar is $R - \chi + O(\chi^2)$. Squaring gives $d_2 = -2R\chi + O(R\chi^2) = -\theta/(2p) + O(\theta^2 p^{-3})$, uniformly for arbitrarily small nonzero θ .

For $j = 5$ use (6.12) with $c_5 = 18$. The same collar argument places its zero at $R - 18/p + O(p^{-2})$, giving the second offset in (7.23). For $0 \leq j < 8$ other than 2, 5, the finite recurrence (6.11) has nonzero limiting value at $k = 2j - 1$. Its derivative trace stays bounded. The collar equation excludes a zero in a fixed positive-width collar of R . Such an index is therefore outside every fixed energy window for large p . The radial spacing excludes a second state in each remaining index.

The first-order operator has no L - $\mathbf{P}_H$ coupling or coupling between the two L states. Their angular distances are at least three, whereas both matrices in (7.18) have angular bandwidth

two. The seed diagonal is

$$\sigma_2 = d_2 + \frac{\theta z_2^2}{pR^2} = \frac{\theta}{2p} + O(\theta^2 p^{-3}),$$

because $(T_\zeta)_{22} = 1/p$ exactly. The other diagonal tends to -72 and its perturbation is $O(\tau/p)$. For $|\lambda| \leq \omega_p$, these two uncorrected diagonal entries have modulus at least $\tau/(4p)$ and 36 , respectively. The remainder block has norm at most $C\theta^2(W_{\text{flow}}^{-1} + p^{-1})p^{-2}$ because $w_2, w_5 \leq 6/p$. A two-dimensional Neumann series gives

$$\|F_L^{-1}\| + \|\partial_\zeta F_L^{-1}\| \leq Cp/\tau.$$

For the derivative use $(F_L^{-1})' = -F_L^{-1}F_L'F_L^{-1}$ and $\|F_L'\| \leq C\tau/p$, which follows from the diagonal structure and the angular derivative bound. In particular,

$$(7.26) \quad \|\mathbf{W}_L F_L^{-1} \mathbf{W}_L\| + \|\mathbf{W}_L (\partial_\zeta F_L^{-1}) \mathbf{W}_L\| \leq C/(p\tau).$$

The second elimination has remainder

$$\widehat{\mathcal{R}} = \mathcal{R}_{HH} - \mathcal{R}_{HL} F_L^{-1} \mathcal{R}_{LH}.$$

Put $a_* = C\theta^2(W_{\text{flow}}^{-1} + p^{-1})$. The weighted new term and its derivative are at most $Ca_*^2/(p\tau)$ by (7.26). For $\tau \leq 20$ this is absorbed in a_* after increasing the p threshold, proving (7.25). Reconstruction gives

$$\|u_L\| \leq C\tau(W_{\text{flow}}^{-1} + p^{-1})\|\mathbf{W}_H v\|.$$

Combining this with (7.22) proves $\|u\| \leq C\|v\|$ and the nonvanishing assertion. The inverse factors $1/\tau$ are paired with the quadratic remainders, giving uniform reconstruction for $0 < \tau \leq 20$. $\square$

Lemma 7.8 (Ordered eigenvalue derivatives). *For the common C^1 closed-form family above, the ordered eigenvalues are locally Lipschitz. At an interior parameter $\varsigma_0 \in (1/64, 1/16)$, suppose an eigenvalue λ has multiplicity m and occupies ordered positions $i, \dots, i+m-1$. Let $\mu_1 \leq \dots \leq \mu_m$ be the eigenvalues of the derivative form restricted to its Hilbert eigenspace. The one-sided derivatives exist and satisfy*

$$(7.27) \quad \lambda'_{i+r-1,+}(\varsigma_0) = \mu_r, \quad \lambda'_{i+r-1,-}(\varsigma_0) = \mu_{m-r+1}, \quad 1 \leq r \leq m.$$

Thus every one-sided derivative is an eigenvalue of the compressed derivative form. An ordered eigenvalue need not have a two-sided derivative at a multiple crossing.

Proof. A fixed positive shift has inverse that is norm C^1 , by the inverse identity on the common form space. Its compact resolvent admits a contour surrounding the isolated finite cluster in question. The contour integral gives a C^1 finite-rank projection, also with values in the form domain. Local unitary transport of this projection identifies its range with a fixed finite-dimensional space. Restricting the resolvent there and inverting that finite matrix gives a C^1 self-adjoint matrix for the original spectral cluster; see also [7, Chapter II, Theorem 5.4 and Theorem 6.8]. At ς_0 it is λI . Its derivative is the restricted derivative form: differentiation of its unitary transport contributes a commutator with λI , which is zero. The matrix expansion is therefore $\lambda I + hB + o(|h|)$, with B having eigenvalues μ_r . The finite-dimensional min-max principle bounds changes of ordered eigenvalues by the operator norm of the remainder. For $h > 0$ the order of $h\mu_r$ is unchanged, and for $h < 0$ it is reversed. Division by h gives (7.27). The same estimate for two nearby parameters gives local Lipschitz continuity. Compactness of the parameter interval gives absolute continuity for each fixed ordered curve, hence a two-sided derivative almost everywhere. $\square$

Theorem 7.9 (Local spectral flow). *For all sufficiently large $p \in \mathcal{P}_{\text{bd}}$, let R, θ be as in Lemma 4.7. There are absolute constants $c_{\text{flow}} > 0$ and $P_{\text{flow}} < \infty$ such that every ordered eigenvalue offset $\lambda_i(\varsigma)$ of $H_\varsigma - R^2$ satisfies*

$$(7.28) \quad \text{sign}(\theta)\lambda'_i(\varsigma) \geq c_{\text{flow}}\tau/p \quad \text{for almost every } \varsigma \text{ with } |\lambda_i(\varsigma)| < \tau/(16p),$$

when $p \geq P_{\text{flow}}$. The constants are uniform in $0 < \tau \leq 20$.

Proof. Fix an eigenvalue λ in the band and a vector u in its entire eigenspace. Put $v = P_H u$. The two exact eliminations show that $v \neq 0$. With v, λ held fixed, differentiation of the reconstructed quadratic form gives

$$(7.29) \quad \langle u, H'_\varsigma u \rangle = \langle v, \partial_\varsigma F_H(\varsigma, \lambda) v \rangle.$$

Indeed, terms differentiating the reconstruction lie in $\mathbf{P}^\perp + L$, and their pairing with the operator vanishes by the eliminated equations. The bounded normalized inverses above make the reconstruction differentiable in the common form norm.

The transport identity gives

$$\partial_\varsigma F_H = \theta Z_H (T_H^{\text{ang}})' Z_H - [D_H, \mathcal{S}'_{HH}] + \partial_\varsigma \widehat{\mathcal{R}}.$$

The first term, after multiplication by $\text{sign } \theta$, is at least $c_1 \tau \|W_H v\|^2$ in quadratic form. Here Lemma 7.2 applies to all the retained indices, and $z_i/R \rightarrow 1$ uniformly in the fixed window; $c_1 > 0$ is absolute. Estimate the transport term on the actual kernel vector, using (7.24): $D_H v = \lambda v - (B_{1,HH} + \widehat{\mathcal{R}})v$. Since $\mathcal{S}'_{HH}$ is skew-adjoint, its real pairing with λv vanishes. Thus

$$\begin{aligned} |\langle v, [D_H, \mathcal{S}'_{HH}] v \rangle| &\leq 2 \|(B_{1,HH} + \widehat{\mathcal{R}})v\| \|\mathcal{S}'_{HH} v\| \\ &\leq C \theta^2 p^{-1} \|W_H v\|^2. \end{aligned}$$

The final remainder is at most $C \theta^2 (W_{\text{flow}}^{-1} + p^{-1}) \|W_H v\|^2$. Choose one constant $C_{\text{flow, err}}$ dominating the transport-error constant and the constants in (7.21) and (7.25). It is independent of W_{flow} , under their stated lower bound on that window. The combined error is at most $C_{\text{flow, err}} \theta^2 (W_{\text{flow}}^{-1} + 2p^{-1}) \|W_H v\|^2$. Choose W_{flow} so large that $20C_{\text{flow, err}}/W_{\text{flow}} < c_1/4$ and the complementary-inverse conditions hold. Next choose p large enough that $40C_{\text{flow, err}}/p < c_1/4$. The other window-dependent requirements are $p^2 \geq W_{\text{flow}}$, $2W_{\text{flow}}/R < 1/10$, and the fixed-index asymptotic threshold in Lemma 7.7. Also impose the near-crossing low-degree threshold in Proposition 4.10. Equations (7.29) and these bounds give

$$(7.30) \quad \text{sign}(\theta) \langle u, H'_\varsigma u \rangle \geq (c_1/2) \tau \|W_H v\|^2.$$

To turn the weighted bound into a $1/p$ margin, put $J_p = \lfloor \sqrt{p}/16 \rfloor$. For large p every index $j < J_p$ other than the removed seed is in the exclusion range of Proposition 4.10. Its ball offset has modulus greater than 32. On those rows of (7.24), invert $D_H - \lambda$, whose inverse norm is at most $1/31$, to get

$$\|v_{j < J_p}\| \leq C \tau \|W_H v\|.$$

On the other indices $w_j \geq c/\sqrt{p}$, giving $\|v_{j \geq J_p}\| \leq C \sqrt{p} \|W_H v\|$. Lemma 7.7 therefore yields $\|W_H v\|^2 \geq c p^{-1} \|u\|^2$. The further resonant low states $j = 8, 11, \dots$ in a large fixed window are included in this argument: they remain in $\mathbf{P}_H$, are covered by its positive angular variation, and are excluded from the near-zero mass by the last row estimate. Substitution into (7.30) proves the claimed derivative lower bound on the entire eigenspace.

By Lemma 7.8, every one-sided derivative of an ordered eigenvalue in the band is an eigenvalue of this compressed derivative form. Its signed lower bound therefore applies at every differentiability point, and almost everywhere on each absolutely continuous curve. This proves (7.28), including multiple crossings. $\square$

8. CHOOSING AN INTEGER SPLIT

The counting window in this section is denoted by W_{count} . It comes from a radial shell comparison and is independent of the wide window W_{flow} used in the spectral-flow proof.

Lemma 8.1 (Avoiding narrow bands on a discrete grid). *Let a compact interval contain at least $p/8$ grid points separated by at least $1/(9p)$. Fix $C_* > 0$ and suppose at most $C_* p^{2/3}$ absolutely continuous real functions can enter $[-1, 1]$, and for each such function λ_i one has $\epsilon_{\text{sgn}} \lambda'_i \geq \mu > 0$ almost everywhere on $|\lambda_i| < \omega$, where $\epsilon_{\text{sgn}} \in \{-1, 1\}$ is fixed and $0 < \omega < 1$. If p is sufficiently large depending on C_* and $\gamma = \mu/(576C_* p^{2/3}) < \omega$, then some grid point satisfies $|\lambda_i| \geq \gamma$ for every i .*

Proof. Clip $\epsilon_{\text{sgn}} \lambda_i$ to $[-\omega, \omega]$. The chain rule for absolutely continuous functions says that its derivative is zero almost everywhere outside the open band and is at least μ inside. On a level set an absolutely continuous function has derivative zero almost everywhere, so the two band endpoints introduce no exceptional contribution. The clipped function is therefore nondecreasing. The preimage of $(-\gamma, \gamma)$ is an interval, and its length is at most $2\gamma/\mu$: integrate the derivative between any two points of that interval. It contains at most $1 + 18p\gamma/\mu$ grid points. The union of all excluded grid points has cardinality at most

$$C_* p^{2/3} (1 + 18p\gamma/\mu) = C_* p^{2/3} + p/32 \leq p/16 < p/8$$

for large p . A remaining point has the asserted property. Functions that never enter $[-1, 1]$ exclude no point. $\square$

Lemma 8.2 (A fixed counting window). *For $p \geq 64$, $R \in [2p - 1, 2p + 4]$, $|\theta| \leq 20$, and every $\varsigma \in I$, the quartic domain lies in the radial shell $B_{1-3/p^2} \subset \Omega_q/R \subset B_{1+3/p^2}$. Every ordered eigenvalue curve that enters the unit energy window about R^2 has its ball counterpart in the fixed window $W_{\text{count}} = 40$. This statement holds also for real block parameters in their weighted Dirichlet realization.*

Proof. The radius differs from one by at most $10/R^2$, using $\|T_\varsigma\|_\infty < 1$. For $p \geq 64$, $10/(2p - 1)^2 \leq 3/p^2$, because $2p^2 - 12p + 3 > 0$. Set $\delta = 3/p^2$. The ordered min-max shell comparison implies that $\lambda_i(\Omega_q/R) \in [R^2 - 1, R^2 + 1]$ forces

$$(1 - \delta)^2(R^2 - 1) \leq \lambda_i(B) \leq (1 + \delta)^2(R^2 + 1).$$

Since $(2p+4)^2+1 \leq 5p^2$ for $p \geq 64$, the upper displacement from R^2 is at most $1+5(6+9/p^2) < 32$; the lower displacement is at most 31. Both are below 40. These are indexwise inequalities for every split, and the ball eigenvalue is split independent. The weighted min-max comparison has the same dilation factors, so the proof includes real block parameters. $\square$

Theorem 8.3 (Integer-split gap). *There are absolute constants $c_q > 0$ and P_q such that for every $p \in \mathcal{P}_{\text{bd}}$ with $p \geq P_q$, there is an integer $a \in [p/2, 3p/4]$ for which the actual quartic domain satisfies*

$$(8.1) \quad \text{dist}(R^2, \text{spec}(-\Delta_D|_{O(a) \times O(2p-a)}; \Omega_q/R)) \geq c_q \tau p^{-5/3}.$$

For each fixed centre accuracy N , the same split gives centre gap

$$(8.2) \quad \gamma_p = c_{\text{gap}} \tau p^{-5/3}, \quad c_{\text{gap}} = c_q/2,$$

for all p above a further threshold depending on N .

Proof. Lemma 8.2 places every ball counterpart of a curve entering $[-1, 1]$ about R^2 in the fixed window $W_{\text{count}} = 40$, independently of ς, θ . Theorem 3.8 therefore bounds the number of possible ordered indices by $C_* p^{2/3}$, where one may fix $C_* = \max(1, C_{40})$. This count concerns

the ball spectrum, which is independent of ς ; the single ball count bounds all curves that can enter the band.

The integers $a \in [p/2, 3p/4]$ provide at least $p/4 - 1$ points $\varsigma_a = (a/(2p) - 1/2)^2$ in I . For consecutive admissible integers, put $d_a = 1/2 - a/(2p)$. Both d_a, d_{a+1} are at least $1/8$, so

$$\varsigma_a - \varsigma_{a+1} = (d_a - d_{a+1})(d_a + d_{a+1}) = \frac{d_a + d_{a+1}}{2p} \geq \frac{1}{8p}.$$

In particular the weaker constants $p/8$ and $1/(9p)$ in Lemma 8.1 hold for large p . Apply that lemma with

$$\epsilon_{\text{sgn}} = \text{sign } \theta, \quad \mu = c_{\text{flow}} \tau / p, \quad \omega = \tau / (16p).$$

The flow hypothesis is Theorem 7.9. For large p the resulting $\gamma = (c_{\text{flow}}/(576C_*))\tau p^{-5/3}$ is smaller than ω . Thus a grid point has the full invariant gap (8.1), with $c_q = c_{\text{flow}}/(576C_*)$. At that grid point the weighted realization is the Euclidean invariant realization by Proposition 7.3. Finally Proposition 6.5 transfers half of the gap to every fixed-order centre after increasing its threshold. $\square$

9. COEFFICIENT SPACES AND CLAMPED POISSON ESTIMATES

From now on $p \in \frac{1}{2}\mathbb{Z}$ and the integer split are fixed. Constants in operator estimates are uniform over $p \geq 64$ and admissible splits unless a subscript explicitly permits dependence on the centre order. The angular coefficient radius is fixed at $\rho_w = 2$ throughout the Newton argument. Larger radii will be used only to estimate the known centre.

9.1. The source, shape, and clamped subspaces. Define the degree weight

$$(9.1) \quad \varpi_p(\ell) = (1 + \ell/p^2)^p, \quad \ell \geq 0.$$

It is increasing and obeys

$$(9.2) \quad \varpi_p(\ell_1 + \ell_2) \leq \varpi_p(\ell_1)\varpi_p(\ell_2), \quad \frac{\varpi_p(\ell + 2)}{\varpi_p(\ell)} \leq e^{2/p}, \quad \varpi_p(\ell) \leq e^{\ell/p}.$$

These follow from $1 + u + v \leq (1 + u)(1 + v)$ and $1 + u \leq e^u$.

Definition 9.1 (The fixed Banach spaces). For the polynomial basis (2.18), let $\mathcal{C} = \mathcal{C}_{p,2}$ be the completion with norm

$$(9.3) \quad \|f\|_{\mathcal{C}} = \sum_{j,s \geq 0} 2^j \varpi_p(2j + 2s) |f_{j,s}|.$$

Its moment spaces have norms

$$(9.4) \quad \|f\|_{\mathcal{C}^{[k]}} = \sum_{j,s \geq 0} 2^j \varpi_p(2j + 2s) (1 + 2j + 2s)^k |f_{j,s}|.$$

Let $\mathcal{G}_p = \{g \in \mathcal{C} : g_{j,0} = 0 \text{ for every } j\}$ and let $\mathcal{E}_1$ be the space of all boundary coefficient sequences with finite norm

$$(9.5) \quad h = \sum_{j \geq 0} h_j G_j, \quad \|h\|_{\mathcal{E}_1} = \sum_{j \geq 0} (1 + j) 2^j \varpi_p(2j) |h_j|.$$

Write $\mathcal{H}_{\text{ext}} h = \sum_j h_j S_j$ for its harmonic extension, and $\mathcal{X}_p = \mathcal{G}_p \times \mathcal{E}_1$ with the sum norm. The Newton spaces are real; their complexifications are used for analytic maps and estimates. Dot products in algebraic differential identities are extended complex-bilinearly; Hilbert inner products and norms retain their Hermitian meaning.

Lemma 9.2 (Completions and shape regularity). *The space $\mathcal{C}$ embeds injectively into invariant normalized $L^2(B)$ with norm at most one. Polynomial partial sums are dense. For each fixed k , harmonic extension sends $\mathcal{E}_1$ continuously into $\mathcal{C}^k(\overline{B})$, with constants depending on k and the dimension. The first two boundary derivative bounds are independent of p . In particular*

$$(9.6) \quad \|h\|_{C^2(S^{2p-1})} \leq C\|h\|_{\mathcal{E}_1}, \quad \|D((\mathcal{H}_{\text{ext}}h)y)\|_{\text{op},\infty} \leq C\|h\|_{\mathcal{E}_1}.$$

The boundary function h is real analytic.

Proof. By (2.20), every basis function has L^2 norm at most one. The triangle inequality gives the continuous embedding on finite sums and hence on their completion. Its injectivity follows by taking the inner product with each orthogonal basis function: all coefficients of a zero L^2 limit vanish. Weighted ℓ^1 completeness and truncation give the remaining Banach-space assertions.

Lemma 2.3 bounds the first two boundary derivatives by $2j$ and $4j^2$. The ratios of these quantities to $(1+j)2^j\varpi_p(2j)$ are uniformly bounded. Summing proves the first estimate in (9.6). Radial homogeneity gives $\partial_\rho S_j = 2j\rho^{2j-1}G_j$; the tangential bound gives $|\nabla S_j| \leq 2\sqrt{2}j$ in the ball. The product rule proves the second estimate. For the full extension, choose any fixed $1 < R_* < \sqrt{2}$. Homogeneity and the endpoint bound give $|S_j(y)| \leq R_*^{2j}$ for real $|y| \leq R_*$. Thus the series for $\mathcal{H}_{\text{ext}}h$ converges uniformly on each compact subset of $B_{\sqrt{2}}$, with supremum there controlled by the shape norm. Its finite sums are harmonic polynomials. Passing to the limit in the mean-value identity shows that the limit is harmonic in that larger ball. The Poisson formula on a ball of fixed radius about each point of $\overline{B}$ can therefore be differentiated any fixed number of times. It gives uniform convergence of all those derivatives on $\overline{B}$, with constants depending on the derivative order and fixed dimension. It also proves real analyticity of the extension on a neighbourhood of $\overline{B}$. The dimension-independent first two boundary bounds are the separate estimates above.

To see analyticity of the boundary directly, expand G_j in Chebyshev polynomials on $[0, 1]$. Its real supremum is at most one, so each coefficient has absolute value at most two. On any Bernstein ellipse of parameter $1 < \rho_{\text{ell}} < 2$ this gives $|G_j(z)| \leq 2(j+1)\rho_{\text{ell}}^j$. The series for h converges uniformly on that ellipse, since $\rho_{\text{ell}} < 2$, and defines a holomorphic function there. Composing with the polynomial $x = |X|^2$ on the real sphere proves real analyticity of the boundary function. $\square$

9.2. The positive multiplier module. For endpoint-normalized Jacobi polynomials with parameters $\alpha, \beta > -1$, put $u = \alpha + \beta + 1$ and $v = \alpha - \beta$. Gasper's theorem [4, Theorem 1] gives nonnegative linearization coefficients provided

$$(9.7) \quad \alpha \geq \beta, \quad u(u+5)(u+3)^2 \geq (u^2 - 7u - 24)v^2.$$

The normalization is at the endpoint corresponding to the larger Jacobi exponent α . In the two-block parameters $\alpha_b = a/2 \leq \beta_b = p - a/2$, this is $x = 1$, the endpoint of the factor $(1-x)^{\beta_b-1}$. Here $\alpha = \beta_b - 1 \geq \beta = \alpha_b - 1$, $u = p - 1$, and $0 \leq v = \beta_b - \alpha_b \leq p/2$. For $p \geq 64$ the margin in (9.7) is at least

$$\frac{3p^4 + 37p^3 + 64p^2 - 16p - 64}{4} > 0.$$

Thus in our normalization

$$(9.8) \quad G_i G_j = \sum_{|i-j| \leq k \leq i+j} c_{ij}^k G_k, \quad c_{ij}^k \geq 0, \quad \sum_k c_{ij}^k = 1.$$

The upper support follows from polynomial degree; the lower support follows from orthogonality, since a polynomial of degree below i has zero inner product with G_i . The sum equals one by evaluating at $x = 1$.

Lemma 9.3 (Positive radial stencils). *For $\beta \geq 0$ and $s \geq 0$,*

$$(9.9) \quad J_s^\beta = \frac{s + \beta + 1}{2s + \beta + 1} J_s^{\beta+1} + \frac{s}{2s + \beta + 1} J_{s-1}^{\beta+1},$$

$$(9.10) \quad t J_s^{\beta+1} = \frac{s + \beta + 1}{2s + \beta + 2} J_s^\beta + \frac{s + 1}{2s + \beta + 2} J_{s+1}^\beta.$$

The term with index -1 in the first identity is zero. All coefficients are nonnegative and sum to one. Multiplication by t in a fixed radial basis also has three nonnegative coefficients of sum one.

Proof. Insert the finite binomial formula (2.19). Comparing each power of t and using $\binom{z+1}{s} = \binom{z}{s} + \binom{z}{s-1}$ gives (9.9) and (9.10). Their coefficients have the displayed signs and sums. For multiplication by t in a fixed basis, orthogonality makes its matrix tridiagonal. The upper coefficient is the positive ratio of leading coefficients; symmetry and the positive squared norms give a positive lower coefficient. The diagonal coefficient is $\int t (J_s^\beta)^2 t^\beta dt / \int (J_s^\beta)^2 t^\beta dt$, which is nonnegative. Evaluation at $t = 1$ gives their sum one. $\square$

Proposition 9.4 (Solid multipliers and the shape algebra). *For integers $i, h \geq 0$, multiplication by $S_i t^h$ is bounded on $\mathcal{C}$ and*

$$(9.11) \quad \|M_{S_i t^h}\|_{\mathcal{C} \rightarrow \mathcal{C}} \leq 2^i \varpi_p(2i + 2h).$$

The boundary space $\mathcal{E}_1$ is a Banach algebra, with $\|h_1 h_2\|_{\mathcal{E}_1} \leq \|h_1\|_{\mathcal{E}_1} \|h_2\|_{\mathcal{E}_1}$. In particular

$$(9.12) \quad \|M_{\mathcal{H}_{\text{ext}} h}\| \leq \|h\|_{\mathcal{E}_1}, \quad \|M_{\mathbf{E} \mathcal{H}_{\text{ext}} h}\| \leq 2 \|h\|_{\mathcal{E}_1}.$$

Here $\mathbf{E} = y \cdot \nabla$.

Proof. A product $S_i t^h \Psi_{j,s}$ has, by (9.8), output profiles $t^{i+j-k+h} J_s^{\beta_j}$ with positive angular coefficients of sum one. If $k \geq j$, raise the radial parameter by $2(k - j)$ using (9.9), then multiply by the remaining powers of t . If $k < j$, use $2(j - k)$ powers for lowering by (9.10). There are enough powers because $i + j - k \geq 2(j - k)$ is equivalent to $k \geq j - i$, part of the triangle support. The remaining powers use the fixed-parameter positive multiplication stencil. Every conversion has coefficient sum one. The output total degree is at most the input degree plus $2i + 2h$, and $2^k \leq 2^{i+j}$. Equation (9.2) proves (9.11), first on basis columns and then by absolute summation and completion.

For boundary products use the same angular support, $1 + k \leq (1 + i)(1 + j)$, and $\varpi_p(2k) \leq \varpi_p(2i) \varpi_p(2j)$. This proves the algebra inequality. The harmonic multiplier bounds follow by summing (9.11) and using $\mathbf{E} S_i = 2i S_i$. These coefficient multipliers equal pointwise multiplication on $\mathcal{C}$: equality holds on finite source sums, which converge in $\mathcal{C}$ and hence in L^2 , while multiplication by a fixed solid polynomial is continuous in L^2 . For a shape extension its finite sums converge uniformly on the ball and in multiplier norm, so both limits give the same pointwise multiplier. $\square$

9.3. Exact Poisson columns and traces. Let $K_D = \Delta_D^{-1}$ on the unit ball, so $K_D = -H_0^{-1}$. Its sign is part of this convention. The same operator restricted to $\mathcal{G}_p$ will occasionally be denoted by K .

Lemma 9.5 (Poisson columns). *Fix an angular input j , and put $B = 2j$, $\beta = B + p - 1$, $b_0 = \beta + 1$, and $d_s = \beta + 2s$. For $s \geq 1$,*

$$(9.13) \quad K_D \Psi_{j,s} = S_j k_s(t),$$

$$k_s = \frac{J_{s-1}^\beta}{4d_s(d_s+1)} - \frac{J_s^\beta}{2d_s(d_s+2)} + \frac{J_{s+1}^\beta}{4(d_s+1)(d_s+2)}.$$

For $s = 0$,

$$(9.14) \quad k_0 = \frac{t-1}{4b_0} = \frac{J_1^\beta - J_0^\beta}{4b_0(b_0+1)}.$$

For $s \geq 1$ the derivative identities are

$$(9.15) \quad k'_s = \frac{J_s^{\beta+1} - J_{s-1}^{\beta+1}}{4(d_s+1)},$$

$$(9.16) \quad \beta k_s + t k'_s = \frac{J_{s+1}^{\beta-1} - J_s^{\beta-1}}{4(d_s+1)}.$$

For $s = 0$ their replacements are $k'_0 = 1/(4b_0)$ and $\beta k_0 + t k'_0 = J_1^{\beta-1}/(4b_0)$.

Proof. Write $J_s^\beta(t) = \sum_{m=0}^s a_{s,m} t^m$, where (2.19) gives

$$a_{s,m} = (-1)^{s-m} \binom{s}{m} \binom{s+\beta+m}{s}.$$

The coefficient equation for $4\{t k''_s + (\beta+1)k'_s\} = J_s^\beta$ is

$$[t^{m+1}]k_s = \frac{a_{s,m}}{4(m+1)(\beta+m+1)}.$$

For $1 \leq m \leq s$, the adjacent-layer ratios are

$$\frac{a_{s-1,m}}{a_{s,m}} = -\frac{s-m}{s+\beta+m}, \quad \frac{a_{s+1,m}}{a_{s,m}} = -\frac{s+\beta+m+1}{s+1-m},$$

whereas

$$\frac{a_{s,m-1}}{a_{s,m}} = -\frac{m(\beta+m)}{(s+1-m)(s+\beta+m)}.$$

With $d = d_s = \beta + 2s$, the coefficient in the proposed formula, divided by $a_{s,m}$, is therefore

$$\begin{aligned} & -\frac{s-m}{4d(d+1)(s+\beta+m)} - \frac{1}{2d(d+2)} - \frac{s+\beta+m+1}{4(d+1)(d+2)(s+1-m)} \\ & = -\frac{1}{4(s+1-m)(s+\beta+m)}. \end{aligned}$$

The equality follows by multiplication by the common denominator and substitution of $d = \beta + 2s$; its right side is exactly $a_{s,m-1}/\{4m(\beta+m)a_{s,m}\}$. For the top coefficient use

$$\frac{a_{s+1,s+1}}{a_{s,s}} = \frac{(d+2)(d+1)}{(s+1)(s+\beta+1)}.$$

Finally the three column coefficients sum to zero, fixing the constant coefficient by $k_s(1) = 0$. This proves (9.13) for every $s \geq 1$.

For (9.15), the coefficient of t^m in $J_s^{\beta+1} - J_{s-1}^{\beta+1}$, divided by $a_{s,m}$, equals

$$\frac{s+\beta+m+1}{\beta+m+1} + \frac{s-m}{\beta+m+1} = \frac{d+1}{\beta+m+1}.$$

This is exactly $4(d+1)[t^m]k'_s/a_{s,m}$. For (9.16), at $m \geq 1$ the left coefficient is $(\beta+m)[t^m]k_s = a_{s,m-1}/(4m)$. The two binomial ratios in the right numerator give

$$\frac{[t^m](J_{s+1}^{\beta-1} - J_s^{\beta-1})}{a_{s,m-1}} = \frac{s+\beta+m}{m} + \frac{s+1-m}{m} = \frac{d+1}{m}.$$

Their nonconstant coefficients agree. The previous derivative identity gives $k'_s(1) = 0$; both sides of (9.16) vanish at one, so their constants agree as well. For $s = 0$, the elementary expression (9.14) solves the equation and gives both replacement derivative identities directly. In particular this layer has the denominator $4b_0$ in (9.14). By (2.13), these polynomial functions solve the Dirichlet Poisson problem. Uniqueness identifies them with $K_D\Psi_{j,s}$. $\square$

Proposition 9.6 (Poisson bounds and exact clamping). *On all of $\mathcal{C}$, including the layer $s = 0$,*

$$(9.17) \quad \|K_D\| \leq \frac{3}{p(p+1)}, \quad \|EK_D\| \leq C/p, \quad \|E^2K_D\| \leq C.$$

More precisely, its weighted basis columns satisfy, for all $j, s \geq 0$,

$$(9.18) \quad \begin{aligned} \|K_D\Psi_{j,s}\|_C &\leq \frac{C}{(p+2j+2s)^2} \|\Psi_{j,s}\|_C, \\ \|EK_D\Psi_{j,s}\|_C &\leq \frac{C}{p-1+2j+2s} \|\Psi_{j,s}\|_C. \end{aligned}$$

The normal trace is the bounded map

$$(9.19) \quad \partial_\nu K_D f = \sum_{j \geq 0} \frac{f_{j,0}}{2(2j+p)} G_j, \quad \|\partial_\nu K_D\|_{C \rightarrow \mathcal{E}_1} \leq \frac{1}{2}.$$

Consequently $U = 1 + K_D g$ has Dirichlet trace one and normal trace zero exactly when $g \in \mathcal{G}_p$.

Proof. The absolute sums of the columns in (9.13) and (9.14) are $1/(d_s(d_s+2))$ and $1/(2b_0(b_0+1))$. The only upward radial shift increases degree by two, whose weight ratio is at most $e^{2/p} < 2$. This proves the first bound. On one profile,

$$EK_D = Bk_s + 2tk'_s, \quad E^2K_D = B^2k_s + 4(1-p)tk'_s + tJ_s^\beta.$$

The second equality uses the Poisson equation to eliminate k''_s . The derivative column in parameter $\beta+1$ has sum at most $1/(2(d_s+1))$; multiplication by t lowers it positively to parameter β . The separate $s = 0$ column gives the same bound up to an absolute constant. Since $B \leq d_s$ and $d_s \geq p-1$, these formulas prove the other two operator bounds and (9.18).

At the boundary $S_j = G_j$ and $\partial_\nu(S_j k_s) = 2jk_s(1)G_j + 2k'_s(1)G_j$. Only $s = 0$ contributes, giving (9.19). Its weighted trace bound follows from $(1+j)/(2(2j+p)) \leq 1/2$ and $\varpi_p(2j) \leq \varpi_p(2j+2s)$. For a general coefficient source, approximate by finite sums. The embedding in L^2 and Dirichlet H^2 regularity give $K_D f_n \rightarrow K_D f$ in $H^2(B)$. The displayed trace maps converge in $\mathcal{E}_1$ and thus in every fixed boundary Sobolev norm, by exponential angular decay. The Sobolev trace theorem identifies this limit with the actual normal trace of the H^2 solution. Dirichlet traces are zero by Poisson construction. Finally orthogonality of the G_j shows that the trace vanishes if and only if each $f_{j,0}$ vanishes. This proves the asserted characterization on the completed spaces. $\square$

9.4. Raising coupling and the clamped gain.

Lemma 9.7 (Contraction under parameter raising). *For $\beta \geq 0$ and integers $s \geq 1$, $m \geq 0$, the coefficient vectors of J_s^β and J_{s-1}^β in parameter $\beta + m$ are probability vectors: their entries are nonnegative and sum to one by (9.9). Their ℓ^1 distance is at most*

$$(9.20) \quad 2e^2 \frac{z}{z+m}, \quad z = 2s + \beta + 1.$$

The bound is uniform in all its parameters.

Proof. One raising step in (9.9) keeps index σ or lowers it by one, with down probability $q_\sigma(b) = \sigma/(2\sigma + b + 1)$. Couple the chains started at $s, s-1$ with one uniform random number at each step. Since q_σ increases in σ , the indices remain adjacent or merge, and after merging remain equal. Before merging, at step h their upper index is at most s and their parameter is $b = \beta + h$. Their merging probability is at least

$$q_\sigma(b) - q_{\sigma-1}(b) = \frac{b+1}{(2\sigma+b+1)(2\sigma+b-1)} \geq \frac{1}{z+h} - \frac{2s}{(z+h)^2}.$$

This also holds when the lower index is zero. The probability of no merger by step m is at most

$$\exp\left(-\sum_{h < m} \frac{1}{z+h} + 2s \sum_{h < m} \frac{1}{(z+h)^2}\right) \leq e^2 \frac{z}{z+m}.$$

Here the first sum is at least $\log((z+m)/z)$, and the second multiplied by $2s$ is at most $2s(z^{-1} + z^{-2}) \leq 2$. The total variation distance of the two distributions is no greater than the nonmerger probability. Their ℓ^1 distance is twice total variation, giving (9.20). Every subsequent positive stencil of sum one is an ℓ^1 contraction, so it preserves this estimate. $\square$

Lemma 9.8 (Clamped multiplication gain). *For integers $c, h \geq 0$ put $T = S_c t^h$ and $D = 2c + 2h$. On clamped sources, that is on $\mathcal{G}_p$,*

$$(9.21) \quad \|M_T \mathbf{E}K\|_{\mathcal{G}_p \rightarrow \mathcal{C}} \leq C \frac{2^c \varpi_p(D)}{p+D}.$$

The constant is independent of c, h , the source indices, p and the admissible split.

Proof. Use a source $\Psi_{j,s}$ with $s \geq 1$, and put $B = 2j$, $\beta = B + p - 1$, $d_s = \beta + 2s$. For an angular output of degree $L = 2k$ set

$$n_\downarrow = h + (2c + B - L)/2, \quad n_\uparrow = h + (2c - B + L)/2.$$

There are $n_\downarrow$ powers of t . First raise the source parameter by $n_\uparrow$, then lower it with those $n_\downarrow$ powers; the net change is $n_\uparrow - n_\downarrow = L - B$. The triangle support guarantees $n_\downarrow, n_\uparrow \geq 0$. If $D \leq 8(B+p)$, the ordinary multiplier bound and the column bound C/d_s for $\mathbf{E}K$ prove the claim, since $p+D \leq 9(B+p) \leq 9(d_s+1)$.

Suppose $D > 8(B+p)$. Then $n_\uparrow \geq D/4$. If $2c < D/2$, one has $h > D/4$ and $n_\uparrow \geq h$. If $2c \geq D/2$, the triangle inequality gives $n_\uparrow \geq h + 2c - B > D/4$. Since $s \geq 1$, the Poisson column has the zero-sum form

$$k_s = \frac{J_{s-1}^\beta - J_s^\beta}{4d_s(d_s+1)} + \frac{J_{s+1}^\beta - J_s^\beta}{4(d_s+1)(d_s+2)}.$$

After $n_\uparrow$ raises, Lemma 9.7 bounds its coefficient sum by $C/(d_s(d_s+n_\uparrow))$. The term $2tk'_s$ begins with the adjacent difference in (9.15), in parameter $\beta + 1$. Raise it by $n_\uparrow$ and then lower with $n_\downarrow + 1$ powers, including its extra t . Its coefficient sum is at most $C/(d_s + n_\uparrow)$. Since $B \leq d_s$, the combination $Bk_s + 2tk'_s$ is bounded by $C/(d_s + n_\uparrow) \leq C/(p+D)$.

All output degrees are at most the input degree plus $D+2$. The extra factor $\varpi_p(2)$ is bounded by two, and angular weights are at most 2^{c+j} . Summing the positive angular coefficients in (9.8), and using (9.2), proves (9.21). $\square$

9.5. Gradient estimates with one shape moment.

Lemma 9.9 (Gradient profiles on every source layer). *Let $A = 2i > 0$, $B = 2j$, $\ell_{\text{out}} = 2k$, and $h = (A + B - \ell_{\text{out}})/2$ for an angular product output in (9.8). Put $Z_{\text{grad}} = h + \ell_{\text{out}} + p - 1$. For $K_D \Psi_{j,s} = S_j k_s$ the two output profiles of $\nabla S_i \cdot \nabla K_D$ and $\nabla S_i \cdot \nabla E K_D$, with the coefficient c_{ij}^k suppressed, are respectively*

$$(9.22) \quad 2hZ_{\text{grad}}t^{h-1}k_s + 2At^h k'_s,$$

$$(9.23) \quad At^h J_s^\beta + (4hZ_{\text{grad}} - 2AB - 4A(p-1))t^h k'_s + 2BhZ_{\text{grad}}t^{h-1}k_s.$$

The term with $h = 0$ and a negative power of t is absent. Consequently, including $s = 0$,

$$(9.24) \quad \|\nabla S_i \cdot \nabla K_D\|_{\mathcal{C} \rightarrow \mathcal{C}} \leq CA2^i \varpi_p(A)/p,$$

$$(9.25) \quad \|\nabla S_i \cdot \nabla E K_D\|_{\mathcal{C} \rightarrow \mathcal{C}} \leq CA2^i \varpi_p(A).$$

Proof. Since both solid factors are harmonic, $2\nabla S_i \cdot \nabla S_j = \Delta(S_i S_j)$. Use $S_i S_j = \sum c_{ij}^k S_k t^h$ and (2.13); its coefficient is $4hZ_{\text{grad}}$. The derivative of the radial profile contributes $2At^h k'_s$, proving (9.22). Apply that formula to $Bk_s + 2tk'_s$. The Poisson equation gives $(Bk_s + 2tk'_s)' = J_s^\beta/2 - (B+2p-2)k'_s$; substitution proves (9.23). The triangle support gives $0 \leq h \leq \min(A, B)$ and

$$(9.26) \quad hZ_{\text{grad}} = h(A + B - h + p - 1) \leq A(B + p - 1).$$

For $A \leq B$ the left side is increasing up to $h = A$. For $A \geq B$ use its value at $h = B$ and $B(A + p - 1) \leq A(B + p - 1)$.

In every nonsaturated output the positive parameter conversions of Lemma 9.3 apply separately to the two terms in (9.22). When $\ell_{\text{out}} < B$, the required lowerings are $B - \ell_{\text{out}}$ for the k_s term and $B - \ell_{\text{out}} + 1$ for the derivative term, whose parameter is $\beta + 1$. The available powers are $h - 1$ and h , respectively. They suffice whenever $\ell_{\text{out}} \geq B - A + 2$; the degrees are even. When $\ell_{\text{out}} = B$, the k_s term keeps its parameter. The derivative term starts in parameter $\beta + 1$ and needs one lowering; one of its $h = i \geq 1$ powers of t supplies it. When $\ell_{\text{out}} > B$, raise the k_s term by $\ell_{\text{out}} - B$ and the derivative term by $\ell_{\text{out}} - B - 1 \geq 1$, followed by their nonnegative remaining powers of t . The case $h = 0$ omits the first term. Thus the only remaining case is $\ell_{\text{out}} = B - A$, $h = A$, where the two terms combine exactly to

$$2At^{A-1}(\beta k_s + tk'_s).$$

Use (9.16) before the remaining $A - 1$ lowerings. This supplies the one missing lowering step. The separate harmonic-column identity in Lemma 9.5 proves the same statement at $s = 0$.

The absolute Poisson column sums are $O((B + p + 2s)^{-2})$, and the two first-derivative columns have sums $O((B + p + 2s)^{-1})$. Together with (9.26), these give a per-output bound $CA/(B+p)$ for (9.22). In (9.23) the three terms have bounds CA , $4CA$, and $2CA$ after positive conversion. In the saturated case use their combined expression

$$At^A J_s^\beta + 2ABt^{A-1}(\beta k_s + tk'_s),$$

which gives the same bound, also at $s = 0$. In that layer the coefficient norm of k_0 is $1/(2b_0(b_0 + 1))$, which supplies the factor $1/(B+p)$. All final degrees are at most input degree plus A . The positive angular coefficients sum to one and $2^k \leq 2^{i+j}$. Applying (9.2) proves (9.24)–(9.25). $\square$

Proposition 9.10 (Grouped shape derivatives). *For $H_i = \mathcal{H}_{\text{ext}} h_i$, the following bounds hold, with absolute constants:*

$$(9.27) \quad \|M_{\nabla H_1 \cdot \nabla H_2}\|_{\mathcal{C} \rightarrow \mathcal{C}} \leq Cp \|h_1\|_{\mathcal{E}_1} \|h_2\|_{\mathcal{E}_1},$$

$$(9.28) \quad \|M_{\mathbf{E}(\mathbf{E}-1)H_1} \mathbf{E}K\|_{\mathcal{G}_p \rightarrow \mathcal{C}} \leq C \|h_1\|_{\mathcal{E}_1},$$

$$(9.29) \quad \|M_{\nabla H_1 \cdot \nabla \mathbf{E}H_2} \mathbf{E}K\|_{\mathcal{G}_p \rightarrow \mathcal{C}} \leq C \|h_1\|_{\mathcal{E}_1} \|h_2\|_{\mathcal{E}_1}.$$

The last two bounds are on clamped sources in $\mathcal{G}_p$.

Proof. For a pair of harmonics of degrees $A = 2i, B = 2j$, the product of gradients has output coefficient $2hZ_{\text{grad}}$ and solid profile $S_k t^{h-1}$. It is zero if $h = 0$; otherwise its total degree is $A + B - 2$. From (9.26), $2hZ_{\text{grad}} \leq Cp(1+i)(1+j)$. The module bound and positive angular summation prove (9.27). For the second inequality apply Lemma 9.8 to the harmonic degree A , with coefficient $A(A-1)$. The quotient $A(A-1)/(p+A)$ is at most A , leaving one shape moment. For the last inequality its pairwise coefficient is $2BhZ_{\text{grad}}$. Apply the same clamped gain at degree $D = A + B - 2$. For $h \geq 1$, $Z_{\text{grad}} = A + B - h + p - 1 \leq p + A + B - 2$, so

$$\frac{2BhZ_{\text{grad}}}{p + A + B - 2} \leq 2B \min(A, B) \leq 2AB.$$

The grading and angular weights are bounded by the product of the two shape weights. Summation gives (9.29). $\square$

9.6. Double-trace lifts and source moments.

Lemma 9.11 (Exact double-trace lift). *For each angular index define*

$$(9.30) \quad L_j^D = S_j[1 - j(t-1)], \quad L_j^N = \frac{1}{2}S_j(t-1).$$

Their Dirichlet and radial normal traces are $(G_j, 0)$ and $(0, G_j)$, respectively. Their coefficient bounds and Laplacians are

$$(9.31) \quad \|L_j^D\|_{\mathcal{C}} \leq 3 \cdot 2^j \varpi_p(2j+2), \quad \|L_j^N\|_{\mathcal{C}} \leq \frac{2^j \varpi_p(2j+2)}{2j+p+1},$$

$$(9.32) \quad \Delta L_j^D = -2j(4j+2p)S_j, \quad \Delta L_j^N = (4j+2p)S_j.$$

For the defects of Corollary 6.4, let $f_N = h_N/R$, $q_N = 1 + \mathcal{H}_{\text{ext}} f_N$ and $d_N = q_N + \mathbf{E}q_N$. Define

$$(9.33) \quad N_N^{\text{ref}} = R(d_N|_{\partial B})N_N^{\text{phys}}, \quad \ell_N = \sum_j (D_{N,j} L_j^D + N_{N,j}^{\text{ref}} L_j^N).$$

For every fixed moment k and $p \geq P_{N,k}$,

$$(9.34) \quad \|\ell_N\|_{\mathcal{C}^{[k]}} \leq C_{N,k} \theta^2 p^{-N}, \quad \|\Delta \ell_N\|_{\mathcal{C}^{[k]}} \leq C_{N,k} \theta^2 p^{2-N}.$$

Proof. Differentiating (9.30) at $t = 1$ gives their traces. The identity $t-1 = (J_1^{\beta_j} - J_0^{\beta_j})/(\beta_j + 2)$ places both lifts in radial layers zero and one and gives (9.31). Formula (2.13) proves (9.32). The reference normal trace in (9.33) follows by differentiating the physical composition $u_N(Rq_N(y)y)$ in the radial direction on the sphere: $D(q_N y)\nu = d_N \nu$.

The multiplier $d_N|_{\partial B}$ is a fixed-degree polynomial with uniformly bounded coefficients at any fixed angular radius. Thus (6.23) bounds its product with the physical defect, losing only the factor $R = O(p)$. The factor $(2j+p+1)^{-1}$ in the normal lift cancels that factor. Choose a slightly larger fixed angular radius in (6.23) to absorb $\varpi_p(2j+2) \leq e^{(2j+2)/p}$ and the prescribed angular moments. This proves the first estimate in (9.34). The Laplacians in (9.32), including the extra p in the reference normal trace, cost at most Cp^2 after absorbing fixed powers of j by the same radius choice. This proves the second estimate. The sums converge in all stipulated

moment norms and their traces agree with the classical lifts by the same coefficient identities and smooth convergence. $\square$

Lemma 9.12 (Derivative columns and moments). *For $\beta \geq 0$, $s \geq 1$, and $0 \leq k < s$,*

$$(9.35) \quad \partial_t J_s^\beta = \sum_{k < s} d_{s,k} J_k^\beta, \quad d_{s,k} = (\beta + 2k + 1) \left(1 - (-1)^{s+k} \frac{\binom{k+\beta}{k}}{\binom{s+\beta}{s}} \right) \geq 0.$$

For every $0 \leq m \leq s$ the endpoint derivative is

$$(9.36) \quad (J_s^\beta)^{(m)}(1) = \frac{(s)_{\text{fall},m}(s + \beta + 1)_m}{m!}.$$

In particular, the first derivative column has sum $s(s + \beta + 1)$ and the second has sum $s(s - 1)(s + \beta + 1)(s + \beta + 2)/2$. Consequently

$$(9.37) \quad \|Eu\|_{\mathcal{C}^{[k]}} \leq Cp\|u\|_{\mathcal{C}^{[k+2]}}.$$

The Laplacian has positive radial columns of sum at most $C(p + 2j + 2s)^4$ and does not raise total degree. For each fixed $k \geq 0$,

$$(9.38) \quad \|\Delta u\|_{\mathcal{C}^{[k]}} \leq C_k \{p^4 \|u\|_{\mathcal{C}^{[k]}} + \|u\|_{\mathcal{C}^{[k+4]}}\}.$$

Proof. For $\beta > 0$, integration by parts gives, since $k < s$,

$$\int_0^1 t^\beta (J_s^\beta)' J_k^\beta dt = 1 - \beta J_k^\beta(0) \int_0^1 t^{\beta-1} J_s^\beta dt.$$

To obtain this expression, subtract $J_k^\beta(0)$ in the integrated derivative term; its remaining factor is t times a polynomial of degree below s , hence orthogonal to J_s^β . Rodrigues' formula and s integrations by parts give the remaining integral as $(-1)^s B(\beta, s + 1)$. Divide by the squared norm $(\beta + 2k + 1)^{-1}$ and use the endpoint value at zero to obtain (9.35). The formula at $\beta = 0$ is its finite continuous limit. Monotonicity in k of $\binom{k+\beta}{k}$ proves nonnegativity. Every further differentiation preserves coefficient positivity. For the endpoint values, write $J_s^\beta(1 + z) = \sum_m \mathfrak{d}_m z^m$ and use its differential equation

$$t(1 - t)J'' + \{\beta + 1 - (\beta + 2)t\}J' + s(s + \beta + 1)J = 0.$$

Its coefficient of z^m gives $\mathfrak{d}_{m+1} = (s - m)(s + \beta + 1 + m)\mathfrak{d}_m/(m + 1)^2$, with $\mathfrak{d}_0 = 1$. Iteration and multiplication by $m!$ prove (9.36). Evaluating the positive columns at one now gives their sums.

On one solid profile $E = B + 2t\partial_t$. The positive t stencil gives column sum $B + 2s(s + \beta + 1) \leq Cp(1 + 2j + 2s)^2$ and no increase in total degree. This proves (9.37). The profile Laplacian is $4(t\partial_t^2 + (\beta + 1)\partial_t)$, so its column sum is $4((J_s^\beta)''(1) + (\beta + 1)(J_s^\beta)'(1))$, bounded by $C(p + 2j + 2s)^4$. Each output degree is at most the input degree $D = 2j + 2s$, so its moment weight $(1 + D_{\text{out}})^k$ is at most $(1 + D)^k$. Multiplying the column bound by this weight and using $(p + D)^4 \leq 8(p^4 + D^4)$ gives (9.38) after absolute summation. $\square$

10. THE NONLINEAR MAP AND THE MATERIAL ISOMORPHISM

The chart in this section uses a harmonic extension on the whole ball. It is different from the homogeneous radial stretching used to derive the spectral flow. Both charts describe the same boundary domain at a fixed centre; only its spectrum is passed from one description to the other.

10.1. A smooth scalar chart on the ball. For a normalized boundary shape $f \in \mathcal{E}_1$ put

$$(10.1) \quad H_f = \mathcal{H}_{\text{ext}} f, \quad q = 1 + H_f, \quad d = q + \mathbb{E}q, \quad \Phi_f(y) = q(y)y.$$

For a field U on the reference ball define

$$(10.2) \quad P_{\Phi_f} \Delta U = [\Delta(U \circ \Phi_f^{-1})] \circ \Phi_f, \quad B_{\Phi_f} = P_{\Phi_f} \Delta + R^2.$$

For a smooth real chart, $B_{\Phi,D}$ denotes the restriction of B_Φ to the invariant Dirichlet graph domain $H^2(B) \cap H_0^1(B)$. Fix the analytic chart radius $\varepsilon_0 = 1/2$ and the working radius $\varepsilon_{\text{work}} = 1/4$. For every real $f \in \mathcal{E}_1$ with $\|f\|_{\mathcal{E}_1} < 1/2$, the map Φ_f is a diffeomorphism of the closed ball onto the region with radial boundary $1 + f$. Indeed $|q - 1| \leq \|f\|_{\mathcal{E}_1}$ and $|d - 1| \leq 2\|f\|_{\mathcal{E}_1} < 1$ on the ball, so $q, d > 0$. On each ray, $\Phi_f(\zeta\omega) = \zeta q(\zeta\omega)\omega$ keeps the direction fixed and has strictly positive radial derivative d . Distinct rays remain distinct and the radial map is one-to-one on each ray. Its Jacobian determinant is $q^{2p-1}d > 0$, including at the origin. The inverse function theorem and the radial endpoint give the global identification. By Lemma 9.2, the harmonic extension and chart are real analytic on a neighbourhood of the closed ball. Both axes and the origin are ordinary points of this chart.

Lemma 10.1 (Exact scalar pullback). *Let $w = \nabla q$, $M_q = w \cdot w$ and*

$$(10.3) \quad A_q = q^{-2} \left\{ I - \frac{y \otimes w + w \otimes y}{d} + \frac{M_q}{d^2} y \otimes y \right\}, \quad C_q = A_q : D^2 q.$$

Then

$$(10.4) \quad \begin{aligned} P_{\Phi_f} \Delta U &= q^{-2} \Delta U - \frac{2}{q^2 d} \nabla q \cdot \nabla \mathbb{E} U + \frac{M_q}{q^2 d^2} \mathbb{E}(\mathbb{E} + 1) U \\ &\quad + \left(\frac{2M_q}{qd^3} - \frac{C_q}{d} \right) \mathbb{E} U, \end{aligned}$$

$$(10.5) \quad C_q = q^{-2} \left\{ -\frac{2(\nabla q \cdot \nabla \mathbb{E} q - M_q)}{d} + \frac{M_q \mathbb{E}(\mathbb{E} - 1)q}{d^2} \right\}.$$

The second identity uses harmonicity of q .

Proof. Since $D\Phi_f = qI + y \otimes w$, its inverse is $q^{-1}(I - y \otimes w/d)$ and $D\Phi_f^{-1} D\Phi_f^{-T} = A_q$. The coordinate change formula for a scalar Laplacian is

$$P_{\Phi_f} \Delta U = A_q : D^2 U - D\Phi_f^{-1}(A_q : D^2 \Phi_f) \cdot \nabla U.$$

Now $A_q : D^2 \Phi_f = yC_q + 2A_q w$, and multiplication by the inverse matrix gives

$$D\Phi_f^{-1}(A_q : D^2 \Phi_f) = \frac{2w}{q^2 d} + y \left\{ \frac{C_q}{d} - \frac{2M_q}{q^2 d^2} - \frac{2M_q}{qd^3} \right\}.$$

On the other hand,

$$A_q : D^2 U = q^{-2} \left\{ \Delta U - \frac{2}{d} (\nabla q \cdot \nabla \mathbb{E} U - \nabla q \cdot \nabla U) + \frac{M_q}{d^2} \mathbb{E}(\mathbb{E} - 1) U \right\}.$$

The terms with $\nabla q \cdot \nabla U$ cancel, giving (10.4). Apply the last formula for $A_q : D^2 U$ to $U = q$ and use $\Delta q = 0$ to get (10.5). All computations are identities of smooth functions; extension to the coefficient spaces is established next. $\square$

Proposition 10.2 (Analytic map with a polynomial second derivative). *For $g \in \mathcal{G}_p$ and $\|f\|_{\mathcal{E}_1} < \varepsilon_0$, define*

$$(10.6) \quad \mathcal{F}(g, f) = B_{\Phi_f}(1 + Kg).$$

This is an analytic map $\mathcal{G}_p \times \{f \in \mathcal{E}_1 : \|f\|_{\mathcal{E}_1} < \varepsilon_0\} \rightarrow \mathcal{C}$, affine in g , and on $\|f\|_{\mathcal{E}_1} \leq \varepsilon_{\text{work}} = 1/4$ it satisfies

$$(10.7) \quad \|D^2\mathcal{F}(g, f)\| \leq Cp(1 + \|g\|_{\mathcal{C}}).$$

For real f , its coefficient extension is exactly the weak scalar pullback of $1 + Kg \in H^2(B)$. Its complexification is the analytic extension of the same algebraic operator formulas.

Proof. The harmonic module estimate gives $\|M_{q-1}\| \leq \|f\|_{\mathcal{E}_1}$. For $d-1 = H_f + \mathbf{E}H_f$, the same estimate applied to each solid coefficient gives

$$\|M_{d-1}\| \leq \sum_{j \geq 0} (1 + 2j)2^j \varpi_p(2j) |f_j| \leq 2 \sum_{j \geq 0} (1 + j)2^j \varpi_p(2j) |f_j| = 2\|f\|_{\mathcal{E}_1}.$$

Their inverse powers are norm-convergent Neumann series on $\|f\|_{\mathcal{E}_1} < 1/2$. On the closed working ball of radius $1/4$, differentiating these series any fixed number of times gives uniformly bounded multilinear multiplier maps. The same series converge uniformly on the closed ball, by (9.6), to the indicated pointwise inverse powers.

Apply (10.4) to $U = 1 + Kg$. All derivatives of the constant vanish. The term $q^{-2}\Delta Kg$ is $q^{-2}g$. Lemma 9.9 controls $\nabla q \cdot \nabla \mathbf{E}K$ with one shape moment. The operator $\mathbf{E}(\mathbf{E} + 1)K$ is bounded by (9.17), and its multiplier M_q is bounded by (9.27) with norm at most $Cp\|f\|_{\mathcal{E}_1}^2$. For the last term group

$$C_q \mathbf{E}K = q^{-2} \{-2d^{-1}(\nabla q \cdot \nabla \mathbf{E}q - M_q)\mathbf{E}K + M_q d^{-2}[\mathbf{E}(\mathbf{E} - 1)q]\mathbf{E}K\}.$$

The first grouped product is bounded by (9.29) and (9.27); the last one by (9.28) followed by the bounded multiplier M_q . The scalar multipliers commute, so the high shape derivative is always placed immediately next to $\mathbf{E}K$ on its clamped source, as in Proposition 9.10. Every numerator is thus a bounded linear, bilinear, or trilinear map from shapes to operators $\mathcal{G}_p \rightarrow \mathcal{C}$, of norm at most Cp . The inverse multiplier series prove analyticity of the operator coefficient $L_f \in \mathcal{B}(\mathcal{G}_p, \mathcal{C})$ in $\mathcal{F}(g, f) = R^2 + L_f g$. Its first two shape derivatives have norm at most Cp on the closed working ball of radius $1/4$. The two mixed derivatives and the pure shape second derivative give (10.7); the pure source second derivative is zero. The term $R^2(1 + Kg)$ is shape independent and contributes no second derivative.

To identify the extension, first use polynomial sources and finite shape sums, where the identity is classical. Approximate any $f \in \mathcal{E}_1$ by finite sums in its norm; the charts converge in C^2 and have uniformly positive q, d on any fixed smaller shape ball. The explicit inverse matrix is therefore uniformly bounded there. For each fixed p , their scalar pullbacks converge as maps $H^2(B) \rightarrow L^2(B)$ by the coordinate formula. Next approximate g in $\mathcal{C}$, hence in L^2 . Dirichlet regularity and uniqueness give convergence of Kg in $H^2(B)$. The bounded coefficient maps converge in $\mathcal{C}$, hence in L^2 , to the same pullback. This proves the assertion on the completed spaces. $\square$

10.2. Coefficient size of the known centre.

Lemma 10.3 (Fixed radial modes in coefficient norm). *Fix the finite mode set of the order- N centre in Theorem 6.2. For each fixed moment k and $p \geq P_{N,k}$, its regular radial modes, rescaled to the unit ball, have $\mathcal{C}^{[k]}$ norms at most $C_{N,k}p^{k+1}$. The bound is uniform in r and $0 < |\theta| \leq 20$.*

Proof. For one fixed j put $\nu = p + 2j - 1$ and write its normalized mode as $\zeta^{1-p} J_\nu(R\zeta) G_j / d_j^{\text{mode}}$. Here d_j^{mode} is either $J_\nu(R)$ for value normalization or $J'_\nu(R) - (p-1)J_\nu(R)/R$ for physical normal normalization. The finite recurrence (6.11) shows that $d_j^{\text{mode}}/J_{p-1}(R)$ has a nonzero analytic limit in every case. In particular the resonant seed is normalized by its normal trace.

In the solid basis the coefficients of this mode are

$$(10.8) \quad c_s = \frac{2(\nu + 1 + 2s)(-1)^s J_{\nu+1+2s}(R)}{R d_j^{\text{mode}}}.$$

To derive them, Rodrigues' formula and s integrations by parts in $t = \zeta^2$ give the integral identity

$$\int_0^1 \zeta^{\nu+1} J_\nu(R\zeta) J_s^\nu(\zeta^2) d\zeta = \frac{(-1)^s}{R} J_{\nu+2s+1}(R).$$

One may evaluate the final integral by the absolutely convergent Bessel power series and the beta integral. The derivative identity used in the integrations is $\partial_t^s(t^{-\nu/2} J_\nu(R\sqrt{t})) = (-R/2)^s t^{-(\nu+s)/2} J_{\nu+s}(R\sqrt{t})$. Dividing by the radial norm in (2.20) proves (10.8). Parseval and the Lommel integral at a general argument give

$$(10.9) \quad \sum_{s \geq 0} \frac{|c_s|^2}{\nu + 1 + 2s} = \frac{J_\nu(R)^2 - J_{\nu-1}(R)J_{\nu+1}(R)}{(d_j^{\text{mode}})^2} \leq C_{j,\text{rad}}.$$

The general Lommel identity follows by differentiating $x^2(J_\nu(x)^2 - J_{\nu-1}(x)J_{\nu+1}(x))/2$ and using the recurrences. The bound follows since every numerator value is a fixed recurrence shift of $J_{p-1}(R)$ and the normalization quotient has a nonzero limit.

For $s \leq S = C'_{j,\text{rad}}p + O_j(1)$, with a fixed $C'_{j,\text{rad}}$ large enough that $\nu + 1 + 2S \geq 2R$, Cauchy–Schwarz in (10.9) gives

$$\sum_{s \leq S} \varpi_p(2j + 2s) |c_s| \leq C_{j,\text{rad}} p.$$

There are $O_j(p)$ terms; their denominator reciprocals in the Cauchy–Schwarz factor are inverted to quantities $O_j(p)$, and the weights are bounded by a constant by (9.2). A fixed degree moment adds at most $C_{j,k} p^k$.

For the remaining tail use the positive ratio bound $J_{m+1}(R)/J_m(R) < R/(m+1)$ when $m \geq 2R$. Here positivity follows from $j_{m,1} > m$. For a proof of the ratio bound, its Riccati equation is $r' = 1 - (2m+1)r/x + r^2$, with $r(x) \sim x/(2m+2)$ at zero. At a first contact with $x/(m+1)$ for $x \leq m$, the difference of derivatives would be $-1 + x^2/(m+1)^2 < 0$, excluding upward contact. Applying the bound twice to (10.8) gives $|c_{s+1}/c_s| < R^2/(m(m+1)) \leq 1/4$ for $m = \nu + 1 + 2s \geq 2R$. The neighbouring grading ratio is at most $e^{2/p}$, and the moment ratio tends uniformly to one as p grows for fixed k . Thus the weighted tail is geometric with ratio below $1/2$ for large p , and is bounded by the head estimate. The angular factor 2^j is bounded for the fixed list of modes. $\square$

Proposition 10.4 (The deformed centre and its source). *Let $f_N = h_N/R$, $\Phi_N = \Phi_{f_N}$, and*

$$(10.10) \quad U_* = u_N \circ (R\Phi_N), \quad U_0 = U_* - \ell_N, \quad g_0 = \Delta U_0, \quad z_0 = (g_0, f_N).$$

For every fixed moment k and $p \geq P_{N,k}$, $g_0 \in \mathcal{G}_p$, $U_0 = 1 + Kg_0$, and

$$(10.11) \quad \|U_*\|_{\mathcal{C}^{[k]}} + \|U_0\|_{\mathcal{C}^{[k]}} \leq C_{N,k} p^{k+1}, \quad \|g_0\|_{\mathcal{C}^{[k]}} \leq C_{N,k} p^{k+5}.$$

For $p \geq P_{N,0}$, on the fixed product region

$$(10.12) \quad \{g \in \mathcal{G}_p : \|g - g_0\|_{\mathcal{C}} \leq 1\} \times \{f \in \mathcal{E}_1 : \|f\|_{\mathcal{E}_1} \leq 1/4\},$$

$$(10.13) \quad \sup \|D^2 \mathcal{F}\| \leq C_N p^6.$$

Proof. For each known centre mode, fix p and put $\nu = p + 2j - 1$. Its radial factor in the solid basis is the entire function of the complex squared-radius variable t ,

$$F_{\nu,j}(t) = \frac{1}{d_j^{\text{mode}}} \sum_{h \geq 0} \frac{(-1)^h (R/2)^{\nu+2h}}{h! \Gamma(\nu + h + 1)} t^h, \quad t \in \mathbb{C},$$

where Γ is Euler's gamma function [1, Eq. 10.2.2]. For real $t > 0$, this is $t^{-\nu/2} J_\nu(R\sqrt{t})/d_j^{\text{mode}}$. The radial expansion in Lemma 10.3 is $\sum_{s \geq 0} c_s J_s^\nu(t)$ on $0 \leq t \leq 1$ in weighted L^2 . For $m = \nu + 1 + 2s \geq 2R$, the ratio bound proved in that lemma gives $|c_{s+1}| \leq R^2 |c_s|/[m(m+1)]$. Since $m \geq 2s$, iteration from a fixed $s_0 \geq 1$ with $\nu + 1 + 2s_0 \geq 2R$ gives

$$|c_s| \leq |c_{s_0}| (R^2/4)^{s-s_0} \left(\frac{(s_0-1)!}{(s-1)!} \right)^2, \quad s \geq s_0.$$

For every fixed $A > 0$, Taylor's formula at one and (9.36) give

$$\sup_{|t-1| \leq A} |J_s^\nu(t)| \leq \sum_{h=0}^s \frac{[As(2s+\nu+1)]^h}{(h!)^2} \leq \exp\{2\sqrt{As(2s+\nu+1)}\}.$$

For fixed p, j, A , this envelope is at most exponential in s , so the factorial bound makes the radial series locally uniformly convergent on $\mathbb{C}$. Its sum is entire. On $[0, 1]$ uniform convergence and the original L^2 expansion identify that sum with $F_{\nu,j}$; continuity turns the almost-everywhere equality into equality on the interval. The identity theorem gives equality on $\mathbb{C}$, including at the deformed radial arguments.

Write $q = 1 + \mathcal{H}_{\text{ext}} f_N$ and $e = q^2 - 1$. The finite centre shape and the module estimate give $\|M_e\| \leq C_N \tau p^{-2} =: \epsilon_{\text{cmp},p}$. For one radial basis polynomial the exact Taylor identity is

$$J_s^\beta(q^2 t) = \sum_{m=0}^s \frac{e^m}{m!} t^m \partial_t^m J_s^\beta(t).$$

The derivative columns and the positive t stencils show that $t^m \partial_t^m J_s^\beta$, whose degree is no greater than s , has coefficient sum $(s)_{\text{fall},m} (s + \beta + 1)_m / m!$. It follows that its relative composition norm is at most

$$(10.14) \quad \sum_{m=0}^s \epsilon_{\text{cmp},p}^m \frac{(s)_{\text{fall},m} (s + \beta + 1)_m}{(m!)^2} \leq \exp\{2\sqrt{\epsilon_{\text{cmp},p} s (2s + \beta + 1)}\}.$$

Indeed bound the product by $[s(2s + \beta + 1)]^m$ and use $\sum z^m / (m!)^2 \leq e^{2\sqrt{z}}$, which follows by comparing with the even terms of the exponential using $\binom{2m}{m} \leq 4^m$. The extra solid factor is q^{2j} , a bounded multiplier because j belongs to the fixed finite set of centre modes.

On the $s = O_N(p)$ head the right side of (10.14) is bounded by C_N . On the tail it is at most $\exp(C_N(s+p)/p)$. Multiply the geometric Bessel tail in Lemma 10.3 by this envelope and by the weight ratio $e^{2/p}$. For large p the resulting geometric ratio is still below $1/2$. Thus each composed mode series converges in $\mathcal{C}$ with norm at most $C_N p$. The finite polynomial chart increases each total degree by at most a fixed multiple depending on N . Therefore each output moment is bounded by a constant times the corresponding input moment; the same argument gives convergence and the bounds in all fixed moments k . For this fixed p , the radial entire-function identity makes the composed partial sums converge uniformly on the closed ball to $S_j(qy) F_{\nu,j}(q^2 t)$: the arguments $q^2 t$ range over a compact set, and $S_j(qy)$ is a fixed polynomial. The polynomial partial sums also converge in $\mathcal{C}$ by the preceding coefficient estimate, hence in $L^2(B)$ by Lemma 9.2. Uniform convergence on the closed ball gives the same L^2 limit. Uniqueness of that limit and injectivity of $\mathcal{C} \hookrightarrow L^2(B)$ identify the coefficient sum with the

composed mode. Adding the finite list of centre modes proves the asserted identification and bounds for U_* . The finite field amplitudes are uniformly bounded by the centre construction. Subtracting the lift uses (9.34) and proves the assertion for U_0 . The Laplacian bound (9.38), with moments included, then gives $\|g_0\|_{\mathcal{C}^{[k]}} \leq C_{N,k} p^{k+5}$.

By construction the two traces of U_0 are exactly 1, 0. Dirichlet Poisson uniqueness yields $U_0 = 1 + K_D g_0$, and (9.19) gives $g_0 \in \mathcal{G}_p$. Finally (10.7) and the p^5 source bound give (10.13) on the product region (10.12). Its exponent does not depend on N . $\square$

10.3. Residuals in the same coefficient space.

Proposition 10.5 (The residual and its material derivative). *At the centre z_0 of (10.10), put $E_0 = \mathcal{F}(z_0)$. For every fixed moment k and $p \geq P_{N,k}$,*

$$(10.15) \quad \|E_0\|_{\mathcal{C}^{[k]}} \leq C_{N,k} \theta^2 p^{2-N}, \quad \|\mathbb{E}E_0\|_{\mathcal{C}^{[k]}} \leq C_{N,k} \theta^2 p^{3-N}.$$

For a shape variation $\eta \in \mathcal{E}_1$, define

$$(10.16) \quad \mathcal{V}_{\text{res}}(\delta g, \eta) = (\mathcal{H}_{\text{ext}} \eta) d_0^{-1} \mathbb{E}E_0, \quad d_0 = 1 + \mathcal{H}_{\text{ext}} f_N + \mathbb{E} \mathcal{H}_{\text{ext}} f_N.$$

Then, for $p \geq P_N$,

$$(10.17) \quad \|\mathcal{V}_{\text{res}}\|_{\mathcal{X}_p \rightarrow \mathcal{C}} \leq C_N \theta^2 p^{3-N}.$$

Proof. The interior equation of the finite centre is exact, so $B_{\Phi_N} U_* = 0$ and $E_0 = -B_{\Phi_N} \ell_N$. The lift occupies only radial layers zero and one. On these layers the operators Δ , $\mathbb{E}^2$ and $\nabla a \cdot \nabla \mathbb{E}$, for any fixed solid polynomial a , have coefficient bounds $C_a p^2$ times a fixed number of angular moments. On layer one,

$$\mathbb{E} \Psi_{j,1} = (2j+2) \Psi_{j,1} + 2(\beta_j+1) \Psi_{j,0}.$$

On a solid monomial $S_l t^h$ of bounded radial power, $\Delta(S_l t^h) = 4h(h+2l+p-1)S_l t^{h-1}$ costs one factor $p+l$. Writing $J_1^{\beta_j} = (\beta_j+2)t - (\beta_j+1)$ adds at most one such factor. Multiplication by the fixed polynomial a has bounded angular shifts and bounded radial powers. Finally $2\nabla a \cdot \nabla U = \Delta(aU) - a\Delta U - (\Delta a)U$. In the displayed Euler formula the large coefficient $p+j$ occurs on the harmonic layer, whose Laplacian product costs only one additional factor $p+j$; the layer-one coefficient is only $2j+2$. The product therefore costs at most $C_a(p+j)^2$. The direct Euler bound also gives Cp^2 after two fixed moments.

At the fixed finite chart, $q_0 - 1, d_0 - 1$ have coefficient multiplier norm $O_N(\tau p^{-2})$. Put

$$\epsilon_{\text{rec}} = \max\{\|M_{q_0-1}\|_{\mathcal{C} \rightarrow \mathcal{C}}, \|M_{d_0-1}\|_{\mathcal{C} \rightarrow \mathcal{C}}\},$$

and increase the threshold until $\epsilon_{\text{rec}} < 1/2$. Their inverse powers preserve all fixed moments: multiplication by a polynomial of degree D_N increases degree by at most D_N , and $\sum_{v \geq 0} (1 + vD_N)^k \epsilon_{\text{rec}}^v < \infty$ for $\epsilon_{\text{rec}} < 1/2$. The constants may depend on N, k but not on p, r, θ . Apply the two-layer estimates to every term of (10.4) and then these inverse multipliers. Equation (9.34) and $R^2 = O(p^2)$ give the first estimate in (10.15). Apply (9.37), using that first estimate at moment $k+2$, to obtain the second. The module bound for $\mathcal{H}_{\text{ext}} \eta$ and the fixed inverse multiplier d_0^{-1} then prove (10.17). $\square$

10.4. The exact material map and its inverse. The radial harmonic chart specifies the interior extension as well as the boundary displacement.

Theorem 10.6 (Material isomorphism). *Fix $N \geq 4$ and take p sufficiently large for the centre above. Put*

$$(10.18) \quad D_p = -\partial_{\nu\nu} U_0|_{\partial B}, \quad \mathfrak{f}_0 = \frac{\mathbb{E}U_0}{d_0}, \quad T_\eta = (\mathcal{H}_{\text{ext}} \eta) \mathfrak{f}_0.$$

The map

$$(10.19) \quad \mathcal{Q}_{\text{mat}}(\delta g, \eta) = \Delta(K\delta g - T_\eta) : \mathcal{X}_p \longrightarrow \mathcal{C}$$

is a bounded isomorphism, with

$$(10.20) \quad \|\mathcal{Q}_{\text{mat}}\| \leq C_N p^8, \quad \|\mathcal{Q}_{\text{mat}}^{-1}\| \leq C_N p^6.$$

If $J_D = B_{\Phi_N, D} K_D$, then the exact factorization is

$$(10.21) \quad D\mathcal{F}(z_0) = J_D \mathcal{Q}_{\text{mat}} + \mathcal{V}_{\text{res}}.$$

All constants are uniform in the admissible split and $0 < |\theta| \leq 20$; the exponents are independent of N .

Proof. The centre has constant Dirichlet trace and zero gradient, so its boundary Hessian is purely normal-normal. Evaluating its equation at the boundary gives

$$(10.22) \quad D_p = \frac{R^2 - E_0|_{\partial B}}{\nu^T A_{q_0} \nu}.$$

Indeed all drift terms multiply the zero gradient, and $D^2 U_0 = (\partial_{\nu\nu} U_0) \nu \otimes \nu$ there. The finite centre polynomial and the boundary algebra show $d_0|_{\partial B} = 1 + O_N(\tau p^{-2})$ and $\nu^T A_{q_0} \nu = 1 + O_N(\tau p^{-2})$ in $\mathcal{E}_1$. For the latter assertion insert (10.3); the product formula in the proof of (9.27), with the extra boundary weight $1 + k \leq (1+i)(1+j)$ on each angular product, gives

$$\|M_{q_0}|_{\partial B}\|_{\mathcal{E}_1} \leq C_p \left(\sum_j (1+j)^2 2^j \varpi_p(2j) |f_{N,j}| \right)^2 \leq C_N \tau^2 p^{-3}.$$

Here all fixed moments of the finite centre shape are $O_N(\tau p^{-2})$ by (6.7). The trace of E_0 in $\mathcal{E}_1$ is bounded by its first source moment, since $J_s^{\beta j}(1) = 1$. To identify this coefficient trace, take polynomial sums F_m of E_0 and put $h_m = F_m|_{\partial B}$. The first source moment gives $h_m \rightarrow h$ in $\mathcal{E}_1$, while the fourth source moment in Proposition 10.5 and (9.38) gives $\Delta F_m \rightarrow \Delta E_0$ in L^2 . Since $F_m - \mathcal{H}_{\text{ext}} h_m$ has zero Dirichlet trace,

$$F_m - \mathcal{H}_{\text{ext}} h_m = K_D \Delta F_m.$$

At fixed p , harmonic extension and Dirichlet Poisson regularity make both terms on the right of $F_m = \mathcal{H}_{\text{ext}} h_m + K_D \Delta F_m$ converge in H^2 . Their L^2 limit is E_0 , hence $E_0 = \mathcal{H}_{\text{ext}} h + K_D \Delta E_0 \in H^2$ and its Sobolev trace is h . Thus (10.15) and the boundary algebra give

$$(10.23) \quad \|D_p^{-1}\|_{\mathcal{E}_1} \leq C_N p^{-2}$$

by the convergent inverse series in (10.22). In particular D_p has no zero.

The moment bounds for the fixed centre and its reciprocal d_0 give $\|\mathbf{f}_0\|_{\mathcal{C}^{[k]}} \leq C_{N,k} p^{k+4}$ by (9.37). Using (9.38) with its sum of two powers yields

$$(10.24) \quad \|\Delta \mathbf{f}_0\|_{\mathcal{C}} \leq C_N p^8.$$

Since $\mathbf{f}_0$ has zero Dirichlet trace, Dirichlet uniqueness implies $\mathbf{f}_0 = K_D \Delta \mathbf{f}_0$. Harmonicity of $\mathcal{H}_{\text{ext}} \eta$ and (9.24) therefore give

$$(10.25) \quad \|\Delta T_\eta\|_{\mathcal{C}} \leq \|M_{\mathcal{H}_{\text{ext}} \eta} \Delta \mathbf{f}_0\|_{\mathcal{C}} + 2\|\nabla(\mathcal{H}_{\text{ext}} \eta) \cdot \nabla K_D \Delta \mathbf{f}_0\|_{\mathcal{C}} \leq C_N p^8 \|\eta\|_{\mathcal{E}_1}.$$

One moment of the unknown shape enters. This proves boundedness of (10.19).

For an arbitrary source $\psi \in \mathcal{C}$ set

$$(10.26) \quad W = K_D \psi, \quad N_W = \partial_\nu W, \quad \eta = (d_0|_{\partial B}) N_W / D_p, \quad \delta g = \psi + \Delta T_\eta.$$

Equations (9.19) and (10.23) imply $\|\eta\|_{\mathcal{E}_1} \leq C_N p^{-2} \|\psi\|_{\mathcal{C}}$; then (10.25) gives $\|\delta g\|_{\mathcal{C}} \leq (1 + C_N p^6) \|\psi\|_{\mathcal{C}}$. At the boundary,

$$T_\eta = 0, \quad \partial_\nu T_\eta = -D_p \eta / d_0.$$

This follows by differentiating $f_0 = (EU_0)/d_0$: $EU_0 = 0$ there and $\partial_\nu EU_0 = \partial_{\nu\nu} U_0 = -D_p$. Hence $W + T_\eta$ has both traces zero. It is the Dirichlet Poisson solution with source δg ; (9.19) implies $\delta g \in \mathcal{G}_p$ and $K\delta g = W + T_\eta$. This proves that (10.26) is a right inverse to $\mathcal{Q}_{\text{mat}}$ and gives its p^6 bound. Conversely, for any input to $\mathcal{Q}_{\text{mat}}$, the field $W = K\delta g - T_\eta$ has zero Dirichlet trace and is recovered uniquely from its Laplacian. Its normal trace is $D_p \eta / d_0$, so (10.26) recovers η and then δg . This proves the left inverse as well.

It remains to prove (10.21). For the chart variation $\Phi_N + t(\mathcal{H}_{\text{ext}}\eta)y$, its reference material vector is $v_\eta = D\Phi_N^{-1}((\mathcal{H}_{\text{ext}}\eta)y) = y(\mathcal{H}_{\text{ext}}\eta)/d_0$. Differentiation of the two compositions in (10.2) gives the exact scalar commutator identity

$$\delta(P_\Phi \Delta)U_0 = -P_{\Phi_N} \Delta(v_\eta \cdot \nabla U_0) + v_\eta \cdot \nabla(P_{\Phi_N} \Delta U_0).$$

It follows either by differentiating the inverse chart or by applying the chain rule first to a smooth physical test function. Since $v_\eta \cdot \nabla U_0 = T_\eta$, adding the frequency term and the source variation yields

$$D\mathcal{F}(z_0)(\delta g, \eta) = B_{\Phi_N}(K\delta g - T_\eta) + (\mathcal{H}_{\text{ext}}\eta)d_0^{-1}EE_0.$$

The field in parentheses has zero Dirichlet trace, so it equals $K_D \mathcal{Q}_{\text{mat}}(\delta g, \eta)$. This proves (10.21) on finite variations. To pass to the completion, fix p and the smooth real centre chart. Dirichlet regularity on the ball gives a bounded map $K_D : L^2(B) \rightarrow H^2(B) \cap H_0^1(B)$; the smooth coefficients in (10.4) give a bounded map $B_{\Phi_N, D} : H^2(B) \cap H_0^1(B) \rightarrow L^2(B)$. These continuity bounds use only ball regularity and the fixed chart, independently of the Dirichlet inverse estimates below. For $\dot{z} \in \mathcal{X}_p$, take finite variations $\dot{z}_m \rightarrow \dot{z}$ in $\mathcal{X}_p$. The already proved boundedness of $\mathcal{Q}_{\text{mat}}$ gives $\mathcal{Q}_{\text{mat}} \dot{z}_m \rightarrow \mathcal{Q}_{\text{mat}} \dot{z}$ in $\mathcal{C}$, hence in $L^2(B)$ by Lemma 9.2. Consequently $J_D \mathcal{Q}_{\text{mat}} \dot{z}_m \rightarrow B_{\Phi_N, D} K_D \mathcal{Q}_{\text{mat}} \dot{z}$ in $L^2(B)$. The finite-variation identity also says $J_D \mathcal{Q}_{\text{mat}} \dot{z}_m = (D\mathcal{F}(z_0) - \mathcal{V}_{\text{res}})\dot{z}_m$. Its right side converges in $\mathcal{C}$ to $(D\mathcal{F}(z_0) - \mathcal{V}_{\text{res}})\dot{z}$, by Proposition 10.2 and (10.17). The continuous injective embedding $\mathcal{C} \hookrightarrow L^2(B)$ identifies the two limits. Thus $J_D \mathcal{Q}_{\text{mat}} \dot{z}$ belongs to $\mathcal{C}$ and satisfies (10.21) for every $\dot{z} \in \mathcal{X}_p$. Surjectivity of $\mathcal{Q}_{\text{mat}}$ also gives boundedness of J_D on all of $\mathcal{C}$, with $J_D = (D\mathcal{F}(z_0) - \mathcal{V}_{\text{res}})\mathcal{Q}_{\text{mat}}^{-1}$. $\square$

11. MIXED NORMS AND THE EXACT RELATIVE GRAPH BOUND

The intermediate space keeps the radial variable Hilbertian and the angular coefficients summable and preserves radial spectral projections in endpoint-normalized angular coordinates.

Definition 11.1 (Mixed space and weighted matrix norms). Let $\mathcal{M}$ be the space of all families $(u_j)_{j \geq 0}$ with $u_j \in L^2((0, 1), pt^{\beta_j} dt)$ and finite norm below, represented as $u = \sum_j S_j u_j(t)$. Put

$$(11.1) \quad \|u\|_{\mathcal{M}} = \sum_j 2^j \|u_j\|_{\beta_j}, \quad \|v\|_{\beta}^2 = p \int_0^1 t^\beta |v(t)|^2 dt.$$

The actual invariant Hilbert space $\mathcal{H}$ has norm

$$(11.2) \quad \|u\|_{\mathcal{H}}^2 = \sum_j \eta_j^2 \|u_j\|_{\beta_j}^2.$$

For an angular block matrix T define, at fixed $\sigma > 0$,

$$(11.3) \quad \|T\|_{\mathcal{M},\sigma} = \sup_j \sum_i 2^{i-j} e^{\sigma|i-j|} \|T_{ij}\|_{\beta_j \rightarrow \beta_i},$$

$$(11.4) \quad \|T\|_{\mathcal{H},\sigma} = \max \left\{ \sup_i \sum_j e^{\sigma|i-j|} \|T_{ij}^H\|, \sup_j \sum_i e^{\sigma|i-j|} \|T_{ij}^H\| \right\}, \quad T_{ij}^H = \frac{\eta_i}{\eta_j} T_{ij}.$$

The block norms in the second line use the same radial Hilbert spaces; only the angular basis has been normalized.

Lemma 11.2 (Embedding and matrix algebra). *There are norm-at-most-one embeddings $\mathcal{C} \hookrightarrow \mathcal{M} \hookrightarrow \mathcal{H}$. The space $\mathcal{M}$ is complete, and finite polynomial sums are dense in all three spaces. The two matrix norms in (11.3)–(11.4) are submultiplicative, have identity norm one, and dominate their respective operator norms. The spaces of matrices with finite such norms are complete. For a block-diagonal operator both reduce to the supremum of its radial block norms.*

Proof. The radial squared basis norm is $p/(p+2j+2s) \leq 1$. The triangle inequality gives the first embedding from (9.3); the inequality $\eta_j \leq 1$ and $\ell^2 \leq \ell^1$ give the second. Finite angular truncations and polynomial density for each positive radial weight give density in $\mathcal{M}$ and $\mathcal{H}$; density in $\mathcal{C}$ is part of its definition. Completeness of $\mathcal{M}$ follows from its description as a weighted ℓ^1 sum of the complete radial Hilbert spaces: a Cauchy family has a limit in every radial block, and Fatou's inequality bounds the sum of the blockwise differences by its Cauchy bound. For products of blocks use $e^{\sigma|i-j|} \leq e^{\sigma|i-k|} e^{\sigma|k-j|}$ and $2^{i-j} = 2^{i-k} 2^{k-j}$, then sum the positive block norm bounds. The ratios η_i/η_j telescope through the intermediate index. This proves submultiplicativity for both row and column sums. The mixed operator bound follows from summing columns; the Hilbert bound is the block Schur inequality, obtained by Cauchy–Schwarz with these positive row and column sums. For completeness, a Cauchy sequence in either matrix norm is Cauchy in every block operator norm. Take its blockwise limits. Fatou's inequality for the positive row and column sums bounds the norm of each difference from that limit by the corresponding Cauchy bound, uniformly over the exterior index. The sequence therefore converges in the matrix norm, and the preceding operator bounds identify the limit with a bounded operator on the indicated space. For diagonal blocks all off-diagonal weights are one and only one term remains in each sum. $\square$

Lemma 11.3 (Solid multipliers in the two matrix norms). *For integers $i, h \geq 0$,*

$$(11.5) \quad \|M_{S_i t^h}\|_{\mathcal{M},\sigma} \leq 2^i e^{\sigma i}, \quad \|M_{S_i t^h}\|_{\mathcal{H},\sigma} \leq (2i+1) e^{\sigma i}.$$

Proof. An angular product output has factor $t^{i+j-k+h} u_j$. Its squared radial weight has exponent

$$\beta_k + 2(i+j-k+h) = \beta_j + 2i + 2h \geq \beta_j.$$

Multiplication by that power of t is a contraction between the indicated radial spaces. The positive angular coefficients sum to one, and $|k-j| \leq i$; this proves the mixed bound. In the true Hilbert space multiplication has norm at most one because $|S_i t^h| \leq 1$ on the ball. Every angular block is an orthogonal compression and has norm at most one. Its bandwidth is i , so each weighted row and column contains at most $2i+1$ terms, proving the Hilbert bound. $\square$

Lemma 11.4 (Poisson and gradient estimates in mixed and Hilbert norms). *On all source layers one has*

$$(11.6) \quad \|K_D\| \leq C p^{-2}, \quad \|\mathbf{E} K_D\| \leq C p^{-1}, \quad \|\mathbf{E}^2 K_D\| \leq C$$

in both matrix norms. For $A = 2i > 0$,

$$(11.7) \quad \|\nabla S_i \cdot \nabla K_D\|_{\mathcal{M},\sigma} \leq CA 2^i e^{\sigma i}/p, \quad \|\nabla S_i \cdot \nabla \mathbf{E} K_D\|_{\mathcal{M},\sigma} \leq CA 2^i e^{\sigma i},$$

$$(11.8) \quad \|\nabla S_i \cdot \nabla K_D\|_{\mathcal{H},\sigma} \leq CA(2i+1)e^{\sigma i}/p, \quad \|\nabla S_i \cdot \nabla \mathbf{E} K_D\|_{\mathcal{H},\sigma} \leq CA(2i+1)e^{\sigma i}.$$

The constants are absolute for fixed σ .

Proof. In angular degree $B = 2j$, let $\mathfrak{k}$ solve $4(t\mathfrak{k}'' + (\beta+1)\mathfrak{k}') = f$, $\mathfrak{k}(1) = 0$, $\beta = B + p - 1$. The radial spectral lower bound gives $\|\mathfrak{k}\|_\beta \leq C(B+p)^{-2}\|f\|_\beta$. Multiplication of the equation by $\bar{\mathfrak{k}}t^\beta$ and integration by parts give

$$4\|\mathfrak{k}'\|_{\beta+1}^2 = -\operatorname{Re}\langle f, \mathfrak{k} \rangle_\beta, \quad \|\mathfrak{k}'\|_{\beta+1} \leq C(B+p)^{-1}\|f\|_\beta.$$

These equalities first hold for polynomial sources; their extensions follow from the Dirichlet form and Poisson uniqueness. The profile identities $\mathbf{E}K_D = B\mathfrak{k} + 2t\mathfrak{k}'$ and $\mathbf{E}^2K_D = B^2\mathfrak{k} + 4(1-p)t\mathfrak{k}' + tf$ then give (11.6) on each block, hence in both matrix norms.

For the gradient operators use the two profiles (9.22)–(9.23) with the general source f in place of J_s^β and $\mathfrak{k}$ in place of k_s . When $h \geq 1$, the output squared weight for $t^{h-1}\mathfrak{k}$ is $t^{\beta+A-2}$, which is bounded above by t^β since $A \geq 2$. The corresponding weight for $t^h\mathfrak{k}'$ is $t^{\beta+A} \leq t^{\beta+1}$. The source term $t^h f$ is also a contraction between its two radial spaces. Equation (9.26) and the preceding energy bounds give $CA/(B+p)$ for the first profile and CA for the second. There is no negative-power term when $h = 0$. Positive angular summation and the bandwidth weight prove (11.7).

In Hilbert coordinates, homogeneity and Lemma 2.3 give $\|\nabla S_i\|_\infty \leq \sqrt{2}A$. For $u = K_D f$, the Dirichlet spectral lower bound gives $\|\nabla u\|_2 \leq (p-1)^{-1}\|f\|_2$. On the ball one also has

$$(11.9) \quad \|D^2 u\|_2 \leq \|\Delta u\|_2 = \|f\|_2.$$

For a smooth real Dirichlet function, integrate the divergence of $(\Delta u)\nabla u - D^2 u \nabla u$. Its interior divergence is $(\Delta u)^2 - |D^2 u|^2$. At the boundary $\nabla u = (\partial_\nu u)\nu$ and $\Delta u = \partial_{\nu\nu} u + H_{\partial B}\partial_\nu u$, so the boundary integral is $\int_{\partial B} H_{\partial B} |\partial_\nu u|^2 \geq 0$. This proves (11.9); use real and imaginary parts and H^2 approximation for the general case. Since $\nabla \mathbf{E} u = \nabla u + (D^2 u)y$, the unweighted Hilbert bounds for the two gradient operators are CA/p and CA . Their angular bandwidth is i , as is also visible in the exact profiles. Compressing to blocks and summing at most $2i+1$ weighted entries per row and column proves (11.8). This argument covers the harmonic source layer as well as all the others. $\square$

Lemma 11.5 (Closed mixed graph realization). *Let H_0 have its Hilbert Dirichlet realization. Its realization on $\mathcal{M}$ has domain*

$$\operatorname{Dom}_{\mathcal{M}}(H_0) = K_D \mathcal{M} = \{u \in \operatorname{Dom}_{\mathcal{H}}(H_0) : H_0 u \in \mathcal{M}\}.$$

It is closed in $\mathcal{M}$ and its graph norm is equivalent, for fixed p , to the norm of its source. Its realization on $\mathcal{C}$ has the identical description with $\mathcal{C}$ in place of $\mathcal{M}$. Let V be defined on $\operatorname{Dom}_{\mathcal{H}}(H_0)$, and suppose $\|V H_0^{-1}\| < 1$ both on $\mathcal{H}$ and on $\mathcal{M}$. Then $H_0 + V$ restricts to a closed operator on $\operatorname{Dom}_{\mathcal{M}}(H_0)$, and

$$\{u \in \operatorname{Dom}_{\mathcal{H}}(H_0) : (H_0 + V)u \in \mathcal{M}\} = \operatorname{Dom}_{\mathcal{M}}(H_0).$$

The same conclusions hold with $\mathcal{C}$ in place of $\mathcal{M}$ when the relative norm is below one on both $\mathcal{H}$ and $\mathcal{C}$.

Proof. The Poisson estimates give a bounded map $K_D : \mathcal{M} \rightarrow \mathcal{M}$, and the Hilbert identification gives $H_0 K_D f = -f$. Thus every element in its range is in the stated Hilbert domain and has source in $\mathcal{M}$. Conversely, if u lies in that Hilbert domain and $H_0 u = f \in \mathcal{M}$, Dirichlet

uniqueness gives $u = -K_D f$, proving equality of the descriptions. The norm $\|H_0 u\|_{\mathcal{M}}$ is a complete norm on the domain, since K_D is a bijection from the complete source space onto it. The Poisson bound makes it equivalent to $\|u\|_{\mathcal{M}} + \|H_0 u\|_{\mathcal{M}}$. If $u_n \rightarrow u$ and $H_0 u_n \rightarrow f$ in $\mathcal{M}$, the embeddings imply the same convergence in $\mathcal{H}$. Closedness of the Hilbert operator gives $u \in \text{Dom}_{\mathcal{H}}(H_0)$ and $H_0 u = f$; hence the description just proved puts u in the mixed domain. This proves closedness directly. The coefficient-space proof uses the coefficient Poisson bound and the same Hilbert embedding, so every step remains valid.

In each indicated source space, the factor $I + V H_0^{-1}$ is bounded and boundedly invertible in the source space. The product $(H_0 + V) = (I + V H_0^{-1}) H_0$ therefore has an equivalent graph norm on the same complete domain and is closed. Adding a bounded scalar shift preserves closedness there. For a source in $\mathcal{M}$, the partial sums of the Neumann series for $(I + V H_0^{-1})^{-1}$ agree with the Hilbert partial sums. Convergence in $\mathcal{M}$ implies convergence in $\mathcal{H}$, so the two inverses coincide on these sources. If $u \in \text{Dom}_{\mathcal{H}}(H_0)$ and $(H_0 + V)u = f \in \mathcal{M}$, Hilbert uniqueness gives $u = H_0^{-1}(I + V H_0^{-1})^{-1}f \in \text{Dom}_{\mathcal{M}}(H_0)$. The reverse inclusion follows by applying the mixed-space operator. This proves the displayed equality and the restriction property. Replacing $\mathcal{M}$ by $\mathcal{C}$ gives the coefficient version. The same argument applies to a bounded angular-diagonal radial spectral projection, which commutes with H_0 and preserves the graph domain. $\square$

11.1. A centre-size bound independent of the accuracy order.

Lemma 11.6 (Order-independent leading size). *For a fixed centre order N , expand $f_N = \sum_j f_{N,j} G_j$ and put*

$$(11.10) \quad \delta = \sum_j (1+j)^{10} (2e^2)^j \varpi_p(2j) |f_{N,j}|.$$

There is an absolute C_0 such that for every fixed N ,

$$(11.11) \quad \delta \leq C_0 \tau p^{-2} \quad (p \geq P_N).$$

Only the threshold depends on N . This stronger norm is used for the known finite centre; the unknown Newton shape retains the space $\mathcal{E}_1$ with one moment and angular radius two.

Proof. Equation (6.7) gives

$$f_N = -\frac{\theta}{2R^2 p g_2} \mathbf{H}_2 + O_N(\tau p^{-4})$$

in every fixed polynomial coefficient norm. The leading term has only index two. Its coefficient is bounded by $C \tau p^{-2}$ independently of N , since $p g_2 > 9/16$ and $\mathbf{H}_2(1) \leq 1$. Its weight in (11.10) is an absolute constant for $p \geq 64$. For the fixed jet the remainder has bounded degree depending on N , and the finite monic-to- G_j connection is uniformly bounded for large p , because $\mathbf{H}_j(1) \rightarrow (1-r)^j > 0$ for each fixed j . Thus its strong norm is at most $C_N \tau p^{-4}$. Choose C_0 to be the leading bound plus one, and increase P_N until $C_N p^{-2} \leq 1$. This proves (11.11). $\square$

11.2. Scalar perturbation and density conjugation. Use the fixed centre chart $\Phi_N = qy$, $d = q + \mathbf{E}q$, and define

$$(11.12) \quad H_{\Phi} = -P_{\Phi_N} \Delta, \quad m = (\det D\Phi_N)^{1/2} = q^{p-1/2} d^{1/2}, \quad \widehat{H}_{\Phi} = m H_{\Phi} m^{-1}, \quad \mathcal{V} = \widehat{H}_{\Phi} - H_0.$$

Here $\widehat{H}_{\Phi}$ is the unitary Dirichlet realization on $\mathcal{H}$. The target volume uses the same fixed normalization as unit-ball volume, so multiplication by m is exactly the unitary density factor.

The scalar graph operator is $J_D = (P_{\Phi_N} \Delta + R^2)K_D$. All full-source graph estimates in this section are at this fixed finite centre with the strong norm (11.10). The Newton iteration uses its frozen inverse, while the nonlinear map is estimated on $\mathcal{G}_p \times \mathcal{E}_1$.

Proposition 11.7 (The Hilbert relative estimate). *For the real finite-centre chart in (11.12), let δ be (11.10). There are absolute constants $C, \varepsilon_H > 0$ such that, if $p \geq 64$ and $p\delta \leq \varepsilon_H$, then*

$$(11.13) \quad \|\mathcal{V}K_0^{-1}\|_{\mathcal{H},1} \leq C(\delta + p\delta^2).$$

The scalar graph difference also satisfies

$$(11.14) \quad \|(P_{\Phi_N} \Delta - \Delta)K_D\|_{\mathcal{H},1} \leq C(\delta + \delta^2).$$

These constants do not depend on N , the split, or the number of nonzero centre modes. All invariant radial source layers are included.

Proof. Use the orthonormal angular coordinates of (11.4). Its ingredients are the endpoint supremum bound, finite angular bandwidth, and ball Hilbert estimates.

1. *Banded operators.* Let P_j^H be the orthogonal projection onto angular degree $2j$. If an operator T has angular bandwidth D , each block $P_i^H T P_j^H$ has norm at most $\|T\|_{\mathcal{H} \rightarrow \mathcal{H}}$, and each row and column has at most $2D + 1$ nonzero blocks. Therefore

$$(11.15) \quad \|T\|_{\mathcal{H},1} \leq (2D + 1)e^D \|T\|_{\mathcal{H} \rightarrow \mathcal{H}}.$$

This bounds both weighted row and column sums.

2. *Multipliers.* For an invariant polynomial multiplier a of Jacobi degree (maximum angular index) at most D , orthogonality gives bandwidth D and multiplication on L^2 has norm at most $\|a\|_\infty$. Hence

$$\|M_a\|_{\mathcal{H},1} \leq (2D + 1)e^D \|a\|_\infty.$$

In particular $|S_i t^h| \leq 1$ gives $(2i + 1)e^i$ for that multiplier. Summing the centre coefficients shows that multiplication by $\mathfrak{h} = q - 1$, $E\mathfrak{h}$, and $E(E - 1)\mathfrak{h}$ has norm at most $C\delta$. The fixed moments in (11.10) absorb all the displayed degree factors. The endpoint bound used here is for $\alpha_b \leq \beta_b$ and $G_i(1) = 1$, as fixed in (2.6).

3. *Products of gradients.* For $A = 2i$ the solid harmonic has $|\nabla S_i| \leq \sqrt{2}A$, by radial homogeneity and the Bernstein bound on great circles. Thus $|\nabla S_i \cdot \nabla S_j| \leq 2(2i)(2j)$. Its angular bandwidth as a multiplier is at most $i + j$. Equation (11.15) gives

$$(11.16) \quad \|M_{\nabla S_i \cdot \nabla S_j}\|_{\mathcal{H},1} \leq 2(2i)(2j)(2i + 2j + 1)e^{i+j}.$$

After summing the shape coefficients, this bounds the multipliers $M_q = \nabla q \cdot \nabla q$, $\nabla q \cdot \nabla E q$, $\nabla d \cdot \nabla d$, and $\nabla q \cdot \nabla d$ by $C\delta^2$. An Euler derivative on a shape adds only its angular degree, which is included in the ten moments. This Hilbert multiplier bound is $C\delta^2$.

4. *Diagonal Poisson operators.* For $u = K_D f$, the ball spectral estimate and its energy identity give

$$\|u\|_2 \leq (p - 1)^{-2} \|f\|_2, \quad \|\nabla u\|_2 \leq (p - 1)^{-1} \|f\|_2.$$

The boundary identity (11.9) gives $\|D^2 u\|_2 \leq \|f\|_2$ on the convex ball. Since $E u = y \cdot \nabla u$ and $E^2 u = y^T D^2 u y + y \cdot \nabla u$, the three Poisson operators have norms at most $(p - 1)^{-2}$, $2/(p - 1)$ and 4. They are angular diagonal, so the same bounds hold in $\|\cdot\|_{\mathcal{H},1}$ on all source layers.

5. *Gradient-Poisson operators.* The preceding energy estimate gives $\|\nabla S_i \cdot \nabla K_D\|_{\mathcal{H} \rightarrow \mathcal{H}} \leq \sqrt{2}(2i)/(p - 1)$. Using $\nabla E u = \nabla u + (D^2 u)y$ gives $\|\nabla S_i \cdot \nabla E K_D\|_{\mathcal{H} \rightarrow \mathcal{H}} \leq \sqrt{2}(2i)(1 + (p - 1)^{-1})$. Their exact profiles (9.22)–(9.23) show angular bandwidth i . Applying (11.15) and summing the centre coefficients proves

$$\|\nabla \mathfrak{h} \cdot \nabla K_D\|_{\mathcal{H},1} \leq C\delta/p, \quad \|\nabla \mathfrak{h} \cdot \nabla E K_D\|_{\mathcal{H},1} \leq C\delta.$$

The same estimates hold with $\mathfrak{h}$ replaced by $\mathbf{E}\mathfrak{h}$, and hence for $d - 1 = \mathfrak{h} + \mathbf{E}\mathfrak{h}$.

6. *Inverse multipliers and density.* The Hilbert matrix norm is a Banach algebra with unit norm one by Lemma 11.2. For $C\delta < 1$, the Neumann series for q^{-1}, d^{-1} converge in it; they also converge uniformly on the ball to the pointwise multipliers. Their fixed inverse powers are bounded. The absolute binomial majorant for $m^{\pm 1}$ gives

$$(11.17) \quad \|M_{m^{\pm 1}}\|_{\mathcal{H},1} \leq (1 - C\delta)^{-p-1} \leq 2$$

when the absolute constant ε_H is sufficiently small. This choice is independent of the finite centre order.

7. *The scalar graph difference.* Compose (10.4) with K_D and subtract the identity. The first term costs $C\delta$ by the inverse multiplier bound, and the second costs $C\delta$ by step 5. The third costs $C\delta^2$ by steps 3 and 4. Formula (10.5) bounds its contracted Hessian multiplier by $C\delta^2(1 + \delta)$; it is followed by $\mathbf{E}K_D$, of norm C/p . The remaining gradient-square term is bounded in the same way. This proves (11.14).

8. *The density commutator.* Both q and d are harmonic. With $\alpha = p - 1/2$,

$$\frac{\nabla m^{-1}}{m^{-1}} = -\alpha q^{-1} \nabla q - \frac{1}{2} d^{-1} \nabla d,$$

$$\frac{\Delta m^{-1}}{m^{-1}} = \alpha(\alpha + 1) q^{-2} \nabla q \cdot \nabla q + \frac{3}{4} d^{-2} \nabla d \cdot \nabla d + \alpha q^{-1} d^{-1} \nabla q \cdot \nabla d.$$

The gradient part of $[\Delta, m^{-1}]K_D$ is bounded by $Cp(\delta/p) = C\delta$, using step 5 and the inverse multipliers. Its potential part is bounded by $Cp^2\delta^2(p-1)^{-2} \leq C\delta^2$, using steps 3 and 4. Thus

$$(11.18) \quad \|[\Delta, m^{-1}]K_D\|_{\mathcal{H},1} \leq C(\delta + \delta^2).$$

The potential starts at quadratic order because q, d are harmonic.

9. *Assembly on the graph domain.* Put $X = (H_\Phi - H_0)H_0^{-1} = (P_{\Phi_N}\Delta - \Delta)K_D$. Multiplication by m^{-1} preserves the Dirichlet boundary condition, and the preceding commutator estimate bounds $H_0 m^{-1} H_0^{-1}$ by an absolute constant. Hence

$$\mathcal{V}H_0^{-1} = mX(H_0 m^{-1} H_0^{-1}) + m[H_0, m^{-1}]H_0^{-1}$$

has Hilbert weighted norm at most $C(\delta + \delta^2)$, which is at most $C(\delta + p\delta^2)$. Finally $H_0 K_0^{-1}$ is an angular-diagonal radial spectral contraction, so multiplying by it proves (11.13). The identities hold first on the smooth Dirichlet graph core; the bounds extend them to the Hilbert graph realization. $\square$

Theorem 11.8 (Exact relative graph bounds). *Fix $\sigma = 1$. There is an absolute C independent of N such that, for every fixed centre order N and all $p \geq P_N$,*

$$(11.19) \quad \|\mathcal{V}K_0^{-1}\|_{\mathcal{M},1} + \|\mathcal{V}K_0^{-1}\|_{\mathcal{H},1} \leq C\tau p^{-2}, \quad K_0 = H_0 + R^2,$$

$$(11.20) \quad \|(P_{\Phi_N}\Delta - \Delta)K_D\|_{\mathcal{C} \rightarrow \mathcal{C}} + \|(P_{\Phi_N}\Delta - \Delta)K_D\|_{\mathcal{M} \rightarrow \mathcal{M}} \leq C\tau p^{-2}.$$

The density multipliers and their graph conjugates also satisfy, for the same $p \geq P_N$,

$$(11.21) \quad \begin{aligned} & \max\{\|M_{m^{\pm 1}}\|_{\mathcal{M},1}, \|M_{m^{\pm 1}}\|_{\mathcal{H},1}\} \leq 2, \\ & \max\{\|H_0 M_{m^{-1}} H_0^{-1}\|_{\mathcal{M},1}, \|H_0 M_{m^{-1}} H_0^{-1}\|_{\mathcal{H},1}\} \leq C. \end{aligned}$$

All these bounds include every source layer. The operator $\widehat{H}_\Phi$ has domain $H^2(B) \cap H_0^1(B)$ in its invariant sector and has the actual centre-domain spectrum.

Proof. Proposition 11.7 supplies the Hilbert half. For the mixed and coefficient bounds, use endpoint-normalized angular coordinates. We keep the size δ in (11.10) throughout the calculation. In $\mathcal{C}$ and both weighted matrix algebras, solid multipliers and their first two Euler derivatives are controlled by their coefficient sums with the corresponding degree factors. The ten moments and the factor $(2e^2)^j$ dominate the angular weights, the bandwidth factor $2j + 1$, and these fixed derivatives. Proposition 9.4 and Lemmas 11.3 and 11.4 thus give

$$\|M_{q-1}\| + \|M_{d-1}\| \leq C\delta, \quad \|\nabla(q-1) \cdot \nabla K_D\| \leq C\delta/p, \quad \|\nabla(q-1) \cdot \nabla \mathbf{E}K_D\| \leq C\delta.$$

For the coefficient-space version use Lemma 9.9, which explicitly includes $s = 0$. The same statements hold with a fixed Euler derivative on the harmonic shape.

Products of two harmonic gradients have multiplier norm $Cp\delta^2$. In the mixed and coefficient spaces this follows from the exact product coefficient $2h(h+2k+p-1)c_{ij}^k$ and its bound by $Cp(1+i)(1+j)$. In the Hilbert matrix algebra use the pointwise product bound Cij , bandwidth at most $i+j$, and the row/column estimate in Lemma 11.3; the available moments absorb the resulting polynomial factors. The coarser bound $Cp\delta^2$ is therefore valid there as well. Products with one Euler derivative on either factor obey the same bound in terms of δ . All inverse powers of q, d are convergent series in these multiplier algebras when $C\delta < 1/2$.

Subtract Δ in (10.4) and compose with K_D . The four terms have bounds, respectively,

$$(11.22) \quad C\delta, \quad C\delta, \quad Cp\delta^2, \quad C\delta^2(1+\delta).$$

For the last bound, (10.5) gives multipliers $Cp\delta^2$ and $Cp\delta^3$, and $\mathbf{E}K_D$ costs C/p . The third term uses the bounded operator $\mathbf{E}(\mathbf{E}+1)K_D$. This proves a scalar graph bound $C(\delta + p\delta^2)$ in each of the three realizations, including the two weighted matrix norms.

The density requires its own estimate. Both q and d are harmonic, since $\Delta \mathbf{E}q = (\mathbf{E}+2)\Delta q = 0$. Write $\alpha = p - 1/2$. Direct differentiation gives

$$(11.23) \quad \frac{\nabla m^{-1}}{m^{-1}} = -\alpha q^{-1} \nabla q - \frac{1}{2} d^{-1} \nabla d,$$

$$(11.24) \quad \frac{\Delta m^{-1}}{m^{-1}} = \alpha(\alpha+1)q^{-2}|\nabla q|^2 + \frac{3}{4}d^{-2}|\nabla d|^2 + \alpha q^{-1}d^{-1}\nabla q \cdot \nabla d.$$

The second formula is quadratic in the shape gradients. The binomial series for $m^{\pm 1}$ has norms at most $\exp(Cp\delta)$, hence at most two after increasing the threshold using (11.11). The commutator

$$[\Delta, m^{-1}]K_D = 2\nabla m^{-1} \cdot \nabla K_D + (\Delta m^{-1})K_D$$

has norm at most $Cp(\delta/p) + Cp^2(p\delta^2)p^{-2} = C(\delta + p\delta^2)$ in both weighted matrix algebras. In particular $H_0 m^{-1} H_0^{-1}$ is bounded by an absolute constant there.

Put $X = (H_\Phi - H_0)H_0^{-1} = (P_{\Phi_N} \Delta - \Delta)K_D$. The exact factorization

$$\mathcal{V}H_0^{-1} = mX(H_0 m^{-1} H_0^{-1}) + m[H_0, m^{-1}]H_0^{-1}$$

therefore has norm at most $C(\delta + p\delta^2)$. The remaining factor $H_0 K_0^{-1}$ is a block-diagonal radial Hilbert contraction in both matrix norms. This proves the corresponding estimate for $\mathcal{V}K_0^{-1}$. Finally insert (11.11), with $\tau \leq 20$. The operator constants are absolute; the threshold P_N absorbs the finite-jet remainder. This proves (11.19) and (11.20) with N -independent constants, as well as (11.21).

We also identify their domains. For this finite smooth chart, composition and multiplication by the positive smooth density map the Dirichlet H^2 domain to itself. The physical smooth Dirichlet realization therefore gives the asserted domain and unitary spectrum. Equivalently the small relative graph bound in $\mathcal{H}$ preserves the ball domain by the closed-operator perturbation theorem. All equalities above first hold for Poisson images of polynomial sources.

Those images are graph dense in the Dirichlet domain, because polynomial sources are dense in L^2 and $K_D : L^2 \rightarrow H^2 \cap H_0^1$ is continuous. Thus the bounded graph extensions are the same geometric operators. By Lemma 11.5, on $\mathcal{M}$ their graph domain is $\{K_D f : f \in \mathcal{M}\}$ with its source graph norm; the embedding into $\mathcal{H}$ and Poisson uniqueness identify it with the restriction of that realization. The identical argument applies to $\mathcal{C}$. $\square$

12. LOCAL COUNTS AND CLUSTER INVERSION

The window in this section is an arbitrary fixed number $W_{\text{loc}} > 0$. The local count depends on this number, but neither on the centre order nor on the detuning or split.

Theorem 12.1 (Local logarithmic count). *There is an integer $M_{W_{\text{loc}}} \geq 1$ with the following quantifier order: for every fixed $D > 0$ there is $P_{W_{\text{loc}}, D} < \infty$ such that every angular interval of length $D \log p$ contains at most $M_{W_{\text{loc}}}$ states of $\mathcal{W}_p(W_{\text{loc}})$ whenever $p \geq P_{W_{\text{loc}}, D}$ and $R \in [2p - 1, 2p + 4]$. In particular the count bound is independent of D .*

Proof. In the oscillatory region $R - \nu \geq p^{2/3}/2$, Lemma 3.6 assigns to each state a lattice point with $|m - \varphi_p(j)| \leq C_{W_{\text{loc}}}/p$. Dividing (3.16) by π gives the stronger bound $C_{W_{\text{loc}}}/(\pi p)$; we use the same constant in the weaker bound above. The threshold includes those of Lemmas 3.5 and 3.6 for this window and the condition $2C_{W_{\text{loc}}}/p < 1$. Its curvature satisfies

$$c/p \leq \varphi_p''(j) \leq Cp^{-5/6}.$$

The lower bound uses $\kappa \leq Cp$; the upper uses $\kappa \geq cp^{5/6}$. On an interval of length $D \log p$,

$$\frac{1}{2}Cp^{-5/6}(D \log p)^3 + 2C_{W_{\text{loc}}}D \log p/p < 1$$

for large p . Lemma 3.7 therefore bounds the number of states by $2 + 4\sqrt{C_{W_{\text{loc}}}/c}$. This number is independent of D ; only the threshold of the last inequality depends on D .

In the turning region $\nu \geq R - 2p^{2/3}$, suppose two states have angular distance at most $D \log p$. Their original roots are within $O_{W_{\text{loc}}}(p^{-1})$ of R . Transport the root of the larger order to the smaller order ν_0 , keeping its radial index fixed. Lemma 3.2 moves it by at most $\pi D \log p$. Every intervening root stays of order p . If the two radial indices were equal, the lower derivative bound in that lemma would make the original roots differ by at least $2(1 - o(1))$, because the angular indices differ by at least one. That contradicts their original window width $O(p^{-1})$. Thus there are two distinct zeros of J_{ν_0} in an interval of length $O_{W_{\text{loc}}, D}(\log p)$ about R .

Throughout that interval the coefficient in

$$v'' + \left(1 - \frac{\nu_0^2 - 1/4}{x^2}\right)v = 0, \quad v(x) = \sqrt{x}J_{\nu_0}(x),$$

is at most $(2 + o(1))p^{-1/3}$, uniformly for the fixed window and interval constant, because $\nu_0 \geq R - 2p^{2/3}$ and $x = R + O(\log p)$. Between the two zeros, integration and the Dirichlet Poincaré inequality give $\pi^2/\ell^2 \leq Cp^{-1/3}$ for their separation ℓ . Hence $\ell \geq (\pi/\sqrt{2} + o(1))p^{1/6}$, in particular $\ell \geq 2.2p^{1/6}$ for large p , contradicting its logarithmic upper bound for large p . The turning interval therefore contains at most one state.

The two regions overlap by an order interval of length $3p^{2/3}/2$. An angular interval of length $D \log p$ has order length $2D \log p$, which is smaller than this overlap for large p . If it is not entirely in the oscillatory region, all its possible window points are in the turning region. There is at most one radial state per angular index by Lemma 3.5. Rounding up the oscillatory count and increasing it to at least one proves the theorem. $\square$

Corollary 12.2 (Bounded cluster cardinality). *Fix $D > 0$ and connect consecutive angular indices of $\mathcal{W}_p(W_{\text{loc}})$ when their separation is at most $D \log p$. For all sufficiently large p , every resulting cluster has at most $M_{W_{\text{loc}}}$ states, and distinct clusters are separated by more than $D \log p$. A cluster may have diameter as large as $(M_{W_{\text{loc}}} - 1)D \log p$. The same cardinality bound holds for every subset of the window.*

Proof. If a cluster had $M_{W_{\text{loc}}} + 1$ states, its first that many states would span at most $M_{W_{\text{loc}}} D \log p$. Apply Theorem 12.1 with the fixed constant $M_{W_{\text{loc}}} D$ in place of D . Its bound is still $M_{W_{\text{loc}}}$, a contradiction. The separation and diameter statements follow from the definition. Removing states can split clusters but cannot join two clusters of the larger set, proving the assertion for subsets. $\square$

Lemma 12.3 (Inversion with arbitrary diagonal Hilbert weights). *Fix positive constants $c, C, q_{\text{gap}}, \sigma$ and an integer $M \geq 1$. Let D satisfy $\sigma D > q_{\text{gap}} M + q_{\text{gap}} + 2$. Let $\tau > 0$ and let F be a finite matrix whose indices carry angular degrees, partitioned into clusters of at most M indices separated by more than $D \log p$. Assume that:*

- (i) *F is self-adjoint for a positive diagonal Hilbert inner product and $\|Fv\|_H \geq \gamma_F \|v\|_H$, where $\gamma_F = c\tau p^{-q_{\text{gap}}}$;*
- (ii) *in a weighted coefficient ℓ^1 norm, $\|F\|_1 \leq C\tau$;*
- (iii) *deleting all inter-cluster entries has error at most $C\tau p^{-\sigma D}$ in both that coefficient norm and the specified Hilbert operator norm.*

Then, for sufficiently large p depending only on $c, C, q_{\text{gap}}, \sigma, M, D$,

$$(12.1) \quad \|F^{-1}\|_1 \leq C' \tau^{-1} p^{q_{\text{gap}} M}.$$

The assertion is uniform as $\tau \downarrow 0$.

Proof. Let F_b keep the diagonal cluster blocks. The symmetric deletion of paired entries preserves self-adjointness in the diagonal Hilbert structure. The Hilbert error is at most $\gamma_F/2$ for large p , since $\sigma D > q_{\text{gap}}$. Thus

$$\|F_b v\|_H \geq (\gamma_F - C\tau p^{-\sigma D}) \|v\|_H \geq (\gamma_F/2) \|v\|_H.$$

Every eigenvalue of every cluster block is real and has modulus at least $\gamma_F/2$.

Conjugate a cluster block of size $m \leq M$ by its coefficient weights, so its ℓ^1 norm is the unweighted column norm. All its entries have modulus at most $C\tau$. The determinant is invariant under this conjugation and has modulus at least $(\gamma_F/2)^m$. Expansion of each cofactor as a sum of $(m-1)!$ products gives its absolute value at most $(m-1)!(C\tau)^{m-1}$. For $m=1$ the empty product is one. Cramer's formula and summing each inverse column give

$$\|F_{b,m}^{-1}\|_1 \leq m!(C\tau)^{m-1} (2/\gamma_F)^m \leq C_{M,c,C\tau}^{-1} p^{q_{\text{gap}} M}.$$

The block diagonal inverse has the maximum of these norms. Its product with the discarded coefficient error is at most $C' p^{q_{\text{gap}} M - \sigma D}$, below $1/2$ for large p . A Neumann series restores that error and proves (12.1). In both absorptions the factor τ cancels. The inverse has exactly one remaining factor τ^{-1} . For an empty index set the bound is vacuous. $\square$

The next section uses a window scaled by τ , and applies the local count to a fixed larger window containing it.

13. THE FULL INVERSE IN COEFFICIENT NORM

Fix any centre order N . All constants and exponents in this section are fixed independently of that order. Only the large- p threshold may depend on it. We assume the actual centre gap (8.2), supplied by Theorem 8.3. Write

$$\mathcal{A} = \widehat{H}_\Phi - R^2, \quad \mathcal{D}_0 = H_0 - R^2, \quad K_0 = H_0 + R^2.$$

The actual scalar Dirichlet graph operator is $J_D = B_{\Phi_N, D} K_D$ and includes the harmonic source layer and the seed.

13.1. A detuning-scaled window. Let $C_{\text{rel}} \geq 1$ be a constant in (11.19). Fix $E_{\text{inv}} \geq 64C_{\text{rel}}$, and define

$$(13.1) \quad \Pi = \mathbf{1}_{[-E_{\text{inv}}\tau, E_{\text{inv}}\tau]}(H_0 - R^2), \quad Q = I - \Pi.$$

This is the full ball spectral projection, including every radial state in the interval. It is distinct from the fixed flow-window projection. The local count will use $W_{\text{loc}} = 20E_{\text{inv}}$.

Lemma 13.1 (Inverse on the full complement). *For $E_{\text{inv}} \geq 64C_{\text{rel}}$ and all p above a threshold depending on $C_{\text{rel}}, E_{\text{inv}}, N$, the operator $\mathcal{A}_Q = Q\mathcal{A}Q$ is invertible on its graph domain, and*

$$(13.2) \quad \|K_0\mathcal{A}_Q^{-1}Q\|_{\mathcal{M},1} + \|K_0\mathcal{A}_Q^{-1}Q\|_{\mathcal{H},1} \leq Cp^2/(E_{\text{inv}}\tau).$$

The two inverses coincide on their common sources.

Proof. The projections in (13.1) are diagonal in angular degree and orthogonal in each radial Hilbert space. Their norms in both matrix algebras and both underlying spaces are at most one. Put $D_Q = Q\mathcal{D}_0Q$. Radial spectral calculus gives

$$(13.3) \quad \|K_0D_Q^{-1}Q\| \leq 1 + \frac{2R^2}{E_{\text{inv}}\tau}$$

in all four norms. Since $R/p \leq 33/16$ for $p \geq 64$, its product with (11.19) is at most

$$\frac{20C_{\text{rel}}}{p^2} + \frac{1089C_{\text{rel}}}{128E_{\text{inv}}}.$$

The chosen window and $p^2 \geq 80C_{\text{rel}}$ make this at most $1/4 + 1089/8192 < 1/2$. The window is fixed before this threshold is imposed. Factor

$$\mathcal{A}_Q = (I + Q\mathcal{V}K_0^{-1}K_0D_Q^{-1}Q)D_Q.$$

The norm-convergent Neumann series in each matrix algebra gives

$$\|K_0\mathcal{A}_Q^{-1}Q\| \leq 2 \left(1 + \frac{2R^2}{E_{\text{inv}}\tau} \right).$$

Increasing the threshold to include $p^2 \geq 40E_{\text{inv}}$ absorbs the constant term into $Cp^2/(E_{\text{inv}}\tau)$, proving (13.2). It maps sources into the graph domain because of its leading D_Q^{-1} factor and the bound after multiplication by K_0 . By the graph realizations of Lemma 11.5, the embedding $\mathcal{M} \subset \mathcal{H}$, and uniqueness of the Hilbert inverse, the two constructions agree on common sources. $\square$

Proposition 13.2 (The Schur matrix and its actual gap). *Define the exact Schur matrix at spectral parameter zero by*

$$(13.4) \quad F_{\text{eff}} = \Pi\mathcal{A}\Pi - \Pi\mathcal{A}Q\mathcal{A}_Q^{-1}Q\mathcal{A}\Pi.$$

For large p , each angular index occurs at most once in its index set. In radial unit vectors with the original angular G_j , its Hilbert inner product has diagonal weights η_j^2 . It is self-adjoint for that inner product, satisfies

$$(13.5) \quad \|F_{\text{eff}}\|_{\mathcal{M},1} + \|F_{\text{eff}}\|_{\mathcal{H},1} \leq C_{E_{\text{inv}}} \tau,$$

and has the Hilbert gap

$$(13.6) \quad \|F_{\text{eff}} v\|_{\mathcal{H}} \geq \gamma_p \|v\|_{\mathcal{H}}.$$

Proof. The containing fixed window is $20E_{\text{inv}}$. Lemma 3.5 gives one radial state per angular index. The normalized radial basis therefore identifies the mixed norm on this finite space with $\ell^1(2^j)$ exactly. Self-adjointness of (13.4) follows from that of $\mathcal{A}$ and the complementary Hilbert inverse. The stated diagonal Hilbert weights follow from (11.2).

On Π , K_0 has norm at most Cp^2 . Equation (11.19) gives $\|\Pi \mathcal{V} \Pi\| \leq C\tau$ in both weighted norms. The feedback factors as

$$(\Pi \mathcal{V} K_0^{-1}) Q (K_0 \mathcal{A}_Q^{-1} Q) (\mathcal{V} K_0^{-1}) (K_0 \Pi).$$

Its four factors have sizes $C\tau/p^2$, $Cp^2/(E_{\text{inv}}\tau)$, $C\tau/p^2$, and Cp^2 , respectively. Its norm is at most $C\tau/E_{\text{inv}}$. The unperturbed diagonal has norm at most $E_{\text{inv}}\tau$. Submultiplicativity from Lemma 11.2 proves (13.5), including exponential angular decay of the feedback.

For $v \in \Pi \mathcal{H}$ set $u = v - \mathcal{A}_Q^{-1} Q \mathcal{A} \Pi v$. It belongs to the graph domain and obeys $Q \mathcal{A} u = 0$, $\Pi \mathcal{A} u = F_{\text{eff}} v$. Orthogonality gives $\|u\|_{\mathcal{H}} \geq \|v\|_{\mathcal{H}}$. The actual centre-domain gap implies $\|\mathcal{A} u\|_{\mathcal{H}} \geq \gamma_p \|u\|_{\mathcal{H}}$. These equalities prove (13.6) for the exact Schur matrix. $\square$

Corollary 13.3 (Finite-matrix coefficient inverse). *Let $M = M_{20E_{\text{inv}}}$ be the local-count bound from Theorem 12.1. Then*

$$(13.7) \quad \|F_{\text{eff}}^{-1}\|_{\ell^1(2^j)} \leq C\tau^{-1} p^{5M/3}$$

for all sufficiently large p . Its constant and exponent are independent of the centre order.

Proof. Take clusters at separation $D_{\text{sep}} \log p$ in the actual window. Corollary 12.2, applied to its fixed superset of width $20E_{\text{inv}}$, bounds their cardinalities by M , independently of D_{sep} . Deleting inter-cluster entries in (13.5) costs at most $C\tau p^{-D_{\text{sep}}}$ in the coefficient norm by weighted column sums and in the Hilbert norm by its weighted row and column sums. Apply Lemma 12.3 with $q_{\text{gap}} = 5/3$, $\sigma = 1$, and $c = c_{\text{gap}}$. Choose once and for all $D_{\text{sep}} > 5M/3 + 5$. This proves (13.7) and its stated independence. $\square$

13.2. Recovering the full radial graph.

Proposition 13.4 (Mixed-space inverse). *The full operator $\mathcal{A}$ and scalar graph J_D have inverses on the mixed space, with*

$$(13.8) \quad \|K_0 \mathcal{A}^{-1}\|_{\mathcal{M} \rightarrow \mathcal{M}} + \|J_D^{-1}\|_{\mathcal{M} \rightarrow \mathcal{M}} \leq C\tau^{-1} p^{2+5M/3}.$$

These inverses agree with their Hilbert realizations.

Proof. For a source $f \in \mathcal{M}$, the exact block solution is

$$v = F_{\text{eff}}^{-1} (\Pi f - \Pi \mathcal{V} Q \mathcal{A}_Q^{-1} Q f), \quad Qu = \mathcal{A}_Q^{-1} (Q f - Q \mathcal{V} \Pi v), \quad u = v + Qu.$$

The first off-diagonal factor has norm at most C/E_{inv} , by (11.19) and (13.2). The input $Q \mathcal{V} \Pi$ has norm at most $C\tau$, and $K_0 \Pi$ costs Cp^2 . Combining these bounds with (13.7) gives $\|K_0 u\| \leq C\tau^{-1} p^{2+5M/3} \|f\|$. By Lemma 11.5, the reconstruction lies in the mixed graph domain and solves the full equation, whose Hilbert solution is unique by the centre gap. This proves the first inverse and identifies it with the Hilbert one.

For the scalar equation $B_{\Phi_N}W = f$, the unitary equation is $\mathcal{A}(mW) = -mf$. Equation (11.21) bounds multiplication by m and $H_0m^{-1}H_0^{-1}$ on $\mathcal{M}$. Since $H_0K_0^{-1}$ is an angular-diagonal radial contraction, it also bounds $H_0m^{-1}K_0^{-1}$ by an absolute constant. Thus

$$J_D^{-1}f = \Delta W = (H_0m^{-1}K_0^{-1})(K_0\mathcal{A}^{-1})(mf),$$

which proves the second bound in (13.8). Its signs use $\Delta = -H_0$ and $mW = -\mathcal{A}^{-1}mf$. Both inverse identities follow from the graph-domain equation and Hilbert uniqueness. Every radial and angular source, including the harmonic layer and the seed, is part of this reconstruction. $\square$

13.3. Finite head, infinite tail, and membership.

Lemma 13.5 (Radial tail and finite-head conversion). *For this lemma the scalar hypotheses on the fixed real finite-centre chart are $R \in [2p - 1, 2p + 4]$ and*

$$(13.9) \quad \|(P_{\Phi_N}\Delta - \Delta)K_D\|_{\mathcal{C} \rightarrow \mathcal{C}} + \|(P_{\Phi_N}\Delta - \Delta)K_D\|_{\mathcal{M} \rightarrow \mathcal{M}} \leq \frac{1}{4}.$$

There is a fixed K_{head} independent of the centre order such that, with coefficient projections

$$\Pi_T = \mathbf{1}_{\{j+s \leq K_{\text{head}}p\}}, \quad Q_T = I - \Pi_T,$$

the compressed operator $Q_T J_D Q_T$ has inverse of norm at most two in both $Q_T \mathcal{C}$ and $Q_T \mathcal{M}$ for large p . Moreover

$$(13.10) \quad \|\Pi_T g\|_{\mathcal{C}} \leq C_{K_{\text{head}}} \sqrt{p} \|g\|_{\mathcal{M}}.$$

Proof. The Poisson columns in Lemma 9.5 have radial bandwidth one and satisfy the column bound (9.18), including the harmonic column. On the indicated tail this is at most $C/((1 + 2K_{\text{head}})p - 1)^2$. The ratio of neighbouring coefficient weights is at most $e^{2/p}$. Therefore

$$\|R^2 Q_T K_D Q_T\|_{\mathcal{C} \rightarrow \mathcal{C}} \leq C/(1 + 2K_{\text{head}})^2.$$

For the mixed bound, normalize the radial polynomials by their squared norms $n_{j,s} = p/(p + 2j + 2s)$. The neighbouring ratios of $n_{j,s}$ and their inverses are bounded by two. After this normalization the tridiagonal Poisson matrix has row and column sums bounded by the same quantity up to an absolute factor: adjacent denominators differ by two and are comparable. Removing entries by Q_T does not increase those absolute sums. The radial Schur inequality in each angular block gives $\|R^2 Q_T K_D Q_T\|_{\mathcal{M} \rightarrow \mathcal{M}} \leq C/(1 + 2K_{\text{head}})^2$ as well.

Choose the fixed cutoff so that both bounds are below $1/4$. The hypothesis (13.9) bounds the norm of $Q_T(P_{\Phi_N}\Delta - \Delta)K_D Q_T$ by $1/4$ in both spaces. Since

$$Q_T J_D Q_T = I_{Q_T} + R^2 Q_T K_D Q_T + Q_T(P_{\Phi_N}\Delta - \Delta)K_D Q_T,$$

a Neumann series proves both tail inverses and their norm bound. The two series agree on common sources.

On the head, $\varpi_p(2j + 2s) \leq e^{2K_{\text{head}}}$ and $n_{j,s} \geq (1 + 2K_{\text{head}})^{-1}$. Each angular block contains at most $K_{\text{head}}p + 1$ radial coefficients. Cauchy-Schwarz gives

$$\sum_{s \leq K_{\text{head}}p - j} \varpi_p(2j + 2s) |g_{j,s}| \leq e^{2K_{\text{head}}} \sqrt{(K_{\text{head}}p + 1)(1 + 2K_{\text{head}})} \left(\sum_s n_{j,s} |g_{j,s}|^2 \right)^{1/2}.$$

Multiply by 2^j and sum in j to prove (13.10). The common angular coefficient structure gives the loss $\sqrt{p}$. $\square$

Theorem 13.6 (Two-sided coefficient inverse). *For every fixed centre order N and all sufficiently large p , J_D is a bounded isomorphism of $\mathcal{C}$, and*

$$(13.11) \quad \|J_D^{-1}\|_{\mathcal{C} \rightarrow \mathcal{C}} \leq C\tau^{-1}p^{5/2+5M/3} \leq C\frac{p^{b_{\text{inv}}}}{\gamma_p}, \quad b_{\text{inv}} = 2M + 3.$$

The constants and exponent are independent of N ; its threshold may depend on N . The estimates are uniform in $0 < \tau \leq 20$ and all admissible splits with the stated gap.

Proof. Equation (11.20) of Theorem 11.8 bounds the sum in (13.9) by $C\tau p^{-2} \leq 1/4$ for $0 < \tau \leq T$ and p above the fixed-order threshold, after increasing it depending on the fixed T . Together with $R \in [2p - 1, 2p + 4]$, this supplies the hypotheses of Lemma 13.5; here $T = 20$, and arbitrary fixed T is covered by Lemma 17.3. For $f \in \mathcal{C}$, first solve $J_D g = f$ in $\mathcal{M}$ by Proposition 13.4. The head $\Pi_T g$ is finite dimensional and belongs to $\mathcal{C}$, with bound (13.10). The full J_D is bounded on $\mathcal{C}$ by an absolute constant: use $J_D = I + R^2 K_D + (P_{\Phi_N} \Delta - \Delta) K_D$, (9.17), $R^2 = O(p^2)$, and (11.20). Thus the tail equation with source $Q_T f - Q_T J_D \Pi_T g$ has a solution $z \in Q_T \mathcal{C}$ and

$$\|z\|_{\mathcal{C}} \leq C(\|f\|_{\mathcal{C}} + \|\Pi_T g\|_{\mathcal{C}}).$$

By the embedding into $\mathcal{M}$ and uniqueness of its tail equation, this z is exactly $Q_T g$. This proves membership in $\mathcal{C}$ before applying a norm estimate to the full solution. Hence

$$\|g\|_{\mathcal{C}} \leq C\|f\|_{\mathcal{C}} + C\sqrt{p}\|g\|_{\mathcal{M}} \leq C\tau^{-1}p^{5/2+5M/3}\|f\|_{\mathcal{C}}.$$

The last bound absorbs the first term since $M \geq 1$ and $\tau \leq 20$. The constructed right inverse is injective on $\mathcal{C}$ by the Hilbert realization and its gap, so both inverse identities hold. Finally $5/2 + 5M/3 - 5/3 = 5M/3 + 5/6 \leq 2M + 3$. Using (8.2) gives the second inequality in (13.11). Every constant used in the matrix, reconstruction and tail steps was fixed before the centre order. $\square$

14. THE NEWTON THRESHOLD AND THE EXACT ZERO

The closing argument uses a threshold for the residual and two polynomial operator bounds in the same spaces. It requires the actual integer-split gap already obtained.

Theorem 14.1 (Polynomial Newton threshold). *Suppose fixed exponents $b_{\text{inv}}, m_{\text{mat}}, d_{\text{nl}} \geq 0$ have been chosen independently of the centre accuracy N . Assume, uniformly in $0 < \tau = |\theta| \leq 20$ and the admissible splits, for all sufficiently large p , that in the real Banach spaces $\mathcal{X}_p, \mathcal{C}$ of Definition 9.1 a centre z_0 satisfies*

$$\begin{aligned} D\mathcal{F}(z_0) &= J_D \mathcal{Q}_{\text{mat}} + \mathcal{V}_{\text{res}}, & \|\mathcal{F}(z_0)\| &\leq C_N \theta^2 p^{2-N}, \\ \|\mathcal{V}_{\text{res}}\| &\leq C_N \theta^2 p^{3-N}, & \|J_D^{-1}\| &\leq C_N p^{b_{\text{inv}}} / (c_{\text{gap}} \tau p^{-5/3}), \\ \|\mathcal{Q}_{\text{mat}}^{-1}\| &\leq C_N p^{m_{\text{mat}}}, & \sup \|D^2 \mathcal{F}\| &\leq C_N p^{d_{\text{nl}}}. \end{aligned}$$

Here J_D and $\mathcal{Q}_{\text{mat}}$ are bounded isomorphisms, and the last bound holds on the valid closed centre ball considered below, on which $\mathcal{F}$ is C^2 . Put

$$(14.1) \quad B_{\text{Newt}} = b_{\text{inv}} + m_{\text{mat}} + 5/3, \quad L_{\text{Newt}} = B_{\text{Newt}} + d_{\text{nl}} + 4.$$

Assume that the accuracy order satisfies

$$(14.2) \quad N \geq \lceil 2B_{\text{Newt}} + d_{\text{nl}} + 8 \rceil.$$

For p above a threshold depending on the fixed constants and N , the closed ball of radius $\mathfrak{r}_p = \tau p^{-L_{\text{Newt}}}$ about z_0 contains a real zero of $\mathcal{F}$, provided that ball is inside the domain of the map. The threshold is uniform for $0 < \tau \leq 20$.

Proof. Let $\mathsf{T}_{\text{Newt}} = \mathcal{Q}_{\text{mat}}^{-1} J_D^{-1}$. This isomorphism $\mathcal{C} \rightarrow \mathcal{X}_p$ satisfies $\|\mathsf{T}_{\text{Newt}}\| \leq C_N p^{B_{\text{Newt}}}/\tau$. The three quantities controlling the contraction are

$$(14.3) \quad \frac{\|\mathsf{T}_{\text{Newt}} \mathcal{F}(z_0)\|}{\mathfrak{r}_p} \leq C_N p^{B_{\text{Newt}} + L_{\text{Newt}} + 2 - N} \leq C_N p^{-2},$$

$$(14.4) \quad \|I - \mathsf{T}_{\text{Newt}} D\mathcal{F}(z_0)\| \leq C_N \tau p^{B_{\text{Newt}} + 3 - N} \leq 20 C_N p^{-5},$$

$$(14.5) \quad \sup \|\mathsf{T}_{\text{Newt}} D^2 \mathcal{F}\| \mathfrak{r}_p \leq C_N p^{B_{\text{Newt}} + d_{\text{nl}} - L_{\text{Newt}}} = C_N p^{-4}.$$

The exact factorization gives $I - \mathsf{T}_{\text{Newt}} D\mathcal{F}(z_0) = -\mathsf{T}_{\text{Newt}} \mathcal{V}_{\text{res}}$, proving the second estimate. The exponents follow from (14.1) and (14.2). Increase p until each quantity is at most $1/4$. For $\mathcal{T}(z) = z - \mathsf{T}_{\text{Newt}} \mathcal{F}(z)$, Taylor's integral formula bounds its displacement from the centre on the closed ball by $\mathfrak{r}_p/4 + \mathfrak{r}_p/4 + \mathfrak{r}_p/8 < \mathfrak{r}_p$. The same formula for its derivative gives Lipschitz constant at most $1/4 + 1/4 = 1/2$. The ball is complete, so Banach's contraction theorem supplies a fixed point. Since T_{Newt} is injective, its fixed-point equation implies $\mathcal{F}(z) = 0$. All the smallness conditions are uniform for $0 < \tau \leq 20$, as the displayed powers show. $\square$

Corollary 14.2 (Instantiation for the centre family). *Fix the constants in the following order:*

$$(14.6) \quad \begin{aligned} C_0 &\longrightarrow C_{\text{rel}} \longrightarrow E_{\text{inv}} \longrightarrow M = M_{20E_{\text{inv}}} \\ &\longrightarrow (D_{\text{sep}}, K_{\text{head}}) \longrightarrow b_{\text{inv}} = 2M + 3 \longrightarrow N = 4M + 40. \end{aligned}$$

Here C_0 and C_{rel} are those of (11.11) and (11.19). For every sufficiently large $p \in \mathcal{P}_{\text{bd}}$, choose a split from Theorem 8.3 and the order- N centre at that split. There is an exact real zero $z_* = (g_*, f_*)$ with

$$(14.7) \quad \|z_* - z_0\|_{\mathcal{X}_p} \leq \tau p^{-L_{\text{Newt}}}, \quad B_{\text{Newt}} = 2M + 32/3, \quad L_{\text{Newt}} = 2M + 62/3.$$

Proof. The centre gap follows from Theorem 8.3 and Proposition 6.5 at the now fixed order. The coefficient inverse has exponent b_{inv} by Theorem 13.6. The material inverse and second derivative have exponents $m_{\text{mat}} = d_{\text{nl}} = 6$ by (10.20) and (10.13). The residuals and factorization are (10.15), (10.17), and (10.21). Thus all hypotheses of Theorem 14.1 hold in exactly $\mathcal{X}_p = \mathcal{G}_p \times \mathcal{E}_1$ and $\mathcal{C}_{p,2}$. The choice $N = 4M + 40$ exceeds $\lceil 2B_{\text{Newt}} + 14 \rceil$. The three gate exponents in this instance are

$$B_{\text{Newt}} + L_{\text{Newt}} + 2 - N = -20/3, \quad B_{\text{Newt}} + 3 - N = -2M - 79/3, \quad B_{\text{Newt}} + 6 - L_{\text{Newt}} = -4.$$

The second gate retains its factor τ ; the first and third have no residual negative power of it. Since $\|f_N\|_{\mathcal{E}_1} = O_N(\tau p^{-2})$ and $\mathfrak{r}_p \rightarrow 0$ uniformly for $\tau \leq 20$, the closed ball is eventually in the product region (10.12): its source displacement is at most one and its total shape norm is at most $1/4$. Increase the threshold to meet all preceding estimates for this one fixed N and apply the threshold theorem. The spaces, centre and operators are real by Definition 9.1 and their defining formulas, so the contraction acts on the real closed ball. $\square$

15. GEOMETRY, CLASSICAL SOLUTIONS, AND THE POMPEIU CONCLUSION

Lemma 15.1 (Analyticity and full-dimensional strict convexity). *Let $f \in \mathcal{E}_1$ be real with $\|f\|_{\infty} + \|\nabla_S^2 f\|_{\text{op}, \infty} < 1/4$. The radial graph $\{(1 + f(\omega))\omega : \omega \in S^{2p-1}\}$ is a closed analytic hypersurface bounding a bounded strictly convex domain. It is invariant under $O(a) \times O(b)$. The conclusion concerns all ambient directions, including the block-orbit directions and the axes.*

Proof. By Lemma 9.2, f is holomorphic in a complex neighbourhood of $x \in [0, 1]$. On the real sphere $x = |X|^2$ is a polynomial. The radial parametrization is therefore analytic, including

at $x = 0, 1$. Its radius is positive and its radial derivative is nonsingular, so it is an embedded closed hypersurface. Its symmetry follows from dependence on x alone.

Put $r_f = 1 + f$ and let g_S be the sphere metric. Differentiating the radial parametrization and its outward normal gives the second fundamental form with positive denominator and numerator

$$(15.1) \quad r_f^2 g_S + 2 df \otimes df - r_f \nabla_S^2 f.$$

Since $3/4 < r_f < 5/4$, its quadratic form on every unit tangent vector is at least $9/16 - 5/16 = 1/4$. This proves positivity in the full tangent space. To make the block directions explicit, at $0 < x < 1$ a unit X -orbit tangent vector has $\nabla_S^2 x(v, v) = 2(1 - x)$ and $dx(v) = 0$; a unit Y -orbit tangent vector has $\nabla_S^2 x(v, v) = -2x$. Their numerators in (15.1) are $r_f^2 - 2r_f(1 - x)f'$ and $r_f^2 + 2r_f x f'$. For the unit meridian direction the numerator is

$$r_f^2 + 8x(1 - x)(f')^2 - r_f\{4x(1 - x)f'' + 2(1 - 2x)f'\}.$$

The coordinate-free positive bound already controls all three expressions and their limits at the axes. The global convexity theorem for compact hypersurfaces with positive second fundamental form (Hadamard's theorem, in the form given by [20]; see also [11]) now makes the enclosed radial domain strictly convex. Its hypotheses hold: the hypersurface is smooth, compact, connected and embedded, and every principal curvature is positive. The convex boundary contains no line segment, since an interior point of such a segment would have zero normal curvature in its direction. Thus the convexity is strict. Boundedness follows from its bounded radius. $\square$

Lemma 15.2 (The non-ball witness survives the correction). *At the zero supplied by Corollary 14.2, the normalized shape f_* satisfies the hypotheses of Lemma 15.1 for large p , and its domain is not a ball.*

Proof. The fixed centre has normalized C^2 size at most $CC_0\tau p^{-2}$ by (9.6), $\|f_N\|_{\mathcal{E}_1} \leq \delta$, and (11.11). The correction has normalized C^2 size at most $C\tau p^{-L_{\text{Newt}}}$ by (9.6) and (14.7). Their sum is below $1/4$ for large p , uniformly for $\tau \leq 20$.

Consider the physical monic seed functional $\mathfrak{W}_p(f) = R[\mathbf{H}_2]f = Rf_2/\mathbf{H}_2(1)$. Its dual norm on $\mathcal{E}_1$ is at most Cp , since $R = O(p)$ and $\mathbf{H}_2(1) = (\beta_b)_2/(p+1)_2$ is bounded away from zero uniformly in the split. At the centre,

$$\mathfrak{W}_p(f_N) = -\frac{\theta}{16r(1-r)p} + O_N(\tau p^{-2}),$$

so its absolute value is at least $c\tau/p$ for large p . The correction changes it by at most $Cp\tau p^{-L_{\text{Newt}}} = o(\tau/p)$ since $L_{\text{Newt}} > 2$. It remains nonzero. Any ball invariant under both orthogonal block groups has centre fixed by both groups, hence centre zero. Its radial function would be constant and would have zero monic degree-two angular coefficient. The nonzero functional excludes such a ball. $\square$

Proposition 15.3 (Identification of the zero with a classical solution). *For a real zero (g, f) of (10.6) in the small chart, set $U = 1 + Kg$, $\widehat{\Omega} = \Phi_f(B)$, and $w = U \circ \Phi_f^{-1}$. Then w is a classical solution of*

$$(15.2) \quad \Delta w + R^2 w = 0 \quad \text{in } \widehat{\Omega}, \quad w = 1, \quad \nabla w = 0 \quad \text{on } \partial \widehat{\Omega}.$$

It is smooth up to the boundary and real analytic there and in the interior. For $\Omega = R\widehat{\Omega}$ and $u(x) = w(x/R)$, one has $\Delta u + u = 0$, boundary value one and zero boundary gradient. The field is nonconstant.

Proof. The source g is an actual L^2 function by Lemma 9.2. Its Poisson field belongs to $H^2(B) \cap H_0^1(B)$, so $U \in H^2(B)$ and its Dirichlet trace is one. The completed-space trace identity (9.19) makes its normal trace zero because $g \in \mathcal{G}_p$. The tangential trace of its gradient is the tangential derivative of its constant Dirichlet trace, hence zero in the Sobolev trace sense. Equivalently, finite clamped Poisson sums have zero full-gradient trace and converge to U in H^2 ; continuity of the gradient trace gives the same conclusion on the completion. Thus the whole gradient trace is zero. The map Φ_f is a smooth diffeomorphism of closed domains; composition transports U to an H^2 field on $\widehat{\Omega}$. Proposition 10.2 identifies its coefficient equation with the weak physical equation in (15.2). The trace chain rule $\nabla w \circ \Phi_f = D\Phi_f^{-T} \nabla U$ transports the zero full-gradient trace as well.

Dirichlet elliptic regularity on a smooth bounded domain applies to $\Delta w = -R^2 w$ and the constant boundary data. Starting from H^2 , each application improves regularity by two derivatives; iteration gives all Sobolev orders, hence C^∞ up to the boundary in the fixed dimension; see [8, Chs. 6, 8, 9]. The boundary is analytic by Lemma 9.2, the operator has analytic constant coefficients, and its Dirichlet data are analytic. Analytic interior regularity [9] and analytic Dirichlet boundary regularity [10] give real analyticity up to that boundary. The scalar Laplacian is strongly elliptic, and the Dirichlet condition is its complementing boundary condition; the lower-order term R^2 has analytic coefficients. Its now classical gradient trace agrees with the zero Sobolev trace. This proves (15.2) pointwise.

The dilation gives $\Delta_x u = R^{-2}(\Delta w)(x/R) = -u$ and preserves both boundary conditions. The only constant solution of $\Delta u + u = 0$ is zero, which is incompatible with boundary value one. Hence u is nonconstant. $\square$

Remark 15.4 (The symmetry sector). The Dirichlet inverse throughout the construction is only in the $O(a) \times O(b)$ invariant sector. At the exact solution, Cartesian derivatives of u satisfy the frequency-one equation and have zero Dirichlet trace. At least one is nonzero, so the full ambient Dirichlet operator has eigenvalue one. Since u is invariant, its X -derivatives transform as the standard $O(a)$ representation and its Y -derivatives as the standard $O(b)$ representation. Averaging any of these derivatives over the corresponding reflection group gives zero. They therefore have zero invariant projection. Within the invariant sector the graph inverse used above includes every mode, in particular the degree-four seed and the harmonic source layer.

Proposition 15.5 (Failure of the Pompeiu property). *Every bounded domain carrying a classical field constructed above fails the Pompeiu property.*

Proof. For any $\xi \in \mathbb{R}^{2p}$ with $|\xi| = 1$, put $v(x) = e^{i\xi \cdot x}$. Both u and v solve the frequency-one Helmholtz equation. Green's second identity gives

$$0 = \int_{\partial\Omega} (u \partial_\nu v - v \partial_\nu u) = \int_{\partial\Omega} \partial_\nu v = \int_{\Omega} \Delta v = - \int_{\Omega} e^{i\xi \cdot x} dx.$$

Thus the Fourier transform of the indicator vanishes on the unit sphere. Fix a unit vector ξ_0 . For every orthogonal map O and translation vector x_{tr} ,

$$\int_{x_{\text{tr}} + O\Omega} e^{i\xi_0 \cdot x} dx = e^{i\xi_0 \cdot x_{\text{tr}}} \int_{\Omega} e^{i(O^T \xi_0) \cdot y} dy = 0.$$

Taking real parts provides the nonzero continuous real function $\cos(\xi_0 \cdot x)$ whose integrals over all rigid copies of Ω vanish. This is the failure of the Pompeiu property. $\square$

Corollary 15.6 (The bounded base construction). *For every sufficiently large $p \in \mathcal{P}_{\text{bd}}$, some integer $a \in [p/2, 3p/4]$ gives a bounded strictly convex analytic non-ball domain with $O(a) \times$*

$O(2p - a)$ symmetry and the data (1.1). The construction uses $0 < \tau \leq 20$ and the cofactor order $N = 4M + 40$.

Proof. Lemma 5.2 supplies unbounded inputs of both parities; Lemma 4.7 supplies their radii and $0 < \tau < 20$. Theorem 8.3 selects an integer split for every sufficiently large member. Fix the order of constants (14.6) and the resulting one finite accuracy order N . The centre exists at every selected split by Theorem 6.2; its gap transfers by Proposition 6.5. Theorem 13.6, Theorem 10.6, and the second-derivative and residual bounds give the exact zero in Corollary 14.2. Lemmas 15.1 and 15.2 give its analytic strictly convex non-ball domain with the required block symmetry. Proposition 15.3 gives the classical frequency-one field, and Proposition 15.5 gives the Pompeiu assertion. There are only finitely many thresholds after the constants and N have been fixed; all are uniform in the split and in the nonzero detuning range. Their maximum discards only finitely many members of $\mathcal{P}_{\text{bd}}$. Its remaining members have unbounded distinct dimensions $2p$, as required. $\square$

16. CONSTANTS

After the constants and centre order have been fixed, the proof uses the maximum of finitely many thresholds. The exponents are

$$b_{\text{inv}} = 2M + 3, \quad B_{\text{Newt}} = 2M + 32/3, \quad L_{\text{Newt}} = 2M + 62/3, \quad N = 4M + 40.$$

The constants in the finite-centre remainder, the material inverse, and the second derivative depend on this fixed N . All estimates use this one fixed order.

The main cost of the coefficient inverse is the cofactor exponent $5M/3$. Its cluster count is for the fixed containing window $W_{\text{loc}} = 20E_{\text{inv}}$. The local-count proof is used at angular length $MD_{\text{sep}} \log p$. It requires, among other conditions,

$$(16.1) \quad \frac{1}{2}Cp^{-5/6}(MD_{\text{sep}} \log p)^3 + \frac{2C_{20E_{\text{inv}}}}{p}MD_{\text{sep}} \log p < 1,$$

$$(16.2) \quad c_t p^{1/6} > \pi MD_{\text{sep}} \log p + O(E_{\text{inv}}/p).$$

In the turning comparison any fixed $0 < c_t < \pi/\sqrt{2}$ is available after increasing its threshold, by the proof of Theorem 12.1. The determinant step also requires $D_{\text{sep}} > 5M/3 + 5$. The inverse window is fixed by $E_{\text{inv}} \geq 64C_{\text{rel}}$. Its local-count bound is

$$M = \left\lceil 2 + 4\sqrt{C_{20E_{\text{inv}}}/c} \right\rceil,$$

with the phase-strip constant and curvature lower constant from Theorem 12.1. Its integer M is fixed before D_{sep} and N , in the order (14.6).

The final threshold includes the Bessel and arithmetic estimates, the flow and integer-grid gap, the centre construction and gap transfer, the relative graph bounds, the local count at angular length $MD_{\text{sep}} \log p$, the cofactor absorption, the head-tail bounds, the three Newton gates, and the geometric smallness conditions. Each threshold is finite for the fixed constants and order; their maximum is uniform in the admissible split and in $0 < \tau \leq 20$.

17. BOUNDED DETUNING, INTERIOR SPLITS, AND RESIDUE CLASSES

Fix $0 < \varepsilon_{\text{sp}} < 1/4$ and write

$$I_{\text{sp}} = [\varepsilon_{\text{sp}}^2, (1/2 - \varepsilon_{\text{sp}})^2], \quad \varepsilon_{\text{sp}} \leq r \leq 1/2 - \varepsilon_{\text{sp}}.$$

Constants in this section may depend on this interval and on a fixed upper bound T for τ . The coefficient spaces and their angular radius two remain those of Definition 9.1.

Lemma 17.1 (Low-state transfer on wider windows). *Fix $C_* > 0$. For every sufficiently large primary crossing p_0 , uniformly for $0 < C_{\text{ap}} \leq C_* \log p_0$ and $|p - p_0| \leq C_{\text{ap}}/p_0$, every even degree $\ell \geq 0$ with $\ell - 1 \leq \sqrt{p_0}/4$, apart from the degree-four seed, has its ball Dirichlet spectrum outside $[R(p)^2 - 32, R(p)^2 + 32]$. Here $R(p) = j_{p,k_0}$ is the Neumann zero continued from $R_{\text{alg}}(p_0)$, and the degree- ℓ spectrum is $\{j_{p+\ell-1,k}^2 : k \geq 1\}$, the ball spectrum in dimension $2p$ at half-lattice inputs. The seed's other radial states are outside this window as well. The threshold depends on C_* . This includes every fixed $C_{\text{ap}} > 0$ after increasing its threshold.*

Proof. Suppose a nonseed low degree ℓ has a zero $z(p)$ with $|z(p)^2 - R(p)^2| \leq 32$. Continue it and the Neumann zero along the interval to p_0 . Equation (3.4) first gives $|z(t) - R(t)| \leq C(1 + C_{\text{ap}})/p_0$ there, so both radii are comparable to $2p_0$ and both order-to-radius ratios belong to $[1/3, 2/3]$. Put

$$f_{\text{ord}}(x) = \arccos x / \sqrt{1 - x^2}.$$

Its derivative is bounded on that compact interval. Equation (3.3) therefore gives

$$|z'(t) - R'(t)| \leq C(\ell + 1)/p_0 + C(1 + C_{\text{ap}})/p_0^2.$$

Integration, followed by multiplication by $z + R$, bounds the change in squared-energy offset by

$$(17.1) \quad CC_{\text{ap}}(\ell + 1)/p_0 + CC_{\text{ap}}(1 + C_{\text{ap}})/p_0^2.$$

For $\ell = O(\sqrt{p_0})$ this tends to zero in this range. The offset at p_0 is then less than 64, contradicting the first conclusion of Proposition 4.10. The other radial seed states are excluded by their frequency spacing and the same continuation estimate. $\square$

Lemma 17.2 (Extension of the split interval). *The reduction, centre, flow, coefficient-space, material, mixed, localization and cofactor estimates above hold on I_{sp} with the replacements in the following table. Every constant may depend on ε_{sp} ; their powers of p and their order of choice are unchanged. The weighted reduction, centre and flow statements allow real r on this interval. The Euclidean coefficient, material, mixed and closing estimates are applied at integer block sizes a, b with $a + b = 2p$. For the $\tau \leq 20$ counting estimate one may use*

$$K_{\text{sp}} = \frac{4}{\varepsilon_{\text{sp}}(1 - \varepsilon_{\text{sp}})}, \quad W_{\text{sp}} = 40K_{\text{sp}}.$$

Statement	Where the interval enters	Replacement
2.1, 2.3, 7.3	Positive beta parameters, endpoint orientation, axis cutoffs	$\alpha_b, \beta_b \geq \varepsilon_{\text{sp}}p > 1$ and $\beta_b \geq \alpha_b$; integer blocks for the Euclidean identification.
6.1	$pg_2 > 9/16$, $\ H_2\ _{\infty} < 9/16$, $\ T_{\varsigma}\ _{\infty} < 1$	$pg_2 \geq 2\varepsilon_{\text{sp}}(1 - \varepsilon_{\text{sp}})$, $pg_2 < 1$, $\ H_2\ _{\infty} \leq 1$, $\ T_{\varsigma}\ _{\infty} \leq K_{\text{sp}}$.
7.1	Derivatives of $\sqrt{\varsigma}$, lower bound on $\mathfrak{D}_j$	$\varsigma \geq \varepsilon_{\text{sp}}^2$, $\mathfrak{D}_j \geq c_{\text{sp}}(p + j)^2$; constants C_{sp} in the same weighted estimates.
7.2	Adjacent coefficient ratio; the upper bound $g_2 < 1/p$	$\alpha_b, \beta_b \geq 16$ suffices for the same ratio bounds; $\mathcal{N}_j, \mathcal{E}_j$ and their certificates are independent of ς .
6.2, 6.4	Compact parameter set and divisors $r(1 - r)$	Use $r \in [\varepsilon_{\text{sp}}, 1/2 - \varepsilon_{\text{sp}}]$; the same finite recursion and each prescribed complex disc.

Statement	Where the interval enters	Replacement
7.4–7.9	Quartic multiplier bounds and positive angular margin	C_{sp} replaces numerical interval constants; then choose W_{flow} and the threshold.
8.2	Supremum of T_ς	Shell width $3K_{\text{sp}}/p^2$ and window W_{sp} for sufficiently large p .
8.1, 8.3	Grid size and spacing	$\#\{a : 2\varepsilon_{\text{sp}}p \leq a \leq (1 - 2\varepsilon_{\text{sp}})p\}$ and spacing at least $\varepsilon_{\text{sp}}/p$.
(9.7), Section 9	Gasper margin for $\beta_b - \alpha_b \leq p/2$	Set $u = p - 1$, $v = \beta_b - \alpha_b \leq u$; the lower margin is $18u^3 + 63u^2 + 45u > 0$.
11.6, 11.8	The bounds for pg_2 and H_2	C_0, C_{rel} depend on the split interval; choose E_{inv} , then M , thereafter.
10.6, 15.2	Normal Hessian and the physical seed coefficient	$\ D_p^{-1}\ _{\mathcal{E}_1} \leq C_{N,\text{sp}}p^{-2}$ and $H_2(1) \geq c_{\text{sp}} > 0$; the seed is $-\theta/(16r(1-r)p) + O_{N,\text{sp}}(\tau p^{-2})$.

Proof. The rational identity (6.3) is valid for all these real parameters. Its leading term is $pg_2 \rightarrow 4r(1-r)$ uniformly on the compact interval, and its negative ς coefficient gives $pg_2 < 1$ as before. The two endpoint values of H_2 tend uniformly to r^2 and $(1-r)^2$; its negative minimum is $O(p^{-1})$. This proves the displayed replacement bounds and $H_2(1) \geq c_{\text{sp}}$. The formulas for $\mathfrak{D}_j$, $\mathfrak{a}_j$, and $\mathfrak{b}_j$ in Lemma 7.1 then give every entry and derivative estimate with C_{sp} .

For positivity, impose $p\varepsilon_{\text{sp}} \geq 16$. The two adjacent-ratio computations in Lemma 7.2 use only $\alpha_b, \beta_b \geq 16$, $j \geq 7$, and $2j + p \geq 78$. Its numerators (7.7), (7.8) and (7.9) have no ς . The last lower bound uses $g_2 < 1/p$, which was retained. The same positive variation therefore holds. In the form proof all series converge when $C_{\text{sp}}\tau/p^2$ is small; its two Schur reductions then give the stated flow estimates with the changed constants.

For the counting row, the normalized radius displacement is at most $10K_{\text{sp}}/R^2 \leq 3K_{\text{sp}}/p^2$ for $\tau \leq 20$. The min–max calculation of Lemma 8.2 then gives a containing window $40K_{\text{sp}}$ for all sufficiently large p depending on K_{sp} . For consecutive integers a , integrating $|d\varsigma/da| = |r - 1/2|/p$ gives spacing at least $\varepsilon_{\text{sp}}/p$. Use this spacing in the clipped-curve count. For the one-split gap, take $\gamma = c'_{\text{sp}}\tau p^{-5/3}$. The clipped-curve count is at most $C_{\text{sp}}p^{2/3} + C_{\text{sp}}c'_{\text{sp}}p$. Choose $c'_{\text{sp}} > 0$ so the second term is less than half the leading grid cardinality, then increase p to absorb the first. This gives the $p^{-5/3}$ gap on the extended interval as well.

For the algebra put $u = \alpha_b + \beta_b - 1 = p - 1$ and $v = \beta_b - \alpha_b$. If $u^2 - 7u - 24 \leq 0$, (9.7) holds directly. Otherwise its left-minus-right side is minimized on $0 \leq v \leq u$ at $v = u$, where it equals $18u^3 + 63u^2 + 45u$. Positive products, endpoint bounds and real radial stencils follow. The material and mixed proofs subsequently use these products and the strong norm of the finite centre; their numerical use of $9/16$ is replaced by the first two rows. The recurrence for that centre divides only by nonzero units, $-1/8$, c_j and powers of $r(1-r)$. These divisors stay uniformly away from zero on the compact interval. Thus all listed estimates follow in their stated order. $\square$

Lemma 17.3 (Arbitrary fixed detuning bound). *Fix $0 < C_{\text{ap}}, T < \infty$ and the split interval above. For half-lattice inputs with $|p - p_0| \leq C_{\text{ap}}/p_0$ and $0 < \tau \leq T$, the centre, flow, cofactor inverse, and Newton statements have the parameterized forms below. Fix $5/3 < q_{\text{gap}} < 2$; the*

base case $q_{\text{gap}} = 5/3$ is included with its original bounds. In the inverse and closing rows assume a selected quartic gap $\kappa\tau p^{-q_{\text{gap}}}$, with fixed $\kappa > 0$. Use $\varsigma \in I_{\text{sp}}$, equivalently $r \in [\varepsilon_{\text{sp}}, 1/2 - \varepsilon_{\text{sp}}]$; material and closing statements use integer blocks. Constants and thresholds may depend on these fixed data. Choose $W_{\text{flow}}(T)$ above the base lower bound and the required multiples of T and of the inverse angular margin. After C_0, C_{rel} choose E_{inv} , then $W_{\text{loc}} = TE_{\text{inv}}$ and $M = M_{TE_{\text{inv}}}$, then separation and the head cutoff, and finally the centre order. The last column gives the justification for each substitution.

Statement	Parameter replaced	Parameterized bound	Justification
4.7; 4.10	Scaled distance 31/10, base detuning bound	Use $ p - p_0 \leq C_{\text{ap}}/p_0$ and $0 < \tau \leq T$; the radius and detuning comparison are (20.1). Low exclusion retains its original energy bounds.	The exact quotient (4.13) and the real zero derivative comparison hold on this fixed window; 17.1 gives the continuation error (17.1).
6.2; 6.4	$ \theta \leq 20$, base split interval	(6.6) and (6.23) are $C_{N,T,\text{sp}}\theta^2 p^{-N}$; (6.7) has physical error $C_{N,T,\text{sp}}\tau p^{-3}$.	In (6.9) the Z derivative at $(\varepsilon, Z) = (0, 4)$ is 8 for every θ ; the recursion divides by units, $-1/8$, c_j and $r(1 - r)$.
7.4; 7.6	$\tau \leq 20$, absolute flow window	Relative bounds $C_{\text{sp}}\tau p^{-2}$ and $C_{\text{sp}}\tau^2 p^{-3}$; (7.21) is $C_{\text{sp}}\tau^2(W_{\text{flow}}(T))^{-1} + p^{-1}$.	The series is $\sum h^6(C_{\text{sp}}\tau/p^2)^h$; the complement ratio is $C_{\text{sp}}\tau/W_{\text{flow}}(T)$. The derivative adds $C_{\text{sp}}\tau^3/W_{\text{flow}}(T)^2$.
7.7; 7.8; 7.9	$\tau \leq 20$, $\varsigma_0 \in (1/64, 1/16)$	Use $\varsigma_0 \in \text{int } I_{\text{sp}}$; the low inverse is $C_{T,\text{sp}}/(p\tau)$. For $ \lambda < \tau/(16p)$ the signed slope is at least $c_{T,\text{sp}}\tau/p$.	The proof of 7.7 adds the first-order term to give seed diagonal $\theta/(2p) + O_T(\theta^2 p^{-3})$; (7.23) keeps the index-five value $-72 + O_T(p^{-1})$. Final norm recovery in 7.9 costs $C_{T,\text{sp}}\sqrt{p}$.
8.2	$ \theta \leq 20$, $W_{\text{count}} = 40$	Shell width $C_{\text{sp}}\tau/p^2$ and containing count window $C_{\text{sp}}(1 + T)$.	(6.5) gives displacement $\tau\ T_\zeta\ _\infty/(2R^2)$; indexwise shell bounds multiply $O(p^2)$ eigenvalues by $1 + O_{\text{sp}}(\tau/p^2)$.
8.1; 8.3	Base split grid and $\tau \leq 20$	The interior grid has $\asymp_{\text{sp}} p$ points and spacing at least $\varepsilon_{\text{sp}}/p$; one split has gap $c_{T,\text{sp}}\tau p^{-5/3}$.	The clipped-curve count is $C_{T,\text{sp}}p^{2/3}(1 + C_{T,\text{sp}}p\gamma/\mu)$, with $\mu = c_{T,\text{sp}}\tau/p$; choose the gap constant before the threshold.

Statement	Parameter replaced	Parameterized bound	Justification
11.6; 11.8	Centre-size and relative constants on the base compact set	$\delta \leq C_{0,\text{sp}}\tau p^{-2}$; (11.20) and (11.19) are $C_{\text{rel},\text{sp}}\tau p^{-2}$. These constants precede N .	The leading quartic size is $C_{\text{sp}}\tau p^{-2}$; the finite-order remainder $C_{N,T,\text{sp}}\tau p^{-4}$ is absorbed by the threshold. The scalar graph line is $C(\delta + p\delta^2)$.
10.4; 10.5	Fixed-centre constants for $\tau \leq 20$	(10.11) retains powers p^{k+1} and p^{k+5} with $C_{N,k,T,\text{sp}}$. Residual and material defect bounds are $C_{N,T,\text{sp}}\theta^2 p^{2-N}$ and $C_{N,T,\text{sp}}\theta^2 p^{3-N}$.	The known-mode head uses the finite factor $e^{C_N\sqrt{T}}$; the factorial radial tail stays geometric. The two-layer lift and (9.38) give the residual powers.
10.6; (10.13)	Fixed-centre constants for $\tau \leq 20$	$\ \mathcal{Q}_{\text{mat}}^{-1}\ \leq C_{N,T,\text{sp}}p^6$ and $\sup\ D^2\mathcal{F}\ \leq C_{N,T,\text{sp}}p^6$ on (10.12).	(10.26) uses $\ D_p^{-1}\ \leq C_{N,T,\text{sp}}p^{-2}$ and $\ \Delta T_\eta\ \leq C_{N,T,\text{sp}}p^8\ \eta\ $; (10.7) costs $Cp(1 + \ g\)$.
(13.1); 13.1; 13.2	Containing window $20E_{\text{inv}}$	The projection stays $\mathbf{1}_{[-E_{\text{inv}}\tau, E_{\text{inv}}\tau]}(H_0 - R^2)$; its fixed superset has width TE_{inv} . The Schur norm is $C_{E_{\text{inv}},T,\text{sp}}\tau$.	(13.3) and (11.19) give complementary absorption; the four feedback factors give $C\tau/E_{\text{inv}}$.
12.3	$\tau \leq 20$, gap exponent	For every $\tau > 0$, gap $c\tau p^{-q_{\text{gap}}}$ and $\sigma D > q_{\text{gap}}M + q_{\text{gap}} + 2$ give $\ F^{-1}\ _1 \leq C\tau^{-1}p^{q_{\text{gap}}M}$.	The cofactor line is $m!(C\tau)^{m-1}(2/\gamma_F)^m$; τ cancels in both error absorptions.
13.3	$M_{20E_{\text{inv}}}, 5M/3, D_{\text{sep}} > 5M/3+5$	$M = M_{TE_{\text{inv}}}$, $\ F_{\text{eff}}^{-1}\ _1 \leq C\tau^{-1}p^{q_{\text{gap}}M}$; choose $\sigma D_{\text{sep}} > q_{\text{gap}}M + q_{\text{gap}} + 2$.	12.2 applies to the fixed superset; deleting inter-cluster entries costs $C\tau p^{-\sigma D_{\text{sep}}}$, followed by the cofactor row.
13.4	Exponent $2 + 5M/3$	(13.8) has right side $C\tau^{-1}p^{2+q_{\text{gap}}M}$.	The block solution costs C/E_{inv} off diagonal, $C\tau$ into the window and Cp^2 for $K_0\Pi$; density graph multipliers are bounded.
13.6	$\tau \leq 20$, exponent $5/2 + 5M/3$	$\ J_D^{-1}\ _C \leq C\tau^{-1}p^{5/2+q_{\text{gap}}M} \leq Cp^{2M+3}/\gamma_p$, with $\gamma_p = c_{\text{gap}}\tau p^{-q_{\text{gap}}}$.	(13.10) costs $\sqrt{p}$; (11.20) gives (13.9). Compare $5/2 + q_{\text{gap}}(M-1) \leq 2M+3$.
6.5	Gap exponent $5/3, \tau \leq 20$	Quartic gap $c_q\tau p^{-q_{\text{gap}}}$ gives centre gap $(c_q/2)\tau p^{-q_{\text{gap}}}$.	The min-max displacement is $C_{N,T,\text{sp}}\tau p^{-2}$; require $p^{2-q_{\text{gap}}} > 2C_{N,T,\text{sp}}/c_q$.

Statement	Parameter replaced	Parameterized bound	Justification
14.1; 14.2	$\tau \leq 20$, gap exponent $5/3$	Inverse hypothesis $C_N p^{b_{\text{inv}}} / (c_{\text{gap}} \tau p^{-q_{\text{gap}}})$; $B = b_{\text{inv}} + m_{\text{mat}} + q_{\text{gap}}$, $L = B + d_{\text{nl}} + 4$, $N \geq [2B + d_{\text{nl}} + 8]$.	(14.3)–(14.5) give $C_N p^{B+L+2-N}$, $C_N \tau p^{B+3-N}$ and $C_N p^{B+d_{\text{nl}}-L}$; only the second uses $\tau \leq T$.
15.1; 15.2	Fixed-centre comparison constants	Normalized C^2 size $C_{N,T,\text{sp}} \tau p^{-2} + O(\tau p^{-L})$; physical seed $-\theta/(16r(1-r)p)$ with error $C_{N,T,\text{sp}} \tau p^{-2} + O(\tau p^{1-L})$.	(15.1) applies when the shape tends to zero in C^2 ; dividing the witness error by τ/p makes it tend to zero.

In the Newton row take $b_{\text{inv}} = 2M + 3$ and $m_{\text{mat}} = d_{\text{nl}} = 6$. Thus $B = B_{\text{bd}} = 2M + 9 + q_{\text{gap}}$, $L = L_{\text{bd}} = B_{\text{bd}} + 10$ and $N = 4M + 40$; the three powers are those in (17.2). Section 16 records the special case $T = 20$, $q_{\text{gap}} = 5/3$. These fixed-detuning families use the cofactor estimate with fixed local cluster cardinality.

Proof. Lemma 17.1 supplies the low-state exclusion. The real root comparison and the exact ratio (4.13) give $R = 2p + 3 + O_{C_{\text{ap}}}(p^{-1})$. For the centre choose a fixed complex θ disc containing $[-T, T]$ before its ε disc. Its radius equation has derivative eight at $(\varepsilon, Z) = (0, 4)$ for every θ on this fixed disc. The finite recursion has the same nonzero leading divisors, and its two defects have the same analytic factor θ^2 . Compactness therefore proves all the stated fixed-order centre bounds with $C_{N,T,\text{sp}}$.

In Lemma 7.4 the dominating series is $\sum h^6(C_{\text{sp}}\tau/p^2)^h$. Choose a fixed W_{flow} larger than the required multiples of T and the inverse angular margin. The complementary series ratio is $C_{\text{sp}}\tau/W_{\text{flow}}$, and its derivative remainder is bounded by $C_{\text{sp}}\tau^2(W_{\text{flow}}^{-1} + p^{-1})$. The extra term $C\tau^3/W_{\text{flow}}^2$ is absorbed by this bound. The seed diagonal is still $\theta/(2p) + O_T(\theta^2 p^{-3})$, the index-five diagonal is $-72 + O_T(p^{-1})$, and the weighted low inverse is $C_T/(p\tau)$. Its two exterior remainders retain their factors τ^2 . Choose the window and then p so the errors are below the positive angular margin. The last step of Theorem 7.9 therefore gives $c_{T,\text{sp}}\tau/p$ in its same narrow band. The shell count uses a fixed window $C_{\text{sp}}(1 + T)$.

The leading quartic strong norm is bounded by $C_{\text{sp}}\tau p^{-2}$ independently of the centre order. For each fixed N its remainder is $C_{N,T,\text{sp}}\tau p^{-4}$. Increasing only the threshold absorbs this remainder into the leading bound, as in Lemma 11.6. Thus C_0, C_{rel} are fixed before N . For the coefficient inverse choose E_{inv} from this relative graph constant. The containing window is now TE_{inv} , so $M = M_{TE_{\text{inv}}}$ is fixed. The determinant proof of Lemma 12.3 retains exactly one power τ^{-1} . Choose separation and head cutoff after M , and the finite centre order after those choices. The material and residual proofs use only the fixed-centre norms and the unchanged clamped estimates. Their constants may depend on T , while their exponents remain six. The three Newton gate estimates and the geometric comparisons retain their powers of τ ; replacing its upper bound twenty by T changes only the threshold. $\square$

Lemma 17.4 (All but a sublinear number of interior splits). *Fix C_{ap}, T , $0 < \eta_{\text{cnt}} < 1/3$, and $q_{\text{gap}} = 2 - \eta_{\text{cnt}}$. At every sufficiently large input of Lemma 17.3, all but $O_{C_{\text{ap}}, T, \text{sp}, \eta_{\text{cnt}}}(p^{2/3 + \eta_{\text{cnt}}})$ integers*

$$2\varepsilon_{\text{sp}}p \leq a \leq (1 - 2\varepsilon_{\text{sp}})p$$

give an actual quartic invariant gap at least $\kappa\tau p^{-q_{\text{gap}}}$, where $\kappa > 0$ depends only on the fixed data.

Proof. The enlarged shell window gives at most $C_{T,\text{sp}}p^{2/3}$ possible ordered curves. Their clipped signed functions are nondecreasing by the absolute-continuity argument of Lemma 8.1, with slope at least $\mu = c_{T,\text{sp}}\tau/p$ inside the narrow flow band. Since the grid spacing is at least $\varepsilon_{\text{sp}}/p$, one curve excludes at most

$$1 + \frac{2p\kappa\tau p^{-q_{\text{gap}}}}{\varepsilon_{\text{sp}}\mu} \leq 1 + Cp^{2-q_{\text{gap}}}$$

points. The gap lies inside the flow band eventually because $q_{\text{gap}} > 1$. Summing the exclusions gives $C(p^{2/3} + p^{8/3-q_{\text{gap}}}) \leq C'p^{2/3+\eta_{\text{cnt}}}$. One can exclude twice the stated radius first, which also covers equality at an endpoint. This proof uses $2p \in \mathbb{Z}$ only to obtain integer second block sizes. $\square$

Proposition 17.5 (Cofactor closure). *Under the hypotheses of Lemma 17.4, every retained split carries an analytic strictly convex non-ball domain and a solution of (1.1) with its prescribed block symmetry. The threshold in p follows these choices. First fix $C_{\text{ap}}, T, \varepsilon_{\text{sp}}, q_{\text{gap}}$. Next fix $C_0, C_{\text{rel}}, E_{\text{inv}}$ and $M = M_{TE_{\text{inv}}}$, then D_{sep} and K_{head} . Finally set*

$$b_{\text{inv}} = 2M + 3, \quad N = 4M + 40, \quad B_{\text{bd}} = 2M + 9 + q_{\text{gap}}, \quad L_{\text{bd}} = B_{\text{bd}} + 10.$$

The closed Newton ball has radius $\tau p^{-L_{\text{bd}}}$.

Proof. Fix the data in the order stated, choosing separation with $\sigma D_{\text{sep}} > q_{\text{gap}}M + q_{\text{gap}} + 2$ and the head cutoff required by Lemma 13.5. Use the parameterized statements of Lemma 17.3 throughout this chain. The centre-to-quartic normalized displacement is $C_{N,T,\text{sp}}\tau p^{-4}$. The indexwise min-max proof of Proposition 6.5 changes energies by $C_{N,T,\text{sp}}\tau p^{-2} = o(\tau p^{-q_{\text{gap}}})$. Thus the centre has half the selected gap.

Use Lemma 12.3 with this gap exponent. The Schur reconstruction and radial conversion in Section 13 give

$$\|J_D^{-1}\| \leq C\tau^{-1}p^{5/2+q_{\text{gap}}M} \leq Cp^{2M+3}/(\tau p^{-q_{\text{gap}}}),$$

since $5/2 + q_{\text{gap}}(M-1) \leq 2M+3$. The separation inequality makes the off-cluster restoration small. The scalar estimate (11.20) supplies (13.9) for fixed T and sufficiently large p , as in the proof of Theorem 13.6. The material inverse and nonlinear bound still have exponent six, so $T_{\text{Newt}} = Q_{\text{mat}}^{-1}J_D^{-1}$ has norm at most $C_N\tau^{-1}p^{B_{\text{bd}}}$.

Apply the contraction proof of Theorem 14.1 with these explicit exponents. The three gate powers are

$$\begin{aligned} B_{\text{bd}} + L_{\text{bd}} + 2 - N &= 2q_{\text{gap}} - 10, \\ (17.2) \quad B_{\text{bd}} + 3 - N &= -2M - 28 + q_{\text{gap}}, \\ B_{\text{bd}} + 6 - L_{\text{bd}} &= -4. \end{aligned}$$

The second carries the additional factor $\tau \leq T$; all three tend to zero uniformly in $0 < \tau \leq T$. The working ball lies in (10.12) for large p . The actual inverse identities in the same spaces give the exact real zero. The geometric and classical-solution arguments of Section 15 use the replacement constants in Lemma 17.2. Its normalized C^2 size is $O_{N,T,\text{sp}}(\tau p^{-2}) + O(\tau p^{-L_{\text{bd}}}) \rightarrow 0$, so Lemma 15.1 gives strict convexity. The leading physical quartic coefficient has size comparable to τ/p ; both the centre remainder and the physical Newton correction are smaller after division by that quantity. This proves non-sphericity as well as strict convexity, and completes the cofactor chain uniformly for $0 < \tau \leq T$. $\square$

Lemma 17.6 (Orthogonal symmetry and similarity classes). *For a differentiable nonconstant radial graph about zero invariant under $O(a) \times O(n-a)$, its orthogonal symmetry Lie algebra is $\mathfrak{so}(a) \oplus \mathfrak{so}(n-a)$. Every bounded strictly convex non-ball domain with real-analytic boundary*

and this invariance is such a radial graph. Consequently any two such domains at distinct splits below $n/2$ are non-similar. Define $\mathcal{N}_{\text{sim}}(n)$ as the number of integers $0 < a < n/2$ for which there exists a bounded strictly convex non-ball domain in $\mathbb{R}^n$ with real-analytic boundary, invariant under $O(a) \times O(n-a)$ and carrying (1.1). Choosing any one such domain at each split gives this many similarity classes, independently of the choices. Suppose that, for every fixed $0 < \varepsilon_{\text{sp}} < 1/4$ and $0 < \eta_{\text{cnt}} < 1/3$, all but $O(n^{2/3+\eta_{\text{cnt}}})$ integer splits in $[\varepsilon_{\text{sp}}n, (1/2 - \varepsilon_{\text{sp}})n]$ give such domains. Then along that family's dimensions,

$$\mathcal{N}_{\text{sim}}(n) \geq (1/2 - 2\varepsilon_{\text{sp}})n - Cn^{2/3+\eta_{\text{cnt}}}, \quad \liminf_{n \rightarrow \infty} \frac{\mathcal{N}_{\text{sim}}(n)}{n} \geq \frac{1}{2}.$$

The limit inferior is taken through that family's dimensions. The upper bound $\mathcal{N}_{\text{sim}}(n) \leq n/2$ gives $\mathcal{N}_{\text{sim}}(n)/n \rightarrow 1/2$. Thus the number of integer splits $0 < a < n/2$ carrying no such domain is $o(n)$ along the family.

Proof. A bounded convex domain Ω with the stated invariance is centrally symmetric and contains zero in its interior. Each ray meets the boundary exactly once, and at $y \in \partial\Omega$ its supporting tangent hyperplane gives $\langle y, \nu(y) \rangle > 0$ for the outward unit normal $\nu(y)$ because zero is interior; hence every ray is transverse and radial projection is an analytic diffeomorphism onto the sphere. For a non-ball domain its radius is nonconstant. Write the radius on the sphere as $F(x)$, $x = |X|^2$. Since x is a submersion for $0 < x < 1$, F is differentiable there; the mean value theorem and continuity at the endpoints give $x_0 \in (0, 1)$ with $F'(x_0) \neq 0$. A skew matrix has diagonal blocks in the two block Lie algebras and off-diagonal blocks $M, -M^t$. Its infinitesimal action on F is $2F'(x)X^tMY$. Vanishing for all X, Y with squared norms $x_0, 1-x_0$ forces $M = 0$, by varying their directions. The block diagonal matrices are symmetries, proving the equality.

Every such bounded domain has barycentre zero by central symmetry. A similarity between two has zero translation and conjugates their orthogonal symmetry Lie algebras. The dimension of the latter is $\{a(a-1) + (n-a)(n-a-1)\}/2$, whose change from a to $a+1$ is $2a+1-n < 0$ on the selected range. Thus different splits give different similarity classes. The assumed exceptional-split bound and integer counting give the displayed lower bound. Fixing any $\varepsilon_{\text{sp}} > 0$ and then taking the lower limit proves a bound $1/2 - 2\varepsilon_{\text{sp}}$. Taking the supremum over these fixed choices proves the limit inferior. The same count leaves at most $2\varepsilon_{\text{sp}}n + Cn^{2/3+\eta_{\text{cnt}}}$ splits without a constructed domain. Divide by n , take the upper limit along the family, and then let the fixed margin tend to zero to obtain $o(n)$ such splits. $\square$

Theorem 17.7 (Dimensions in prescribed residue classes). *For every positive integer L_{ap} and every integer b_{ap} , there is an unbounded sequence of dimensions (n_k) , indexed by the continued-fraction convergent index k , with $n_k \equiv b_{\text{ap}} \pmod{4L_{\text{ap}}}$. For every sufficiently large k , some $m = m_k \geq m_0$ satisfies*

$$|n_k/2 - \widehat{p}_m| < 32L_{\text{ap}}^2/\widehat{p}_m, \quad 0 < \tau < 256L_{\text{ap}}^2.$$

Every sufficiently large term has the domains and multiplicity of Proposition 17.5. The starting index and thresholds may depend on the fixed $L_{\text{ap}}, b_{\text{ap}}, \varepsilon_{\text{sp}}, \eta_{\text{cnt}}$. In particular every residue modulo L_{ap} occurs infinitely often. For even L_{ap} the residue determines the parity; for odd L_{ap} both parities occur infinitely often in each residue class modulo L_{ap} .

Proof. Set $M_{\text{ap}} = 4L_{\text{ap}}$. Consecutive convergent vectors of α_{ar} form an invertible matrix modulo M_{ap} . Choose representatives $A, B \in \{0, \dots, M_{\text{ap}} - 1\}$ satisfying

$$P = AP_k^{\text{cf}} + BP_{k-1}^{\text{cf}} \equiv b_{\text{ap}} + 3 \pmod{M_{\text{ap}}}, \quad Q = AQ_k^{\text{cf}} + BQ_{k-1}^{\text{cf}} \equiv 1 \pmod{M_{\text{ap}}}.$$

They cannot both vanish, so $Q \geq Q_{k-1}^{\text{cf}} > 0$. In (4.2), for $T_k^{\text{cf}} \geq A/B$ the expression is at most $B(A+B)$; when $1 < T_k^{\text{cf}} < A/B$ its maximum is attained at $T_k^{\text{cf}} = 1$ and then is at most $|A-B|\max\{A, (A+B)/2\}$. The cases $A=0$ or $B=0$ follow directly. Both bounds are at most $2(M_{\text{ap}}-1)^2$. Set $m = (Q-1)/4$, $n = P-3$, and $p = n/2$. The crossing expansion gives

$$\limsup \widehat{p}_m |p - \widehat{p}_m| \leq \frac{\alpha_{\text{ar}}}{2} (M_{\text{ap}} - 1)^2 + d_{\text{ar}} < 32L_{\text{ap}}^2.$$

For the last strict inequality use $\alpha_{\text{ar}} < 363/157$ and $d_{\text{ar}} < 1$. The dimensions tend to infinity and have the prescribed residue. Equation (20.1), with $C_{\text{ap}} = 32L_{\text{ap}}^2$, gives eventually $\tau < (16\delta_0 + o(1))32L_{\text{ap}}^2 < 256L_{\text{ap}}^2$; non-coincidence gives $\tau > 0$. Apply Lemma 17.3 and Proposition 17.5 with these fixed constants. Taking $\varepsilon_{\text{sp}} = 1/8$, the sublinear exclusions leave a split in $r \in [1/4, 3/8]$ as well. For a residue modulo an odd L_{ap} , its lifts b_{ap} and $b_{\text{ap}} + L_{\text{ap}}$ have opposite parities, giving the final assertion. $\square$

Corollary 17.8 (Almost every fixed split ratio). *Use the explicit continued-fraction selections in Definition 4.3, Lemma 5.2, and Theorem 17.7, retaining their convergent index k . There is a set $\mathcal{R}_{\text{split}} \subset (0, 1/2)$ of full Lebesgue measure such that, for every $r_0 \in \mathcal{R}_{\text{split}}$, each of these countably many selected dimension families (n_k) eventually has a constructed Schiffer domain with split*

$$a_k = \lfloor n_k r_0 \rfloor.$$

The starting index may depend on r_0 and the selected family.

Proof. Fix one selected family and a compact ratio interval inside $(0, 1/2)$. Choose a slightly larger compact interval for Lemma 17.4; rounding changes the ratio by at most $1/n_k$, so it remains in that interval eventually. Take $\eta_{\text{cnt}} = 1/6$. At dimension n_k at most $Cn_k^{5/6}$ splits are excluded, with C depending on this family's fixed detuning bound and the compact interval. Each integer split corresponds under $a = \lfloor n_k r \rfloor$ to an interval of r of length $1/n_k$. The set of ratios giving an excluded split therefore has measure at most $Cn_k^{-1/6}$.

The selected denominator is at least Q_{k-3}^{cf} for the first two selections and at least Q_{k-1}^{cf} for the residue-class selection. Continued-fraction denominators satisfy $Q_{k+2}^{\text{cf}} \geq Q_{k+1}^{\text{cf}} + Q_k^{\text{cf}} \geq 2Q_k^{\text{cf}}$. Since the selected dimensions are asymptotic to α_{ar} times their selected denominators, $n_k \geq c2^{k/2}$ eventually. Hence $\sum_k n_k^{-1/6} < \infty$. The measure of the union of the excluded-ratio events after index K is bounded by the tail of this convergent series and tends to zero. Almost every ratio has only finitely many exclusions.

Take a countable compact exhaustion of $(0, 1/2)$ and then the intersection of these full-measure sets over the two primary selections and all fixed moduli and residues. Their exceptional sets have total measure zero. Proposition 17.5 gives the domain at every eventual retained split, proving the assertion with the stated order of quantifiers. $\square$

18. SHARPER CLAMPED AND MATERIAL ESTIMATES

The spaces in this section remain $\mathcal{C}$ and $\mathcal{X}_p = \mathcal{G}_p \times \mathcal{E}_1$. The centre and the fixed material field have the membership supplied by Section 10. The estimates below follow from their exact equations. For fixed wider parameter ranges use the extensions in Lemmas 17.2 and 17.3.

Lemma 18.1 (The harmonic gradient coefficient sum). *For endpoint-normalized solid harmonics S_i, S_j ,*

$$(18.1) \quad \nabla S_i \cdot \nabla S_j = \sum_{k < i+j} 2h(h+2k+p-1)c_{ij}^k S_k t^{h-1}, \quad h = i+j-k, \quad \sum_{k < i+j} 2h(h+2k+p-1)c_{ij}^k = 4ij.$$

Consequently, for $H_u = \mathcal{H}_{\text{ext}} h_u$,

$$(18.2) \quad \|M_{\nabla H_1 \cdot \nabla H_2}\|_{C \rightarrow C} \leq 4 \|h_1\|_{\mathcal{E}_1} \|h_2\|_{\mathcal{E}_1}.$$

Proof. The product rule for two harmonic polynomials gives $2\nabla S_i \cdot \nabla S_j = \Delta(S_i S_j)$. Apply (2.13) to each term of the positive solid product to obtain the first identity. At a unit point of the $x = 1$ block axis, $S_i = S_j = 1$ and their tangential gradients vanish. Their full gradients are $2iy$ and $2jy$. Evaluation therefore gives the second identity; the coefficients are all nonnegative.

Every multiplier $S_k t^{h-1}$ has total degree $2i + 2j - 2$ and norm at most $2^k \varpi_p(2i + 2j - 2)$ by Proposition 9.4. This is at most $2^{i+j} \varpi_p(2i) \varpi_p(2j)$. The positive coefficient sum is $4ij \leq 4(1+i)(1+j)$. Sum the absolute shape coefficients with their $\mathcal{E}_1$ weights to get (18.2). A constant harmonic has zero gradient, and completion gives the assertion for all shapes. $\square$

Proposition 18.2 (Sharp centre and material bounds). *Under the root and centre hypotheses of Lemma 20.2, for each fixed centre order $N \geq 4$, uniformly in $0 < \tau \leq T \leq \log p$ and on the working region (10.12),*

$$(18.3) \quad \|g_0\|_C \leq p^3 \mathcal{B}_N(T), \quad \|\mathcal{Q}_{\text{mat}}^{-1}\| \leq p^2 \mathcal{B}_N(T), \quad \sup \|D^2 \mathcal{F}\| \leq p^3 \mathcal{B}_N(T).$$

Here $\mathcal{B}_N(T) = C_N(1 + T)^{d_N^{\text{tame}}} e^{C_N \sqrt{T}}$ is the envelope of Lemma 20.2, enlarged if necessary; C_N, d_N^{tame} depend on the fixed order and compact split interval. For fixed T it is a constant. The inverse identities are those of Theorem 10.6.

Proof. In the proof of Proposition 10.2, replace (9.27) by (18.2). Every other grouped estimate there has a constant independent of p . In particular (9.28) and (9.29) are applied to clamped sources, as their statements require. The inverse multiplier series and their first two derivatives are bounded on the working ball. Thus

$$(18.4) \quad \|D_f \mathcal{F}(g, f)\| \leq C \|g\|_C, \quad \|D^2 \mathcal{F}(g, f)\| \leq C(1 + \|g\|_C).$$

The frequency term is affine in g and independent of the shape.

Put $\mathcal{R}_{\text{sc}} = (P_{\Phi_N} \Delta - \Delta) K_D$. Equation (11.20) gives $\|\mathcal{R}_{\text{sc}}\| \leq C \tau p^{-2} < 1/2$ at the fixed centre. Since $U_0 = 1 + K g_0$, its residual equation is exactly

$$E_0 = (I + \mathcal{R}_{\text{sc}}) g_0 + R^2 U_0.$$

Use $\|g_0\| \leq 2\|(I + \mathcal{R}_{\text{sc}}) g_0\|$ and $\|U_0\| \leq C_N p$ from (10.11). This yields $\|g_0\| \leq C_N p^3$ and hence the last bound in (18.3) by (18.4). This is norm absorption on the already defined clamped source.

The fixed material field is $\mathbf{f}_0 = d_0^{-1} \mathbf{E} U_0 = d_0^{-1} \mathbf{E} K g_0$. The all-source estimate for $\mathbf{E} K_D$ in (9.17) gives $\|\mathbf{f}_0\| \leq C_N p^2$. Its zero Dirichlet trace and $\Delta \mathbf{f}_0 \in \mathcal{C}$, already proved in (10.24), give $\mathbf{f}_0 = K_D \Delta \mathbf{f}_0$. Apply the material identity to the constant shape variation one:

$$B_{\Phi_N} \mathbf{f}_0 = d_0^{-1} \mathbf{E} E_0 - D_f \mathcal{F}(z_0)[1].$$

The residual bound and (18.4) bound this by $C_N p^3$. On the other hand it equals $(I + \mathcal{R}_{\text{sc}}) \Delta \mathbf{f}_0 + R^2 \mathbf{f}_0$. The same norm absorption proves

$$(18.5) \quad \|\Delta \mathbf{f}_0\| \leq C_N p^4.$$

For an unknown shape η ,

$$\Delta T_\eta = (\mathcal{H}_{\text{ext}} \eta) \Delta \mathbf{f}_0 + 2 \nabla (\mathcal{H}_{\text{ext}} \eta) \cdot \nabla K_D \Delta \mathbf{f}_0.$$

The module estimate and the all-source gradient bound (9.24) give $\|\Delta T_\eta\| \leq C_N p^4 \|\eta\|_{\mathcal{E}_1}$. The exact reconstruction (10.26) has $\|\eta\|_{\mathcal{E}_1} \leq C_N p^{-2} \|\psi\|_C$, by its normal trace and the normal Hessian bound. It follows that $\|\delta g\| + \|\eta\| \leq C_N p^2 \|\psi\|$. Both inverse identities were proved before estimating this expression, so this proves the middle bound in (18.3). The dependence

on T in these same steps is given by row 12 and the clamped-source and material calculations in the proof of Lemma 20.2, yielding the stated envelope. $\square$

Lemma 18.3 (Quadratic graph remainder at a polynomial chart). *Let H be a real invariant harmonic polynomial of fixed angular degree, with size δ in a fixed multiplier norm including its first four Euler derivatives. For $p\delta$ sufficiently small put $q = 1 + H$, $d = q + \mathbf{E}q$, $m = q^{p-1/2}d^{1/2}$, and denote its unitary Dirichlet operator by $\widehat{H}_H$. Its first variation is*

$$V_1(H) = 2H\Delta + 2\nabla H \cdot \nabla \mathbf{E} + 2\nabla l_{\text{lin}} \cdot \nabla, \quad l_{\text{lin}} = pH + \frac{1}{2}\mathbf{E}H.$$

In each of the mixed and Hilbert matrix norms at fixed exponential weight,

$$(18.6) \quad \|(\widehat{H}_H - H_0 - V_1(H))K_0^{-1}\| \leq C\delta^2.$$

The constant may depend on the fixed degree and weights. This is an all-source estimate at a fixed polynomial chart.

Proof. Lemma 18.1 also gives a p -independent multiplier bound for fixed-degree gradient products in $\mathcal{M}$: each positive solid multiplier is bounded by Lemma 11.3. In $\mathcal{H}$ use its fixed bandwidth and the pointwise gradient bound as in (11.16). These bounds apply to H and each of its fixed Euler derivatives. With $l = \log m$, harmonicity of q, d gives

$$\Delta l = -(p - \frac{1}{2})q^{-2}|\nabla H|^2 - \frac{1}{2}d^{-2}|\nabla(1 + \mathbf{E})H|^2.$$

Thus $\Delta l = O(p\delta^2)$, $|\nabla l|^2 = O(p^2\delta^2)$, and $\mathbf{E}l, \mathbf{E}^2l = O(p\delta)$ in the multiplier norms. Subtracting l_{lin} in its gradient leaves harmonic gradients times coefficients of size $O(p\delta)$; after ∇K_D the result is $O(\delta^2)$. The exact conjugations are

$$\begin{aligned} m\Delta m^{-1} &= \Delta - 2\nabla l \cdot \nabla + |\nabla l|^2 - \Delta l, \\ m(\nabla H \cdot \nabla \mathbf{E})m^{-1} &= \nabla H \cdot \nabla \mathbf{E} - (\nabla H \cdot \nabla l)\mathbf{E} - (\mathbf{E}l)\nabla H \cdot \nabla \\ &\quad + (\mathbf{E}l)(\nabla H \cdot \nabla l) - \nabla H \cdot \nabla \mathbf{E}l, \\ m\mathbf{E}(\mathbf{E} + 1)m^{-1} &= \mathbf{E}(\mathbf{E} + 1) - 2(\mathbf{E}l)\mathbf{E} + (\mathbf{E}l)^2 - \mathbf{E}^2l - \mathbf{E}l, \\ m\mathbf{E}m^{-1} &= \mathbf{E} - \mathbf{E}l. \end{aligned}$$

Insert these in the negative scalar pullback (10.4). Its linear part is exactly $V_1(H)$; there is no linear potential because $\Delta l_{\text{lin}} = 0$. The following table records every remainder after composition with K_D ; all Poisson and gradient bounds are the all-layer bounds in Lemma 11.4.

Remainder	Bound after K_D
Quadratic part of $q^{-2}\Delta$	$C\delta^2$
Nonlinear density-gradient terms	$Cp\delta^2 \cdot p^{-1}$
Density potential	$Cp^2\delta^2 \cdot p^{-2}$
Nonlinear coefficient of $\nabla H \cdot \nabla \mathbf{E}$	$C\delta \cdot \delta$
$(\nabla H \cdot \nabla l)\mathbf{E}$ and $(\mathbf{E}l)\nabla H \cdot \nabla$	$Cp\delta^2 \cdot p^{-1}$
Their zero-order terms	$C(p^2\delta^3 + p\delta^2)p^{-2}$
$ \nabla H ^2\mathbf{E}(\mathbf{E} + 1)$	$C\delta^2$
Its density corrections	$C(\delta^3 + \delta^4 + \delta^3/p)$
The scalar $\mathbf{E}$ coefficient involving C_q	$C\delta^2/p$
Its density potential	$Cp\delta^3 \cdot p^{-2}$

Formula (10.5) makes C_q quadratic in the fixed harmonic gradients and Euler derivatives. The inverse multipliers have convergent series in both matrix algebras. Every row is bounded by $C\delta^2$, including the possible p^2 potentials. The identities first hold on the polynomial source

core and extend by the graph bounds. Multiplying the K_D version by the angular-diagonal contraction $H_0 K_0^{-1}$ proves (18.6). $\square$

19. PHASE GEOMETRY AND MIXED CONDITIONING

Fix the compact split interval of Section 17 and an integer split a, b in it, with $a + b = 2p \in \mathbb{N}$. In this section $R \in [2p - 1, 2p + 4]$ is the continued primary Neumann zero, $0 < \tau \leq \log p$, and $|p - p_0| = O(\log p/p)$. All assertions are for sufficiently large p , with constants independent of τ in that range. Fix $1 < q_{\text{gap}} < 2$ and $0 < \xi_{\text{inv}} < 2 - q_{\text{gap}}$. Assume that the actual quartic domain has invariant gap at least

$$\gamma = \kappa_{\text{gap}} \tau p^{-q_{\text{gap}}}, \quad \kappa_{\text{gap}} > 0 \text{ fixed.}$$

The low-state structure is that of Lemma 17.1 and the finite recurrence (6.11).

19.1. One radial pole at each angular index.

Lemma 19.1 (Radial projection and complement). *For $0 \leq j \leq \lfloor 2p \rfloor$, choose the positive zero z_j of J_{ν_j} closest in frequency to R , choosing the smaller root in a tie. Let Π_{rad} retain its unit radial vector and let $Q_{\text{rad}} = I - \Pi_{\text{rad}}$. These are contractions in $\mathcal{M}$ and $\mathcal{H}$. Put $d_j^{\text{pole}} = z_j^2 - R^2$. Then*

$$(19.1) \quad \|K_0\{Q_{\text{rad}}(H_0 - R^2)Q_{\text{rad}}\}^{-1}Q_{\text{rad}}\| \leq Cp, \quad \|K_0\Pi_{\text{rad}}\| \leq Cp^2$$

in both spaces and their angular-diagonal matrix norms. On the retained space the mixed norm is exactly $\ell^1(2^j)$; the Hilbert metric is $\text{diag}(\eta_j^2)$.

Proof. The projections are radial orthogonal projections in each angular block. Lemma 11.2 proves the contraction statements. Spacing greater than π implies that every nonretained zero in a retained block is at distance at least $\pi/2$ from R . For $j > 2p$ one has $\nu_j > 5p - 1$, so the centrifugal lower bound puts its entire spectrum above $(5p - 1)^2$. The scalar spectral ratio $(z^2 + R^2)/|z^2 - R^2|$ is therefore at most Cp on the complement.

The radial sine trial on $(1/2, 1)$ in (2.17) bounds the first eigenvalue by $4\nu_j^2 + 4\pi^2$. Its first zero is thus at most $2\nu_j + 7$. If $R \leq 2\nu_j$, comparison with that first zero gives $z_j \leq \max\{2R, 2\nu_j + 7\} = O(p)$. If $R > 2\nu_j$, the differential equation for $\sqrt{x}J_{\nu_j}(x)$ has frequency square $1 - (\nu_j^2 - 1/4)/x^2 \geq 3/4$ for $x \geq R$. Comparison with a sine on an interval of length $2\pi/\sqrt{3}$ gives a zero within a bounded distance beyond R . For a direct proof, put $\kappa_{\text{cmp}} = \sqrt{3}/2$ and $w(x) = \sin(\kappa_{\text{cmp}}(x - a))$ on $[a, a + \pi/\kappa_{\text{cmp}}]$. If a solution v of $v'' + Q(x)v = 0$, $Q \geq \kappa_{\text{cmp}}^2$, had no zero there, choose its sign positive. Then $(v'w - vw')' = -(Q - \kappa_{\text{cmp}}^2)vw \leq 0$, whereas the difference of the endpoint values is $\kappa_{\text{cmp}}(v(a) + v(a + \pi/\kappa_{\text{cmp}})) > 0$. This contradiction proves the comparison. Again the closest zero has size Cp . Thus its K_0 eigenvalue is at most Cp^2 . The norm identifications follow from the radial normalization and (11.2). $\square$

Lemma 19.2 (Linear quartic reduction). *Let $H_q = z_q S_2$ be the harmonic extension of f_q , so $|z_q| \leq C\tau p^{-2}$, and put $L_q = H_0 + V_1(H_q) - R^2$. The radial complementary inverse exists and its exact pole Schur matrix is*

$$(19.2) \quad \mathcal{S}_{\text{pole}} = \text{diag}(d_j^{\text{pole}}) + \mathcal{B}_{\text{pole},2} + p^{-1}\mathcal{B}_{\text{pole},4} + \mathcal{R}_{\text{pole},6},$$

where the first two bands have bandwidths two and four and

$$(19.3) \quad \|\mathcal{B}_{\text{pole},2}\| \leq C\tau, \quad \|\mathcal{B}_{\text{pole},4}\| \leq C\tau^2, \quad \|\mathcal{R}_{\text{pole},6}\| \leq C\tau^3 p^{-2}$$

in the coefficient column norm and Hilbert row/column norms. Define the second-order interaction truncation by

$$\mathcal{S}_{\text{pole}}^{(0)} = \text{diag}(d_j^{\text{pole}}) + \mathcal{B}_{\text{pole},2} + p^{-1}\mathcal{B}_{\text{pole},4}.$$

It deletes the entire remainder, which contains all paths with at least three interactions and need not have bandwidth six. The exact matrix and this bandwidth-four truncation have Hilbert gap at least $c\gamma$.

Proof. For this degree-four harmonic chart the first variation gives

$$V_1(H_q)K_D = 2z_q M_{S_2} + 2z_q \nabla S_2 \cdot \nabla E K_D + (2p+4)z_q \nabla S_2 \cdot \nabla K_D.$$

The exact radial profiles in Lemma 9.9 factor each angular block into $c_{2,j}^i$ times a radial operator of norm at most $C|z_q|$. Lemmas 11.3 and 11.4 bound its relative graph norm by $C\tau p^{-2}$ in both matrix algebras. By (19.1), the radial complementary Neumann ratio is $C\tau/p$. Direct projection has size $C\tau$; one complementary inverse gives size $C\tau^2/p$ and bandwidth four. Each further inverse/perturbation factor contributes $C\tau/p$. Summing the convergent series proves (19.3), including all complementary radial states.

For the spectral comparison the exact unitary form is

$$\int A_q(\nabla u - u \nabla \log m) \cdot \overline{(\nabla u - u \nabla \log m)}.$$

Its linear density term integrates to $\int \Delta l_{\text{lin}} |u|^2 = 0$. The linear form has principal coefficient $(1 - 2H_q)I - y \otimes \nabla H_q - \nabla H_q \otimes y$ and is uniformly comparable to the ball form. The remaining form error is bounded by $C|z_q|^2(\|\nabla u\|_2^2 + p^2\|u\|_2^2)$. The ball Poincaré bound absorbs the second term into the first. Indexwise min-max therefore changes eigenvalues near R^2 by at most $C\tau^2 p^{-2} = o(\gamma)$, since $\tau p^{d_{\text{gap}}-2} \rightarrow 0$ uniformly here. Thus L_q has at least half the actual quartic gap. For a pole vector reconstruct the radial complementary component; the full image is exactly its Schur image and its Hilbert norm is at least that of the pole vector. This gives the Schur gap as in Proposition 13.2. Deleting $R_{\text{pole},6}$ changes that gap by $C\tau^3 p^{-2} = o(\gamma)$, proving the last assertion. $\square$

19.2. The two phase estimates. Use the corrected phase

$$(19.4) \quad \varphi_{\text{Bes}}(\nu, x) = \Xi(\nu, x) - \arctan b(\nu, x)$$

with the functions in (3.15). Chebyshev polynomials of the second kind are denoted by U_k^{Ch} , to distinguish them from the Debye polynomials U_k .

Lemma 19.3 (Two-sided phase and energy comparison). *Fix $0 < c_* < 1$. For $p-1 \leq \nu \leq c_* R$, let z be the zero of J_ν nearest to R . If $d_\varphi = \text{dist}(\varphi_{\text{Bes}}(\nu, R), \pi(\mathbb{Z} + 1/2))$, then for all sufficiently large p ,*

$$(19.5) \quad cp(d_\varphi - Cp^{-2}) \leq |z^2 - R^2| \leq Cp(d_\varphi + Cp^{-2}).$$

The constants depend only on c_ .*

Proof. Put $c_1 = \sqrt{1 - c_*^2}/2$ and choose a fixed interval $|x - R| \leq X_*$, $X_* = 2\pi/c_1$. There $\kappa \geq cp$. Lemma 3.4 at order one gives

$$\frac{J_\nu(x)}{\sqrt{2/\pi\kappa^{-1/2}}} = \sqrt{1 + b^2} \cos \varphi_{\text{Bes}}(\nu, x) + e(x), \quad |e(x)| \leq Cp^{-2}.$$

The explicit phase derivatives give $c_1 \leq \partial_x \varphi_{\text{Bes}} \leq 1$ after increasing p . At a zero the distance of its phase from $\pi(\mathbb{Z} + 1/2)$ is at most Cp^{-2} . Conversely, on either side of a lattice phase at phase distance $2Cp^{-2}$, enlarge C to dominate the error above. The two values have opposite

signs, so the intermediate value theorem supplies a zero between them. Every lattice phase whose distance from both endpoints of the phase interval is greater than $2Cp^{-2}$ has such a zero inside the argument interval. Since $\partial_x \varphi_{\text{Bes}} \geq c_1$ and $X_* = 2\pi/c_1$, the phase interval contains $[\varphi_{\text{Bes}}(\nu, R) - 2\pi, \varphi_{\text{Bes}}(\nu, R) + 2\pi]$. The lattice phase nearest to $\varphi_{\text{Bes}}(\nu, R)$ is at distance at most $\pi/2$, so it has a fixed positive endpoint margin and satisfies this condition for all sufficiently large p .

Choose the lattice phase nearest to the phase at R and a zero z' associated to it. The derivative lower bound gives $|z' - R| \leq (d_\varphi + Cp^{-2})/c_1$. Thus the nearest frequency zero z also lies in the chosen interval and obeys that upper bound. For its lower bound, its phase is within Cp^{-2} of some lattice phase, whence $d_\varphi \leq |\varphi_{\text{Bes}}(\nu, z) - \varphi_{\text{Bes}}(\nu, R)| + Cp^{-2} \leq |z - R| + Cp^{-2}$. Finally $cp \leq z + R \leq Cp$ on this interval. Multiplication proves (19.5). This proof uses the order-one finite expansion for all $x/\nu > 1$ bounded away from one, including the ratios below $3/2$ required for the high critical regions. $\square$

Lemma 19.4 (Finite-shift resonance through Lommel polynomials). *Fix $0 < \alpha_{\text{loc}} < 1$. If retained indices $j, j + d$, $1 \leq d \leq 4$, both satisfy $|d_j^{\text{pole}}|, |d_{j+d}^{\text{pole}}| \leq p^{1-\alpha_{\text{loc}}}$, then*

$$|U_{2d-1}^{\text{Ch}}(\nu_j/R)| \leq Cp^{-\alpha_{\text{loc}}}.$$

The ratio ν_j/R is within $Cp^{-\alpha_{\text{loc}}}$ of

$$(19.6) \quad \mathcal{T}_{\text{crit}} = \{1/2, 1/\sqrt{2}, \sqrt{3}/2, \cos(\pi/8)\}.$$

In particular $d = 1$ is impossible for large p .

Proof. Write $\nu = \nu_j$, $\mu = \nu + 2d$, and let z, z' be the two retained zeros. Their distances from R are at most $Cp^{-\alpha_{\text{loc}}}$. At $J_\nu(z) = 0$, one has $J_{\nu+1}(z) \neq 0$ by ODE uniqueness. Dividing the Bessel recurrence by this nonzero value gives the finite Lommel recurrence, with initial values zero and one at shifts zero and one. This identity holds for every real order. For the finitely many shifts at most nine its recurrence coefficients are $2\nu/z + O(p^{-1})$. The recurrence $U_{k+1}^{\text{Ch}}(t) = 2tU_k^{\text{Ch}}(t) - U_{k-1}^{\text{Ch}}(t)$ therefore gives, by induction,

$$\frac{J_{\nu+m}(z)}{J_{\nu+1}(z)} = U_{m-1}^{\text{Ch}}(\nu/z) + O(p^{-1}), \quad 1 \leq m \leq 9.$$

The ratios and the derivative ratio of J_μ are bounded by an absolute constant there, using the derivative recurrence. On the interval joining z, z' , the coefficients of the system for (J_μ, J'_μ) are uniformly bounded: $x \asymp p$ and $\mu < z' \leq R + Cp^{-\alpha_{\text{loc}}}$. Its integral equation and Gronwall's inequality give $\sup |J'_\mu| \leq C|J_{\nu+1}(z)|$ on that interval. Since $J_\mu(z') = 0$, integration proves $|J_\mu(z)| \leq Cp^{-\alpha_{\text{loc}}}|J_{\nu+1}(z)|$. Divide and use the preceding recurrence; replacing z by R changes its polynomial argument by $O(p^{-1-\alpha_{\text{loc}}})$. This proves the bound, including near the turning region.

The zeros of U_{2d-1}^{Ch} are the simple numbers $\cos(h\pi/(2d))$, $1 \leq h < 2d$; this follows from $U_k^{\text{Ch}}(\cos \beta) = \sin((k+1)\beta)/\sin \beta$. Here $\nu/R \geq 1/2 - o(1)$ and is at most $1 + o(1)$. The roots in this range, for $d = 1, 2, 3, 4$, are respectively none, $1/\sqrt{2}$, $1/2$, $\sqrt{3}/2$, and $1/\sqrt{2}, \cos(\pi/8)$. Simplicity and $U_{2d-1}^{\text{Ch}}(1) = 2d$ prove the remaining assertions. $\square$

19.3. Core geometry and Green propagation. Choose $\alpha_{\text{loc}} > 0$ with $q_{\text{gap}} + \xi_{\text{inv}} < 2 - 2\alpha_{\text{loc}}$. Set $W_{\text{cond}} = C_{\text{cond}}(1 + \tau)$, with a fixed sufficiently large C_{cond} , and

$$\mathcal{I}_{\text{win}} = \{j : |d_j^{\text{pole}}| \leq W_{\text{cond}}\}.$$

The angular complement of this set in the bandwidth-four matrix $\mathcal{S}_{\text{pole}}^{(0)}$ has every principal inverse bounded by $2/W_{\text{cond}}$. Indeed its band perturbation has norm $C\tau$ and division by the unperturbed diagonal has norm at most $1/W_{\text{cond}}$, also after any coordinate restriction. Let $\mathcal{F}_{\text{win}}$ be the Schur complement of $\mathcal{S}_{\text{pole}}^{(0)}$ onto $\mathcal{I}_{\text{win}}$. It is self-adjoint for $\text{diag}(\eta_j^2)$ and has Hilbert gap $c\gamma$ by Lemma 19.2 and angular complementary reconstruction. The remainder $\mathcal{R}_{\text{pole},6}$ is restored after the window inverse in the proof of Theorem 19.9.

Lemma 19.5 (Critical strips and the true minima). *For $c_{\text{cr}} \in \mathcal{T}_{\text{crit}}$ let*

$$\mathcal{N}_{c_{\text{cr}}} = \{j \in \mathbb{R} : |p + 2j - 1 - Rc_{\text{cr}}| \leq 2p^{1-\alpha_{\text{loc}}/2}\}.$$

These neighbourhoods are disjoint for large p . Write $c_{\text{cr}} = \cos(\ell_{c_{\text{cr}}}\pi/(2d_{c_{\text{cr}}}))$ with reduced pairs $(\ell_{c_{\text{cr}}}, d_{c_{\text{cr}}}) = (2, 3), (1, 2), (1, 3), (1, 4)$, respectively, and put

$$f_{c_{\text{cr}}}(j) = \varphi_{\text{Bes}}(p + 2j - 1, R) + \ell_{c_{\text{cr}}}\pi j/d_{c_{\text{cr}}}.$$

On $\mathcal{N}_{c_{\text{cr}}}$ its second derivative is between c_{curv}/p and C_{curv}/p , where $c_{\text{curv}}, C_{\text{curv}} > 0$ are fixed, and $|f'_{c_{\text{cr}}}| \leq Cp^{-\alpha_{\text{loc}}/2}$. Its true minimum is attained at

$$(19.7) \quad j_{c_{\text{cr}}}^* = \{Rc_{\text{cr}} - (p - 1)\}/2 + O(p^{-1}).$$

Connect successive window vertices in each high neighbourhood $c_{\text{cr}} > 1/2$ if their distance is at most $L_{\text{core}}(p) = K_{\text{core}} \log p / \log \log p$. Each resulting component has a common phase level and residue and diameter at most

$$(19.8) \quad L_{\text{core}}(p) + C\sqrt{W_{\text{cond}} + 1}.$$

Its indices satisfy $j \geq c_2p$ for a fixed $c_2 > 0$. The low core consists of window indices $j \leq C\sqrt{W_{\text{cond}} + 1}$, all congruent to two modulo three. Every other low-neighbourhood window vertex has $j \geq c\sqrt{p}$ and is isolated at scale $L_{\text{core}}(p)$. For the nonseed indices of the low core one also has the uniform formula

$$(19.9) \quad d_j^{\text{pole}} = -\frac{4}{3}(2j - 1)(2j - 4) + O((1 + \log p)^2/p),$$

$$j \equiv 2 \pmod{3}, \quad 5 \leq j \leq C\sqrt{W_{\text{cond}} + 1}.$$

For $j = 5, 8, 11$, these leading offsets are $-72, -240, -504$, respectively.

Proof. The derivatives in the proof of Lemma 3.6 give $f''_{c_{\text{cr}}} = 4/\kappa + O(p^{-3})$. At $j_{c_{\text{cr}}} = \{Rc_{\text{cr}} - (p - 1)\}/2$ the derivative of the leading phase cancels the added linear term, while the corrected derivative is $-2b_\nu/(1 + b^2) < 0$, of size $O(p^{-2})$. Its strictly positive second derivative therefore puts its unique minimum at distance $O(p^{-1})$, proving (19.7).

By Lemma 19.3, window vertices lie within $C(W_{\text{cond}} + 1)/p$ of a level of the lattice $\pi(\mathbb{Z} + 1/2) + \ell_{c_{\text{cr}}}\pi j/d_{c_{\text{cr}}}$. Levels are spaced by $\pi/d_{c_{\text{cr}}}$. An edge of length at most L_{core} changes $f_{c_{\text{cr}}}$ by $o(1)$, so its endpoints have the same level; reduction of $(\ell_{c_{\text{cr}}}, d_{c_{\text{cr}}})$ makes their residues equal. Every component thus lies in one strip $|f_{c_{\text{cr}}} - g| \leq C(W_{\text{cond}} + 1)/p$. If $f'' \geq m > 0$, either monotone side of $|f - g| \leq \delta$ has length at most $2\sqrt{\delta/m}$: its variation over length h is at least $mh^2/2$ and at most 2δ . A connected chain either stays on one side or has an edge of length at most L_{core} joining the two sides. This proves (19.8). High neighbourhoods have $j \asymp p$ by their definitions.

For $c_{\text{cr}} = 1/2$ the seed $j = 2$ anchors the first level, since $d_2^{\text{pole}} = -\theta/(2p) + O(\theta^2 p^{-3})$. For $j \leq c\sqrt{p}$ the phase cannot reach another level. The curvature bound then gives $j \leq C\sqrt{W_{\text{cond}} + 1}$, and the first-level residue is $j \equiv 2 \pmod{3}$. Beyond this first group, a window vertex has $j \geq c\sqrt{p}$. There $f'_{c_{\text{cr}}} \geq cj/p$; moving by an integer between one and L_{core} changes its level by more than $2C(W_{\text{cond}} + 1)/p$ and less than $\pi/3 - 2C(W_{\text{cond}} + 1)/p$. Hence such vertices

are isolated at that scale. To obtain (19.9), apply Lemma 4.9 at the associated primary crossing p_0 . For $j \equiv 2 \pmod{3}$, $j \geq 5$, its value/normal ratio is $c_j/p_0 + O((j+1)^4/p_0^2)$: the numerator has the stated first-order recurrence coefficient, and the normal denominator is its nonzero limiting sign plus $O((j+1)^2/p_0)$. The collar ODE is uniformly bounded for $j = O(\sqrt{\log p})$; its nearby zero therefore has squared offset $-4c_j + O((j+1)^4/p_0)$. Continue this zero and the Neumann zero to p . Their frequency difference stays $O(\log p/p)$ by the global order derivative bound, so spacing identifies the continued zero with the retained pole. Repeating the refined derivative comparison in the proof of Lemma 17.1 gives $|z'(t) - R'(t)| \leq C(j+1)/p + C(\log p)/p^2$. The interval has length $O(\log p/p)$; the change of squared offset is consequently at most $C(\log p)(j+1)/p + C(\log p)^2/p^2$. Since $j \leq C\sqrt{W_{\text{cond}} + 1} = O(\sqrt{\log p})$, these errors give (19.9). $\square$

Lemma 19.6 (Bin-elimination Green bound). *Let a matrix on a finite subset of integer sites be $\text{diag}(d_j) + B$, where B has bandwidth four and norm at most $b_* \geq 0$ in a weighted coordinate ℓ^1 norm or a diagonal Hilbert norm. Suppose $W > 0$, $|d_j| > W$, $b_*/W < 1/4$, and all coordinate compressions are contractions. Assume the same band bound on principal restrictions. Divide the sites into consecutive bins of width four. Every principal inverse has norm at most $2/W$. If a run of m intervening bins has all $|d_j| \geq L > 0$, with $L \geq 4(b_* + 2b_*^2/W)$, then its bin-to-bin Green propagation has bound*

$$(19.10) \quad \frac{C}{W} (2b_*/L)^m.$$

Without the stronger diagonals there is exponential decay in the number of intervening bins. Empty bins disconnect the matrix.

Proof. Diagonal division and a Neumann series bound every principal inverse by $1/(W - b_*) \leq 2/W$. The bin matrix is block tridiagonal. Eliminate bins from the right. The inverse of each right Schur block is the compression of the corresponding tail inverse, so has norm at most $2/W$. The diagonal block at a strong bin differs from the restriction of $\text{diag}(d_j)$ by at most $b_* + 2b_*^2/W$. Another diagonal Neumann series therefore bounds its inverse by $2/L$. On a homogeneous row, backward elimination gives exactly $u_b = -G_b T_{b,b-1} u_{b-1}$, with $\|T_{b,b-1}\| \leq b_*$. Iteration across the strong run gives $(2b_*/L)^m$. For a source bin, eliminate on both sides; its resulting solution operator is a compression of the full inverse and costs $2/W$. This proves (19.10). Replacing L by the ordinary Schur bound gives ratio $2b_*/W < 1/2$ on other bins. The argument is valid separately in either specified norm; no change of coordinates occurs inside the propagation estimate. $\square$

Lemma 19.7 (Valuations of low angular paths). *Uniformly for $j \leq C\sqrt{1 + \tau} + 4$, the quartic angular coefficients satisfy*

$$(19.11) \quad \begin{aligned} c_{2,j}^{j+2} &= O(1), & c_{2,j}^{j+1}, c_{2,j}^j &= O((j+1)/p), \\ c_{2,j}^{j-1}, c_{2,j}^{j-2} &= O((j+1)^2/p^2). \end{aligned}$$

Every path of quartic interactions and angular-diagonal radial resolvents carries the product of its angular coefficients. This factorization applies to the feedback band $p^{-1}\mathcal{B}_{\text{pole},4}$ and to every further complementary path. A two-interaction path starting at a low index j in the stated range has an extra factor $C(j+1)/p$ in its angular coefficient product in each of the two cases of net displacement $+3$ and -3 .

Proof. Put

$$u_j = \frac{H_{j+1}(1)}{H_j(1)}, \quad v_j = \alpha_j^2 \frac{H_{j-1}(1)}{H_j(1)}, \quad a_{\text{quad}} = \frac{2(\alpha_b + 1)}{p + 2},$$

with $\mathbf{v}_0 = 0$. The normalized recurrence is $xG_j = \mathbf{u}_j G_{j+1} + \mathbf{b}_j G_j + \mathbf{v}_j G_{j-1}$. Two applications give the exact coefficients

$$\begin{aligned} H_2(1)c_{2,j}^{j+2} &= \mathbf{u}_j \mathbf{u}_{j+1}, \\ H_2(1)c_{2,j}^{j-2} &= \mathbf{v}_j \mathbf{v}_{j-1}, \\ H_2(1)c_{2,j}^{j+1} &= \mathbf{u}_j(\mathbf{b}_j + \mathbf{b}_{j+1} - a_{\text{quad}}), \\ H_2(1)c_{2,j}^{j-1} &= \mathbf{v}_j(\mathbf{b}_j + \mathbf{b}_{j-1} - a_{\text{quad}}), \\ H_2(1)c_{2,j}^j &= \mathbf{u}_j \mathbf{v}_{j+1} + \mathbf{v}_j \mathbf{u}_{j-1} + H_2(\mathbf{b}_j). \end{aligned}$$

Terms outside the nonnegative index range are zero. The formulas (2.9)–(2.10) and (2.6) give explicitly

$$\begin{aligned} \mathbf{u}_j &= \frac{(p(1-r) + j)(p + j - 1)}{(p + 2j)(p + 2j - 1)}, \\ \mathbf{v}_j &= \frac{j(pr + j - 1)}{(p + 2j - 1)(p + 2j - 2)}, \\ \mathbf{b}_j - r &= \frac{2j(p + j - 1)(1 - 2r)}{(p + 2j)(p + 2j - 2)}, \quad a_{\text{quad}} - 2r = \frac{2(1 - 2r)}{p + 2}. \end{aligned}$$

Their denominators are comparable to the displayed powers of p uniformly for $j \leq C\sqrt{1 + \tau} + 4$. In particular $\mathbf{u}_j = O(1)$, $\mathbf{v}_j = O((j + 1)/p)$, $\mathbf{b}_j - r = O((j + 1)/p)$, and $a_{\text{quad}} - 2r = O(p^{-1})$. The endpoint ratios are bounded above and below on this range, and $H_2(1) \geq c_{\text{sp}} > 0$. Writing (6.1) in powers of $x - r$ gives $H_2(\mathbf{b}_j) = O((j + 1)^2/p^2 + p^{-1})$. Since $j = O(\sqrt{\log p})$, every displayed coefficient has exactly the bound in (19.11).

The exact block formula in the proof of Lemma 19.2 factors one interaction as $c_{2,j}^i$ times its radial operator. An angular-diagonal radial resolvent preserves this scalar factor. Induction on the number of interactions therefore gives the product on any path, followed by the product of its bounded radial norms. The Hilbert endpoint ratio η_i/η_j telescopes along the path. Two steps of lengths at most two with net displacement three must include a one-step edge. Equation (19.11) gives the asserted extra factor. Positive coefficient sums and bounded bandwidth convert these bounds to column estimates, without a factor for the number of low indices. $\square$

Lemma 19.8 (The seven off-core block types). *Take the high components of Lemma 19.5, its low core, and all remaining window vertices as singleton cores. For sufficiently large fixed K_{core} , in both the coefficient operator norm and the Hilbert operator norm,*

$$(19.12) \quad \|\text{Off}_{\text{cores}} \mathcal{F}_{\text{win}}\| \leq C\tau(1 + \tau)^2 p^{-2+2\alpha_{\text{loc}}}.$$

The low-core internal off-diagonal part has the stronger bound $C\tau(1 + \tau)^2 p^{-2}$ in both norms.

Proof. Call an outside singleton one outside all $\mathcal{N}_{c_{\text{cr}}}$, and a low singleton one in $\mathcal{N}_{1/2}$ beyond its low core. The following seven types exhaust all pairs of distinct cores; the last row takes precedence when the neighbourhoods differ.

Block type	Estimate and mechanism
Outside–outside	$C\tau(1+\tau)^2p^{-2+2\alpha_{\text{loc}}}$; two attenuated endpoints.
Outside–low singleton	Same bound; the low endpoint has the stronger attenuation $C\tau/p$.
Low singleton–low singleton	Same bound; two primary edges cannot join distinct such vertices.
Outside–high component	Faster than any chosen power, including pairs crossing a critical-neighbourhood boundary.
High component–high component in one neighbourhood	Convex phase barrier; $C\tau p^{-B_*}$ for any prescribed fixed B_* .
Low core–other core in its neighbourhood or outside	Generic Green decay across distance at least $c\sqrt{p}$.
Cores in different critical neighbourhoods	Generic Green decay across distance at least cp .

Here are the operator estimates behind the table. An outside vertex has every complementary neighbour within four sites at energy distance greater than $p^{1-\alpha_{\text{loc}}}$, by Lemma 19.4; otherwise it would lie in the smaller critical neighbourhood of radius $Cp^{1-\alpha_{\text{loc}}}$. Thus either endpoint map, with its adjacent diagonal inverse, has norm $C\tau p^{-1+\alpha_{\text{loc}}}$. At a low singleton, for $h = \pm 1, \pm 2$ the phase change to $j + h$ is $-2h\pi/3 + o(1)$, because $j/p \rightarrow 0$ in the low neighbourhood and $\partial_\nu \varphi_{\text{Bes}} = -\arccos(\nu/R) + O(p^{-2})$. Thus primary neighbours have phase separated from the zero lattice by a fixed amount and denominators at least cp . Feedback edges already have size $C\tau^2/p$, and their denominator is at least W_{cond} . They give the endpoint bound $C\tau/p$. All these maps have fixed bandwidth, so the estimates hold on whole columns and, in Hilbert coordinates, on rows as well.

No direct band-four entry connects two distinct singleton cores. Nor can two primary edges connect them: the endpoints would be two window vertices within four sites and hence in the same critical component. A length-two path between distinct singletons therefore includes a feedback edge. With one such edge put the diagonal inverse at the primary endpoint, giving $C\tau^3 p^{-2+\alpha_{\text{loc}}}$; with two use $C\tau^4/(p^2 W_{\text{cond}})$. For all paths of length at least three, the resolvent identity places a diagonal inverse at each endpoint and a bounded Neumann inverse between them. Their global bound is $C\tau^3 p^{-2+2\alpha_{\text{loc}}}$. This proves the first three rows as operator norms, with no sum over the total number of vertices.

For two distinct high components in one neighbourhood, let their nearest endpoints be $u < v$, with $D = v - u > L_{\text{core}}$, and set

$$h = \pi/d_{\text{cr}}, \quad \delta = C(W_{\text{cond}} + 1)/p.$$

Here h is the spacing of the union of the phase levels of all integer residues. If their phase levels differ, there is an intervening phase interval a fixed distance from all levels. Since $|f'_{\text{cr}}| \leq Cp^{-\alpha_{\text{loc}}/2}$, this supplies at least $cp^{\alpha_{\text{loc}}/2}$ consecutive sites with denominator cp by (19.5). If both endpoints have level g , define their excursion by $\epsilon = g - \min_{[u,v]} f_{\text{cr}}$. The convexity inequality gives $g - f_{\text{cr}}(x) \geq cD^2/p - \delta$ on $[u + D/4, v - D/4]$. If $\epsilon \geq h/4$, continuity supplies the phase interval $[g - h/4, g - h/8]$, a fixed distance from every level; use the preceding barrier on its preimage. If $\epsilon < h/4$, the middle half has matching-residue denominators at least cD^2 , and all other residues have denominators at least cp . This branch forces $D \leq C\sqrt{p}$, so, after decreasing its fixed constant, take $L = cD^2 \leq cp$. Since $D > L_{\text{core}}$ and $L_{\text{core}}^2/(1+\tau) \rightarrow \infty$, one has $L \geq 4(b_* + 2b_*^2/W_{\text{cond}})$ with $b_* = C\tau$. The middle half contains at least $D/8 - 2$ complete bins of width four. Their diagonals exceed W_{cond} for large p . Lemma 19.6 bounds propagation

by $\exp\{-cD \log(D^2/[C(1+\tau)])\}$. Uniformly for $\tau \leq \log p$,

$$\log \frac{L_{\text{core}}(p)^2}{C(1+\tau)} \geq \frac{1}{2} \log \log p$$

eventually. Choose K_{core} to give any fixed power p^{-B_*-2} . There are only $O(p)$ retained vertices; summing the already weighted entry bounds gives all coefficient columns and Hilbert rows and columns with the desired power. Each such Schur term has two exterior couplings. Together with the source factor $1/W_{\text{cond}}$ their size is at most $C\tau^2/W_{\text{cond}} \leq C\tau$, giving the detuning factor in the high-component rows of the table.

At a boundary of $\mathcal{N}_{\text{cer}}$, $|f'_{\text{cer}}|$ is comparable to $p^{-\alpha_{\text{loc}}/2}$. Two window points on opposite sides of that boundary cannot have the same level at integer distance between one and $cp^{\alpha_{\text{loc}}/2}$: their phase change exceeds $C(W_{\text{cond}} + 1)/p$. Different levels require at least that distance as well. Thus generic exponential Green decay proves the outside-high row, including these boundary pairs. Distinct neighbourhoods are separated by cp . The low core is separated from other vertices of its neighbourhood by $c\sqrt{p}$. These facts prove the last two rows.

It remains to bound the internal low-core off-diagonal part. Its vertices have residue two modulo three, so there is no direct primary edge between distinct vertices. The direct feedback of displacement three has size $C\tau^2(j+1)p^{-2}$ by Lemma 19.7. A primary endpoint lands on a nonresonant index with denominator cp , uniformly for $j \leq C\sqrt{1+\tau}$. Indeed $f_{1/2}(j) - f_{1/2}(2) = O((j+1)^2/p) = O(W_{\text{cond}}/p)$, while either of the other two residues has its zero lattice shifted by $\pi/3$ from this level. Lemma 19.3 therefore supplies that lower bound also for the neighbouring indices. A feedback endpoint of displacement one, two or four also lands on a nonresonant index; displacement three retains the extra coefficient $(j+1)/p$ even if its denominator is only W_{cond} . Two-primary-edge paths of net displacement three gain the same extra coefficient and a cp denominator. A zero-displacement step stays at its window index and is therefore absent from an endpoint map into the angular complement. Consequently each endpoint map has coefficient norm at most $C\tau/p$. A two-edge term containing a feedback edge is bounded by $C\tau^3p^{-2} + C\tau^4/(p^2W_{\text{cond}})$, placing the inverse at the primary endpoint when one is present. The resolvent identity bounds the sum of paths with at least three edges by $C\tau^3p^{-2}$, using the two endpoint bounds and a bounded middle Neumann inverse. Bounded-band column sums therefore give $C\tau(1+\tau)^2p^{-2}$ for the entire low-core off-diagonal matrix. It is self-adjoint in a positive diagonal metric. Its Hilbert operator norm equals its spectral radius, which is at most its coefficient operator norm. This supplies the Hilbert bound without any factor for the growing low-core cardinality. $\square$

Theorem 19.9 (Mixed conditioning and restoration). *Under the hypotheses of this section the linear quartic operator has a two-sided mixed graph inverse with*

$$(19.13) \quad \|K_0 L_q^{-1}\|_{\mathcal{M} \rightarrow \mathcal{M}} \leq Cp^{2+\xi_{\text{inv}}}/\gamma.$$

For a fixed order N , suppose its true centre has bounded degree, strong size $C\tau p^{-2}$, and

$$(19.14) \quad \|f_N - f_q\|_{\text{strong}} \leq \tau p^{-4} \mathfrak{E}_N(p), \quad \mathfrak{E}_N(p) = p^{o(1)}$$

uniformly in the parameters. Then its actual scalar graph is a bounded isomorphism on $\mathcal{C}$ and

$$(19.15) \quad \|J_D^{-1}\|_{\mathcal{C} \rightarrow \mathcal{C}} \leq C_N p^{5/2+\xi_{\text{inv}}}/\gamma.$$

The threshold may depend on the compact split interval, $N, q_{\text{gap}}, \xi_{\text{inv}}, \kappa_{\text{gap}}$, and the envelope in (19.14).

Proof. Delete cross-core entries of $\mathcal{F}_{\text{win}}$. Their Hilbert norm is $o(\gamma)$ by (19.12), so every core retains Hilbert gap $c\gamma$. In a high core, the exact ratio

$$(19.16) \quad \frac{\eta_{j+1}^2}{\eta_j^2} = \frac{2j+p-1}{2j+p+1} \frac{(pr+j)(j+1)}{(p(1-r)+j)(p-1+j)}$$

is bounded above and below by fixed positive constants since $j \geq c_2p$ and $j = O(p)$. In coordinates $\tilde{u}_j = 2^j u_j$, the coefficient norm is unweighted ℓ^1 and the Hilbert weights are $\eta_j/2^j$. Their consecutive ratios have the same boundedness property. The diagonal similarity over the core diameter, followed by ℓ^2 - ℓ^1 comparison over its cardinality, therefore costs at most

$$\exp\{CL_{\text{core}}(p) + C\sqrt{W_{\text{cond}} + 1}\} \sqrt{L_{\text{core}}(p) + C\sqrt{W_{\text{cond}} + 1}} + 1 = p^{o(1)}.$$

For a singleton the cost is one. On the low core, delete its internal off-diagonal part in the Hilbert norm. Each diagonal entry then has modulus at least $c\gamma$. Restoring that part by a coefficient diagonal Neumann series gives inverse C/γ , because $(1+\tau)^2 p^{q_{\text{gap}}-2} \rightarrow 0$.

The direct sum of all core inverses has coefficient norm at most $Cp^{\xi_{\text{inv}}}/\gamma$, since the displayed subpolynomial factor is at most $p^{\xi_{\text{inv}}/2}$ for sufficiently large p . Its product with the off-core error is bounded by $C(1+\tau)^2 p^{q_{\text{gap}}+\xi_{\text{inv}}-2+2\alpha_{\text{loc}}} = o(1)$. Thus $\mathcal{F}_{\text{win}}^{-1}$ has that same bound. Recovering the angular complement costs bounded factors, since τ/W_{cond} is small. Restoring $R_{\text{pole},6}$ costs $C\tau^2 p^{q_{\text{gap}}+\xi_{\text{inv}}-2} = o(1)$. Hence the exact pole Schur inverse has the same coefficient bound.

The radial complementary reconstruction has reduced right side of norm at most $(1+C\tau/p)\|f\|_{\mathcal{M}}$. The pole component costs $Cp^{\xi_{\text{inv}}}\|f\|/\gamma$; its K_0 norm costs a further Cp^2 . The complementary graph norm is bounded by $Cp(\|f\| + C\tau\|\Pi_{\text{rad}}u\|)$. This proves (19.13) on the entire graph domain, with all radial states included. Lemma 11.5 applies to the small relative perturbation $V_1(H_q)$; the bounded scalar shift leaves its graph domain unchanged. The reconstructions converge in that graph norm. The embeddings and uniqueness identify this inverse with the Hilbert inverse on mixed sources, and prove both inverse identities.

For the actual centre put $\Phi = \Phi_N$. Apply Lemma 18.3 with $H = \mathcal{H}_{\text{ext}}f_N$ and use (19.14). They give

$$\|(\hat{H}_{\Phi} - H_0 - V_1(H_q))K_0^{-1}\| \leq C_N \tau p^{-4}(\mathfrak{E}_N(p) + \tau) = \tau p^{-4+o(1)}.$$

Multiply by the mixed graph inverse before any radial coefficient conversion. Its product is $p^{q_{\text{gap}}+\xi_{\text{inv}}-2+o(1)} = o(1)$. A Neumann series restores the true centre on $\mathcal{M}$. On $\mathcal{H}$, its linear graph inverse satisfies $\|K_0 L_q^{-1}\| \leq Cp^2/\gamma$. Indeed, for $L_q u = f$ write $K_0 u = f + 2R^2 u - V_1(H_q)u$, use the Hilbert gap $\|u\| \leq C\gamma^{-1}\|f\|$, and absorb $\|V_1(H_q)K_0^{-1}\| < 1/2$. The same centre perturbation therefore has Hilbert product $p^{q_{\text{gap}}-2+o(1)} = o(1)$. The restored Hilbert and mixed inverses agree on mixed sources by the embedding and Hilbert uniqueness. The density and graph-conjugation proof of (11.21) uses only $p\delta$ small; its binomial bound is $\exp(Cp\delta) = \exp(O(\tau/p))$, hence bounded. The scalar return therefore preserves the mixed inverse exponent. The scalar estimates in the proof of Theorem 11.8 bound the sum of the $\mathcal{C}$ and $\mathcal{M}$ relative norms by $C(\delta + p\delta^2) \leq C\tau p^{-2}$. Thus (13.9) holds for large p . The finite-head/infinite-tail proof of Lemma 13.5 now costs exactly $\sqrt{p}$. Its tail is solved in $\mathcal{C}$ and its head is converted from the already constructed mixed solution. This proves coefficient membership, both inverse identities, and (19.15). All perturbations retain a factor τ ; only the final inverse has a factor τ^{-1} . $\square$

20. UNIFORM DEPENDENCE ON THE DETUNING BOUND

Lemma 20.1 (Root continuation and detuning on logarithmic windows). *Fix $C_* > 0$. For every sufficiently large primary crossing $p_0 = \hat{p}_m$, the Neumann zero continues throughout*

$|p - p_0| \leq C_{\text{ap}}/p_0$, uniformly for $0 < C_{\text{ap}} \leq C_* \log p_0$. In this range,

$$R - R_{\text{alg}} = -(2\delta_0 + O(p^{-1}))(p - p_0), \quad 2R^2 - 4(p+1)(p+2) = 4p^2(1 + O(p^{-1})),$$

and

$$(20.1) \quad \theta = (16\delta_0 + O(p^{-1}))p_0(p - p_0), \quad R = 2p + 3 + O((1 + C_{\text{ap}})/p).$$

The constants and threshold depend only on C_* .

Proof. Simplicity and (3.4) continue the zero across this interval with uniformly bounded derivative. Since $R(p_0) = 2p_0 + 3 + O(p_0^{-1})$ and $|p - p_0| = O(\log p_0/p_0)$, throughout it $p/R = 1/2 + O(p^{-1})$. The comparison function in (3.3) is smooth near $1/2$, giving $R' = 2\pi/(3\sqrt{3}) + O(p^{-1})$, whereas $R'_{\text{alg}} = 2 + O(p^{-2})$. Integrating their difference retains the factor $p - p_0$. The physical trace denominator equals $-2(p+1)R^{-3}J_{p-1}(R)\{2R^2 - 4(p+1)(p+2)\} \neq 0$, since $J_{p-1}(R) = J'_p(R) \neq 0$ by simplicity. Using $R^2_{\text{alg}} = 4(p+1)(p+2)$ in (4.13) and multiplying by $4Rp$ gives (20.1). $\square$

Fix the compact split interval, a finite centre accuracy N , and every finite derivative order and complex disc required for that centre. Constants in this section depend on these choices. Coefficient and mixed operator estimates are at integer block sizes; the intervening flow parameters use the weighted realization of Proposition 7.3. They are independent of p and of $0 < \tau \leq T \leq \log p$. We may replace T by $\max(1, T)$ when forming an envelope.

Lemma 20.2 (Detuning-dependent estimates). *Suppose $|p - p_0| \leq C_{\text{ap}}/p_0$, where $C_{\text{ap}} \leq C_* \log p_0$ with fixed C_* , and the continued zero has $0 < \tau \leq T \leq \log p$. There are finite integers k_N, d_N^{tame} and constants $c_N, C_N > 0$ such that the finite-centre estimates hold on*

$$|\varepsilon| \leq c_N(1+T)^{-k_N}, \quad \mathfrak{P}_N(T) = (1+T)^{d_N^{\text{tame}}}, \quad \mathcal{B}_N(T) = C_N \mathfrak{P}_N(T) e^{C_N \sqrt{T}}.$$

Increasing the fixed exponents and constants when necessary, every bound in the following table holds. Here $E = K_{\text{flow}}(1+\tau)$, $W = C_{\text{cond}}(1+\tau)$, and δ is the strong centre size. The references in the first column specify the statement and the quantity whose bound is being extended.

	Quantity and statement	Uniform bound / small parameter
1	Root continuation, 4.7	$R - R_{\text{alg}} = -(2\delta_0 + O(p^{-1}))(p - p_0)$.
2	Denominator in (4.13)	$2R^2 - 4(p+1)(p+2) = 4p^2(1 + O(p^{-1}))$.
3	Low-state transfer, 17.1	$CC_{\text{ap}}(\ell+1)/p + CC_{\text{ap}}(1 + C_{\text{ap}})/p^2$.
4	Radius equation, (6.9)	Analytic in (ε, w) , $w = \theta\varepsilon^2$; $ w \leq T/p^2$.
5	Finite centre jet, 6.2	Each fixed ε coefficient is polynomial in θ ; common ε radius $c_N(1+T)^{-k_N}$.
6	Collar and complex angular disc, (6.13)	Fixed discs; physical displacement $C\tau/p$.
7	Two physical defects, (6.6)	$C_N \theta^2 p^{-N} \mathfrak{P}_N(T)$, also in the weighted coefficient norms of 6.4.
8	Quartic comparison, (6.7)	Physical error $C_N \tau p^{-3} \mathfrak{P}_N(T)$; normalized error $C_N \tau p^{-4} \mathfrak{P}_N(T)$.
9	Density, (11.21)	$\ m^{\pm 1}\ \leq e^{Cp\delta}$, $\delta \leq C\tau/p^2$, so $p\delta \leq CT/p$.
10	Radial modes, 10.3	$C_{N,k} p^{k+1}$; normalization units remain separated from zero.
11	Known-mode composition, (10.14)	Head factor $e^{C_N \sqrt{T}}$; tail step factor $e^{C_N \sqrt{T}/p}$ against Bessel ratio $1/4$.

	Quantity and statement	Uniform bound / small parameter
12	Clamped centre and material, 10.4, 18.2	$\ U_0\ \leq p\mathcal{B}_N(T)$; $\ g_0\ \leq p^3\mathcal{B}_N(T)$; $\ \mathcal{Q}_{\text{mat}}^{-1}\ \leq p^2\mathcal{B}_N(T)$; $\sup\ D^2\mathcal{F}\ \leq p^3\mathcal{B}_N(T)$.
13	Relative flow form, 7.4	$C\tau p^{-2}$ and $C\tau^2 p^{-3}$; series ratio $C\tau/p^2$.
14	Flow complement, 7.6	Choose $E = K_{\text{flow}}(1 + \tau)$; ratio $C\tau/E$.
15	Schur derivative, (7.21)	$C\tau^2(E^{-1} + p^{-1})$; $\tau^3/E^2 \leq \tau^2/K_{\text{flow}}E$.
16	Seed diagonal, 7.7	$\theta/(2p) + O(\theta^2 p^{-3})$; weighted low remainder $C\tau^2(E^{-1} + p^{-1})p^{-2}$.
17	Index-five diagonal, (7.23)	$-72 + O((1 + T)/p)$; primary perturbation $O(\tau/p)$.
18	Low feedback, (7.26)	For $a_* = C\tau^2(E^{-1} + p^{-1})$, bound $Ca_*/(p\tau)$; ratio to τ tends to zero.
19	Transport, (7.19)	$C\tau^2/p$ in the weighted derivative form; ratio to the main margin $O(\tau/p)$.
20	Norm recovery, 7.9	$\ u\ \leq C(\sqrt{p} + \tau)\ Wv\ $; slope at least $c\tau/p$.
21	Phase/count, 3.5–3.8	Phase thickness $C(1 + W)/p$; count $C\sqrt{1 + W}p^{2/3}$ for $W \leq C \log p$.
22	Shell and split count, 8.2, 17.4	Shell $C\tau/p^2$; count window $C(1 + \tau)$; bad splits $C\sqrt{1 + \tau}(p^{2/3} + p^{8/3 - q_{\text{gap}}})$.
23	Radial conditioning, 19.1	Complementary graph Cp , Neumann ratio $C\tau/p$.
24	Linear spectral comparison, 19.2	Error $C\tau^2 p^{-2}$; divided by γ it is $C\tau p^{q_{\text{gap}} - 2}$.
25	Angular conditioning window	$W = C_{\text{cond}}(1 + \tau)$; every principal inverse $2/W$.
26	High core, 19.5	Diameter $L_{\text{core}} + C\sqrt{W + 1}$; condition factor $e^{C(L_{\text{core}} + \sqrt{W + 1})} \sqrt{L_{\text{core}} + C\sqrt{W + 1} + 1}$.
27	High barrier, 19.8	$\log(L_{\text{core}}^2/[C(1 + \tau)]) \geq \frac{1}{2} \log \log p$; choose K_{core} before p .
28	Low core, 19.7	$j \leq C\sqrt{1 + \tau}$; one-step coefficient $C(j + 1)/p$, downward coefficients $C(j + 1)^2/p^2$.
29	Low-core off-diagonal, 19.8	$C\tau(1 + \tau)^2 p^{-2}$ in coefficient and Hilbert norms.
30	Singleton off-core, (19.12)	$C\tau(1 + \tau)^2 p^{-2 + 2\alpha_{\text{loc}}}$.
31	True-centre graph, 18.3	$C_N \tau p^{-4}(\mathfrak{P}_N(T) + \tau)$; restore in $\mathcal{M}$ before coefficient conversion.
32	Radial conversion, 13.5	One factor $\sqrt{p}$; tail inverse uniformly bounded.
33	Convexity, 15.1	Normalized C^2 shape $C\tau p^{-2}$ plus Newton correction.
34	Non-ball witness, 15.2	Leading physical coefficient $-\theta/(16r(1 - r)p)$; centre error $C_N \tau p^{-2} \mathfrak{P}_N(T)$.

The double lift, residual, and material-defect bounds are

(20.2)

$$\|\ell_N\| \leq C_N \theta^2 p^{-N} \mathfrak{P}_N(T), \quad \|\mathcal{F}(z_0)\| \leq C_N \theta^2 p^{2-N} \mathfrak{P}_N(T), \quad \|\mathcal{V}_{\text{res}}\| \leq C_N \theta^2 p^{3-N} \mathfrak{P}_N(T).$$

The normal Hessian satisfies $\|D_p^{-1}\|_{\mathcal{E}_1} \leq C_N p^{-2} \mathcal{B}_N(T)$. The scalar fixed-centre graph difference is $C\tau p^{-2}$ and the unitary relative bound is $C(\delta + p\delta^2)$. Every $o(1)$ obtained from these envelopes is uniform in $0 < \tau \leq \log p$. The local cluster/cofactor bound is used only with fixed T ; the growing-window inverse is Theorem 19.9.

Proof. We give the dependence calculations in the order of their inputs.

Roots and low states. Lemma 20.1 proves rows 1–2 and (20.1). Row 3 is (17.1) in Lemma 17.1.

Finite jets and their remainders. The radius equation is analytic in (ε, w) with $w = \theta\varepsilon^2$. Coefficient extraction in the finite recurrence, the collar ODE, the monic products, and the update (6.20) shows inductively that every fixed ε coefficient is a polynomial in θ . All leading divisors are independent of θ : they are the normalization units, $-1/8$, c_j , and nonzero functions of r . In the physical amplitudes and displacement, every monomial of positive θ degree has positive ε order: the parameterizations are $a_2 = \theta\varepsilon\mathfrak{A}$, $a_j = \theta^2\varepsilon^3Z_j$ for $j \in \mathcal{I}_{\text{res}}$, $j \geq 5$, $\chi = \theta\varepsilon^2/(4\rho)$, and $w = \theta\varepsilon^2$, as in (6.16) and (6.10). Every term of the defect equations has positive total degree in the amplitudes and displacement, since the ball solution is exact; θ enters through w , χ and the resonant amplitudes, so the unit eliminations preserve positive ε order. For this fixed finite jet choose k_N larger than the finitely many ratios of positive θ degrees to positive ε orders. On $|\varepsilon| \leq c_N(1+T)^{-k_N}$, $|\theta| \leq 2(T+1)$, its amplitudes and displacement are uniformly small by choosing c_N . The variable w , the collar denominators $\rho + \varepsilon y$, and every finite normalization remain in fixed compact analytic sets. The same choice works on the prescribed fixed complex angular disc.

The exact defects of the evaluated finite jet vanish through the required order in ε . They have an analytic factor θ^2 by the identical field/boundary linear seed, as proved in Theorem 6.2. Divide by this factor; the maximum principle on the larger θ circle bounds the quotient uniformly on the smaller disc. The Cauchy remainder estimate in ε costs at most a fixed power $(1+T)^{d_N^{\text{tame}}}$. The same argument applied to $(h_N - h_q)/\theta$ uses its vanishing first three ε coefficients and gives row 8. For $T \leq \log p$ the condition $p^{-1}(1+T)^{k_N} \rightarrow 0$ puts the real input inside the chosen disc. Lemma 6.3 transfers the bounds to every fixed coefficient moment and radius. This proves rows 4–8.

The first term of the normalized shape has size $C\tau p^{-2}$, with a constant independent of the fixed order. Its remainder is $C_N\tau p^{-4}\mathfrak{P}_N(T)$; its ratio to the leading size tends to zero. Thus $\delta \leq C\tau p^{-2}$ in the strong norm. The scalar estimates in the proof of Theorem 11.8 are $C(\delta + p\delta^2)$, with no fixed- T compactness constant. Since $p\delta = O(T/p) \rightarrow 0$, the density series give row 9 and the stated graph bounds.

The mode-normalization ratios in Lemma 10.3 are finite recurrence functions of (ε, w) and tend to the same nonzero constants. Its Parseval head bound and the real-order tail ratio therefore hold uniformly. For composition, the argument of the square root in (10.14) is $C_N\tau p^{-2}s(2s + \beta + 1)$. It is $O_N(T)$ on the $s = O_N(p)$ head, and on the tail gives an extra geometric step $e^{C_N\sqrt{T}/p}$, absorbed by the ratio $1/4$. The finite amplitudes stay bounded on the real input. This proves rows 10–11 and the membership and moment bounds of Proposition 10.4 with envelope $\mathcal{B}_N(T)$. The exact lift columns and their Laplacians give (20.2) by precisely the two-layer calculation of Proposition 10.5. The inverse multiplier series preserve every fixed moment since $C\tau/p^2 \rightarrow 0$. The normal Hessian identity (10.22) now gives its stated inverse bound. Repeating the two norm absorptions in Proposition 18.2 yields row 12; finite products of its envelopes are included by increasing d_N^{tame}, C_N . These arguments use composition only for the known finite modes, clamped gains only on $\mathcal{G}_p$, and all-layer Poisson estimates for $\mathfrak{f}_0$.

Uniform flow. The series in Lemma 7.4 is dominated by $\sum h^6(C\tau/p^2)^h$, proving row 13 with an absolute constant for the fixed split interval. In the complementary proof choose $E = K_{\text{flow}}(1+\tau)$ independently of ς . Then $C\tau/E \leq C/K_{\text{flow}}$ and $E/p^2 \rightarrow 0$. The differentiated complementary inverse adds $C\tau^3/E^2$, bounded as in row 15. Thus rows 14–15 hold without differentiating a moving window projection. The finite recurrence and the collar root argument

give rows 16–17 with their displayed factors of θ . In the low feedback calculation

$$\frac{a_*^2}{p\tau} \leq C \left(\frac{\tau}{pK_{\text{flow}}^2} + \frac{\tau^2}{p^2K_{\text{flow}}} + \frac{\tau^3}{p^3} \right) = o(\tau).$$

The transport error is $C\tau^2/p = o(\tau)$. Choose K_{flow} to absorb $C\tau/K_{\text{flow}}$ into the positive angular margin and then take p large. Low-state transfer from row 3 gives the same 32 exclusion on indices below $\lfloor \sqrt{p}/16 \rfloor$. The norm recovery in the last paragraph of Theorem 7.9 is $\|u\| \leq C(\sqrt{p} + \tau)\|Wv\|$; $\tau \leq \log p = o(\sqrt{p})$ proves rows 18–20.

Counting. The proof of Lemma 3.5 needs only $2W/R < 1/10$ and a frequency interval shorter than π . These hold for $W \leq C \log p$. In Lemma 3.6, the Debye error is independent of W and replacing the zero by R costs at most CW/p . Its thickness is therefore $C(1+W)/p$. On each dyadic band the determinant error in Lemma 3.7 has its same fixed curvature term and additional error $O(Wp^{-2/3}) = o(1)$. Its number of collinear points is at most $C\sqrt{1+W}$. Summing the dyadic intervals as in Theorem 3.8 gives row 21; the turning strip has only $O(p^{2/3})$ indices uniformly in this range of W . The shell width $C\tau/p^2$ gives count window $C(1+\tau)$ by indexwise min–max. The grid proof of Lemma 17.4 then gives row 22 with its explicit square-root factor.

Conditioning and geometry. Rows 23–32 are the estimates proved in Section 19, which kept $\tau \leq \log p$ throughout. Row 8 supplies its true-centre hypothesis with $\mathfrak{E}_N(p) = C_N \mathfrak{P}_N(\log p)$. The high-core diameter in row 26 is linear in $\sqrt{W}$; its exponential is therefore subpolynomial. Its barrier gain is row 27. The three low-core facts used in rows 28–29 are: its indices have residue two modulo three; every primary edge lands on a nonresonant index with denominator cp ; a feedback edge that lands on another resonant index has displacement three and the extra coefficient $(j+1)/p$. These are the statements proved in Lemmas 19.5 and 19.7. The full off-diagonal estimates are operator norms as in Lemma 19.8. The factor in row 31 is multiplied by the mixed graph inverse, yielding $p^{q_{\text{gap}} + \xi_{\text{inv}} - 2 + o(1)} \rightarrow 0$. The final radial conversion costs just row 32. Rows 33–34 now follow from the fixed-order shape comparisons and the dimension-independent C^2 embedding.

Finally

$$\mathcal{B}_N(\log p) = \exp\{O_N(\log \log p + \sqrt{\log p})\} = p^{o(1)}.$$

The other envelopes in the table are fixed powers of $\log p$ or have the form

$$\exp\{O(\log p / \log \log p + \sqrt{\log p})\}.$$

The strict power margins absorb all of them. The two factors in (20.2) are θ^2 , the inverse has only τ^{-1} , and the radius and witness carry τ . No positive lower bound for τ has been used. $\square$

Corollary 20.3 (Uniform logarithmic flow and grid count). *For the inputs above, the signed slope is at least $c\tau/p$ in $|\lambda| < \tau/(16p)$. For any fixed $5/3 < q_{\text{gap}} < 2$ and $\kappa_{\text{gap}} > 0$, all but*

$$(20.3) \quad C\sqrt{1+\tau}(p^{2/3} + p^{8/3-q_{\text{gap}}})$$

integer splits $2\varepsilon_{\text{sp}}p \leq a \leq (1-2\varepsilon_{\text{sp}})p$ have actual quartic gap at least $\kappa_{\text{gap}}\tau p^{-q_{\text{gap}}}$. The constant and threshold depend on the fixed split interval, q_{gap} , κ_{gap} , and the fixed data of Lemma 20.2.

Proof. Rows 13–22 of Lemma 20.2 establish the flow and count hypotheses. Lemma 17.2 gives the integer-grid spacing at least $\varepsilon_{\text{sp}}/p$. Apply the clipped-curve argument of Lemma 17.4 with its explicitly stated number of curves. The expression in (20.3) is $o(p)$ because $q_{\text{gap}} > 5/3$. $\square$

21. THE LOGARITHMIC FAMILY AND FIXED-ORDER CLOSURE

Definition 21.1 (The logarithmic input family). With m_0 as in Definition 4.4, put

$$\mathcal{P}_{\log} = \left\{ p \in \frac{1}{2}\mathbb{Z} : p > 1, \quad \exists m \geq m_0 : |p - \hat{p}_m| \leq \frac{\log p}{8p} \right\}.$$

The root R and detuning are obtained by continuation from this crossing. For sufficiently large members the crossing is unique, and Lemma 20.1 gives $\tau \leq (2\delta_0 + o(1)) \log p < \log p$. Non-coincidence gives $\tau > 0$. Every fixed-width family $\mathcal{P}_{\text{fix}}(C_{\text{ap}})$ is contained in $\mathcal{P}_{\log}$ from some point on.

Lemma 21.2 (Multipliers giving unbounded detuning). *In each parity, $\mathcal{P}_{\log}$ has an unbounded selection with $\hat{p}_m |p - \hat{p}_m| \rightarrow \infty$ and $\tau \rightarrow \infty$.*

Proof. The congruence and shifted congruence lemmas give infinitely many pairs (P, Q) with $Q \equiv 1 \pmod{4}$, either desired parity of P , and $0 < e = Q|Q\alpha_{\text{ar}} - P| < 9/2$. For each sufficiently large Q choose the largest positive $l \equiv 1 \pmod{4}$ with

$$l^2 e \leq \frac{1}{10} \log(lQ).$$

There is such a multiplier eventually, and the set is finite since $l^2 / \log(lQ) \rightarrow \infty$ at fixed Q . Moreover $l \rightarrow \infty$: every fixed admissible multiplier eventually satisfies the inequality because $e < 9/2$. Failure at $l + 4$ gives the explicit two-sided bound

$$(21.1) \quad \left(\frac{l}{l+4} \right)^2 \frac{\log(lQ)}{10} < l^2 e \leq \frac{\log(lQ)}{10}.$$

Set $P' = lP$, $Q' = lQ$, $p' = (P' - 3)/2$ and let p'_0 be the crossing with index $(Q' - 1)/4$. Then $Q' \equiv 1 \pmod{4}$ and the parity of $2p'$ is preserved. The crossing expansion gives

$$p'_0(p' - p'_0) = -\frac{\alpha_{\text{ar}}}{4} l^2 Q(Q\alpha_{\text{ar}} - P) + d_{\text{ar}} + o(1).$$

Since $l/(l+4) \geq 1/5$, it follows that

$$\frac{\alpha_{\text{ar}}}{1000} \log Q' - C \leq p'_0 |p' - p'_0| \leq \frac{\alpha_{\text{ar}}}{40} \log Q' + C.$$

Here $p'/Q' \rightarrow \alpha_{\text{ar}}/2$ and $p'_0/p' \rightarrow 1$. As $\alpha_{\text{ar}}/40 < 1/8$, the upper bound puts these inputs in $\mathcal{P}_{\log}$ eventually; the lower bound tends to infinity. Equation (20.1) then gives $\tau \rightarrow \infty$. Successive primary crossings are separated by $2\alpha_{\text{ar}} + o(1)$, so another crossing cannot place these same inputs in a fixed-width family. $\square$

Theorem 21.3 (Newton closure at order twenty). *Fix $5/3 < q_{\text{gap}} < 2$ and $0 < \xi_{\text{inv}} < 2 - q_{\text{gap}}$, a compact split interval, and a positive gap constant. At every sufficiently large logarithmic input whose quartic domain has gap at least $\gamma = \kappa_{\text{gap}} \tau p^{-q_{\text{gap}}}$, the order-20 centre has an exact real zero within*

$$\|z_* - z_0\|_{\mathcal{X}_p} \leq \tau p^{-11}.$$

The zero gives a bounded strictly convex non-ball analytic domain, invariant under $O(a) \times O(2p - a)$, with the frequency-one Schiffer data. The lower bound $q_{\text{gap}} > 5/3$ matches the counting range of Corollary 20.3; Section 19 assumes $1 < q_{\text{gap}} < 2$ and $q_{\text{gap}} + \xi_{\text{inv}} < 2$.

Proof. Fix the parameters in the statement. Choose $\alpha_{\text{loc}} > 0$ with $q_{\text{gap}} + \xi_{\text{inv}} < 2 - 2\alpha_{\text{loc}}$, as in Theorem 19.9. Fix the angular window constant and then K_{core} to control the global Green sums. Fix the radial head cutoff. Take $N = 20$ and fix the complex angular disc and coefficient moments needed by the centre, clamping and material estimates. These choices all precede the

threshold in p . The subpolynomial conditioning factor in that proof satisfies $p^{o(1)} \leq p^{\xi_{\text{inv}}/2}$ for sufficiently large p .

Lemma 20.2 supplies the true-centre hypothesis of Theorem 19.9 and the same-space residual and material factorization. Its row 12 supplies the sharp material and nonlinear bounds uniformly in this range. They give

$$\|\mathsf{T}_{\text{Newt}}\| \leq \tau^{-1} p^{B_{\log} + o(1)}, \quad B_{\log} = \frac{9}{2} + q_{\text{gap}} + \xi_{\text{inv}} < \frac{13}{2}, \quad \sup \|D^2 \mathcal{F}\| \leq p^{3+o(1)},$$

where $\mathsf{T}_{\text{Newt}} = \mathcal{Q}_{\text{mat}}^{-1} J_D^{-1}$ is a bounded isomorphism. With $N = 20$, $d_{\text{nl}} = 3$ and $L' = 11$, the bound $B_{\log} < 13/2$ gives the three gate exponents

$$B_{\log} - 7 < -1/2, \quad B_{\log} - 17 < -21/2, \quad B_{\log} - 8 < -3/2.$$

The strict margins absorb the displayed $o(1)$ losses. On the closed ball of radius $\mathfrak{r}_p = \tau p^{-11}$, the three contraction quantities are bounded by

$$(21.2) \quad \|\mathsf{T}_{\text{Newt}} \mathcal{F}(z_0)\| / \mathfrak{r}_p \leq p^{B_{\log} - 7 + o(1)} \leq p^{-1/2 + o(1)},$$

$$(21.3) \quad \|I - \mathsf{T}_{\text{Newt}} D \mathcal{F}(z_0)\| \leq \tau p^{B_{\log} - 17 + o(1)} \leq (\log p) p^{-21/2 + o(1)},$$

$$(21.4) \quad \sup \|\mathsf{T}_{\text{Newt}} D^2 \mathcal{F}\| \mathfrak{r}_p \leq p^{B_{\log} - 8 + o(1)} \leq p^{-3/2 + o(1)}.$$

All tend to zero uniformly in $0 < \tau \leq \log p$. The source displacement is below one and the total shape norm below $1/4$, so the entire closed ball is in the working region. Increase p until each displayed contraction quantity is at most $1/4$. Taylor's integral formula bounds the displacement of $z \mapsto z - \mathsf{T}_{\text{Newt}} \mathcal{F}(z)$ by $\mathfrak{r}_p/4 + \mathfrak{r}_p/4 + \mathfrak{r}_p/8 < \mathfrak{r}_p$ and its Lipschitz constant by $1/4 + 1/4 = 1/2$. Completeness and injectivity of T_{Newt} give an exact real zero.

Its normalized C^2 shape is $O(\tau p^{-2}) + O(\tau p^{-11})$, which tends to zero. The physical seed coefficient is $-\theta/(16r(1-r)p) + O(\tau p^{-2} \mathfrak{P}_{20}(\log p))$; the correction changes it by at most $C\tau p^{-10}$. Division by τ/p makes both errors tend to zero. Thus the curvature and non-ball arguments in Section 15 apply. The completed-space identification in Proposition 15.3 supplies the classical field and its two boundary traces in the integer dimension $2p$. $\square$

Proposition 21.4 (Assembly of the logarithmic family). *For every fixed $0 < \varepsilon_{\text{sp}} < 1/4$ and $0 < \eta_{\text{cnt}} < 1/3$, every sufficiently large $p \in \mathcal{P}_{\log}$ has the domains of Theorem 21.3 at all but $O_{\text{sp}, \eta_{\text{cnt}}}(p^{2/3 + \eta_{\text{cnt}}})$ integer splits $2\varepsilon_{\text{sp}}p \leq a \leq (1 - 2\varepsilon_{\text{sp}})p$. The threshold and constant depend on the fixed $\varepsilon_{\text{sp}}, \eta_{\text{cnt}}$. Distinct retained splits give pairwise non-similar domains. The similarity-class lower limit is at least $1/2$ after division by $n = 2p$, along this family's dimensions.*

Proof. Choose

$$q_{\text{gap}} = 2 - \eta_{\text{cnt}}/2, \quad \xi_{\text{inv}} = \eta_{\text{cnt}}/4, \quad \alpha_{\text{loc}} = \eta_{\text{cnt}}/16.$$

These obey all strict inequalities in the conditioning argument. Definition 21.1 and Lemma 20.2 supply its arithmetic, low-state and uniform-constant hypotheses. Corollary 20.3 gives the selected quartic gaps and at most $C\sqrt{\log p} p^{2/3 + \eta_{\text{cnt}}/2} = O(p^{2/3 + \eta_{\text{cnt}}})$ exceptions. For every retained split apply Theorem 21.3 at the single fixed accuracy twenty. All its thresholds are uniform over the compact split interval. Lemma 17.6 proves non-similarity and the lower-limit conclusion by the same integer count and then by letting the fixed margin tend to zero. $\square$

Remark 21.5 (The logarithmic window constant). The coefficient $1/8$ in Definition 21.1 gives $\tau \leq (2\delta_0 + o(1)) \log p < \log p$ through Lemma 20.1, since $16\delta_0 < 8$. In Lemma 21.2, the inequality $\alpha_{\text{ar}}/40 < 1/8$ places the multiplier sequence in that window. The flow, conditioning and Newton estimates use the resulting bound $\tau \leq \log p$.

22. SHAPE AND SCOPE

Proposition 22.1 (Quartic shape and Hermite limit). *Fix a construction with a compact interior split interval and a fixed centre order. For its domains write the physical boundary as $\varrho = R + h(x)$ and set $v_r = r(1 - r)$. In either family, uniformly in every fixed C_x^k norm,*

$$(22.1) \quad h(x) = -\frac{\theta}{16v_r p} \mathbf{H}_2(x) + O(\tau p^{-2}), \quad \frac{p^2}{\theta} \left(\frac{\varrho(x)}{R} - 1 \right) = -\frac{(x-r)^2}{32v_r} + O(p^{-1}).$$

For y in a fixed compact interval the concentration-scale limit is

$$(22.2) \quad \frac{p^2}{\theta} h(r + \sqrt{v_r/p} y) = -\frac{\text{He}_2(y)}{16} + O(p^{-1/2}), \quad \text{He}_2(y) = y^2 - 1.$$

The constants may depend on the fixed centre order in the bounded family; the logarithmic family uses order twenty.

Proof. For fixed bounded detuning, (6.7) and the cofactor Newton radius give

$$h = -(2\chi/g_2)\mathbf{H}_2 + O(\tau p^{-3}).$$

For logarithmic detuning the same formula holds with remainder $C\tau p^{-3}\mathfrak{P}_{20}(\log p)$ by row 8 of Lemma 20.2; the physical Newton correction is $O(\tau p^{-10})$. The ellipse estimate in the proof of Lemma 9.2 is $|G_j(z)| \leq 2(j+1)\rho_{\text{ell}}^j$ for fixed $1 < \rho_{\text{ell}} < 2$. Cauchy's formula on a smaller ellipse therefore bounds each fixed x derivative by $C_k(j+1)\rho_{\text{ell}}^j$, absorbed by the coefficient weight $(j+1)2^j$. Thus these comparisons hold in every fixed C_x^k norm. Now $\chi = \theta/(4Rp)$, $R = 2p + 3 + O(\log p/p)$ and $g_2 = 4v_r/p + O(p^{-2})$. Fixed powers of $\log p$ divided by p tend to zero. These estimates prove (22.1).

The exact identity following from (6.1) is

$$\mathbf{H}_2(x) = (x-r)^2 - \frac{2(1-2r)}{p+2}(x-r) - \frac{v_r}{p+1}.$$

At $x = r + \sqrt{v_r/p} y$ it equals $v_r p^{-1}(y^2 - 1) + O(p^{-3/2})$ on compact y intervals. The sharper quartic comparison above, multiplied by p^2/θ , has error at most $C\mathfrak{P}_{20}(\log p)/p = O(p^{-1/2})$ in the logarithmic family, and $O(p^{-1})$ in the bounded family. This proves (22.2). Every error before division contains τ , so the assertions are uniform as $\tau \downarrow 0$. $\square$

Remark 22.2 (Trace differentiation on two zero curves). Fix an odd integer $m \geq 3$. Suppose two actual simple Bessel zero curves $R_N(p) = j_{p,k_N}$ and $R_D(p) = j_{p+m,k_D}$ meet at (p_0, R_0) , where $p_0 \rightarrow \infty$ and $R_0/p_0 \rightarrow \sec \beta_{\text{tr}}$, $0 < \beta_{\text{tr}} < \pi/2$. The indices k_N, k_D are fixed on each local continuation and may vary with the chosen crossing. Define along the Neumann curve

$$\chi_m(p) = \frac{J_{p+m}(R_N(p))}{J'_{p+m}(R_N(p)) - (p-1)J_{p+m}(R_N(p))/R_N(p)}.$$

Then

$$(22.3) \quad \chi'_m(p_0) = R'_N(p_0) - R'_D(p_0), \quad \frac{d}{dp}(4R_N(p)p\chi_m(p)) \Big|_{p=p_0} = \left(\frac{4m(\tan \beta_{\text{tr}} - \beta_{\text{tr}}) \cos \beta_{\text{tr}}}{\sin^3 \beta_{\text{tr}}} + o(1) \right) p_0.$$

For the primary quartic crossing this coefficient is $16\delta_0$. The detuning comparison in Lemma 4.7 follows from the exact quotient (4.13).

Proof. At a Dirichlet zero the denominator of the trace quotient equals $J'_{p+m}(R_D(p)) \neq 0$. Its partial derivative with respect to its radius argument is therefore one. Differentiating its zero value along the actual Dirichlet curve gives partial order derivative $-R'_D(p_0)$. The chain rule on R_N proves the first identity. By (3.3),

$$R'_N - R'_D = f_{\text{ord}}(p_0/R_0) - f_{\text{ord}}((p_0 + m)/R_0) + O(p_0^{-2}).$$

For $x = \cos \beta_{\text{tr}}$ one has $f'_{\text{ord}}(x) = -\cos \beta_{\text{tr}}(\tan \beta_{\text{tr}} - \beta_{\text{tr}})/\sin^3 \beta_{\text{tr}}$. Taylor expansion therefore makes the difference $\{m(\tan \beta_{\text{tr}} - \beta_{\text{tr}}) \cos^2 \beta_{\text{tr}}/\sin^3 \beta_{\text{tr}} + o(1)\}/p_0$. Since $\chi_m(p_0) = 0$, differentiating its prefactor $4R_N p$ adds zero. Multiplication by $4R_0 p_0$ proves the second identity. For $m = 3$, $\beta_{\text{tr}} = \pi/3$, it reduces to $16\delta_0$. $\square$

Proposition 22.3 (Density of the arithmetic families). *Each fixed-width candidate family $\mathcal{P}_{\text{fix}}(C_{\text{ap}})$ and the candidate family $\mathcal{P}_{\text{log}}$ has natural density zero in either parity lattice. Their dimension sets have density zero in $\mathbb{N}$.*

Proof. The crossing expansion shows that membership implies

$$\left\| \frac{c_{\text{ar}}}{\pi} p + b_{\text{ph}} \right\|_{\mathbb{R}/\mathbb{Z}} \leq C \frac{1 + \log p}{p}$$

with $b_{\text{ph}} = 3\sqrt{3}/(2\pi) - 3/4$; a fixed-width family has the stronger numerator C . The frequency $c_{\text{ar}}/\pi = \sqrt{3}/\pi - 1/3$ is irrational by transcendence of π . On either lattice $p = k$ or $k + 1/2$, each nonconstant integer Fourier mode of this rotation has a bounded geometric partial sum, so its average tends to zero. Continuous trigonometric majorants and minorants of a fixed arc [12] then give its frequency equal to its length. For any fixed $0 < \zeta < 1/2$, all sufficiently late candidate points lie in the fixed arc of radius ζ because $(1 + \log p)/p \rightarrow 0$. Their upper density is at most 2ζ . Letting ζ tend to zero proves the assertion on both lattices; combining them and rescaling by two proves the dimension statement. $\square$

The existence assertions use the quartic angular seed. Extending this construction to other angular seeds requires a centre construction, a signed-flow estimate for its wider angular interaction, and a closure in the resulting amplitude scale. The density statement above describes the listed arithmetic families; it leaves the set of all dimensions supporting Schiffer domains undetermined.

Lemma 22.4 (The spectral cap is inactive). *For every sufficiently large input in either family and every candidate split in a fixed compact interval, the actual quartic invariant Dirichlet gap is less than R^2 . Thus*

$$\min\{R^2, \text{dist}(R^2, \text{spec } H_a)\} = \text{dist}(R^2, \text{spec } H_a),$$

where H_a is the quartic Dirichlet operator in unit-ball normalization.

Proof. In the radial angular-zero sector of the ball, the sine trial on $(1/2, 1)$ gives $\lambda_1(B) \leq 4(p-1)^2 + 4\pi^2$. The quartic shell has relative width $\delta \leq C_{\text{sp}}\tau/p^2$. Min-max and dilation imply

$$0 < \lambda_1(H_a) \leq 4(p-1)^2 + 4\pi^2 + C_{\text{sp}}\tau < (2p-1)^2 \leq R^2$$

for fixed T or $\tau \leq \log p$ and sufficiently large p . The gap is at most $R^2 - \lambda_1(H_a) < R^2$, proving the claim. $\square$

Remark 22.5 (Choosing a split from strict gap enclosures). For the finite set of candidate splits, first enclose the strict conditions $\eta_{\text{alg}} < 1/2$ of Lemma C.1. They hold for all candidates at sufficiently large inputs. Lemma C.2 then supplies rational enclosures $[L_a, U_a]$ of their capped gaps with widths tending to zero. At each precision set $U = \max_a U_a$ and retain the least a

with $L_a > 0$ and $2L_a > U$, when one exists. For sufficiently large family inputs, let G be the maximum actual gap. Lemma 22.4 makes the caps inactive. Whenever $G > 0$, this selection terminates: an index attaining G has $L_a \rightarrow G$, while $U \rightarrow G$. Its output has actual gap at least $L_a > G/2$. For fixed-detuning inputs Theorem 8.3, with the fixed-parameter extension, gives $G \geq c_T \tau p^{-5/3}$. For logarithmic inputs Corollary 20.3 gives, for each fixed $5/3 < q_{\text{gap}} < 2$, $G \geq c_q \tau p^{-q_{\text{gap}}}$. Either bound supplies the positive gap and the appropriate closure. This strict-comparison rule needs no decision of equality between two spectral gaps.

APPENDIX A. NOTATION INDEX

The split parameter is ς ; the radial Jacobi index is s . Monic angular polynomials use the font $\mathbf{H}$, while H_0 is an operator. Physical radii use ϱ ; the squared reference radius is t and the nonsquared one is ζ . Constants written as C may increase within a proof. A subscript records permitted dependence. Unless that dependence is stated, constants in the bounded base chain are independent of p , the split, and $0 < |\theta| \leq 20$. Section 17 allows dependence on its fixed compact split interval and detuning bound; Lemma 20.2 gives the explicit envelopes for the logarithmic chain.

Symbol	Meaning and first definition
$n = 2p$	Ambient dimension; $p \in \frac{1}{2}\mathbb{Z}$ and $n \in \mathbb{N}$ in the final construction.
$\mathcal{P}, (p_m)$	Selected integer set and its increasing enumeration; Definition 4.3.
$\widehat{p}_m, p_0$	Real primary crossing orders for $m \geq m_0$; p_0 is associated to the current half-lattice input.
$P_j^{\text{cf}}, Q_j^{\text{cf}}$	Continued-fraction numerator and denominator; selected pairs without an index are (P, Q) .
T_j^{cf}, ρ_j	Continued-fraction tail $[a_{j+1}; a_{j+2}, \dots]$ and denominator ratio $Q_{j-1}^{\text{cf}}/Q_j^{\text{cf}}$ in (4.2).
$c_{\text{ar}}, \alpha_{\text{ar}}, d_{\text{ar}}, \delta_0$	Arithmetic and detuning constants in (4.4).
$R_{\text{alg}}(v), A_m$	Algebraic crossing radius and phase centre in (4.5).
a, b	Integer block sizes, $a + b = 2p$; the base range is $p/2 \leq a \leq 3p/4$, extended in Section 17.
α_b, β_b	Beta parameters $a/2, b/2$.
r, ς	Split ratio $a/(2p)$ and flow parameter $(r - 1/2)^2$.
X, Y	Cartesian block vectors in $\mathbb{R}^a, \mathbb{R}^b$.
x, t, ζ, ϱ	Angular coordinate $ X ^2/ Y ^2$, reference squared radius, reference radius, and physical radius.
q_X, q_Y	The invariant polynomials $ X ^2, Y ^2$.
μ_r	Normalized beta measure, (2.4).
$\mathbf{H}_j, G_j, P_j$	Monic, endpoint-normalized and orthonormal angular Jacobi polynomials; (2.5).
η_j	Angular Hilbert norm of G_j ; (2.7).
$\mathbf{a}_j, \mathbf{b}_j$	Jacobi multiplication recurrence coefficients; (2.9)–(2.10).
$\mathbf{a}_k, \mathbf{b}_k, \mathbf{e}_k$	Leading sequence, first correction and remainder of Lemma 4.9.
$Q_{\text{e}}, Q_{\text{o}}, E(A, B)$	Even and odd denominators, and the error product, in the proof of Lemma 4.1.
S_j	Solid harmonic $t^j G_j(x)$ of degree $2j$.
Λ, λ_j	Positive angular operator and eigenvalues $4j(j + p - 1)$.

Symbol	Meaning and first definition
ν_j, β_j	The common number $p - 1 + 2j$, used respectively as Bessel order and radial Jacobi parameter.
$J_\nu, Y_\nu, H_\nu^{(1)}$	Ordinary Bessel functions of the first and second kind, and their outgoing Hankel combination.
K_ν	Modified Bessel function of the second kind; its roman font distinguishes it from the graph operator K_0 .
$j_{\nu,k}, z_{j,k}$	Positive Bessel zeros and the separated ball radii $j_{p-1+2j,k}$.
$J_s^\beta, \Psi_{j,s}$	Radial Jacobi polynomial and full invariant polynomial basis, (2.18).
$A_{\text{Deb}}, \mathcal{A}_\nu$	Scalar Debye amplitude and its normalization; (3.5), Lemma 3.3.
R_ν^{Deb}	Finite Debye remainder bound; (3.8).
$\Xi(\nu, z), b(\nu, z)$	Corrected-phase ingredients in (3.15).
$\varphi_p(j)$	Lattice phase graph in (3.17).
R	Continued radial Neumann zero $J_p(R) = 0$; error $O(p^{-1})$ at fixed detuning and $O(\log p/p)$ in the logarithmic family.
χ, θ, τ	Seed trace ratio, signed detuning $4Rp\chi$, and magnitude $ \theta $.
$\mathfrak{c}_{ij}^k, c_{ij}^k$	Monic and endpoint-normalized angular product coefficients, respectively.
g_2	Monic quartic self-coupling $[\text{H}_2]\text{H}_2^2$.
h_q, f_q	Physical and normalized quartic boundary displacement; (6.5).
h_N, f_N	Physical finite-centre displacement and its normalized version h_N/R .
N, ε	Fixed accuracy order and the formal small parameter $1/p$.
$R_{\text{an}}, \rho', \rho_{\text{ell}}$	Prescribed analytic defect disc radius, stronger coefficient radius $\rho' > 2$, and boundary analyticity ellipse parameter $1 < \rho_{\text{ell}} < 2$.
ρ, Z, S_0	Scaled centre radius R/p , its square, and the polynomial in (6.9).
$\mathfrak{A}, \mathfrak{A}_v, a_j$	Formal scaled seed amplitude, its coefficients, and finite-mode field amplitudes.
$\mathcal{I}_{\text{res}}, c_j$	Fixed resonant index set $\{2, 5, 8, \dots\}$ and positive nonseed divisors in (6.12).
$T_\varsigma, \epsilon_q, \rho_\varsigma$	Quartic angular multiplier, $-\theta/(2R^2)$, and normalized trial radius $1 + \epsilon_q T_\varsigma$.
W, w_j	Flow degree-weight matrix and $(j+1)/(p+j+1)$.
D, E	Mass-unitary radial generator $\zeta \partial_\zeta + 1/2$, and ordinary Euler derivative $y \cdot \nabla$.
H_ς	Exact unitary quartic Dirichlet operator in the flow representation.
$\mathfrak{q}_\varsigma$	Quartic Dirichlet form; (7.11).
$\mathfrak{g}_\gamma, \mathcal{G}(\gamma)$	Quadratic correction form and its weak form operator; Proposition 7.3.
$c_j^{\text{diag}}, d_j^{\text{diag}}$	Rational diagonal coefficients in (7.5).
$\mathcal{N}_j, \mathcal{E}_j$	Angular variation numerators; (7.7), (7.8).
$j_{\text{pos}}, v, D_{\text{pos}}$	The shifts $j - 7$ in Lemma 7.2, $j - 8$ in Appendix B, and its positive denominator.

Symbol	Meaning and first definition
P, P_H, L	Flow-window spectral projection, its high block, and the two-state block $j = 2, 5$.
T_P^{ang}	Angular principal submatrix, without radial overlaps.
S_ς	Actual compressed transport matrix, including radial overlaps.
D_P, Z_P, D_H, Z_H	Window energy offsets and normalized boundary frequencies, and their high-block restrictions; Lemmas 7.5, 7.7.
$\mathfrak{C}_\varsigma$	Normalized flow perturbation $K_0^{-1/2}(H_\varsigma - H_0)K_0^{-1/2}$; proof of Lemma 7.6.
$\mathfrak{U}_0, \mathfrak{U}$	Unperturbed and exact normalized complementary inverses in that proof.
$\mathcal{R}_{2,\varsigma}, \mathcal{R}_\varsigma$	Quadratic form remainder and exact flow Schur remainder; Lemmas 7.4, 7.6.
F_P, F_H	Exact flow Schur matrices, at spectral parameter λ .
ω_p	Local-flow half-bandwidth $\tau/(16p)$.
ϵ_{sgn}	Sign of the eigenvalue flow, fixed to $\text{sign } \theta$ in the grid argument.
γ_p, c_{gap}	Centre Dirichlet gap $c_{\text{gap}}\tau p^{-5/3}$, with $c_{\text{gap}} = c_q/2$.
$\mathcal{C}, \mathcal{C}^{[k]}, \varpi_p$	Source coefficient space, its degree moments, and grade weight $(1 + \ell/p^2)^p$.
ρ_w	Fixed angular coefficient radius two in Section 9.
$\mathcal{G}_p, \mathcal{E}_1, \mathcal{X}_p$	Clamped source subspace, one-moment shape space, and their product.
$\mathcal{H}_{\text{ext}}$	Harmonic extension from the sphere by $G_j \mapsto S_j$.
$\mathcal{M}, \mathcal{H}$	Mixed radial Hilbert/angular coefficient space and true invariant L^2 space.
$\mathcal{H}_{\text{flow}}$	Mass-unitary representation of $\mathcal{H}$ with radial measure $d\zeta$.
σ	Exponential angular matrix-norm parameter; fixed to one in the inverse proof.
K_D, K	Dirichlet inverse of Δ , and its restriction to clamped sources.
d_s, b_0, k_s	Poisson denominators $\beta + 2s, \beta + 1$ and radial column profile; Lemma 9.5.
$\ell_{\text{out}}, Z_{\text{grad}}$	Output harmonic degree and factor $h + \ell_{\text{out}} + p - 1$; Lemma 9.9.
L_j^D, L_j^N	Unit Dirichlet and normal double-trace lift columns; (9.30).
$\mathfrak{k}, \mathfrak{h}$	General radial Poisson profile in Lemma 11.4, and $q - 1$ in Proposition 11.7.
δ	Strong centre size; (11.10).
H_0, K_0	Positive ball Dirichlet Laplacian and shift $H_0 + R^2$.
Φ_f, q, d	Scalar harmonic chart and its factors in (10.1).
$P_\Phi \Delta, B_\Phi, B_{\Phi,D}$	Scalar pullback Laplacian, its Helmholtz shift, and its restriction to the Dirichlet graph domain; (10.2) and following paragraph.
J_D	Scalar Dirichlet graph operator $B_{\Phi_N,D} K_D$ at the fixed centre.
$\varepsilon_0, \varepsilon_{\text{work}}$	Analytic chart radius 1/2 and closed working shape radius 1/4.
A_q, M_q, C_q	Pullback metric, gradient square, and contracted Hessian; (10.3).
$m, H_\Phi, \widehat{H}_\Phi, \mathcal{V}$	Jacobian square root, scalar Dirichlet pullback, its unitary realization, and $\widehat{H}_\Phi - H_0$.
D_N, N_N^{phys}	Physical boundary defects of a finite centre.

Symbol	Meaning and first definition
N_N^{ref}, ℓ_N	Reference normal defect and exact double-trace lift.
U_*, U_0, g_0, z_0	Pulled-back finite field, clamped field, its source and centre pair; (10.10).
d_j^{mode}	Value or normal mode normalization; Lemma 10.3.
$\epsilon_{\text{cmp}, p}$	Known-centre composition multiplier size in (10.14).
ϵ_{rec}	Fixed-centre inverse-series ratio in the proof of Proposition 10.5.
$\mathcal{F}, E_0, \mathcal{V}_{\text{res}}$	Nonlinear coefficient map, centre residual and material residual map $\mathcal{X}_p \rightarrow \mathcal{C}$; $\mathcal{V} = \hat{H}_\Phi - H_0$ instead acts on the Dirichlet graph domain.
D_p, f_0, T_η	Normal Hessian, fixed material field and material shape variation; (10.18).
$\mathcal{Q}_{\text{mat}}$	Exact material isomorphism, (10.19).
$\mathcal{A}, \mathcal{D}_0$	Centred unitary and ball operators $\hat{H}_\Phi - R^2, H_0 - R^2$.
$\Pi, Q, \mathcal{A}_Q$	Inverse-window projection, its complement, and exact complementary operator.
F_{eff}	Exact Schur matrix for the coefficient inverse, at spectral parameter zero.
M, D_{sep}	Cluster cardinality bound and logarithmic separation constant.
$\Pi_T, Q_T, K_{\text{head}}$	Coefficient head/tail projections and their fixed degree cutoff.
$b_{\text{inv}}, B_{\text{Newt}}, L_{\text{Newt}}$	Inverse and Newton exponents; (14.6), (14.7).
$m_{\text{mat}}, d_{\text{nl}}$	Material-inverse and nonlinear second-derivative exponents in Theorem 14.1; both equal six in the application.
$\mathbb{T}_{\text{Newt}}, \mathfrak{r}_p$	Frozen Newton inverse and contraction radius $\tau p^{-L_{\text{Newt}}}$.
C_0, C_{rel}	Absolute centre-size and relative graph constants; (11.11), (11.19).
E_{inv}, c_t	Fixed inverse-window factor, and fixed turning-spacing margin $0 < c_t < \pi/\sqrt{2}$ in (16.2).
$\mathfrak{W}_p$	Physical monic non-ball witness $R[\mathbf{H}_2]f$.
$\mathcal{P}_{\text{fix}}(C_{\text{ap}}), \mathcal{P}_{\text{bd}}$	Full half-lattice fixed-width input sets, with $\mathcal{P}_{\text{bd}} = \mathcal{P}_{\text{fix}}(31/10)$; Definition 4.4.
$\mathcal{P}_{\text{log}}$	Logarithmic input family, Definition 21.1.
$\epsilon_{\text{sp}}, I_{\text{sp}}$	Fixed interior split margin and its ς interval. The formal centre parameter remains $\epsilon = 1/p$.
$K_{\text{sp}}, W_{\text{sp}}$	Quartic multiplier and enlarged counting-window constants, Lemma 17.2.
C_{ap}, T	Fixed scaled-distance and detuning bounds; T is allowed to grow only under Lemma 20.2.
$L_{\text{ap}}, b_{\text{ap}}, M_{\text{ap}}$	Arithmetic modulus, residue and $4L_{\text{ap}}$, Theorem 17.7.
$\mathcal{R}_{\text{split}}, r_0, n_k$	Full-measure set of fixed ratios, a chosen ratio, and a continued-fraction-indexed selected dimension family; Corollary 17.8.
$q_{\text{gap}}, \xi_{\text{inv}}, \alpha_{\text{loc}}$	Gap exponent, conditioning loss and critical-neighbourhood exponent.
η_{cnt}	Positive loss in the bad-split counting exponent; the material shape variation remains η .
$\gamma, \kappa_{\text{gap}}$	Gap lower bound $\kappa_{\text{gap}} \tau p^{-q_{\text{gap}}}$ and its fixed constant.

Symbol	Meaning and first definition
$B_{\text{bd}}, L_{\text{bd}}, B_{\text{log}}$	Newton inverse/radius exponents for the extended bounded chain and inverse exponent for the logarithmic chain.
$\mathcal{N}_{\text{sim}}(n)$	Number of integer splits $0 < a < n/2$ admitting a bounded strictly convex non-ball analytic domain with the block symmetry and data (1.1); one representative per split gives that many similarity classes by Lemma 17.6.
F_l, G_l	Rational sine–cosine coefficients for half-integer Bessel functions.
$\mathbb{L}_{k,\mu}^{\text{Lom}}, U_k^{\text{Ch}}$	Lommel polynomials and second-kind Chebyshev polynomials; U_k without the superscript remains the Debye polynomial.
φ_{Bes}	Corrected Bessel phase $\Xi - \arctan b$, (19.4).
$\Pi_{\text{rad}}, Q_{\text{rad}}, z_j, d_j^{\text{pole}}$	One-pole-per-angular-block projection, its complement, retained zero, and retained squared-energy offset.
z_q, H_q, L_q	Quartic harmonic coefficient, harmonic lift $z_q S_2$, and linear quartic unitary operator minus R^2 .
$\widehat{H}_H, V_1(H), l, l_{\text{lin}}$	Fixed polynomial-chart unitary operator, its first variation, log density and its linear part; Lemma 18.3.
$\mathcal{R}_{\text{sc}}$	Fixed-centre unshifted scalar graph perturbation in Proposition 18.2.
$\mathcal{S}_{\text{pole}}, \mathcal{S}_{\text{pole}}^{(0)}$	Exact radial-pole Schur matrix and its second-order interaction truncation of bandwidth four.
$\mathcal{B}_{\text{pole},2}, \mathcal{B}_{\text{pole},4}, \mathcal{R}_{\text{pole},6}$	Pole bands and all paths of at least three interactions in (19.2); the remainder need not have bandwidth six.
$W_{\text{cond}}, \mathcal{I}_{\text{win}}, \mathcal{F}_{\text{win}}$	Conditioning angular window width, retained index set, and Schur complement of $\mathcal{S}_{\text{pole}}^{(0)}$ onto that set.
$\mathcal{T}_{\text{crit}}, c_{\text{cr}}$	Four critical phase ratios and a member of this set; c without a subscript remains a generic positive constant.
$\mathcal{N}_{c_{\text{cr}}}, f_{c_{\text{cr}}}$	Critical angular neighbourhood and phase plus its rational slope.
$\ell_{c_{\text{cr}}}, d_{c_{\text{cr}}}, j_{c_{\text{cr}}}, j_{c_{\text{cr}}}^*$	Reduced critical angle pair, uncorrected stationary index, and true corrected-phase minimum.
$h, \mathfrak{e}$	In the same-level high-core argument, the spacing $\pi/d_{c_{\text{cr}}}$ of the union of residue levels and excursion $g - \min_{[u,v]} f_{c_{\text{cr}}}$.
$c_{\text{curv}}, C_{\text{curv}}$	Fixed lower and upper phase-curvature constants in Lemma 19.5.
$K_{\text{core}}, L_{\text{core}}(p)$	Fixed separation constant and $K_{\text{core}} \log p / \log \log p$; unrelated to the normal Hessian D_p .
$\mathbf{u}_j, \mathbf{v}_j, a_{\text{quad}}$	Endpoint-normalized Jacobi recurrence coefficients and $a_{\text{quad}} = 2(\alpha_b + 1)/(p + 2)$, Lemma 19.7.
$\mathfrak{P}_N(T), \mathcal{B}_N(T), d_N^{\text{tame}}, k_N$	Polynomial and exponential-square-root envelopes, their finite exponent, and the finite-jet disc exponent.
$\mathfrak{E}_N(p)$	Subpolynomial strong-centre comparison envelope in (19.14).
v_r, He_2	Variance factor $r(1 - r)$ and monic Hermite polynomial $y^2 - 1$.
$f_{\text{ord}}, R_N, R_D, \chi_m$	Watson comparison function, actual Neumann/Dirichlet zero curves, and their trace ratio.
β_{tr}	Limiting trace-curve angle with $R_0/p_0 \rightarrow \sec \beta_{\text{tr}}$ in Remark 22.2.
$n_{\uparrow}, n_{\downarrow}$	Numbers of radial parameter-raising and lowering steps in Lemma 9.8.
γ_F	Hilbert gap in Lemma 12.3.

Symbol	Meaning and first definition
$F_{\nu,j}, C_{j,\text{rad}}$	Entire fixed-mode radial factor and its radial norm constant in Section 10.
$K_{0,P}, K_{0,Q}$	Restrictions of K_0 to the two flow spectral blocks.
$W_L, F_L, u_L, \mathcal{R}_{HL}, \mathcal{R}_{LH}$	Low flow weight, Schur block, eigenvector block, and off-diagonal remainder blocks.
O, x_{tr}	Orthogonal rigid-motion factor and translation vector in Proposition 15.5.
$H_a, \mathcal{K}_a, \mathcal{R}_{\text{alg},a}$	Quartic unitary Dirichlet operator at split a , its compact positive inverse, and $(H_a - H_0)H_0^{-1}$; Appendix C.
$\mu_{\text{alg}}, \eta_{\text{alg}}$	Absolute quartic coefficient $ z_q $ and computable relative bound $4000p\mu_{\text{alg}}$.
$\Pi_{J,K_{\text{rad}}}^{\text{alg}}, \Lambda_{\text{miss}}$	Rectangular Bessel–Jacobi projection and lower bound for omitted ball eigenvalues.
L_a, U_a, f_R	Rational capped-gap enclosures and the Lipschitz function transferring the compact inverse spectrum to those gaps.
ϵ_{cut}	Endpoint cutoff for the positive-weight integrations in Lemma C.2.

The spectral windows.

Name	Width about R^2	Purpose
W_{flow}	fixed	flow complement and derivative estimates
W_{count}	fixed	count all curves that may enter the flow band
W_{loc}	arbitrary fixed	local lattice count and clusters
$E_{\text{inv}}\tau$	proportional to τ	coefficient inverse Schur matrix
E	$K_{\text{flow}}(1 + \tau)$	uniform logarithmic flow
W_{cond}	$C_{\text{cond}}(1 + \tau)$	mixed-conditioning cores

In the bounded base inverse $W_{\text{loc}} = 20E_{\text{inv}}$; Section 17 uses $W_{\text{loc}} = TE_{\text{inv}}$ for its fixed detuning bound T . The flow and curve-count constants are fixed independently of E_{inv} . The small flow band $\omega_p = \tau/(16p)$ and the resulting gap γ_p shrink with the detuning.

Dependence of constants. Each row below records the uniformity convention used in its statements. A listed threshold may increase to accommodate a finite number of earlier thresholds. Unlisted free parameters are quantified uniformly over the ranges in the corresponding statement. These base-chain rows are uniform for $0 < \tau \leq 20$.

Estimate	Constants may depend on	Threshold may depend on
Uniform Debye bounds	explicit ν, z, L as displayed	none beyond the stated inequalities
Phase strip and global count	fixed window W	W
Logarithmic local count	W_{loc} ; the integer M_W is independent of D	W_{loc}, D
Cluster cardinality	M_W only	$W_{\text{loc}}, M_W D$
Finite centre defects	fixed N , complex neighbourhood U , requested fixed derivative order	the same fixed choices

Estimate	Constants may depend on	Threshold may depend on
Coefficient defect transfer	N, k, ρ'	N, k, ρ' ; uniform in θ and the split
Quartic angular matrices	derivative order $k \leq 2$ and the fixed interval I	$p \geq 64$
Band form estimate	explicit bandwidth factor $(m+1)^3$	none depending on the radial index
Flow complement remainder	absolute constants; independent of W_{flow}	fixed W_{flow} , with $p^2 \geq W_{\text{flow}}$
Local flow margin	absolute constants fixed before p	W_{flow} and low-degree exclusion
Curve count	C_{40} for $W_{\text{count}} = 40$	the threshold of the fixed-window count
Integer-split gap	c_{flow}, C_{40} , both absolute	their fixed values
Centre gap transfer	N, c_q	N, c_q
Source module and clamped estimates	fixed angular radius two; displayed shape degrees only	$p \geq 64$
Higher smooth chart norms	fixed derivative order and fixed dimension	fixed-domain estimates
Centre source moments	N, k	N, k
Material inverse and $D^2\mathcal{F}$	N ; exponents six independent of N	N and the fixed working radius $1/4$
Strong centre-size bound	absolute C_0 , independent of N	N absorbs its finite-degree remainder
Relative graph bounds	C_0 , fixed moment ten and fixed exponential weights	N , plus absolute smallness conditions
Complementary inverse	fixed E_{inv} and relative constant	those constants and N
Cofactor inverse	$M, c_{\text{gap}}, E_{\text{inv}}, D_{\text{sep}}$	the same fixed constants
Head/tail conversion	fixed K_{head}	the relative bound threshold
Full coefficient inverse	preceding absolute constants	N and the fixed preceding constants
Newton ball	fixed $N = 4M + 40$ and preceding constants	their finite maximum
Convexity and non-ball witness	absolute shape embeddings and fixed-order centre constants	N and the Newton thresholds
Classical regularity after the zero	fixed resulting domain and dimension	fixed-domain estimates

The auxiliary real-parameter extensions used to construct centres and vary the split are separate from the Euclidean conclusion. Every final domain is formed at $p \in \frac{1}{2}\mathbb{Z}$ and integer block sizes $a, b = 2p - a$. The accuracy N is fixed before those sufficiently large inputs are considered and remains fixed as p grows.

APPENDIX B. A POLYNOMIAL POSITIVITY IDENTITY

The rational functions $\mathcal{N}_j, \mathcal{E}_j$ in the angular variation lemma obey

$$\frac{\mathcal{N}_j}{2} - \frac{2}{3} \sum_{h=-1}^2 \mathcal{E}_{j+h} = \frac{4p^2}{D_{\text{pos}}(p, j)} \sum_{k=0}^{12} v^k Q_k(u), \quad u = p - 64, \quad v = j - 8.$$

Here the positive denominator is

$$\begin{aligned} D_{\text{pos}}(p, j) = & 3(2j + p)^2(p + 1)(p + 2)^2(p + 4)(p + 5) \\ & \cdot (2j + p - 5)(2j + p - 4)^2(2j + p - 3)(2j + p - 2)^2 \\ & \cdot (2j + p + 1)(2j + p + 2)^2(2j + p + 3). \end{aligned}$$

The following lists give the coefficients of $Q_k(u)$ in increasing powers of u , starting with its constant coefficient. Their lengths for $k = 0, \dots, 12$ are 13, 13, 13, 12, 11, 10, 9, 8, 7, 6, 5, 4, 3, a total of 114 coefficients. Higher powers have coefficient zero. Multiplying the closed forms (7.7) and (7.8) by $D_{\text{pos}}(p, j)$ and substituting $p = 64 + u$, $j = 8 + v$ gives the identity. Every listed coefficient is positive, so the numerator is strictly positive for $u, v \geq 0$.

$k = 0$: 7753073057355915455846400, 1215360332373242887229952, 87272032201419163433472, 3795893130561640498368, 111379277105283072768, 2322600629405384192, 35294561281256064, 393801439328684, 3201816890896, 18499863104, 72102712, 170196, 184.

$k = 1$: 3180378375750299550050304, 473937198500360715148800, 32259215635963195792896, 1325755805929596036096, 36623628139750000400, 716093978925466680, 10156021734940612, 105193508825770, 789036501246, 4175089100, 14771472, 31314, 30.

$k = 2$: 562664300050453696883712, 78738823661925490820352, 5007990979423659644544, 191197382688847752160, 4872666916298666520, 87154787492568924, 1118925565579542, 10351501782387, 68133328557, 308708398, 902196, 1491, 1.

$k = 3$: 57679238717080616484864, 7495244567904683655936, 439582532126405531840, 15344545497976190928, 353819688334375200, 5650365815615444, 63648016435252, 504433412122, 2746860336, 9738624, 20076, 18.

$k = 4$: 3852565146000679166976, 459625726775754912896, 24519727131647080608, 769653952044921056, 15727048908327064, 218338672283856, 2082487959758, 13447668765, 56116245, 136135, 145.

$k = 5$: 177747049322645907456, 19228553890224334656, 918983144616089664, 25449989920330928, 449695578923952, 5252148923508, 40491107304, 198350292, 558912, 688.

$k = 6$: 5833725019983771648, 564050096670576064, 23716561720515776, 566030886527952, 8379892988992, 78720883076, 457623948, 1502480, 2128.

$k = 7$: 137658610846236672, 11688344788927104, 422564419955328, 8425693173440, 99978826496, 705144000, 2732800, 4480.

$k = 8$: 2323153433438208, 169347834087168, 5104046345952, 81336054512, 721895088, 3377952, 6496.

$k = 9$: 27391092756480, 1662337753920, 39942732160, 474290240, 2777600, 6400.

$k = 10$: 214440898560, 10360755904, 184765248, 1437184, 4096.

$k = 11$: 1001816064, 35857920, 414720, 1536.

$k = 12$: 2113536, 48896, 256.

APPENDIX C. COMPUTABLE SPECTRAL GAP ENCLOSURES

This appendix supplies the optional selection rule of Remark 22.5 and is not part of the proof chain of Theorems A and B.

Fix an input p, R, θ and the finite set of integer splits in a compact interval of Section 17. Represent each quartic Dirichlet operator H_a on $\mathcal{H}$ by the harmonic chart

$$q = 1 + z_q S_2, \quad d = 1 + 5z_q S_2, \quad z_q = -\frac{\theta H_2(1)}{2R^2 p g_2}.$$

All parameters of this chart are computable from an isolated Neumann zero R and the rational block parameters. The following bound supplies a strict starting condition for the spectral enclosure.

Lemma C.1 (A computable quartic relative bound). *Assume integer block sizes with $a+b = 2p$, $\beta_b \geq \alpha_b \geq 16$, $p \geq 64$, and $\mu_{\text{alg}} = |z_q| \leq 1/(100p)$. Then the positive unitary quartic operator satisfies*

$$(C.1) \quad \|(H_a - H_0)H_0^{-1}\|_{\mathcal{H} \rightarrow \mathcal{H}} \leq \eta_{\text{alg}}, \quad \eta_{\text{alg}} = 4000p\mu_{\text{alg}}.$$

For each fixed compact split interval, the condition $\eta_{\text{alg}} < 1/2$ holds at every sufficiently large input of either theorem, uniformly over its candidate splits.

Proof. Write $\mu = \mu_{\text{alg}}$ within this proof. The endpoint proof in Lemma 2.3 uses $\beta_b - 1 \geq \alpha_b - 1 \geq 0$ and therefore applies to these parameters. Homogeneity and its derivative bounds, at degree four, give

$$\|S_2\|_{\infty} \leq 1, \quad \|\nabla S_2\|_{\infty} \leq 4\sqrt{2} < 6, \quad \|D^2 S_2\|_{\text{op}, \infty} \leq 32.$$

For the last bound, at a unit point the radial-radial entry is $12G_2$, the radial-tangential entries are $3\nabla_S G_2$, and the tangential block is $\nabla_S^2 G_2 + 4G_2 g_S$. The diagonal blocks have norm at most twenty and the off-diagonal block at most twelve. Homogeneity gives the same bound inside the ball.

Thus $q, d > 0$, $q^{-1}, d^{-1} \leq 2$, and

$$\|D\Phi^{-1}\|_{\text{op}, \infty} \leq 3, \quad \|A_q\|_{\text{op}, \infty} \leq 9, \quad \|A_q - I\|_{\text{op}, \infty} \leq 200\mu.$$

Indeed the inverse is $q^{-1}(I - y \otimes \nabla q/d)$. In (10.3), the last bound follows from

$$|q^{-2} - 1| \leq 8\mu, \quad q^{-2} (2|\nabla q|/d + |\nabla q|^2/d^2) \leq 96\mu + 576\mu^2.$$

The drift vector in the scalar coordinate formula is $-D\Phi^{-1}(2A_q \nabla q + y(A_q : D^2 q))$. Its norm is at most $3(108 + 288n)\mu \leq 2000p\mu$, using $n = 2p$. For $u = H_0^{-1}f$, the ball estimates give $\|D^2 u\|_2 \leq \|f\|_2$ and $\|\nabla u\|_2 \leq (p-1)^{-1}\|f\|_2$. The Frobenius norm of a symmetric n -matrix is at most $\sqrt{n}$ times its operator norm. Consequently, for $H_{\Phi} = -P_{\Phi}\Delta$,

$$(C.2) \quad \|(H_{\Phi} - H_0)H_0^{-1}\| \leq 200\sqrt{2p}\mu + \frac{2000p\mu}{p-1} \leq 500p\mu.$$

Put $m = q^{p-1/2}d^{1/2}$ and $l = \log m$. Since $p\mu \leq 1/100$, the inequalities $\log(1+t) \leq t$ and $-\log(1-t) \leq 2t$ on $0 \leq t \leq 1/2$ give $\|m\|_{\infty}, \|m^{-1}\|_{\infty} \leq 2$. Harmonicity of q, d gives

$$\|\nabla l\|_{\infty} \leq 13p\mu, \quad \|\Delta l\|_{\infty} \leq 200p\mu^2.$$

For example the gradient bound follows from $2(p-1/2)6\mu + 5 \cdot 6\mu \leq 13p\mu$; the Laplacian bound follows from $4p(6\mu)^2 + 2(30\mu)^2 \leq 200p\mu^2$. The identity $\Delta m^{-1} = m^{-1}(|\nabla l|^2 - \Delta l)$ therefore yields

$$\|[H_0, m^{-1}]H_0^{-1}\| \leq 2 \left\{ \frac{26p\mu}{p-1} + \frac{169p^2\mu^2 + 200p\mu^2}{(p-1)^2} \right\} \leq 60\mu.$$

Here $p \geq 64$ and $\mu \leq 1/(100p)$ bound the braces by 30μ . It follows that $\|H_0 m^{-1} H_0^{-1}\| \leq 3$. Finally use $H_a = m H_\Phi m^{-1}$ and the exact factorization

$$(H_a - H_0)H_0^{-1} = m(H_\Phi - H_0)H_0^{-1}(H_0 m^{-1} H_0^{-1}) \\ + m[H_0, m^{-1}]H_0^{-1}.$$

Its norm is at most $2(1500p\mu + 60\mu) \leq 4000p\mu$. All estimates hold on the Dirichlet graph core and extend by its density. In the families under consideration, $\mu_{\text{alg}} \leq C_{\text{sp}}\tau p^{-2}$, hence $\eta_{\text{alg}} \leq C_{\text{sp}}\tau/p \rightarrow 0$ uniformly for fixed detuning bounds and for $\tau \leq \log p$. $\square$

Lemma C.2 (Convergent gap enclosures). *For a computable input satisfying the hypotheses of Lemma C.1 and $\eta_{\text{alg}} < 1/2$, the capped gap*

$$g_a = \min\{R^2, \text{dist}(R^2, \text{spec } H_a)\}$$

has rational lower and upper enclosures of arbitrarily small prescribed width.

Proof. Set $\mathcal{R}_{\text{alg},a} = (H_a - H_0)H_0^{-1}$ and $\mathcal{K}_a = H_a^{-1}$. The strict relative bound gives

$$(C.3) \quad \mathcal{K}_a = H_0^{-1} \sum_{v=0}^{\infty} (-\mathcal{R}_{\text{alg},a})^v, \quad \left\| H_0^{-1} \sum_{v>L} (-\mathcal{R}_{\text{alg},a})^v \right\| \leq \frac{\eta_{\text{alg}}^{L+1}}{(p-1)^2(1-\eta_{\text{alg}})}.$$

The operator $\mathcal{K}_a$ is compact, positive and self-adjoint. Use the orthonormal Bessel–Jacobi basis of H_0 and the rectangular projection $\Pi_{J,K_{\text{rad}}}^{\text{alg}}$ onto $0 \leq j \leq J$, $1 \leq k \leq K_{\text{rad}}$. The radial min–max principle gives

$$z_{j,k}^2 \geq \nu_j^2 - 1/4 + \pi^2 k^2.$$

Thus one may take the omitted-eigenvalue lower bound

$$\Lambda_{\text{miss}} = \min\{\nu_{J+1}^2 - 1/4 + \pi^2, \nu_0^2 - 1/4 + \pi^2(K_{\text{rad}} + 1)^2\}.$$

It tends to infinity as $J, K_{\text{rad}} \rightarrow \infty$. For this projection, abbreviated Π^{alg} , commutation with H_0^{-1} and self-adjointness give

$$(C.4) \quad \|\mathcal{K}_a - \Pi^{\text{alg}} \mathcal{K}_a \Pi^{\text{alg}}\| \leq \frac{2}{(1-\eta_{\text{alg}})\Lambda_{\text{miss}}}.$$

Indeed $(I - \Pi^{\text{alg}})\mathcal{K}_a$ is bounded by the right side without its factor two, using $\mathcal{K}_a = H_0^{-1}(I + \mathcal{R}_{\text{alg},a})^{-1}$. Its adjoint has the same bound, and their sum gives (C.4).

We describe the finite matrix computations in these two truncations. The basis values and derivatives are computable from the convergent Bessel series, finite Jacobi polynomials and their normalizations. The positive roots are simple and separated by more than π . The sine comparison in Lemma 19.1, applied on disjoint length-four intervals beyond 2ν , gives the coarse bound $j_{\nu,k} \leq 2\nu + 8k$. Since the first root exceeds $\nu > 1$, a rational mesh starting at one, with gaps below π , finds every root up to a prescribed index by sign changes. Mesh points can be chosen with certified nonzero values: of two candidate points less than π apart, at least one is not a zero, and successive series enclosures detect that value. Choosing them in small disjoint neighbourhoods of an initial mesh preserves its order and gap bound. The same procedure refines each isolated root interval. Simplicity then gives a nonzero derivative bound after finitely many refinements, supplying its normalization.

For a finite basis sum v , the function $\mathcal{R}_{\text{alg},a}v$ is an explicit smooth invariant function. Its coefficients and its squared norm are integrals in the radial and angular variables with positive weights. Every chart denominator is bounded away from zero by $q \geq 1 - \mu_{\text{alg}}$, $d \geq 1 - 5\mu_{\text{alg}}$. The regular Bessel series times solid harmonics gives computable uniform bounds for the function and its required derivatives on the closed ball: each derivative series has factorial

tails and only finitely many input modes occur. After the coordinate substitution, use an endpoint cutoff $0 < \epsilon_{\text{cut}} < 1/2$. The normalized radial and angular densities are $2p\zeta^{2p-1}$ and $x^{\alpha_b-1}(1-x)^{\beta_b-1}/B(\alpha_b, \beta_b)$. Their omitted endpoint measures are bounded, respectively, by

$$\frac{2p\epsilon_{\text{cut}}^{2p}}{2p}, \quad \frac{\epsilon_{\text{cut}}^{\alpha_b}}{\alpha_b B(\alpha_b, \beta_b)}, \quad \frac{\epsilon_{\text{cut}}^{\beta_b}}{\beta_b B(\alpha_b, \beta_b)}$$

at the origin and the two angular endpoints. Multiplying by the computed supremum bounds controls those tails. On the remaining compact rectangle, derivative bounds give a computable modulus of continuity and rational interval Riemann sums converge from above and below. Here $B(\alpha_b, \beta_b) = \Gamma(\alpha_b)\Gamma(\beta_b)/\Gamma(p)$. For these integer or half-integer parameters the Gamma formulas give convergent rational enclosures; a positive lower enclosure of $B(\alpha_b, \beta_b)$ gives the required upper bound for its reciprocal. Equivalently one can obtain this lower enclosure from its positive integral. Thus the tail constants $2p$ and $1/B(\alpha_b, \beta_b)$ have computable rational upper bounds. This constructs convergent rational enclosures of every required integral.

To approximate the image of a finite vector, use Parseval in this complete basis. For a larger finite projection Π' , its squared residual is

$$\|\mathcal{R}_{\text{alg},a}v\|^2 - \sum_{i \in \Pi'} |\langle \mathcal{R}_{\text{alg},a}v, e_i \rangle|^2.$$

Its upper enclosure eventually falls below any positive target by completeness, increasing the projection and integration precision. This supplies a terminating precision search for each application of $\mathcal{R}_{\text{alg},a}$. An input error is multiplied by at most η_{alg} and added to the next approximation error. There are only finitely many applications in the truncated series (C.3). Taking errors sufficiently small for its finitely many input columns gives a rational approximation to the finite compression in operator norm. A column error at most the requested tolerance divided by the number of columns suffices, by Cauchy–Schwarz. Symmetrization preserves that error bound.

The spectrum of a finite rational symmetric matrix can be enclosed by rational arithmetic. One construction enumerates rational orthogonal matrices formed from coordinate reflections and plane rotations with entries $(1-t^2)/(1+t^2)$ and $2t/(1+t^2)$, $t \in \mathbb{Q}$. This family is dense in the orthogonal group. Continue until the off-diagonal Frobenius norm of the conjugated matrix is below a prescribed tolerance. The spectral theorem guarantees termination, and the diagonal entries then enclose its spectrum with that error. For self-adjoint operators an operator-norm error bounds their spectral Hausdorff distance: outside that neighbourhood the resolvent identity and a Neumann series give invertibility, in both directions. Combining this fact with (C.3)–(C.4) gives arbitrary norm-accurate finite spectral enclosures of $\mathcal{K}_a$, with zero adjoined for the infinite complement.

Finally put

$$f_R(t) = \min\{R^2, |t^{-1} - R^2|\} \quad (t > 0), \quad f_R(t) = R^2 \quad (t \leq 0).$$

It is continuous and Lipschitz with constant $4R^4$: its nonconstant part has $t \geq 1/(2R^2)$ and derivative of absolute value at most $1/t^2$. Also it is Lipschitz with constant one as a function of R^2 , so the input radius can be enclosed at the same time. Spectral inversion gives $g_a = \min f_R(\text{spec } \mathcal{K}_a)$. The preceding finite spectral enclosures therefore give rational bounds on g_a with error tending to zero. Clipping the lower bound at zero preserves the enclosure. This proves the assertion. $\square$

USE OF AI

AI systems, including GPT-6 Astra, GPT-6.1 Sol and Claude Opus 5.5, were used in developing the proofs and preparing the manuscript. The author takes responsibility for the content.

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